

Steiner Waldorf Pedagogy in Schools

A Critical Introduction

Martyn Rawson

ROUTLEDGE

Steiner Waldorf Pedagogy in Schools

This book offers a comprehensive exploration of Steiner or Waldorf pedagogy and practice in schools. Drawing on key research, it traces the origins of Steiner education from the original Waldorf school and shows how this approach has since been adapted and applied in educational settings around the world.

Outlining the educational philosophy of Steiner education, the book considers its unique features, such as its commitment to a pedagogical anthropology that takes the whole developing human being into account – body, mind and spirit – and the developmental approach that arises out of this. It sets out the specific curriculum and teaching approach alongside vignettes of teaching and learning situations adopted in Steiner educational settings to show how the approach works in practice. Offering a critical perspective on this teaching style, Rawson examines the contributions that Steiner education has made in different cultures and looks towards future developments in China and other Asian countries.

Considering all aspects of Steiner education, this book is essential reading for anyone wanting to understand the fundamental elements of this approach and its continuing relevance within the educational landscape.

Martyn Rawson currently works on the Waldorf Master Programme at the Waldorflehrerseminar Kiel and at the Freie Hochschule Stuttgart on the International Master Programme in Germany. He is currently Honorary Professor at the National Tsing Hua University, Taiwan and is an alumni research fellow at Plymouth University. His research focusses on curriculum development in Steiner/Waldorf education, teacher education and learning as well as learning and development in students of all ages.



Taylor & Francis
Taylor & Francis Group
http://taylorandfrancis.com

Steiner Waldorf Pedagogy in Schools

A Critical Introduction

Martyn Rawson

Routledge
Taylor & Francis Group
LONDON AND NEW YORK

First edition published 2021
by Routledge
2 Park Square, Milton Park, Abingdon, Oxon, OX14 4RN

and by Routledge
52 Vanderbilt Avenue, New York, NY 10017

Routledge is an imprint of the Taylor & Francis Group, an informa business

© 2021 Martyn Rawson

The right of Martyn Rawson to be identified as author of this work has been asserted by him in accordance with sections 77 and 78 of the Copyright, Designs and Patents Act 1988.

All rights reserved. No part of this book may be reprinted or reproduced or utilised in any form or by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying and recording, or in any information storage or retrieval system, without permission in writing from the publishers.

Trademark notice: Product or corporate names may be trademarks or registered trademarks, and are used only for identification and explanation without intent to infringe.

British Library Cataloguing-in-Publication Data
A catalogue record for this book is available from the British Library

Library of Congress Cataloging-in-Publication Data
A catalog record for this book has been requested

ISBN: 978-0-367-33397-3 (hbk)
ISBN: 978-0-367-33398-0 (pbk)
ISBN: 978-0-429-31967-9 (ebk)

Typeset in Sabon
by Apex CoVantage, LLC

To my family, friends and all the pupils and students who have taught me over the years.



Taylor & Francis
Taylor & Francis Group
http://taylorandfrancis.com

Contents

Acknowledgements x
The author xii

Introduction 1
Birthdays 1
Good practice – dubious ideology? 3
The aims of this book 4
The author's position 6
Steiner and/or Waldorf? 7
References 7

1 Rudolf Steiner and the origins of his educational ideas 10
Rudolf Steiner: "Stranger in a strange land" 10
Biography and biographical mythos 11
Key stages in a varied career 15
What motivated Steiner? 16
Grand narrative or grand narratives? 17
Key ideas 18
An epistemological and philosophical basis for Waldorf education 18
Knowledge as a productive activity 20
Who thinks? 26
Working with a spiritual perspective 29
Key ideas 32
Steiner's experiences as a teacher and tutor 32
Education as part of the cultural domain, free from political or economic determination 35
The Education of the Child, *1907 36*
Steiner's spiritual anthropology 39
Key ideas 44
References 45

2 Generative principles of teaching and learning in Waldorf education 50
Generative principles and how to work with them 50
Five vignettes of Waldorf practice 51
The lower school main lesson 51
Optics in grade 7 55
A craft project involving grades 5 and 10 57
Art history, grade 9 59
A foreign language lesson in grade 12 62
Generative principles for teaching and learning 64
Principle 1: Waldorf education takes the spiritual dimension seriously 64
Principle 2: Sense of coherence is the basis for healthy learning and well-being 72
Principle 3: The quality of the teacher's preparation influences the quality of learning 75
Principle 4: Children and young people need to be ready to learn, and they need time to learn 79
Principle 5: Learning is a rhythmical process 83
Principle 6: The learning processes are structured over time 94
Principle 7: Block teaching strengthens learning dispositions 98
Principle 8: Direct experience is the basis for learning 102
Principle 9: Activating the imagination through vivid pictorial descriptions and images is another powerful starting point for learning 103
Principle 10: A phenomenological approach enables the organic growth of knowledge 105
Principle 11: The teaching must be artistic 116
Principle 12: The self-activity of the students is essential to learning 119
Principle 13: Good teaching and learning depends on the development of the senses 122
References 126

3 Communities of learning: Generative principles 132
Principle 14: The Waldorf class is a learning community 132
Principle 15: Teachers support their pupils' learning and development by generating active knowledge using assessment for learning 135
Values 137
Self-assessment 137
Individual annual reports 138
Lesson planning and reviewing as assessment for learning 138

Principle 16: The teachers are a collegial learning community with responsibility for the educational leadership of the school 138
Principle 17: The curriculum maps out possible learning situations and learning pathways in relation to the developmental tasks 142
Principle 18: Becoming a Waldorf teacher is a process of transformative learning 152
References 154

4 Waldorf education and the academy: positions, research and outlook 158
Introduction 158
Reception problems 159
Hermeneutics of suspicion or empathy? 161
Critique as polemic 162
Some key perspectives on Waldorf education 164
Alumni research 171
Studies of pupils in school 173
References 175

PART 5
Waldorf schools around the world 182

The literature 182
Overview of the phases 182
After the Wall came down 185
Waldorf-inspired ideas in the world 187
Camphill schools and curative education 187
Emergency education 188
References 190

Conclusions 192
References 196

Subject Index 197
Index of Persons 200

Acknowledgements

Many thanks to Ulrike Sievers for the support, corrections and always helpful feedback. I am also grateful to Jan Swann and Christiane Niemeyer for their help with proofreading. My thanks, too, go to Annamarie Kino at Routledge for having faith in this project and for having the patience to extend my various deadlines.

The reader will notice the frequency of references to Gert Biesta and Johannes Kiersch. This reflects the extent to which these two scholars have influenced the way I understand education. Gert's many challenging ideas have enabled me to locate myself within the wider field of educational philosophy. Johannes, now in his 80s, has long been an inspiration for me within the Waldorf field – his open-mindedness, deep insight, encouragement and inspiration remain a model for me, of how to retain clarity of focus and yet maintain a sense for where this connects to the wider cultural discourse. My understandings of Waldorf education have also been influenced by fellow teachers and now Waldorf scholars Neil Boland, Peter Lutzker, Jost Schieren, Wilfred Sommer, Angelika Wiehl and Michael Zech, all of whom share my commitment to engaging with the mainstream educational discourse. I also wish to acknowledge my longtime friend and cofounder of the York Steiner School, Michael Rose, whose insights and talents have frequently inspired me.

There are many talented and capable people in Waldorf education from whom I have learned. Among those I have worked with are Josie Alwyn, Amanda Bell, Edith Bierman, Kath Bransby, Erhardt Dahl, Nana Göbel, Cathy Hu, Gabriel Knight, Sunita Krishnan, Hsini Li, ZeWu Li, Yi Ling, Magda Maier, Smitha Mallya, Jon McAlice, Trevor Mepham, Janni Nicol, Florian Osswald, Porn Panosot, Margareta van Raemdonck, Claus-Peter Röh, Wolfgang Schad, Rita Schumacher, Ulrike Sievers, Jörg Soetebeer, Stefan Sigler, Wilfred Sommer, Thomas Stöckli, John Thompson, Christof Wiechert and Heinz Zimmermann. On another day, perhaps this list would vary, because there really are a lot of talented people who have been my fellow travellers. Very much outside the Waldorf camp, I would like to mention Pete Kelly, Nick Pratt and Ulrike Hohmann at Plymouth University, from whom I also learned much.

Finally, I acknowledge my Scottish mother, Wilma Morton Rawson, who just recently passed away at the age of 87. She was a state-school teacher and special-needs advisor and was constructively critical of my early naivety as a Waldorf teacher, yet was always hugely supportive of me and my school. She would never have acknowledged her influence, but I hereby insist on acknowledging it! From her I acquired a sense of social justice in education and life, and I continue to strive to live up to her unobtrusive but profoundly lived ethic of care. I also learned from her the importance

of books, reading and lively discourse. She introduced me to Gordon Wells' book *The Meaning Makers*, which led me to Vygotsky, Bruner and ultimately to narrative and socio-cultural perspectives on learning. These ideas have enriched and transformed my understanding of Waldorf education and have enabled me to add a new dimension to this education – a perspective that is still shared by very few of my colleagues.

The author

Martyn Rawson was born in Glasgow in 1954. Because his father was a soldier, he 'enjoyed' a very wide range of schools as the family moved every two years. He read history and English at York University, and much later completed a master's degree and an education doctorate at Plymouth University. He was a cofounder of York Steiner School in 1979 before moving to Stuttgart, where he taught English and art history at the Freie Waldorfschule Uhlandshöhe, the original Waldorf school. He returned to the United Kingdom, teaching in Botton Village School in North Yorkshire and Michael Hall School in Sussex. Starting in 1994, he also worked for the Steiner Waldorf School's fellowship with responsibility for publications and curriculum development. From 1995 until 2010, he was a member of the College of the Pedagogical Section at the Goetheanum and was involved in various research projects related to international curriculum development. In 2003 he returned to Germany to teach at the Freie Hochschule Stuttgart and returned to part-time school teaching in 2007, currently at the Christian Morgenstern School in Hamburg. He has been involved in Waldorf teacher education for the past 25 years, teaching on the Plymouth University International Masters Programme, on the Masters Programme in Waldorf Education at the Freie Hochschule Stuttgart and the Teacher Training Seminar in Kiel and is currently Honorary Professor at the National Tsing Hua University in Taiwan. He has published widely on various aspects of Waldorf education and contributes as schools' advisor and teacher educator in China, Taiwan, India and South Africa.

Introduction

Birthdays

On a radiant sunny Sunday afternoon, 7th September 1919, a ceremony was held in the Great Hall of the City Gardens in Stuttgart, Germany, to mark the opening of the Waldorf School. A thousand people were present, including around 250 pupils and their parents, the teachers and the entire management of Waldorf-Astoria Cigarette Factory and its owner, Mr. Emil Molt, who was the founder of the school. There was recitation of poetry, the music of Bach, Beethoven and Carl Loewe, a Eurythmy performance and speeches from Molt, Rudolf Steiner, the school's educational founder; E.A. Karl Stockmeyer, representing the teachers' college; and a Mr. Saria, a representative of the factory workers, for whose children the school was being founded. After the ceremony, all the participants paraded through the city centre and back up the hill to the new school on the *Uhlandshöhe*, a site including a well-known café with a magnificent view over the city below, which Molt had bought for the school. At the school, each child was given a *Laugenbretzel* – the traditional product of Stuttgart's bakeries – and a box of chocolate, printed with the message 'Welcome to the Waldorf School'. The adults were served coffee, and presumably the children were offered local apple or grape juice. Games were played well into the sunny early evening, and the day was rounded off with a performance of Mozart's *The Magic Flute* in the *Staatstheatre* at the invitation of Molt and his wife, Bertha. Teaching in the school only started on 16th September because the renovations of the buildings were not yet complete. The details of the school opening are known to us from a range of well-researched sources (Frielingsdorf, 2019; Göbel, 2019; Neider & Schukraft, 2011).

Thus began an educational movement that has become, after Montessori schools one of the most widespread alternative forms of education in the world. Its philosophy and practice certainly stand in marked contrast to most other educational approaches due to its commitment to the spiritual dimension in the human being and in the world, its unique pedagogical anthropology and its practice of non-hierarchical self-management and leadership. These are aspects I will be exploring later in the book.

One hundred years later, on 7th September 2019 and the following days, the founding of the Waldorf School and the movement that grew out of it was celebrated with speeches, performances by pupils, a concert in Stuttgart's Liederhalle and an international congress. The audience was greeted in 23 languages, including Stuttgarter Schwäbisch ("I bin Baschdian, I komm aus Schduagert . . ."). Winfried Kretschmann (Green Party), the Minister President of Baden-Württemberg – the federal state of which

Stuttgart is the capital – emphasized that Waldorf education was the most extraordinary and successful development in education in Germany in the last 100 years. He praised the social engagement of the Stuttgart industrialist Emil Molt, who founded the school, to improve social relationships and compared him with another Stuttgart entrepreneur, Robert Bosch. Whilst praising Rudolf Steiner's deeds and ideas, Kretschmann criticized his terminology in relation to race and emphasized the importance of the Stuttgart Declaration in 2007, in which all the Waldorf schools in Germany committed themselves to stand against all forms of nationalism, discrimination and racism. Waldorf education, Kretschmann argued, does "not have an ideology, but a method of teaching that engages the whole human being and can be called a higher school of empathy". He referred to the founding of the school as "a great cultural and pedagogical deed that has shaped the mentality of the city and works positively in the world".

The city of Stuttgart, which today has 14 Waldorf kindergartens and 5 schools, is proud of its Waldorf connection. The long-serving mayor Manfred Rommel (after whom the city's airport is named) often quipped that Stuttgart is famous for its exports to the world – Daimler Benz, Porsche, Bosch and Waldorf education. The federal state of Baden-Württemberg has 59 Waldorf schools, and Germany currently has 254 schools and 11 teacher education centres. The city of Hamburg and its catchment area alone has 15 schools, whilst Berlin has 13. There are around 90,000 students in Waldorf schools, with about 9,000 Waldorf teachers in Germany (see www.waldorfschule.de).

In December 2020 there were 1,958 Waldorf kindergartens in 70 countries and 1,187 Waldorf schools in 64 countries (Paull & Hennig, 2020). This number is steadily growing, and the number of unofficial initiatives is significantly larger, as it is difficult with the resources available to the Waldorf movement to keep track of these developments. The China Waldorf Forum had 71 officially registered schools as of December 2020, and the International Association for Steiner/Waldorf Early Childhood Education had over 400 institutional members in China in 2019 (currently reduced to 230 + due to Corona) (with over 300 Waldorf kindergartens, many of which are growing into schools).

The growth of the Waldorf movement has now been well documented (Göbel, 2019a, 2019b; Frielingsdorf, 2019; Zdrazil, 2019) and illustrates that it has not been a centrifugal movement from a single centre outwards, but rather a network with nodal points radiating out into their local regions, whilst demonstrating a rhizome-like structure with rich cross-fertilization and hybridity. The 'invisible' network comprises a set of generative ideas and people who encounter and translate them in various, often idiosyncratic ways. This expansion also highlights a number of challenges that have actually been there since the beginning but are only really being systematically addressed today. These include the 'translation' of a Middle European idea to other cultural settings and the status of Steiner's Anthroposophy as a basis for Waldorf practice.

Waldorf schools offer *an* alternative, but obviously not *the only* alternative, to mainstream education. They are an example of counter-practice. As Gerd Biesta (1998) points out,

> a counter practice should not be designed out of arrogance that it will be better . . . than what exists. A counter-practice is only different. The critical task of a counter-practice can therefore be to show (to prove, Foucault says) that the way

> things were is only one (limited) possibility. But this step is crucial, as it opens up the possibility 'of no longer being, doing, or thinking what we are, do, or think' (Foucault, 1984, p. 47). In this way, Foucault argues, 'it is seeking to give new impetus . . . to the undefined work of freedom' (1998, 507).

In other words, Waldorf education at least shows that the ways things are in public education or indeed in private (i.e. non-state funded) education, is not the only way they can be. It shows that there is an alternative. Waldorf education contributes to pluralism and diversity at a time when these qualities are in short supply in education. Waldorf is not just a miniature model with a high profile. These schools operate in around 70 countries and therefore represent a counter-practice with a wide range of applications in very different cultures – and all that without a central, directing body or leader.

Though the overall expansion of the movement continues apace, this is no guarantee that individual schools can sustain the momentum. Institutions as social organisms have their developmental phases, like pioneer phases and consolidation phases; they have inner crises; they attract opposition; they lose their vitality, focus and purpose; and finally, they outlive their usefulness. Failure has a high price for all concerned. Bob Dylan's lyric from the song, "It's Alright Ma (I'm Only Bleeding)" – *he not busy being born is busy dying* -– is true of Waldorf schools. To survive, they have to maintain and develop their quality continuously. From a spiritual perspective, they have to continue to grow and transform and channel energy into meaningful things, they have to be places in which people – and especially children and young people – feel spiritually seen, recognized and enabled to develop their unique potential. When they cease to be able to do this, schools lose their purpose. The United Kingdom (UK), where Waldorf schools have been around. The United Kingdom (UK), where Waldorf schools have been around since the 1920's, has recently lost a number of schools, including some of those pioneers and some founded since the 1970's. Some of these schools were unable to cope with the challenge of sustainability and renewal, some failed inspections and were not in a position to make the necessary changes in the time available, though this failure was perhaps symptomatic of a wider inability of leadership to respond to changing circumstances. This does not mean that Waldorf has failed; it means that those schools, for a variety of reasons, failed. The Waldorf movement in the UK (led by the Steiner Waldorf Schools Fellowship) now has a new, highly competent leadership, which will go a long way to ensure that its member schools are doing the right things and doing them well. Of course, when the state makes demands that make Waldorf unworkable, then hard decisions have to be made, because Waldorf in name only, is not Waldorf in reality. However, the message is unavoidably clear; Waldorf only works if it can maintain its own very high standards of pedagogical creativity, clear-sightedness, knowledge creation and ethic of care.

Good practice – dubious ideology?

Despite the growing number of Waldorf-related schools, early-years settings, therapeutic schools and communities and numerous emergency and peace-education projects, not to mention the other Steiner-influenced fields of work in medicine, agriculture,

organizational development, banking and the arts around the world, this movement is generally held at arm's length by the academic world. Professor Heiner Ullrich, a leading academic expert on Waldorf education, notes, "because the scientific discourse on Waldorf schools has become more intense, the otherness of Rudolf Steiner' pedagogy has become clearer but its fruitfulness has also been shown to be even stronger" (2008, p. 224).

In various publications over the years, Ullrich has made significant criticism of Waldorf education, in particular its underlying theory. Ullrich's position – and this is typical for the academic world generally – has been formulated by Professor Christian Rittelmeyer, "you have in many ways a good educational practice but a dubious (anthroposophical) ideology" (2010, p. 8). Johannes Kiersch pointed out some time ago that Waldorf education provokes critique on two grounds, firstly "because it is successful and secondly because it has no adequately secured theory" (1986, p. 543).

External critiques of Waldorf education always circle around the idea that it would be good to separate the obsolete, anachronistic, esoteric und unscientific ideas of the person most centrally identified with this education – Rudolf Steiner – from what is widely regarded as a good education that ever more parents around the world choose for their children. Rittelmeyer (2010), who is neither a Waldorf teacher nor an anthroposophist, refutes this solution, saying, no, one cannot separate the education from its underlying educational philosophy and pedagogical anthropology. But, he adds, this doesn't mean that Waldorf teachers shouldn't use other sources and interpret Steiner in new ways.

That indeed is the crucial question for Waldorf teachers: What is the (right, appropriate, intended, necessary, actual) relationship between practice in a Waldorf school and the ideas of Steiner's Anthroposophy? It is a question that this book seeks to explore and answer. It is perhaps less of a question for parents, who choose schools on the basis of their reputation and what they value for their children. If parents have a choice and can afford to choose – which, of course, many do not – in the long run, they tend to choose and stay with schools that deliver what they value, and that is in most cases not an esoteric philosophy but a good education for their children. A school that doesn't deliver what parents want, doesn't keep those children. It is a competitive 'market'. It is remarkable, however, how many professionals, including academics and state school teachers, choose Waldorf schools for their children and even become Waldorf teachers, in spite of its "dubious ideology" (Barz & Randoll, 2007).

Both the educational success and the dubious ideology are also questions for the educationalists in the academy, for researchers and for university students who study education. Waldorf is one of the few educational alternatives with a comprehensive and very different educational practice that one can study and research in so many different settings and countries. In Germany at least, it has become quite a busy niche for educational researchers. Given the increasing homogeneity of schools generally in terms of their aims and methods, Waldorf is a relatively rare example of other education.

The aims of this book

In 2000, I co-edited and published *The Educational Tasks and Content of the Steiner Waldorf Curriculum* (Rawson & Richter, 2000, republished in 2014 as Avison, Rawson, & Richter). The aim of that publication was to present an account of Waldorf education and its curriculum in a systematic way, in accessible language that people

outside the Waldorf camp and especially educational authorities could relate to. In this book, I take up this challenge again, 20 years later. One of the differences today is that we have considerably more academic literature to draw on about Waldorf education, most of it in German. The earlier book set out to present Waldorf as it is, though it was inevitably an interpretation. This book is much more explicitly a new interpretation. In particular, it departs from most previous presentations of this education in that it offers a series of generative principles that are drawn from the underlying ideas of Steiner, plus 100 years of practice, but also takes more recent education theory into account. These generative principles can be used to generate new practice and evaluate and research existing practice. It offers an interpretation of Waldorf education today – not as it is practiced everywhere today, but rather what constitutes (in my judgement) best practice and the current Waldorf discourse, which has changed significantly over the last 15 years.

The book starts with Rudolf Steiner the man and his ideas about the spiritual dimension of life and about how knowledge can be generated, including knowledge of the world not accessible to our senses – the spiritual world. In Parts 2 and 3, I introduce and explain a series of generative principles that underpin Waldorf practice in terms of teaching and learning, and then I look at Waldorf schools as learning communities. In Part 4, I summarize the existing research on the education and give a brief overview of the international growth of the Waldorf movement.

In order to explain Waldorf education to the reader in the terms of this new discourse, I have to leave much out. I refer to the past only in as much as it directly shapes the present. I discuss Steiner's spiritual ideas only in as much as they relate to pedagogy today. Likewise, the book does without visual images, though so much of Waldorf is visual – from children's books and craft works, to the colour of classroom walls and the shape of their architecture. These have been illustrated in high quality elsewhere, not least through the wealth of publications that appeared in connection with Waldorf 100. In particular I refer to *Insights Waldorf*, sponsored by the Leica Camera company and illustrated by well-known Leica photographers (Freunde & Leica Camera, 2019; and also Göbel & Reinthal, 2019). There are also a number of helpful websites: the Friends of Waldorf Education (www.freunde-waldorf.de), which is also available in English; the Association of Waldorf Schools in Germany (www.waldorfschule.de), or the website of the Association of Waldorf Schools in North America (www.waldorfeducation.org); the Waldorf Resources site of the Pedagogical Section in Dornach, Switzerland (www.waldorf-resources.org); and of course, the Waldorf 100 site in English (www.waldorf-100.org).

I have also resisted the temptation to analyse the current educational climate, with its brisk winds of testing and performativity blowing across the educational landscape and the inhospitable climate of standardization it brings for creative education. Others have done this far better than I could (e.g. Ball, 2012, 2013, 2019; Verhaeghe, 2015; Biesta, 2010, 2012, 2015). This climate change of neoliberal policy technologies is the biggest single threat to the future of Waldorf education and remained in my mind throughout the writing of the book, as the reader will notice here and there.

This book is not a manual for becoming a Waldorf teacher (or parent). That would require a very different kind of book (in preparation for Floris Books) and above all an interactive process. Nor does it detail the curriculum or teaching methods. It refers to these in order to illustrate the generative principles. The book also restricts itself to Waldorf education in schools, which typically teach children and young people from the age

of 6 to 18/19 years. Waldorf early-years and kindergarten education has been more than adequately covered (Nicol, 2016; Nicol & Taplin, 2017; Patzlaff & Sassmannshausen, 2007; Rawson & Rose, 2003). Nor does it address Waldorf special education or its various applications in emergency education, or indeed Waldorf education in state schools.

There are three factors that make understanding and writing about Steiner in English particularly challenging. Steiner lived in another era and spoke German, and most of his pedagogical works are in the form of transcripts of lectures. As Johannes Kiersch (2012) has pointed out, Steiner tried to describe experiences for which there was and still is no common, unambivalent vocabulary. Kiersch suggests that Steiner, despite his extraordinarily creative use of language, was actually linguistically sceptical, permanently wrestling with the impossibility of finding adequate words for what cannot really be expressed in words. He continuously sought to approach ideas from as many different perspectives as possible and to ensure that these concepts remained as fluid as possible. Interestingly, Steiner once commented that in order to 'translate' spiritual truths, it is necessary to dogmatize them, in the sense of constructing a system which fixes the meaning of words and concepts, but that one must rigorously avoid seeing truth in the dogma (Steiner, 2001, cited in Kiersch, 2012, p. 313). Thus, the notion that Steiner wanted to articulate absolutely valid knowledge has to be relativized in the light of this insurmountable gap of alignment between word, meaning and truth – words always have to be interpreted.

These aspects have perhaps contributed to the muted reception of Steiner in the English-speaking academic world. In order to overcome these limitations, I have tried to explain Steiner's key epistemological, philosophical and pedagogical ideas. Of course, such explanations and translations are always also interpretations. Most of the translations from German I have done myself, if there are no published translations that I feel catch the meaning (based on my subjective judgement), and I have also chosen to use the female gendered pronouns she/her where the original German was *Mensch*, i.e. the human being or where male pronouns were used in a general sense (or I have used the symbolic distancing device 'sic').

The author's position

Writing this book after 40 years of teaching in Waldorf schools has provided me with an invaluable opportunity to review my position on practically everything related to this education. In all cases, it has led to a deepening of my understanding. I position myself as a critical insider, in the sense outlined by Kathryn Herr and Gary Anderson in their chapter on the continuum of positionality in action research (2015, pp. 37–59). Writing this book has been a bit like action research within an extensive literature review. Regarding my insider status, I have taught in Waldorf schools in the UK and in Germany since 1979, and I have worked in Waldorf teacher education in both of these countries and as a visiting lecturer and advisor in many other countries in Europe, Africa and Asia, as well as in the United States.

I take a somewhat different stance to Heiner Ullrich (Ullrich, 2015), whose book is also a critical introduction (*Waldorfpädagogik: Eine kritische Einführung*). In a sense, this book also tries to offer answers to some of the questions that Ullrich poses. My position may be summed up as follows: Waldorf education has an interesting, challenging and epistemologically important educational theory. The practice is often highly effective but is not always as good as it could and should be for a variety of reasons. My position is thus that of an insider who is critical in both the sense of

recognizing weaknesses and asking critical questions about assumptions, but also in the sense of describing, contextualizing, reflecting on, analyzing and evaluating both theory and practice – rigorous skills that I learned during my practitioner master's degree and education doctorate at Plymouth University.

Finally, I am not a passive observer but an active participant involved in the further development of this education. My particular contribution is to identify a series of generative principles, based on Steiner's pedagogical anthropology and Waldorf practice, as a set of tools for the evaluation, research and development of practice. To echo Jennifer Gidley's (2016, p. 15) apology for her complex analysis of integral thinking on the evolution of consciousness (including the works of Steiner), I hereby state that this is my hermeneutic interpretation, my narrative, and it is woven into the fabric of my lived experiences and reflects my interests. It leaves much out – maths for example, gets short shrift. I leave the gaps to those more capable than me, to fill in. I can do no more than articulate the fusion of my horizons with a lifetime engagement with this educational practice. This book brings me to a new starting point.

Steiner and/or Waldorf?

Finally, a word about terminology. As I have briefly outlined in the previous section, Rudolf Steiner was the educational founder of the first Waldorf school for the children of the workers in the Waldorf-Astoria Cigarette Factory, whose owner, Emil Molt, invited Steiner to found the school. The school was named after the factory. Steiner himself frequently spoke of Waldorf pedagogy (die *Waldorfpädagogik*), Waldorf ideas and thinking (*der Waldorfgedanke*) and so on. Schools around the world either refer to themselves as Rudolf Steiner schools or Waldorf schools, with some national preferences (e.g. the United States and Germany prefer Waldorf, with Australia, New Zealand and the Scandinavian countries preferring Steiner). In the Netherlands, they are referred to as Vrijeschool (free in the sense of independent schools). Other schools that follow this educational approach have quite different names; for example, my own school, the Christian Morgenstern School in Hamburg, or the Scuola Novalis in Zoppè di San Vendemiano, Italy. In the English-speaking academic and political world, people usually refer to Steiner education.

My personal identification is not with the person Rudolf Steiner, though he is the primary creator of this educational approach and was certainly a remarkable man, but rather with the tradition of practice that has grown from its origin in the Stuttgart Waldorf School, where I taught from 1987–1993. From the beginning of my career, I have been inspired by the idea of a school for the children of factory workers just after the catastrophe of World War I. Therefore, in this book I refer to *Waldorf* education and pedagogy, *Waldorf* teachers constructing *Waldorf* curricula and so on. This includes all the schools in the World List of Waldorf Schools, whether called Waldorf, Steiner or anything else.

References

Avison, K., Rawson, M., & Richter, T. (Eds.). (2014). *The educational tasks and content of the Steiner-Waldorf curriculum*. Edinburgh: Floris Books.

Ball, S. J. (2012). *Global education inc. New policy networks and the neo-liberal imaginary*. London and New York: Routledge.

Ball, S. J. (2013). *The education debate* (2nd ed.). Bristol, UK: Policy Press.
Ball, S. J. (2019). *Foucault as educator*. Berlin: SpringerBriefs in Education.
Barz, H., & Randoll, D. (2007). *Absolventen von Waldorfschulen. Eine empirische Studie zu Bildung und Lebensgestaltung (Alumni/gradutaes from Waldorf schools. An empirical study of education and life stories)*. Wiesbaden: VS Verlag für Sozialwissenschaften.
Biesta, G. J. J. (1998). Say you want a revolution . . . suggestions for the impossible future of critical pedagogy. *Educational Theory*, *48*(4), 499–510.
Biesta, G. J. J. (2010). *Good education in the age of measurement: Ethics, politics and democracy*. Herndon, VA: Paradigm Publishers.
Biesta, G. J. J. (2012). Giving teaching back to education: Responding to the disappearance of the teacher. *Phenomenology and Practice*, *6*(2), 35–49.
Biesta, G. J. J. (2015, December). The duty to resist: Redefining the basics for today's schools. *Research on Steiner Education*, *6*, 1–11.
Foucault, M. (1984). What is enlightenment? In P. Rabinow (Ed.), *The Foucault reader*. New York: Pantheon.
Freunde, D. E. K., & LeicaCamera. (2019). *Insights worldwide- 100 years Waldorf education*. Heidelberg: Kehrer.
Frielingsdorf, V. (2019). *Geschichte der Waldorfpädagogik. Von ihren Ursprung bis zur Gegenwart*. Weinheim Basel: Beltz Verlag.
Gidley, J. M. (2016). *Postformal education. A philosophy for complex futures*. Weinheim Basel: Springer.
Göbel, N. (2019a). *Die Waldorfschule und ihre Menschen Weltweit. Geschichte und Geschichten 1919 bis 2019. Waldorfpädagogik. Ihre Entwicklung und Ausbreitung bis zum Ende des 2. Weltkrieges* (Vol. 1). Stuttgart. Verlag Freies Geistesleben.
Göbel, N. (2019b). The diversity of waldorf schools in the cultures of the world. In T. Zdazil & H. Kullak-Ublick (Eds.), *The understanding of the human being in Waldorf education across the world's diverse cultures* (pp. 17–29). Stuttgart: Edition Waldorf.
Göbel, N., & Reinthal, C. (2019). *100 Jahre Erziehung zur Freiheit. Waldorfpädagogik in den Ländern der Welt*. Stuttgart: Verlag Freies Geistesleben.
Herr, K., & Anderson, G. L. (2015). *The action research dissertation* (2nd ed.). Thousand Oaks, CA: Sage.
Kiersch, J. (1986). Rezension zu 'Heiner Ullrich: Waldorfpädagogik und okkulte Weltanschauung'. *das Goetheanum*, *41*, 314–315.
Kiersch, J. (2012). Spiritualität auf der Suche nach Wirklichkeit- in pädagogischer Hinsicht. In P. Loebell & P. Buck (Eds.), *Spiritualität in Lebensbereichen der Pädagogik. Diskussionsbeiträge zur Bedeuting spiritueller Erfahrungen in den Lebenswelten von Kindern und Jugendlichen* (pp. 325–342). Opladen; Berlin; Toronto: Verlag Barbara Budrich.
Neider, A., & Schukraft, H. (2011). *Rudolf Steiner in Stuttgart*. Stuttgart: Belser, Kunstmuseum Stuttgart.
Nicol, J. (2016). *Bringing the Steiner Waldorf approach to your early years practice* (3rd ed.). London: Routledge.
Nicol, J., & Taplin, J. T. (2017). *Understanding the Steiner Waldorf approach*. London: Taylor & Francis.
Patzlaff, R., & Sassmannshausen, W. (2007). *Developmental signatures. Core values, and practices in Waldorf education for children ages 3–9*. Ghent, NY: Association of Waldorf Schools in North America.
Paull, J., & Hennig, B. (2020). Centenary world maps of the global diffusion of "The School of the Future". *Journal of Social Sciences and Humanities*, *6*(1), 24–33.
Rawson, M., & Richter, T. (2000). *The edcational tasks and content of the Steiner Waldorf curriculum*. Forest Row, UK: Steiner Schools Fellowship Publications.
Rawson, M., & Rose, M. (2003). *Ready to learn: From birth to school readiness*. Stroud: Hawthorn Press.

Rittelmeyer, C. (2010). Vorwort. In H. Paschen (Ed.), *Erziehungswissenschaftliche Zugänge zur Waldorfpädagogik* (pp. 7–10). Wiesbaden: VS Verlag.

Ullrich, H. (2008). *Rudolf Steiner*. London: Bloomsbury.

Ullrich, H. (2015). *Waldorfpädagogik: Eine kritische Einführung*. Weinheim Basel: Beltz Juventa.

Verhaeghe, P. (2015). *What about me? The struggle for identity in a market-based society*. Melbourne and London: Scribe Publications.

Zdrazil, T. (2019). *Freie Waldorfschule in Stuttgart 1919–1925. Rudolf Steiner-das Kollegium-die Pädagogik*. Stuttgart: Verlag Freies Geistesleben.

Part 1 Rudolf Steiner and the origins of his educational ideas

Summary: In this section I introduce Rudolf Steiner, who created Waldorf education. Following a brief account of his life, I explain his early epistemological and philosophical works because these provided a foundation for his subsequent projects, including the development of an educational approach. Returning to his biography, I outline some of Steiner's experiences that influenced his educational ideas and then summarize his first published work on education.

Rudolf Steiner: "Stranger in a strange land"

> "*I stand alone here. There is no one who has the slightest understandings of the things that move and inspire me.*" (from a letter to Rosa Mayreder in 1890) "*Whether people around me understand me or not – I follow my own star,*"
>
> (from a letter to the Specht family from the same year, both cited in Selg, 2014)

It is always a challenge to understand our proximity to and distance from people who lived in other times and places. It is hard to judge how much they were a product of their times and how much their unique individuality shines through. We have to decide how much we can identify with and what remains out of reach, strange, foreign and other. This difficulty applies to all historical figures, and especially to famous people because we know what became of them and because our understanding is coloured by received opinion and the critical reception of their work. Understanding Rudolf Steiner is particularly challenging because recognition of his achievements by the wider cultural world has been grudging. This is partly because of the remarkable range of his works, partly because they still do not fit into conventional categories and, not least, because of the hermetic nature of the movement that grew up among his followers.

To the first generations of his followers, Steiner was the great sage of our times and his works were an endless source of wisdom, like an inexhaustible wellspring of important ideas in many different fields. Others, both during his lifetime and up until today, have ignored, dismissed or discredited these works, mainly because of their 'occult' background and because they (still) do not fit into the current paradigms of science, philosophy, politics, education or the arts. The term *occult* has a dubious reputation, but it actually simply refers to knowledge that we cannot base on what our normal senses mediate. However, it was exactly this aspect that attracted (and still attracts) a lot of capable people to Steiner's ideas. These were and are on the whole modern, educated and critical people, dissatisfied with materialistic answers to the challenges of modernity and who want a modern spirituality that does not look back to romanticism or mysticism, and who also want to use these insights to change the current world.

It took a while before Steiner's message got through, and for much of his career he was respected but treated as an outsider. Once his message started to be understood, and he had understood how best to formulate it, people joined his cause. At this point he started attracting opposition, some of it dangerous, not least among the emergent National Socialist movement. Any informed study of his life must conclude, with Peter Selg (2014), that Steiner was entirely motivated not by personal ambition but by the will to do something useful to ameliorate the evils of his times and those he anticipated. He was modest but determined. He arrived at his insights not through illumination or grace or because he was born into them, but through hard and often lonely work.

As the 20th century descended into totalitarianism, global war, racial genocide, materialism at all levels of society and the ongoing victories of capitalism over socialism, a worldwide, growing ecological disaster, the rise of new global hegemonies and the slow or rapid fall of older ones, it did not seem very likely that Steiner's work would bear fruit and usher in a new era of peace, spiritual enlightenment and social harmony. However, perhaps like Zhou Enlai's apocryphal comment on the success of the French Revolution, it really is too early to judge how fruitful Steiner's ideas have been. Really good ideas do last. Some of Aristotle's ideas – such as the value of practical wisdom or phronesis – or Laozi's ideas of attunement and flow, are still relevant today, and the comparison is not exaggerated. It is certainly too early to pass judgement on the relevance of Steiner's ideas, though as this book hopes to show, in education they do seem highly relevant.

Biography and biographical mythos

Today we have the benefit of a number of well-researched biographies of Steiner and historical accounts of some of his work (in English, Childs, 2003; Lachman, 2007; Gidley, 2011; MacDermott, 2009; Lindenberg, 2013; Selg, 2014 and an interesting series of biographical essays at www.rudolfsteinerweb.com/Rudolf_Steiner_Biography.php). It is, I believe, now possible to begin to appreciate Rudolf Steiner as a person with a human biography. Today, there are critical editions of his works, and more and more people are taking a new look at his ideas in practice. In what follows, I briefly outline Steiner's biography in as much as it sheds light on his educational ideas. These undoubtedly grew out of his life and works and can only really be fully appreciated in this context. Jan Göschel (2012) has developed the notion of biographical mythos (drawing on Aristotle's notion of mythos as plot or storyline in drama and Ricoeur's revival of this term for the meaning-making narrative structure of a life-story) to describe the constructed interpretation of the underlying narrative of a person's life. In this sense I construct a biographical narrative of Steiner's life with from the specific perspective of his educational ideas.

Steiner was born in 1861 in Donje (Lower) Kraljevec, which today is in Croatia and at that time was on the Hungarian side of the Austrian–Hungarian border. People there spoke Croatian. His Catholic baptism certificate is in Croatian and Latin. In 1924, Steiner could apparently still read Hungarian, which was spoken at his secondary school, though his mother tongue was German. Hemleben (1963) describes the region that Steiner grew up in, the Burgenland, as lying at the boundary between Middle and Eastern Europe, not just politically but also culturally. Steiner's parents had needed to move from their beloved homeland in the forests of the *Waldviertel* north of the Danube in Lower Austria, a place of great natural beauty that had

been barely changed by modernity, and to which they returned at the end of their lives. His father had originally been a gamekeeper to the local nobility but had to find a new profession, as he was not allowed to get married. He became a poorly paid telegrapher and later stationmaster. We are told by Hemleben (1963) that he came from "hardy mountain peasant stock", an evocative phrase one might not find in a modern biography. As Hemleben (1963) points out, the boy grew up within the polarity of the healing power of nature and the world of technology with its attractive power, though Steiner declared that he experienced nature as the stronger force (Steiner, 1928, p. 3). Steiner evidently possessed clairvoyant abilities and could experience spiritual aspects in nature and in people. He could apparently move beyond the barriers of the sense world at will. Even as a child, he realized that though such experiences were real for him, others clearly did not experience them, and there was no cultural context for them in his social environment.

The family moved several times to different railway stations, each move bringing him nearer to Vienna. Steiner himself emphasized in his autobiography (1928, p. 2) that he didn't feel he belonged to the place he was born in, nor to the places in which he grew up, which may have enhanced his experience of being an outsider. This multicultural background and his later experiences in cosmopolitan Vienna sensitized him to the problems that would arise when the new, emergent nationalism was linked to ethnic or cultural identity, a problem that is still unresolved today. Quite early on he saw the need for a cultural identity that was broader and more universal than ethnic or national identifications. In his major philosophical work *The Philosophy of Freedom* (1963a, though originally published in 1894), he argued that people should be judged as individuals irrespective of their origins or cultural affiliations, social status or gender.

Lindenberg (1992) opens his earlier biography of Steiner with the sentence "Rudolf Steiner was the child of poor people". From a humble background, he worked his way up the social mountain to become an internationally renowned figure. When Steiner died in 1925, his obituary appeared on the front page of *The New York Times* (Uhrmacher, 1995, p. 401). Thousands of people from all over Europe attended his funeral. He was certainly one of very few people (if any) from his background to gain a doctorate in philosophy (in 1891 at the University of Rostock). He was clearly precocious, reading Kant at 15 years of age and excelling at secondary school, and later at the Technical University in Vienna.

Steiner makes specific reference to his discovery that, though he loved the material world around him, "the reality of the spiritual world was to me as certain as that of the physical" (1928, p. 11). Born into a Catholic family, he naturally attended mass, but it was the ritual of the liturgy rather than the doctrine or the sermons that made the deepest impression. However, it was his discovery of the laws of geometry that provided him with a key to understanding his spiritual experiences. He wrote,

> I said to myself 'the objects and occurrences which the senses perceive are in space. But, just as this space is outside of man (sic), so there exists also within man a sort of soul-space which is the arena of spiritual realities and occurrences'. In my thoughts I could not see anything in the nature of mental images such as man forms within him of actual things, but I saw a spiritual world in this soul-arena. Geometry seemed to me to be a knowledge which man had appeared to have produced but which had, nevertheless, a significance quite independent of man.
>
> (Steiner, 1928, p. 11)

This discovery laid the basis for his later theory of knowledge. Steiner's question was not, "Is there a spiritual dimension to life?" but instead "How do the physical and spiritual dimensions relate to each other?" (Hemleben, 1963, p. 23). He kept such experiences to himself, and it is remarkable that he was clearly able to do this in a healthy way at such a young age. Only at the age of 18 did he meet someone with whom he could share his experiences – a herb-gatherer, Felix Koguzski, whom he met on the train to Vienna and who sold herbs to apothecaries in the city. He became friends with this man and his family. This outwardly simple herb-gatherer was clearly an initiate with profound spiritual insight that he had developed on his own, though he was versed in esoteric literature (Selg, 2009). Steiner describes how he first had to learn to understand the man's spiritual dialect:

> According to the usual conception of "learning", one might say that it would be impossible to 'learn' anything from this man. But, if one possessed in oneself a perception of the spiritual world, one might obtain glimpses very deep into this world through another who had a firm footing there.
>
> (Steiner, 1928, p. 40)

This was a crucial encounter for Steiner's development because it led to him meeting an unknown Master, who initiated him (Steiner, 1966). As Childs (2003, p. 11) notes, Steiner did not mention his experiences of the spiritual world to anyone for some 34 years, with the exception of Koguzski, but he dedicated himself to finding a scientific approach to spiritual knowledge, as he was convinced that this was necessary in modern times.

This experience helped Steiner to formulate a theory of knowledge that the philosophy of his day did not provide, as it was assumed that human knowledge is limited to what the senses provide and what logic can deduce. After painstakingly working his way through Kant's *Critique of Pure Reason* at the age of 16, he realized that he would have to build a philosophical bridge between the world of the senses and the spiritual world that Kant ruled out, as being beyond the boundaries of the human mind. Childs' (2003) chapter title on this phase of Steiner's life is apt – 'Researching in two worlds' – for this is what Steiner did during his scientific studies at university and later during his career as a private tutor, scholar, researcher, journalist, publicist, adult educator, artist and public lecturer.

Lindenberg (2013) entitles the opening chapter of his Steiner biography (in Jon McAlice's translation) 'Stranger' and opens with the sentence "Born a stranger in a strange land." This gives expression not only to the circumstances of his birth, but to the more subtle fact that Steiner, with his special gifts and spiritual insights, was mostly not recognized as such, especially growing up in such a humble background, though this experience followed him during most of his life. He even appears to us from this historical perspective as an outsider, an enigma, an exceptional person in the many social contexts he ended up in.

Steiner himself played down his personal life in his autobiography, stating that it was of no concern of the reader, and until recently his followers have treated this side of his life with discretion verging on hagiographic reverence. We know that he was married twice and had no children and that his first wife, Anna Eunike-Steiner, was a widow with five children with whom he maintained contact over many years. His second wife, Marie von Sivers, who was of German-Russian origin, was trained in

recitation and drama and was heavily involved in esotericism. The first marriage faded as his career became more successful, his life became more peripatetic and his relationship to Marie von Sivers grew, but Steiner waited until Anna died in 1911, and then three years later married Marie von Sivers. He kept in contact with the family members and visited them when he could. It is assumed that both marriages had an element of social propriety, though both relationships were evidently personally very important to Steiner.

The didactic aim of Steiner's autobiography, *The Story of My Life* (1928, first edition in English), which was partly dictated from his sickbed at the end of his life, was to explain his spiritual path and how his Anthroposophy developed, rather than be an autobiography in a personal sense. Steiner's autobiography is to a large extent an account, from his perspective, of his relationships to people and how these influenced him, in particular how ideas merged and separated and what effect this had on him and on the world both were embedded in. This is a profoundly relational view of the world and people's actions and understandings within it.

My sense is that Steiner's biographical mythos involved significant encounters with people, through which he learned to translate and apply his inner experiences into actions in the world. A few examples (from the many mentioned in his biography) include the herb-gatherer Felix Koguzski; the poet and feminist Rosa Mayreder, with whom he was close whilst writing *The Philosophy of Freedom*; Margaret McMillan and her commitment to poor children and nurturing their imagination; or Rosa Luxemburg (Steiner died before he could reflect on the second half of his life in his autobiography), whose vision of the role of education in emancipation inspired him and whose violent death shortly before the founding of the Waldorf School deeply affected him. These relationships and the ideas that they generated were not only of personal biographical importance for the people involved but also changed the cultural lifeworld in significant ways. They helped open up a space for something new to enter history. Steiner is in effect telling the reader, this person helped me see something new, but far more significantly, he is saying, in this meeting of minds, new ideas and new social processes became possible.

In his book *Riddles of Philosphy* (2009), Steiner gives a summary of key Western philosophical ideas. Each philosopher is presented in a positive and convincing light, encouraging empathy and identification in the reader. Then he moves to the next philosopher and the experience in the reader repeats itself. From a sceptical position, one could say that his presentation of each philosophy points the way to his own philosophy. That is not unknown in scholarly works. But I believe that Steiner's message is another one. He is illustrating both that he stands in a long historical tradition and that this tradition also reveals an evolution of consciousness, and the message is that the consciousness we have today is not the only one possible. Steiner (1935) could defend the Darwinist Ernst Haeckel from his critics and yet be a critic of Haeckel's materialism in the same publication. Steiner could criticize Darwinism yet also speak (in a lecture in Berlin on 27th October 1904) in the highest terms of respect for Charles Darwin, whom he characterizes as a careful and profound researcher who acknowledged and revered what, for him, was unknowable. Darwin, as Steiner saw it, relied modestly only on what his own insights told him – which is high praise from Steiner, the author of *The Philosophy of Freedom*, which argues for reliance on one's own powers of thinking (just as Darwin himself admired Alexander von Humboldt for similar reasons). Steiner practiced a hermeneutic methodology of empathic understanding in order to extend the horizon of his own understanding and

that of the reader, in order to then incrementally extend that horizon. I believe that Steiner's way of working involved what the philosopher Hans-Georg Gadamer (2013) later called a fusion of horizons, an iterative process of the coming into being of new understandings that change not only those involved but also the lifeworld. Throughout Steiner's esoteric writings runs the idea that thoughts and feelings are as real as deeds and that we are connected to the lifeworld we are embedded in (which he would have described as our cultural/spiritual environment) through our consciousness and through our deeds.

Key stages in a varied career

It is only possible to sketch the barest of outlines of Steiner's career, and the reader is advised to turn to the several biographies listed earlier. He completed his formal education with a doctorate at the University of Rostock in northern Germany (with the title *Truth and Knowledge: The Fundamentals of a Theory of Cognition with Special Reference to Fichte's Scientific Teachings*) in 1891. Before this and whilst still studying in Vienna at the *Technische Hochschule* (Technical University, 1883–1889), he was entrusted with the editing of Johann von Goethe's scientific works for the publishing house *Kürschners Nationalliteratur*. This work with Goethe led to him publishing a book on Goethe's theory of knowledge and to lecturing on Goethe's aesthetic theory. From 1890 until 1897, Steiner worked at the Goethe-Schiller Archive in Weimar. During this time he published his major philosophical works *Truth and Knowledge* (1963b) and *The Philosophy of Freedom* (1963a), also published in English as *The Philosophy of Spiritual Activity* and *Intuitive Thinking as Spiritual Activity*. He also researched, ordered and published the Friedrich Nietzsche archives, edited and published the complete works of Arthur Schopenhauer in 12 volumes (1894–96) and an eight-volume complete works of the poet Jean Paul, as well as other literary and editorial projects. Steiner was a highly productive editor and philosophical author in this phase of his life.

From 1897 until 1913, Steiner was based in Berlin, where he continued to work as a literary magazine editor, writer and lecturer, including regular weekly contributions to the *Berlin Workers' Education School* in close cooperation with Rosa Luxemburg, the Marxist theorist, pacifist and revolutionary socialist who was at various times a member of Germany's Social Democratic Party, the Independent Social Democratic Party and the Communist Party. During this time, he also worked for the Theosophical Society, becoming the General Secretary of the German Section. Following a range of differences with the Theosophists, not least his refusal to acknowledge the young Jiddu Krishnamurti as the second coming of Christ, Steiner split from the Theosophical Society and concentrated his work within the Anthroposophical Society, which he founded in 1912.

In 1913 he moved to Dornach, in northern Switzerland near Basel, where he designed and supervised the building of the Goetheanum, a large wooden building with a central hall and stage that pioneered organic forms of architecture. This became the centre for his work until the end of his life in 1925. In 1922, the Goetheanum was destroyed by arson and the construction of a new version in sculpted concrete forms – the Second Goetheanum – was started, which stands to this day. Gidley (2011) records that Steiner gave over 5,000 lectures between 1888 and 1924 in 96 European cities in 17 countries, the regularity of which increased to over 400 a year in his last years of

work. He died after a short illness on 30th March 1925 in Dornach, at the relatively young age of 64. His death was unexpected, even by him, as his notes record a number of projects he was intending to carry out on his recovery.

It is worth recalling that Steiner lived during the turn of the 19th to the 20th century in a world of new mass media, populated by an incredibly wide spectrum of people rejecting the traditional status quo, ranging across anarchists, revolutionaries, spiritualists, theosophists, popularizers of science, social Darwinists, politicians promoting an American vision of democratic capitalism (e.g. Woodrow Wilson) and reactionaries including xenophobic nationalists and ultra-conservatives. It was also, of course, a time of radical new ideas in the arts (Symbolism, Fauvism, Primitivism, Post-impressionism, early Picasso, Cubism), as well as major new ideas in science such as Freud's work on the unconscious, Einstein's theory of relativity and the emergence of the philosophy of phenomenology and Weber's foundational works on sociology.

What motivated Steiner?

In 1918, Steiner published a revised edition of his *The Philosophy of Freedom/The Philosophy of Spiritual Activity* (1963a), 25 years after its original publication. In a lecture given shortly afterwards, Steiner (2015) revisited his motivations for writing the book at the time and reflected on its subsequent reception. He recalled speaking at an assembly of workers and their families ("the hall was full of screaming children, babes in arms and even dogs") in Berlin Spandau (it must have been around 1903). Rosa Luxemburg, the socialist campaigner, had just spoken about the workers and science and Steiner followed up with a spontaneous talk, arguing that science could help find new forms of social life out of a spiritual understanding. He said that the audience of workers listened intently to him for over an hour. He reflected that the proletariat (his term) was more receptive to a spiritual outlook than the bourgeoisie or the intellectuals on whom the workers depended for their knowledge and often for their political leadership. This anecdote offers an important insight into Steiner's original motivations and his affirmation of these in the situation of 1918. Society was torn by social injustice and inequality, working people everywhere were oppressed, imperialism and the industrial and military elite had led the world into a catastrophic war and the Bolshevik revolution was now turning into a terrible civil war.

Rationalism alone could not help. What was needed, in his view, was an extended consciousness and with it new knowledge, what Gidley (2016) refers to as postformal knowledge, i.e. that which goes beyond Jean Piaget's highest form of cognition – formal operations, the basis for rational thinking. Gidley describes Steiner's thinking as "post-conventional, innovative and future orientated" (2016). Throughout his public career, Steiner wanted to help people to find solutions based on their own insight, and he was convinced that any fruitful solution to society's problems had to take the spiritual dimension into account in a practical way. Only science can provide a basis for real solutions, but not a positivistic science based solely on empiricism that rejects all other forms of knowledge. One could say that understandings of knowledge have moved on from this simple binary of positivist rationalism as against intuition. Nevertheless, the kind of evidence usually drawn on in the calls for evidence-based education or medicine is often still reduced to quantifiable outcomes based on positivist science that takes little account of more subtle processes. The spiritual dimension still remains partitioned off from science and rational thinking, a source of personal

inspiration for some but nothing to do with science. Despite the variety of thinking available to us today, Steiner's aim to make a science of the spirit still remains remote.

On the testimony of contemporaries who met him, including a reporter for *The Manchester Guardian* (Paull, 2011), Steiner was evidently a charismatic speaker, yet someone who had the remarkable ability to put people, especially children, at ease with his profound empathy and humour. Hardly any of the photographs of Steiner suggest this aspect of his character, but it is well attested by contemporary accounts.

Kaiser (2020) has argued that Steiner should be understood as a storyteller in the sense of offering an authentic opportunity to his listeners to question and understand his narrative accounts. Steiner's preference for the lecture, which he cultivated as an art form over years and varied according to his audience, as his primary mode of presenting his ideas can be understood both as performance and ritual in the sense of Joseph Beuys' performance art. Kaiser cites descriptions by a number of contemporaries, both supporters and critics, who attested to Steiner's talent and power as a speaker but not as a demagogue. The effect of his performances on stage was in a positive sense more akin to the technical skills of a great actor, which is not to say that he was 'merely' playing roles. People testify to his sincerity and lack of egotism – he was not merely performing, but rather enacting a profound form of communication, which seemed to read the mood and responses of the audience and at its best, let them feel uplifted *and* free (Davy, 1993).

Grand narrative or grand narratives?

Bruce Uhrmacher (1995) suggests that Steiner appealed to people seeking a grand narrative, though his 'narrative' is not as grand nor as coherent as some of his biographers make out. Like my comments on Steiner's language skepticism, the same applies to much of his work as represented in the 380 volumes of his complete works. These are perhaps better seen as numerous attempts, at different times for different audiences, to give answers to specific questions – another place, another group of people, other questions, other answers – not because he wasn't certain, and certainly not because he wanted to tell them what they wanted to hear. Rather, the different audiences represented and perhaps needed different perspectives.

Steiner was also highly eclectic in his reading and read in several languages including English (Paull, 2018), and he clearly often took on board what he read. His interests often reflect the different circles he moved in – literature, art, philosophy, language, various sciences, medicine, religion, the 'social question', agriculture and so on. The breadth of his interests and activities was remarkable, as were his achievements in many of these fields, and of course they were all connected through his own spiritual experience. But even if many of his narratives were taken up by his followers as truths, my reading is that they were intended as contributions to an ongoing discourse, rather than readings from a fixed and comprehensive system. Of course, there are also texts in which he makes a systematic presentation of certain ideas and tries to explain each step clearly, but even some of these were revisited, amended and re-explained. For the field of education, as I argue in this book, he described an anthropology including the interactions between body, psyche and spirit, and following on from these a series of generative principles can be formulated.

Uhrmacher (1995) argues that one of Steiner's most important contributions was in integrating body, psyche and spirit in ways that Western thinking had failed to do

since classical times. Western culture, he suggests, has lost its faith in the knowledge that the body and the bodily senses are also paths to spirituality, and has lost touch with higher meaning because it places its faith in rationalism, empiricism and abstraction. Phenomenology and more recently feminism have drawn our attention back to notions of embodiment. Steiner's approach makes it clear that the body is equally important to human development and individuation; indeed, emancipation processes start with bodily emancipation. Uhrmacher argues that,

> by offering a somatic bridge between Self and Other, Anthroposophy in essence will not go away. That is, even if the movement were to die out, the general ideas or underlying spirit would be subsumed under some other group or organization.
>
> (Uhrmacher, 1995, p. 400)

This idea, whose time has come, is like the genie that cannot be put back in the lamp. Steiner, in a sense, makes all three dimensions of the human being – body, psyche and spirit – legitimate fields of inquiry. The idea of the reality of all three dimensions, body, soul and spirit – Steiner's central message – is the new narrative that opens up to many more. All other fundamental ideas flow from this, once we have grasped what it implies. Recognizing that the growing child has to integrate her soul/spiritual part into her body, that the soul emerges and emancipates itself from the living body and that recognizing the self as the emergent locus of agency, reveals what education is actually about.

Key ideas

- Steiner was born with natural clairvoyance. This set him apart from most other people in his environment. He was precocious in his interests and reading, showing considerable interest in philosophy, the arts, science and politics.
- It is significant that he grew up and lived between several worlds – spiritually, socially, politically and culturally – and was an active participant in each of these worlds.
- Like many of his contemporaries, Steiner sought to interpret modernism and offer remedies for the social ills of his time as he saw them by trying to make the spiritual and unconscious dimension of life once more relevant and accessible.
- He identified his own life task as finding a way to expand the knowledge process to include understanding of the spiritual dimension of life and to apply this knowledge to countering materialistic perspectives, addressing the challenges of his time, addressing issues of social health and justice and mapping out paths towards emancipation.

An epistemological and philosophical basis for Waldorf education

Steiner's ideas on education are rooted in his early epistemological and philosophical works (Steiner, 1963a, 1963b, 1968), and there are two aspects of this I wish to draw attention to. The first is the basis on which teachers construct knowledge about their pupils, their pedagogical practice and the whole educational context.

The second point relates to the learning processes of pupils and what it says about children and young people's learning and development. Learning is not merely assimilating prescribed and pre-packaged information but rather the generation of knowledge by the learner. In order to address these points, I will first briefly summarize Steiner's theory of knowledge and the ethics that belong to this theory.

Clearly, this is a complex field, and space does not permit a critical discussion of this epistemology and philosophy. The best I can do is to explain what Steiner's theory of knowledge means in practice and refer to the philosophical discussion of this that has been done elsewhere (in English by Barfield, 1988; Bortoft, 1996; Welburn, 2004; Macdermott, 2009, 2012; Schieren, 2011, 2012, 2019; Dahlin, 2013, 2017; Heusser, 2016 and in German by Schneider, 1982; Gabriel, 2014; Wiehl, 2015; Wagemann, 2016; da Veiga, 2016).

Canon A.P. Shepherd's (1961) biography of Steiner was called *Scientist of the Invisible*. This title was apt because Steiner sought to develop a scientific approach to the non-sense-perceptible world, which he experienced as being in every way as real as the material world. He was a scientific explorer like Humboldt, Darwin or Wallace and he reported on his journeys to what, for Europeans then, were also unknown and new worlds – though for the people who inhabited those 'unknown' worlds, they were, of course, deeply familiar, just as there have always been people who are 'indigenous' to these spiritual dimensions. Also like these famous explorers, Steiner's aim was not just to catalogue what he found but to use his discoveries to try to answer the bigger questions of origins, being, meaning and knowledge.

Steiner eventually used the term *Anthroposophy* to describe his scientific approach to spiritual reality, having previously referred to it as theosophy and spiritual science (*Geisteswissenschaft*). He didn't invent the term *Anthroposophy* – which combines the ancient Greek *anthropos*, the human being and *sophia*, knowledge or wisdom. It had various antecedents going back to Thomas Vaughan in the 17th century. Steiner defined Anthroposophy as "a path of knowledge, to guide the Spiritual in the human being to the Spiritual in the universe" (1973), and it communicates knowledge that is gained in a spiritual way by going beyond the boundaries of everyday consciousness and scientific knowledge based on sense perceptions. Anthroposophy is therefore a science of knowing about the spirit, through the spirit, as higher forms of thinking and consciousness are deemed by Steiner to be spiritual. For knowledge about the sense-perceptible human world, he said, we can rely on the methods of science, which today in the human sciences are far more developed and varied than in Steiner's day.

Steiner sought to bridge the gap between the conventional scientific approach used in the human sciences of his day, such as psychology, anthropology or sociology, to understanding the human being, which he lumped together under the name of anthropology, and Anthroposophy as the science of the spiritual dimension to human beings. Steiner describes the relationship between anthropology and Anthroposophy as complementary rather than antagonistic, pointing out that they approach the same phenomena from two perspectives: one through what the senses offer us and one using intuitive thinking. In practical terms, there may well be limits to our knowledge but in principle, these can be overcome. He argued that the boundaries between them are flexible.

Walach (2017) has argued that even conventional empirical research does not rely only on what the senses reveal to us. In order to make sense of the mass of data using

the familiar method of induction, we need creativity to find the meaningful patterns in order to create a theoretical model:

> the creative process is in fact the most important of all. Without it, we have a heap of meaningless data. Normally we have powerful theories that help us sort the data we collect. But where do they come from? . . . They are 'invented' in the best sense of the word: a creative scientist "finds" the theory in an inventive, creative process.
>
> (Walach, 2017, p. 12)

Walach relates this process to what the philosopher Charles Sanders Peirce called abduction, a process of adopting a satisfactory explanation of a phenomenon, which can then be subsequently tested (Douven, 2017). Walach points out that this closely resembles spiritual insight or experience.

Knowledge as a productive activity

According to Steiner, in his earliest philosophical work, the human being

> is not a passive onlooker in relation to evolution, merely repeating in mental pictures cosmic events taking place without his participation; he is the active co-creator of the world-process, and cognition is the most perfect link in the organism of the universe.
>
> (1963b, p. 297)

Unpacked, this statement (and what follows in the text) contains a number of radical ideas; we are not spectators of a reality 'out there' that is in a complete and stable state, waiting to be discovered, captured, mastered, named and applied. Nor do we merely mirror 'in here', i.e. cognitively, what is 'out there'. By participating in the world through the knowledge process, we change both ourselves and the world. Knowledge is therefore neither objective nor is it entirely constructed by the human mind and therefore entirely subjective. The process of generating reality occurs in cognition, which is partly the activity of the person but involves a part of the person – the I – that is embedded in the world conceived as an integrated organism. Thus, the process is both subjective and objective. Human beings participate in the world and contribute to world processes. The phrase 'perfect link' (*vollendetste Glied*) suggests that the knowledge process is complete at the point of intersection and articulation between the world as given, and the human being as meaning-maker. The act of knowing is a process that brings what is incomplete to completion. What is separated up to this point are appearance (and our experience of this through the senses) and meaning. The intersection that occurs in the knowledge process is between the human I, as spiritual core of being and the world processes, which are spiritual, transcendental or divine in nature. The link is the human I, which is both embedded in the spiritual whole, though usually in an unconscious state, and embodied in the physical organism, with the potential for consciousness. Thus, the human I is the point at which the cosmos, the whole, has the potential of becoming conscious of itself, though in an individualized way. This is a holistic, relational view of the world as a coherent totality.

Knowledge has a historical and evolutionary dimension and is co-constructed. Because it evolves, and because human beings are involved in creating it, the world content as lifeworld also includes the outcomes of socio-cultural processes *and* world processes. Co-creating suggests the existence of a creator other than human beings, namely the world itself as given.

In the course for the teachers in the Waldorf School (1996c), that I refer to as the *Foundations* (from the English title of the book *Foundations of Human Experience*, 1996c), Steiner returned to this idea when he insisted that the human mind is the stage upon which cosmic events unfold. The teacher is not merely a spectator watching the drama but is an active protagonist. What does it mean to be the stage upon which people act and events unfold? In order to understand pedagogical phenomena, the teacher has to participate and experience the events not only from without but from within. It is an intensified form of listening, opening to the other and holding back of judgement, whilst also observing the effect this has on one's own thoughts, feelings and intentions, because this process reveals something of the meaning of the situation. Instead of holding experiences at arm's length in a detached way, the teacher has to generate knowledge that is both objective and subjective, both open to the other and open to one's own responses. It requires an essentially hermeneutic or interpretive approach. Apart from anything else, it also requires the teacher to look at her own role in bringing forth the phenomenon.

If a child is behaving in a way that disturbs the lesson, the first task is for the teacher to reconstruct empathically what it feels like to be the child in that situation, including what effect the teacher's behaviour has on the child. Rather than externalizing the 'problem' and locating it in the child, the pedagogical task is to observe how this plays out in oneself and then within the whole situation. It also means that we are changed by the encounter. The purpose of generating knowledge is not academic, but practical; we hope that the process brings balance back to the situation. In other words, this knowledge process can be a therapeutic one for all involved, through which growth can occur within an ethic of care. We do not construct the child's behaviour – accurate observation will show us that child is really doing something. The behaviour is objective but our response to it is subjective. But our subjectivity can enable us to experience the situation from the child's perspective, at least to some extent. Understanding is a process of entering into an iterative relationship with the object of our interest, knowing that we are part of the context that may be a factor in bringing the phenomenon into being in the first place. We may not be the direct cause, but we are part of the whole context and possibly part of the solution. As the quotation earlier from Steiner goes on to say, high levels of reflexivity are required if we are to base our actions on the insights we gain through knowledge.

This has significant consequences for human agency. As Steiner puts it:

> This insight has the most significant consequences for the laws that underlie our deeds, that is, our moral ideals; these too, are to be considered not as copies of something existing outside us, but as being present solely within us. . . . Our moral ideals are our own free creations. . . . This, of course, is only valid when our power of thinking penetrates – with complete insight – into the motivating impulses of our deeds.
>
> (Steiner, 1963b, p. 296)

Steiner's epistemology is also an ethics. We can act in freedom when we have full insight into our motivations and, one can add, into the likely consequences of our actions.

Steiner's first published works were related to his task of editing Goethe's scientific works and a theory of knowledge implicit in Goethe's world view (1886/1968). The passage quoted comes from the published version of his PhD dissertation (1892/1963b). He followed this up with the publication of *The Philosophy of Freedom* (1894/1963a). In these works, he developed an outline of how knowledge can be generated and what implications this has for our actions. I will briefly summarize this.

The world that appears to our senses is not the full picture. The phenomena we experience also have a non-sense-perceptible dimension by virtue of being embedded in a whole world network of relationships. From the human perspective, the process of generating knowledge is a productive process of engagement in which a partial reality is completed. It is an epistemological position that assumes that the world really exists, though as we experience it, it is without meaning, without context and coherence. Only human beings can give it meaning through their cognitive activity by progressively relating the isolated parts to ever larger contexts within the whole. Steiner says that the sensory world appears to our consciousness in a semi-real form because in the process of sense perception something is suppressed, extinguished or filtered out (Steiner uses various metaphors) by the way our organism is structured, but the I can recreate and restore reality if it develops the relevant powers.

What we normally perceive through the senses is only the appearance of reality, not the full reality of the appearance. What is missing is the understanding of how this portion of the world relates to the whole. This is what makes us spectators; we only see the world from the outside.

> Perception is therefore not something finished, not something self-contained, but one side of the total reality. The other side is the concept. The act of knowledge is the synthesis of perception and concept. Only the perception and the concept together constitute the whole thing.
>
> (Steiner, 1963b, p. 109)

Steiner described the concept "as a principle according to which the disconnected elements become joined into a unity" (1963b, p. 345). The percept is the mental image we make of what our senses mediate to us. A concept is an organizing principle that brings the initially incoherent parts of a percept into focus and coherence. An idea is a larger context and incorporates several concepts. So we can understand the thinking activity associated with the concept as an organizing activity that structures the chaotic experience and brings it into relationship with other ideas. This is akin to Peirce's notion of abduction.

In the fully developed process of knowledge, what is missing in the percept is intuitively supplied by the I. It can do this because as spirit, it is embedded in the world as well as being anchored in our body. The I is active in the psyche in the processes of thinking, feeling and willing. In thinking it is fully conscious, in feeling it is semi-conscious and in willing it is unconscious (though still active).

Steiner describes the process of generating knowledge as follows:

> . . . in contrast to the content of perception given to us from outside, the content of thought shines forth in the inner being of the human being. The manner in

> which content of thought first appears, we call intuition. Intuition is for thinking what observation is for perception. Intuition and perception are the sources of knowledge. An observed object or event is foreign to us as long as we do not have in our inner being the corresponding intuition which completes for us that part of reality which is missing to the perception.
>
> (Steiner, 1963a, pp. 112–113)

Once we have an experience and have constructed a percept of it, we then need to "penetrate with organizing intellect the endless manifoldness of shapes, forces, colours, sounds, etc., that arises before us. We try to become clear about the mutual interdependencies of all the single entities confronting us" (Steiner, 1988, p. 20). All conceptual understanding comes as a result of thinking. Thinking is what mediates knowing. Thinking "approaches the given world-content as an organizing principle" (1963b, p. 349). Thus, according to Steiner, the core act of knowing involves intuitive thinking matching *concepts* derived from the world of ideas to the percepts we have constructed of our sense experiences. Bo Dahlin (2013) sums up the process in a formula: experience + concept = reality. However, this reality is not an absolute truth. Indeed, it is a relative truth, and there are many truths that are valid. In order to understand this, we need to proceed with Steiner's account.

Following Goethe, Steiner saw thinking as the inner experience *within* the experience. He refers to the activity of thinking, using the metaphor of translator, interpreter or intercessor (in German *Dolmetsch*) of experience that gives meaning to the gestures of experience, a process through which, "the mute percept suddenly speaks an intelligible language to us" (1968, p. 52). Thinking transforms the chaotic nature of our relationship to the mental images we have made of the world into systematic understandings:

> this means that thinking approaches the given world-content as an organizing principle. The process takes place as follows: Thinking lifts out certain entities from the totality of the world-whole. In the given nothing is really separate; everything is a connected continuum. Then thinking relates separate entities to each other in accordance with the thought-forms it produces, and also determines the outcome of the relationship.
>
> (1963a, p. 349)

Steiner says that the source of concepts and ideas is the human world of ideas and that their validity is self-evident. He gives the example of the concept of a triangle, which any person can think. As Steiner puts it, "The one undivided concept, triangle, does not become a multiplicity because it is thought by many. For the thinking of the many is itself a unity" (1963a, p. 108). He says that the concept of a triangle is universal and self-explanatory. Triangles in the geometrical sense do not exist in the natural world – they are an idea whose validity is to be found in thinking and not in seeing. Schieren (2016, p. 431) speaks of concepts being self-evidential entities founded ontologically in their own inherent laws and are thus objective in character, in the sense that they are not created by the subject but are qualities inherent in the object in its world context. Ultimately, the validity of concepts is their capacity to be related to other concepts in meaningful and coherent contexts; as Steiner put it, "if any single thought appears in my consciousness, I am not satisfied until it has been brought into harmony with

the rest of my thinking" (1988, p. 47). Again, he makes the point that understanding depends on contextualizing experience; "all knowledge depends on the human being's establishing a correct relationship between two or more elements of reality, and comprehending the result of this" (1963b, p. 349) and gradually to become conscious of this, that we become conscious of the performance of the act of knowing.

In order to think *about* the concepts that we produce and to be able to communicate them, concepts need to be 'clothed' in language. However, Steiner insisted, "what a concept is, cannot be stated in words. Words can do no more than draw attention to our concepts" (Steiner, 1963b, p. 76). That is the point – language shows us concepts, makes them comprehensible, usable and communicable. Language has a twofold nature; it is both sense perceptible (the sounds or shapes of the letters and words) and has non-sense-perceptible meaning and both aspects exist at the same time, which enables meaningful perception at least in its original form, such as how children experience it when they first learn language. We lose the experience of this because we are used to thinking in terms of language as signs that represent something else. We get used to assuming that the words stand for the meaning, that the meaning is somehow behind the appearance. Steiner states, "when I hear a sound, the first thing I do is find a concept that corresponds to this observation" (1963b, p. 78). To this we have to add that finding a concept is at the same time a naming of it, e.g. 'door slamming'. Actually, it is a 'reading' of the experience because once we have learned a language, the sensory part, the meaning and the word(s) belong together.

Once we become aware of the experience as an integrated blend of percept and concept and articulate this in words (or images or symbols), this is 'clothed' in culturally specific 'garments' or symbolic forms and can now be communicated. Language both articulates concepts and structures experiences as objects, phenomena and meanings. For Steiner, each language articulates a specific aspect of the Logos, the universal, fundamental and non-sensory, pre-linguistic language of the world order, a stage of language in which word and meaning are the same. This experience with language is probably under normal circumstances only available to young children as they learn their mother tongue. They learn that experience, word and meaning are one, as the young Helen Keller did when she famously learned that the letters w-a-t-e-r spelled out the meaning of running water on her hand. Ernst Cassirer (1962, p. 33f) cites this example to show how we learn to understand the world through symbolic systems such as language. Indeed, he concludes that relational thinking is dependent on symbolic thought, such as language provides. The philosopher Hans-Georg Gadamer's insight that "being that can be understood is language" (2013, p. 490), makes it clear that meaning assumes understanding and that both are dependent on language. Our everyday experience of language disguises this primary experience of language, and intellectually we quickly assume that words represent something. In generating knowledge in the classroom, for example, finding the right words for our experiences is an important part of the learning experience.

Although the process of generating knowledge is intuitive, the act of knowing is not passive. It is an active, intentional, productive and performative act (da Veiga, 2016). In the willed act of observation, in which attention is directed towards the sensory experience that interests us, mental images arise, though without coherence or meaning. It requires a performative act to match the percepts we have constructed with the appropriate organizing ideas so that they are once more meaningful in relation to the whole. In the act of knowing, the subject individualizes an objective concept and thus enters into a subjective relationship to it. She does so in a particular

social, cultural and historical context. In learning, we slow down and structure this process.

This process of progressively embodying experience from both sources – bodily sense experience of the world and the experiences of the ideas-world – and the meaning that experiences have for us, forms the basis of our learning. Learning is about generating knowledge of the world, but it is also about knowing-in-life and knowing-in-practice. Meaningful and meaning-making activities are knowledgeable practice. Whether doing science, writing an essay, sewing a shirt, carving a wooden bowl or playing volleyball; these are all activities that require the ability to know-in-action. In Waldorf education, the emphasis is not on taking in and assimilating existing knowledge but on generating and applying knowledge. The experiences we have doing this form us, shape how we see the world and who we are. Thus, this way of knowing and generating knowing shapes our identities.

Knowing is an emergent, iterative, ongoing process that never ends because there are no absolute limits, not least because the whole itself is evolving and we too are growing and developing. Steiner imagined that, given the right education, people could go on learning life-long – a novel idea in 1919 (1994). In making meaning and acting out of a consciousness of this meaning, we contribute to and change the lifeworld that we collectively live in. The ideas-world is therefore continuously changing through the never-ending resource of people experiencing and acting and translating these lived experiences into embodied experiences as practical knowing. Through the act of knowing, our relationship to the world changes, and the sum of such changes across many people at a particular time contributes to what lives in a particular culture.

What enables us, for example, to see something in a museum and to recognize it as Egyptian, as opposed to Greek? The answer is that the things people in Ancient Egypt did and made in response to their relationship to the world, including the spiritual world, created their lifeworld or culture, and this cultural lifeworld in turn shaped the way people experienced their world and that influenced what they did and made. This gave their culture a particular signature, which reflected their consciousness. The Marxist idea that the conditions we live in shape our consciousness is here folded back into a feedback loop; what we do (and think, feel and say) contributes to the cultural consciousness and this in turn, especially when learned as children through language and participation in social and cultural practices, shapes how we see the world. When we recognize an artefact as Ancient Egyptian, we have recognized that an artefact reflects a particular concept that is the gesture of the consciousness of that people at that time. An Egyptologist will be able to do this with greater insight and can determine which dynasty a work of art belonged to. We cannot know it entirely, of course, because we do not belong to that culture. Thus, the way knowledge is generated, applied and talked about in a school shapes the consciousness of all participants and manifests in a school culture. Speaking metaphorically, if knowledge is pre-selected, pre-packed, marketed, located quickly by a few clicks and then consumed by learners, this creates a learning culture of acquisition rather than one of participation and forming meaningful relationships. This has far reaching implications for education and society.

That is why Welburn (2004) describes Steiner's theory of knowledge as a theory of relationships because it is a theory of participation. Welburn writes,

> In Steiner, the participation of the knower is not a boundary and a limit, but a key to the nature of what happens and how we know it. . . . For by making friends with the world we come to realize, above all, that it is a world suited to produce

> human life, consciousness, etc. Our own presence in the equation, so to speak, is therefore not a blank, a boundary-line or mere unknown: the nature of the world out there therefore does not lie in a 'given' completely other than ourselves, but is a cypher that can potentially be filled in by an understanding of our own grounding to the wider matrix of events.
>
> (2004, pp. 87–88)

Knowledge always includes the relationship to the knower and cannot simply be reduced to knowledge by the subject *of* an object.

Who thinks?

The act of thinking poses the question as to *who* thinks, because it is "through the thinker that thinking is combined with observation. Human consciousness is the stage upon which concept and observation meet one another and become united . . . [it] is the mediator between thinking and observation" (Steiner, 1963a, pp. 78–79). Thus, the capacity to think defines our nature as self-conscious subjects. As Steiner argues, "the subject does not think because it is subject; rather it appears to itself as subject because it is able to think" (1963a, p. 79). Therefore, the being that thinks is neither subjective nor objective but something more ambivalent, or dynamic. Long before contemporary philosophy and sociology questioned the existence of an essential self, in his theory of knowledge Steiner made the self the interpreter, the intercessor. This makes the self vulnerable but not at the mercy of world, because as an interpreter of experience, the self is also a learner capable of development and in particular, capable of learning from the other. When we encounter the other person, when we are given the percept of the other person, we see them and hear them, we feel their physical and emotional presence, and this immediately sets our thinking in activity. The percept is replaced by my thinking, yet as the percept fades, its effect

> forces me – as a thinking being – to extinguish my own thinking and to put in its place the thinking of what is revealed. And this thinking I grasp as an experience that is like the experience of my own thinking. I have really perceived the thinking of the other . . . and I am aware of my own consciousness as little as I am aware of it in a dreamless sleep.
>
> (1963a, pp. 274–275)

Moments later, we awaken to our own thinking, and communication involves an oscillation of sleeping in one's own consciousness and awakening to the other.

Steiner's understanding of the evolution of consciousness led him to adopt a notion of evolution and cultural development that potentially leads to the autonomous, ethical and socially responsible individual, whose freedom is based on her ability to transcend biological instincts, socialization and cultural influences of all kinds to be able to progressively combine percept with concept in order to generate knowledge as a basis for action, and the ideal is to "live in love of the action and to let live, having understanding for the other person's will" (1963a, p. 181). Steiner adds:

> If human nature were not fundamentally social, no external laws could make it so! Only because individual human beings are one in the spiritual part of their

> being, can they live out their lives side by side. The free person is confident that others who are free belong to the same spiritual world as she does, and that they will meet her in their intentions. The free person does not demand agreement from her fellow human beings, but she expects it, because it lies in human nature . . . Which of us can say that she is really free in all she does? But in each of us exists a higher being in whom the free person comes to expression.
>
> (ibid., pp. 181–182)

When we recognize the other's right to be and accept that this is the precondition for our being, a social order can emerge that enables and empowers the other in respectful mutual codependency. This resonates with Steiner's so-called social ethic that says "a healthy social life is found only when the whole community is given shape in each person's mind, and when the virtue of each individual lives in the consciousness of the community" (Steiner, 2011a; see Selg, 2011 for a fuller account).

It is this aspect that leads Gertrude Hughes (2012) to take the view that Steiner's epistemology and ethics of freedom align with much in feminist philosophy. Feminism has taught us of the dangers of universalizing or essentializing, gendering and 'othering'. Universalizing involves making unreflected statements of what is taken for granted, and essentializing puts people into fixed categories that prevent us from seeing the individual case behind the generalization. Gendering is not only about specifying gender into masculine/feminine but also includes all forms of asymmetrical binaries such as us/them, individual/society, objective/subjective or matter/spirit, in which one of the pair has more power or significance. The third aspect is 'othering', which Hughes characterizes as constituting

> a Self by marginalizing some Other. Perhaps the Other is cast as unusually emotional, sensitive, and caring; perhaps unusually exotic or, alternatively, otherworldly, or enticing; perhaps unusually civilized or, alternatively, unusually sexual. The question to ask is, "unusual compared to what?"
>
> (2012, p. 228)

As she points out, each of these attitudes relates to individualism and involves unexamined assumptions.

Hughes argues that Steiner's epistemology counters these three tendencies by arguing against limits to knowledge and by linking cognition with ethical, individual responsibility and freedom. For Steiner, knowledge is linked to power if we do not generate it ourselves. Thinking for ourselves is the best way of avoiding ideas imposed on us by others, however powerful they are. Adopting orthodox knowledge because it belongs to a hegemonic tradition or because a syllabus requires us to learn it, makes it harder to generate new knowledge ourselves. Feminism has helped to show us that what is taught as fact and knowledge has often arisen through a white, male, Western, rationalizing, colonizing and positivist paradigm. Feminist epistemologies complement post-colonial perspectives and also recognize holistic, ecological, relational and intuitive forms of knowledge – what Gidley (2016) calls postformal knowing. Steiner does not deny that at any given moment, our socio-cultural positionality and history and also our bodily organization may in practice limit our knowledge and therefore our freedom, but not in principle – and that is an important point. His is an activist epistemology because it links freedom with the ability to generate knowledge. His is

not a theoretical philosophy but an attempt to establish a basis for meaningful and ethical action.

Freedom is based on the performative act of knowing. In contrast to individualism, which is associated with "egomaniacal virtuosos, ruthless go-getters, rugged survivors and prevailers" (Hughes, 2012, p. 241), individuality expresses itself in the way a person's conduct is based on her intuitions. Hughes reminds us that Steiner defines individuality in *The Philosophy of Freedom* as the sum of ideas which are effective in us, the concrete content of our intuitions:

> Our thinking unites us with the world; our feeling leads us back to ourselves, and this makes us individuals. . . . Each of us combines particular feelings, and these in the most varying degrees of intensity, with her perceptions. This is what remains over when we have allowed fully for all the determining factors in our milieu.
>
> (Steiner 1963b, pp. 126–127)

Steiner deconstructs the binary of subjective/objective by showing that both are valid and necessary aspects of knowing. Steiner's notion of freedom is the activity of individuals taking active responsibility for their actions based on intuitive insight and by treating individuals as unique rather than representatives of a genus, of a collective group. The binary of individual versus society is dissolved in Steiner's social ethic that defines social health as the extent to which the individual has the good of the whole community in mind in her actions and the extent to which the community recognizes and respects the unique individual. This similarly applies to the dichotomy of equality and difference.

In *The Philosophy of Freedom*, Steiner argues strongly against essentializing and stereotyping, which he refers to as genus versus individuality, especially of women. He writes that

> the tendency to judge according to the genus is at its most stubborn where we are concerned with differences of sex. Almost invariably man sees in woman, and woman in man, too much of the general character of the other sex and too little of what is individual. In practical life this harms men less than women.
>
> (1963a, p. 251)

As Hughes comments, "generic thinking erases individuality". Steiner's notion of individuality means the person striving to overcome the effects of instincts, cravings and desires, socialization, education and religious and external moral imperatives. The paradox is that individuality has the same source as universality, namely the spiritual dimension, which is the source of meaning. As Hughes writes, "Our uniqueness is what we have in common. Paradoxically, the realities and processes by which we individuate are universal ones" (2012, p. 248). Thinking is the activity that unites individuality with what is universal.

It is this aspect that shows us the close linkage between Steiner's theory of knowledge and Waldorf education. The process of individuation is central to both. The individuation path of each person is unique and individual, yet it is shaped by general principles that reflect the universal nature of human beings. All human beings are a single biological species, and medical science can treat people anywhere. As far as we

know, learning is a universal process and the same goes for development, yet each individual does this differently. The pedagogical task is, on the one hand, to keep the general principles of human development in mind, whilst trying to recognize the unique signature of each individual – the biographical mythos (Göschel, 2012) – in order to support the process of individuation.

Generating knowledge is an active process of entering into a relationship with the world that shapes our identity, that generates who we are and that applies to children and young people as much as teachers. In Part 2 of this book, I explore how the theory of learning in Waldorf is based on this approach.

Working with a spiritual perspective

A final aspect of Steiner's theory of knowledge is his suggestion as to how the researcher or teacher can work with the ideas of spiritual science – in other words, how to build an epistemological bridge from a science based on sensory empiricism to a science based on intuitive empiricism. Whilst Steiner argued that there are no limits in principle to knowledge, in practice there are. These limits are there because of the way our bodily organization functions today and because of our socio-cultural assumptions and expectations. In earlier cultures, he suggests, people did not experience themselves as detached spectators in the world, and thus spiritual perception was not unusual. There is evidence (Hay & Nye, 2006) that even today, most if not all young children have spiritual experiences until these are suppressed by the "social destruction of spirituality" (ibid., p. 33). However, the state of what Barfield (1988) calls original participation has been replaced by rationality. Thus, there are in fact boundaries to our knowledge.

However, Steiner speaks about boundary cognition; that is, thinking that approaches and crosses the boundaries of experience founded on sense perception. He suggests that such boundary ideas can be a "pneumato-psychic [i.e. spiritual-psychological MR] stimulus, akin to a tactile experience in the sense world" (Steiner, 2010, p. 29). As he put it (in Barfield's translation),

> the soul is able to undergo, and to diversify, the experience it has with ideas it forms at the boundaries of cognition and to learn from them that those boundaries are simply events that occur when the psyche is stimulated by a touch of the spiritual world.
>
> (ibid)

This, he suggests, is the humblest, simplest form of spiritual experience. One is, as it were, touched by the spirit and changed by the experience. In fact, it is not so much the content of the experience, Steiner says, as the energy or forces within the idea that resonate in the psyche. Such experiences are retained in the mind like seeds, and

> the energy they evince in the psyche by virtue of what they are, quite apart from the fact that they reproduce phenomena, can be allowed to act itself out . . . once the mind has been alerted to the presence of such forces, other ideas and representations may also serve, though not quite so well, for further progress in the direction it has now taken.
>
> (ibid., p. 31)

On the one hand, Steiner's description of spiritual phenomena, such as the body of formative life forces, is an abductive, explanatory tool to help explain, for example, the relationship between the organism and social practice; on the other hand, repeated application of this idea disposes the teacher to experiencing the phenomenon.

Seen in this light, the oppression or denial of children's spiritual experiences is tragic. Were they allowed to grow with the child, such seeds of spirituality would dispose the child to subsequent spiritual experiences and the possibility of finding meaning in these experiences. If we rationalize such experiences away or simply suppress them, we are denying children a basic part of their being and increasingly the likelihood of disillusionment. This occurs not only through secularism and learning materialistic concepts. The evidence of researchers such as Hay and Nye (2006; Miller, 2015) is that spirituality is associated with rich experiences in nature and in meaningful activities that generate a sense of wonder, gratitude, respect and love. Robert Macfarlane's (2015) chapter on Childish, the wonderful language of children and their experience of place in nature, offers vivid evidence of their being in the world in ways adult rationality can barely imagine. From an anthroposophical perspective, one would say the encounter with authentic life and nature as opposed to a digital version of it, of being in the presence of other beings, constitutes an environment for what one could call a natural religiosity. Likewise, the moral life of the people we are with, their deeds and thoughts, are a reality that either fosters spirituality or denies and suppresses it.

For teachers, such experiences with boundary cognition can be had by engaging with ideas from spiritual science, such as those Steiner presented in his lectures to the teachers in the first Waldorf School, referred to here as the Foundations. If the teacher takes such ideas, opens herself to them and poses the question "Does this perspective offer me fruitful insights into the pedagogical phenomena I am interested in?", this may offer a new perspective on a phenomenon that can then subsequently be tested in practice using normal processes of critical reflection and analysis.

Johannes Kiersch (1978) has argued that the notion of boundary ideas is crucial to the development of Waldorf education, as most of its key ideas are not based on conventional scientific thinking but instead on spiritual science. If such ideas are not simply to be based on belief but on experience, then teachers must develop the capacity of boundary knowledge in this sense. Steiner, he says, placed astonishing faith in teachers' powers of rational thinking, yet at the same time offered ideas that are not immediately accessible by rational thinking and empirical methods. He suggests that the ability to expand our perceptual faculties and develop living perceptions based on rationally directed and controlled thought experiments in boundary knowledge can be enhanced through meditation and social and artistic exercises, activities that play an important role in Waldorf teacher education. Another way of putting this, as Kiersch (2010) has also suggested, is to use Steiner's anthroposophical ideas as heuristic constructs; that is, as theoretical models that can be applied to interpret phenomena. The generative principles I have developed in Part 2 are another way of engaging with boundary ideas.

My experience in teacher education has been that a hermeneutic process of engaging with texts containing potential boundary ideas can be of great value (e.g. such as Steiner's Foundations). In a preliminary exercise, the teacher students are asked to represent in words or pictures the assumptions and expectations they have regarding the topic before starting with the text. This in effect is their starting horizon in Gadamer's (2013) sense in his hermeneutic phenomenological method, as applied by Crotty (1998). The students then read the text in an empathic mode of understanding. That

is, they try to 'walk a mile in the author's shoes', trying to withhold all judgement and trying to imagine what meaning the author intends and then putting these ideas into their own words. A second dialogic phase of interpretation is to contextualize the text, set it in relation to other ideas, critiquing it and entering into dialogue and discussion with others about it (Rawson, 2021).

Between the first and second stage, the students are asked to distil the essence of the ideas in the text and contemplate these in a simple contemplation exercise, based on Arthur Zajonc's (2009) method of contemplative meditation or on Otto Scharmer's (2016) Theory U. I have described this elsewhere (Rawson, 2010, 2012, 2014, 2018). The third stage of understanding is for each student to represent, in words, gestures or images, in what sense they feel the engagement with the text has in any way changed them, or their understanding. Crotty (1998) calls this a transactional process of understanding. It brings us to what Gadamer called the fusion of horizons. This process has expanded the boundary and is the starting point for further reflections. Through the hermeneutic process, the student/researcher has become more experienced, and this alters the way she now looks at pedagogical phenomena and the next boundary idea.

The process, as Steiner suggested, functions as "a force in the development of the psyche" (2010, p. 31). Steiner's account is hard to follow, but what he says is that the mental representations we make of such boundary experiences, indeed of any experiences, as processes, what they are in themselves, remain unconscious as long as they retain their original vitality. Here we see the analogy of the seed at work. Such representations have to lose their vitality and die down so that we can become conscious of them. What we recall of an experience in the representation is a pale shadow of the original experience (one need only compare our memory of a delicious meal with the actual taste, or a memory of a person compared to being with her). Each perception is an image of what in reality is a living force that becomes benumbed in our construction of it as an image or representation. Boundary concepts, however, "refuse to be deadened" (ibid., p. 34) and become departure points for spiritual perception. Significant concepts and spiritual ideas retain their seed character and have a long-term effect that I would call a disposition. Spiritual ideas dispose us to have spiritual experiences. Ideas that have become part of us in this sense dispose us to see the world differently. Thus, we learn to see pedagogical phenomena differently. This idea aligns with John Dewey's (1938) notion of continuity of experience by which each genuine experience changes the conditions for subsequent experiences.

The consequence of this thought is that engaging with the ideas in Steiner's spiritual anthropology as outlined in the Foundations lectures, or in ideas such as the body of formative forces, for example, and using these boundary ideas as perspectives when we reflect our practice can expand our understanding, can push and extend the boundaries of what we can understand. This is the basic idea of Waldorf teacher education and ongoing continuing professional development based on reflection and research of practice (Rawson, 2020 a & b). The teacher becomes progressively more experienced, her understanding increases, her noticing of what is relevant increases and she gradually develops what Max van Manen (1991) calls pedagogical tact, or the ability of knowing-in-practice, and what Steiner referred to as pedagogical intuition – the ability to know, in the moment, what is pedagogically meaningful to do.

Steiner (1982) suggested that teachers can develop pedagogical intuition using a meditative method, which involves studying the Foundations (not just literally the Foundations lectures but any aspect of his spiritual pedagogical anthropology), meditating on key aspects and then being alert in real situations for what needs to be

done. The 'seed' effect of internalizing boundary ideas directs our attention to what is salient. The idea is that the energy in the generative idea enables the teacher to make an inner connection to the pedagogical situation and directs her attention to what is relevant in the moment she encounters the pedagogical situation. In Steiner's account, the 'answer' to the pedagogical challenge is there, but we need to direct our attention to key percepts. This somewhat vague explanation has been developed into a practical method of contemplative meditation that can be used to supplement both lesson preparation and practitioner research (Rawson, 2018). What is missing in Steiner's account is critical reflection after the experience of intuition. The core idea is that working with the boundary ideas in this way disposes the teacher to notice more, and this manifests as pedagogical tact or intuition.

Key ideas

- The process of generating knowledge is one in which experiences are constructed as percepts which are then given meaning by matching them with concepts, whose source is the world of ideas.
- The knowledge process is iterative and involves expanding understanding by relating the concepts gained to ever wider contexts of meaning and relationships.
- Concepts are initially intuitive but take symbolic or linguistic form when we try to relate them to other concepts and communicate them.
- Knowing is a process in which we continuously bring ourselves into being.
- This theory of generating knowledge is the basis for how children learn in Waldorf schools.
- We can progressively expand our potential for spiritual understanding by embodying boundary thoughts and applying them heuristically to interpret pedagogical situations.
- Boundary experiences dispose us to having new spiritual experiences and thus becoming more spiritually aware.

Steiner's experiences as a teacher and tutor

Even during his youth, Steiner started gathering experiences as a teacher. At the age of 14, he was already giving tuition to fellow pupils, usually in science subjects but also in Latin, Greek and bookkeeping (Steiner, 1928, p. 69), and this continued throughout his own studies at university and beyond. As Zdrazil (2019, p. 20ff.) points out, Steiner was influenced by Karl Julius Schroer, his tutor at university in Vienna, and studied his book on pedagogy. This contained ideas drawn from the classic German Idealism of Schiller, Goethe and Fichte that emphasized the artistic and creative element in teaching, the key role of imitation, the importance of living and cultivated language, the problem of relying on textbooks and a critique of military-style sports lessons.

Steiner was also impressed by a book by the English nursery school pioneer and reformer Margaret McMillan (1860–1931), called *Education Through the Imagination* (1904), which he evidently read in English (Paull, 2018). McMillan was a Christian Socialist who was a leading member of the newly founded Labour Party, and she campaigned for universal suffrage and children's health. She founded the Open-Air Nursery School & Training Centre in Deptford, which Steiner visited in 1923, and she

cohosted the New Art of Education Conference with Steiner in Ilkley that year. There were a number of aspects in McMillan's approach apart from the emancipatory, social aspect that must have interested Steiner. The first was the basic idea that the imagination and consciousness in children is akin to the consciousness of pre-modern people and that the development of an intellectual, rational consciousness should emerge out of this earlier imaginative consciousness when the child is ready. McMillan writes in the preface to her book, "It is through the Imagination that the Reason begins to declare itself" (xi). She adds that in a healthy, unhurried and appropriate development, "the child begins by seeing fantastic resemblance, which is the first step towards perception of real resemblances – that is to say to reasoning through analogy" (xi-xii). That is, the child initially assumes the reality of the 'fantastic' image and then gradually develops a conceptual understanding through images, metaphors, analogies and comparisons – in short, the full spectrum of conceptual metaphors which, according to Lakoff and Johnson (2003), is how people conceptualize complexity, even in science. McMillan summarizes this by saying, "Between creative imagination and rational research there is a community of nature – each presupposes the faculty for seizing resemblances" (1904, p. xii).

McMillan also emphasizes the importance of stories for stimulating the imagination of the child: "a tale is very often for young children a kind of answer to a pressing question" (1904, p. xii). She articulates the view that myths and tales were the attempts by earlier cultures not merely to create an aesthetic form for their experiences but also an effort to explain the origin of things; "they had a logic of images and not of ideas" (ibid., p. xii). This is an idea we also find in the main work of Ernst Cassirer (1962). It is also behind Steiner's suggestions for a narrative curriculum of tales, legends and myths. McMillan's artistic approach also included the realization that drawing, and in particular the drawing of forms (lines, curves, loops, mirrored patterns), is a precondition for children learning to write and that reading is enhanced by the child's imaginative abilities. Many of these experiences and ideas would resurface in Steiner's book *The Education of the Child* (Steiner, 1996b, published in 1907, see later).

Perhaps Steiner's most important pedagogical experience was as a live-in home tutor for the four sons of the family of the Jewish merchant Ladislaus Specht and his wife Pauline – one of whom, Otto, suffered from hydrocephalus. He was seriously limited in his overall development but particularly in his cognitive development, and even minor mental efforts led to massive headaches and other bodily symptoms. After careful observation, Steiner began to understand how to stimulate the cognitive processes through activating the limbs and accompanying this with rich feelings, without exhausting the cognition. It was essential to extract the essence of a particular topic so that Otto could more easily grasp it. As Steiner frequently pointed out after the founding of the Waldorf School, he had to prepare the lessons with great care, spending, as he put it, three hours preparing for 15 minutes of learning. This experience led to his notion of 'soul economy' in learning, which I will return to later. What Steiner learned through this experience was the interaction between mental and bodily processes and how to observe these in the child. Otto's development was such that he was later able to study medicine and became a doctor. In 1914, he was called up at the start of World War I as a military doctor in the Austrian Army, and he died of typhoid in Poland in 1915 (Steiner, 1928; Kugler, 1994, p. 85).

Steiner taught the four Specht boys from 1884–1890, during which time he came into regular contact with the children's doctor, Dr. Josef Breuer, with whom he shared many scientific, medical and cultural interests. Breuer mentored the younger Sigmund Freud and they are both considered the founders of the Vienna School of psychoanalysis. Like Breuer, Freud also started out researching the physiology of the brain and nervous system, mapping out the relationship between the lungs and nervous system and describing the sense of balance. Breuer and Freud collaborated on the publication in 1895 of a set of case studies entitled *Studies in Hysteria*, which laid the groundwork for psychoanalysis. This biographical note on Steiner gives us a glimpse of the kind of ideas that Steiner was immersed in during his twenties when he was part of the Specht family's social circle of Vienna's well-known artists, writers, musicians and scientists.

Steiner left the Specht family to take up a position in Weimar in the Goethe-Schiller Archive, editing Goethe's scientific writings, whilst writing his PhD, which was later published as *Truth and Science* (1892, also translated as *Truth and Knowledge*) (1963b). During the Weimar years, he continued his work as a private tutor. Later in 1897, when Steiner moved to Berlin and worked as a writer and publicist (he edited the *Magazin für Literatur*), he also taught, as we have seen previously, at the *Workers Education School* (*Arbeiterbildungsschule*) founded by the socialists Rosa Luxemburg and Karl Liebknecht. After a long day of work in the factories, workers came to the school for educational courses from 9 p.m. to 10:30 p.m. Steiner's courses in history, literature, economics and practical rhetoric were so successful that the income funded the whole school (Kannenberg, 2011). In 1901, he and Luxemburg opened a new branch in Berlin Spandau. Steiner said that he understood the desire of the workers for liberation and, like Luxemburg, he saw education as the path.

Unlike the Marxists, he did not see the cultural and spiritual life as merely 'Überbau' – a cultural layer on top of the social strata, not fundamental to social and economic life. Steiner saw the human spirit, rather, as the source of human creativity, economic productivity and liberation. He explained alienation as capitalism's denial of the workers' identification with the fruits of their spiritual nature, and the preserving of this 'privilege' for the bourgeoisie, much as early Marx (1965/1844) did. Writing a critique of Marxists such as Lenin and intellectual socialist leaders in 1919, Steiner (1985) argued that the revolution would simply transfer government by the middle class with the same structures to government by the self-appoined intellectual representatives of the proletariat. Indeed, recent historical analysis of the uprising of the sailors of the Imperial German Fleet and the Revolution in Kiel in November 1918 shows very clearly how the workers' and soldiers' council was taken over by the representatives of the new Social Democrat government in Berlin, Conrad Haußmann, and the SPD (*Sozialdemokratische Partei Deutschlands*) party leader Gustav Noske, who claimed the political leadership of the uprising and thus diffused the revolution (Kinzler & Tillmann, 2018). Noske took over as governor of the city and ordered all soldiers and sailors to return to their homes on permanent leave, thus dissipating the uprising whilst inadvertantly distributing men with revolutionary experience and weapons around Germany.

Steiner argued that if the workers were provided with a comprehensive education, they could individually develop the ability both to understand the existing system and to recognize what was lacking. In other words, his commitment was to enabling the self-development of the person, rather than to expressions of collective resistence led by intellectuals. He had more trust in the innate potential of actual workers to

emancipate themselves because they intuitively grasped the processes of transformation that underlie work and productivity. He wrote in 1919,

> the intellectual leaders and agitators of the proletarian masses are blinded by utopean ideas and theories which derive from a social science still based on the old economic concepts that so urgently need changing . . . [they] have not even the faintest idea that their notions about politics, economics and cultural life are in no way different from the 'bourgeois notions' they are fighting, and that at bottom all they are striving for is to see the old notions realized by a new group.
>
> (Steiner, 1985, p. 87)

Steiner's idea was in a sense more radical. It is little wonder that the other leaders in the Workers Education School in Berlin wanted to get rid of Steiner. That Steiner managed to teach for five years at the *Arbeiterbildungsschule* was due to Luxemburg's support for him. Mücke and Rudolf (1955) report that in a vote as to whether Steiner should continue teaching, 348 workers voted for and only 12 against Steiner, though the leadership contrived to ignore this.

Education as part of the cultural domain, free from political or economic determination

Much later in 1919, Steiner (1977, 1985) developed his notion of a threefold structuring of the primary, interrelated but in principle distinct, domains of society – the social, political and rights domain, the cultural-spiritual domain and the economic domain. The domain of rights should be guided by the principle of democratic equality and justice before the law; the domain of culture, science and spiritual life should be characterized by the principles of freedom, autonomy and individual capacity. The economic domain should be guided by the principle of fraternity, or as Dahlin (2017) suggests, socialism in its original sense of the means of production serving the needs of society as a whole – which, as Steiner argued, does not mean collectivization or a centrally planned economy but in fact requires entrepreneurship. The central idea with regard to education is that it should not be determined by economic interests, nor should the state prescribe what should be taught. He formulated this as follows.

> We should not ask: what does a person need to know or be able to do in order to fit into the existing social order? Instead we should ask: what lives in each human being and what can be developed in him or her? Only then will it be possible to direct the new qualities of each emerging generation into society. Thus society will change and become what these fully developed individuals construct through their engagement with the existing order. The new generation should not merely be made into what the existing social order wants them to be. . . . A healthy relation exists between school and society only when society is kept constantly supplied with new and individual potential of persons whose education has allowed them to develop unhampered. . . . What a person ought to know and be able to do at any particular stage of life must be decided by human nature itself.
>
> (Steiner, 1985, pp. 71–72)

Dahlin (2010, p. 50) has called Steiner's educational philosophy of 'Menschenbildung' (the formation of the human being) fundamentally democratic because it aims to realize the potential of each individual, rather than the state imposing a vision of what the individual should become and know, and because it sees the relationship between the individual and society as mutually formative:

> According to this view a democratic society is characterized by making it possible for each individual to develop his or her own innate potential *and then allowing society to develop in accordance with the abilities and the creativity that is released in this way*. This means that the future development of a truly democratic society is, actually, *unpredictable*. The logical consequence of this idea is that schools are to develop the inherent positive abilities of all children, without considering what the state and/or economical agents currently believe that the nation needs.
>
> (Dahlin, 2010, p. 50)

In the same article cited above, Steiner (1985) argues that the state, civil society and the economy will be the beneficial recipients of this new spiritual productivity in the rising generation but not if the outcomes are prescribed, not if the state specifies what needs to be learned in each life phase. Instead, the curriculum should be orientated to supporting emergent human nature. However, this should not imply that education be remote from real life, as the life of the state and the processes of the economy are also born out of the nature of human beings. Indeed, the opposite is the case, Steiner says; it is the prescriptions and rules and examinations that the state insists on that are detached from social reality! Exams only instrumentalize education to enforce a particular mindset that serves the existing economic and political status quo. Steiner is radical in his critique of the educational policies of his day, arguing that people feel themselves so embedded in the existing social structures that they confuse the notion of a general education with the training of young people to be good servants of the state (Steiner, 1985).

The Education of the Child, 1907

In 1907 Steiner published a small book, *The Education of the Child* (1996b), based on a series of lectures he had given on education and on questions related to what children bring with them as dispositions and gifts and what they acquire in life. This short book builds on his major work *Theosophy* (2011b, published in 1904), in which he had outlined a spiritual anthropology and in particular described the interactions between body, psyche and spirit and the implications of this for human life. The book followed Steiner's intensive involvement with the Theosophical movement, his reading of esoteric literature and his own spiritual research and was intended to demonstrate the practical application of theosophical esoteric perspectives. The full title of *The Education of the Child* in German makes it clear that its ideas are based on esoteric studies, which can be seen in its central idea of three 'births' and the structuring of development into three seven-year phases.

As Michael Zech (2017, 2019) has suggested, the hebdomatic (seven-year) model that Steiner introduced has an esoteric background in alchemy, European mysticism, the Kabbalah and other esoteric traditions. Steiner used it in his description of the

seven life processes (1996a) and in his design of the capitals in the first Goetheanum building, which involve a sevenfold metamorphosis. Zander (2016, 2019) accuses Steiner of plagiarizing an eclectic range of sources. Selg (2009, 2014) takes the view that Steiner himself was an initiate in an esoteric tradition that he was explicitly drawing on in his exoteric work. Kaiser (2020, p. 187) cites a letter from Steiner in 1905, in which he stated his intention to give entirely new and public forms to secret masonic teachings, both in terms of ideas and as enacted ritual. One of the ways he did this was in his writings about education.

Steiner also follows the tradition going back to Classical Greece of distinguishing between body, soul and spirit, drawing on Aristotle's distinction between *pneuma* meaning spirit, *psyche* meaning soul and *soma* meaning body, though he also drew on Theosophical terminology and created some neologisms himself. In this book, I follow Barfield's (2010) suggestion of using the term psyche for what Steiner refers to as the soul because in English, soul has many meanings, some synonymous with spirit. Psyche here refers to the activities within the human mind that we call thinking, feeling and willing and includes the realm of the unconscious. I generally treat psyche as synonymous to mind. The psyche emerges in each developing person. In the first seven years (approximately) it is embryonic and is bound up with bodily development. Then it is 'born'. This metaphorical birth relates to the emancipation of a part of the life processes from their function of shaping and tuning the bodily organism. Once freed from this function, these forces metamorphose into the processes that shape the psyche, a person's independent inner life – what Steiner called in German *Seele*. This leads to the distinction between the lived-in body, the psyche as the inner life of thinking, feeling and willing and the spiritual dimension including states of consciousness and the activity of the I.

The Education of the Child blends esoteric ideas, progressive educational ideas (the influence of Margaret McMillan is noticeable, as Wiehl, 2015 points out), his own educational experience and practical observations (such as the references to simple dolls and puppets and wooden toys with moveable parts). Steiner refers briefly to what he detailed in *Theosophy* regarding the fourfold nature of the developing human being, which distinguishes between the physical body, the life-body, the sentient body and the I and describes the 'birth' of each of these 'bodies' as significant transitions in the life course of the child and young person. The central idea is that the I, as spiritual core of being, works as a force to educate, edify and ennoble the other constituent bodily and psychological processes within the human being, that is a process of self-formation or Bildung.

Steiner's notion of the I is highly complex, even paradoxical in being both individual and universal, both central and peripheral. It helps to think of it in this context as the individual source of agency. It is the subject of the sentence, "I grow and develop". The I is active in the fields of the body, its life processes and in the psyche, individualizing each over the course of the child's development from birth to adulthood, when the process continues beyond the remit of education. This is a process of individuation or self-formation, in which the I interacts with its inherited body, socialization, education, culture and the brute facts of life (i.e. things over which the individual has no control, such as accidents, sickness, poverty, war, natural disaster). The I is not the product of these processes, as most theories of personhood explain today. The I is a separate, pre-existent entity with its own intentions and dispositions. It emerges as individuality through engagement with the other, literally other people but also the world as other, and continuously strives to construct coherent, stable identities.

The unique human I expresses itself, following Steiner, in the way a person seeks to develop her potentialities, or how she relates to self, others and world and how she can exercise agency within the constraints and affordances of the situations she finds herself in. The ethics of care of Waldorf education means teachers recognizing, valuing and supporting this process, in the sense articulated by the philosopher Kwame Appiah, who describes the aims of ethics as answering the question "What is it for a human life to go well?" (2019, p. 177). Waldorf education seeks to help life go well for the pupils. The ecological, developmental approach of Waldorf education recognizes that children undertake this process of *becoming* whilst being embedded in nested 'scapes', starting with the body-scape and the child-scape of the domestic environment and mother tongue, followed by the institutional scapes of kindergarten and school, then the task-scapes and work-scapes of life in the world (Bransby & Rawson, 2020). The metaphor 'scape' comes from landscape, as a humanly constructed concept of space, originally derived from a theme in early modern Dutch painting. Teachers try to read the biographical of individuality in *how* the child and young person engage with the developmental tasks that they encounter in these 'scapes', through the crises that arise, in their relationships and achievements.

Steiner's argument in *The Education of the Child* is that it is possible for the I, as the spiritual core of the human being, through self-development to overcome instincts, unconscious dispositions, drives and desires and to behave in a conscious, socially responsible, reflexive, ethical and altruistic way. The I is thus the agent of self-formation or individuation. The task of education is to support the development of the I by educating the body, the life body of formative forces and sentient soul, in its thinking, feeling and willing. Steiner's perspective at the time he wrote the book was on the spiritual development of the person, which is why he continues the description of spiritual development from the I to higher forms of consciousness, noting that cultural development has also followed this trajectory. He also makes the analogy between human evolution and the development of the child. This parallel is a central part of Steiner's evolutionary and monistic world view. Just as culture emerged out of human evolution, so too does the I bring forth an autonomous self through learning and development. There is no dualism of body and mind, and he refutes the idea that the body/brain at some point in evolution brought forth higher human cognitive abilities that manifest in the emergence of tools, language, complex social structures and symbolic systems such as art (see Rawson, 2003). Steiner's idea, somewhat simplified, is that the spiritual world, in which everything is related, comes to collective cultural consciousness and then increasingly to individual consciousness in each person. In mythic terms, humans gain access to the wisdom of the gods, but are cast out of original participation and lose their implicit sense of meaning.

Owen Barfield (2011, pp. 69–71) makes a similar case, based on his study of language as a source for investigating the evolution of human consciousness, that mind cannot be understood merely as the reaction patterns of a stimulated organism. I would like to cite Barfield's conclusion, though for space reasons, without reiterating his arguments:

> the whole character and history of language cries aloud to us: namely that the prevailing assumption that matter preceded mind in the history of the universe is a historical fallacy, and, alas, one with far-reaching consequences. It becomes clear to us that, both ontogenetically and phylogenetically, subjectivity is never something that was developed out of nothing at some point in space, but is a form

> of consciousness that has *contracted* from the periphery into individual centres. Phylogenetically, it becomes clear that the task of *Homo sapiens*, when he (sic) first appeared as a physical form on earth, was not to evolve a faculty of thought somehow out of nothing, but to transform the unfree wisdom, which he experienced through his organism as given meaning, into the free subjectivity that is correlative only to active thought to the individual activity of thinking.
>
> (Barfield, 2011, p. 79)

This, in a nutshell, is not only the 'task' of cultural evolution, but also that of education: namely to support the emergence of free subjectivity, emancipated from biological, social and cultural determinants. It is this 'plot', telos, mythos, trajectory or direction of each individual biography, that reveals the progression *from-to*, not from savagery to civilization, from lower to higher, but from unfree, because unconscious, states to free and conscious states of mind and action. This is the basis for a good life. Living the good life involves the development of practical wisdom, what Aristotle called phronesis, and, as Schwartz and Sharpe (2011) have contextualized in modern life, this depends on performing a particular social practice well, "figuring out the right way to do the right things in a particular circumstance, with a particular person, at a particular time" (ibid., pp. 3–4).

In *The Education of the Child*, Steiner describes the development of the physical body, the life body and the sentient body, pointing out that the art of education is based on detailed knowledge of human nature, including those non-physical aspects. He says, "to work a machine you must approach it not with phrases and truisms but with real detailed knowledge", the analogy perhaps reflecting his experience of technology in childhood and his subsequent scientific and technical education. An effective education requires knowledge of the various constituent parts of the whole human being, body, psyche and spirit.

Steiner's spiritual anthropology

Because Steiner's description of the constituent parts of the human being are central to Waldorf education, I offer here a brief summary of what in *The Education of the Child* is only alluded to, but which he gave a systematic account of in his book *Theosophy*. The human being can be analysed as comprising three interwoven but ontologically independent domains with their own specific structures (which Steiner refers to using the metaphor *body*): the physical body, the life body and the astral or sentient body, and the I. The physical body is the only domain that is accessible to sense-based empirical study; the other three domains are only accessible to us through spiritual activity in thinking. The I as the spiritual core of the human being permeates these three domains and comes progressively to embodied expression through them. The physical body is what is known to medical science. Steiner drew attention, for example, to the significance of our upright posture, both in terms of human evolution – something that has broadly been confirmed by paleoanthropology in the last 50 years (Lieberman, 2013) – and in terms of the development of the child, as an expression of the I-forces in the physical body.

From an evolutionary perspective, human beings have a common biological ancestry with all other living beings as manifest in the physical body. Throughout the evolution of organisms there has been a trend towards increasing internalization of key functions such as respiration, reproduction and thermoregulation (Rosslenbroich, 2014),

which enables the organism to have a greater degree of independence from its environment. Furthermore, the evolution of primates and our hominid ancestors involved a trend towards uprightness, with human-like feet, legs and later hips, followed by the arms and hands and finally the head and increased brain size (Lieberman, 2013, Schad, 2015). This biological development ultimately afforded the cognitive development of our ancestors that culminated in the emergence of those quintessential human faculties – language, symbolic thinking, art, culture, complex tools and social structures and the mastery of practically all environments on earth (Gamble et al, 2014). As mentioned earlier, however, this process is not one of physical substance evolving consciousness out of itself, but rather of the capacity for consciousness and eventually self-consciousness manifesting in and through the bodily organism; rather, it is a twofold process and co-evolution involving an internalizing and focusing of consciousness (Tomasello, 2019, Rawson, 2003).

Steiner also made a fundamental distinction between the nerve-sense system, the rhythmic system comprising breathing and the blood circulation, and the metabolic (e.g. digestion, sexual organs) and limb system with regard to the body's relationship to the environment. Furthermore, he identified the three primary psychological processes of thinking, feeling and willing as being primarily active in one of the three bodily domains – thinking is associated with the brain/nervous system, feeling with the rhythmic system and willing with the metabolic-limb system. This threefold correspondence can be summarized as follows in Table 1.1.

Table 1.1 The correspondences between the bodily, psychological and spiritual functions

Bodily functions	*Psychological functions*	*Relationship to the world*	*State of consciousness*
Metabolic-limb system	Willing	Embedded in the world	Unconscious/sleeping
Circulation-respiratory – rhythmic system	Feeling	In rhythmic interchange with the world	Semi-conscious/ dreaming
Brain and nervous system	Thinking	Isolated from the world	Waking consciousness

More challenging is Steiner's description of the life-body, which organizes, shapes and forms the body from a genetic blueprint. The metaphor of body suggests a coherent, autopoietic, integrated system of processes and the life-body is the organizing principle at work in the physical body. The life-body is variously referred to by Steiner as the body of formative forces, the etheric body, the habit body, the body of life-processes, the memory body, the bearer of life rhythms and cycles and the temporal body. Each of these terms highlights a particular aspect of the phenomenon in a different context. The life-body is present in all living organisms and is absent in all dead ones. Its ultimate withdrawal or dissolution marks the point of death, when the forces of physical disintegration take the upper hand. From an anthroposophical perspective, the etheric body of life forces does not die but disconnects from the physical, which then disintegrates into its constituent physical elements – water, gasses and minerals. The etheric body then returns to its source in the larger rhythms of life and nature.

Anthropologists and ethnologists (e.g. Ingold, 2000) have documented that many pre-scientific peoples experienced this moment of life departing in humans and animals. It is

part of the cycle of birth and death, which many pre-modern people experienced not as a beginning and an end, but as Ingold (2000, p. 143) puts it, borrowing a phrase from Deleuze and Guattari, a "matter of coming and going rather than starting and finishing". What goes for hunted animals applies to humans; they may no longer be present, but they continue to exist in some form; "co-presence may be temporally bounded, but existence is not. Or to put it another way, the past may be absent from the present, but is not extinguished by it. Death punctuates, but does not terminate, life" (Ingold, 2000, p. 143). Many modern, urban people may also experience death in this way, but may lack the terminology and cultural frame of reference to understanding the experience.

The body of life processes (i.e. the etheric body) organizes and regulates the processes that animate the living body, such as thermoregulation, metabolism, respiration and so on. The term autopoiesis (Rose, 1998) was coined by biologists to explain the self-organization and thermodynamic regulation of an organism over time. The temporal organization that gives life its rhythmical, cyclical structure has been established by the field of chronobiology, which shows that temporal structures are hugely complex, occurring in micro and macro time frames, and are highly flexible (Heusser, 2016). Wolfgang Schad (2003) has claimed that chronobiology is in fact the science of the etheric. Whilst the physical body is essentially a spatial phenomenon, the etheric body (life body) is a temporal phenomenon. Physical and etheric, however, always work in an integrated way in life. Whilst the etheric forces shape the bodily rhythms, it is 'educated' by adopting social habits, dispositions and practical procedures.

In Steiner's anthropology, the body of formative forces 'flows' into the embryo from the environment via the mother's body and after the physical birth, from the natural and social worlds the child is embedded in. In their original form, these forces are the same processes to be found in all organic life, namely regulatory processes relating to respiration, nutrition, growth and reproduction. In human beings these forces are socialized because the human child is always embedded in a human community with culturally specific patterns of behaviour, habits, language and social practices. Originally the body was attuned to natural rhythms (diurnal, lunar, seasonal, etc.) through the life processes, but these are overlain by layers of social and cultural patterns. Though he would probably deny the analogy, the sociologist Pierre Bourdieu's notion of habitus is close to Steiner's idea of an etheric, habit body. Bourdieu links habitus to the biological body as something that has been "durably incorporated in the body in the form of permanent dispositions" (Bourdieu, 1980, cited in Maton, 2012, p. 55). Bourdieu (1992) sees the notion of habitus as a generative principle (I borrowed from him) rather than as something that can be empirically identified. Bourdieu coined the term habitus to resolve the tensions between social regulation of behaviour and individual agency. Habitus is structured by experience of social structures; it structures behaviour and thus influences social practices. Habitus "expresses first the result of an organizing action . . . it also designates a way of being, a habitual state (especially of the body), and in particular, a predisposition, tendency, propensity or inclination" (Bourdieu,1977, cited in Maton, 2012, 50).

Steiner links habits and dispositions to the life body, the structuring processes within the body. Habits and dispositions are embodied behaviours, attitudes, ways of seeing and doing and are learned through experience. The image of embodiment implies something being *put into* the body and leaving an impression. Experiences, which are usually multi-sensory, are 'impressed' into the life body as memories, often in the form of patterns or structuring structures. The life body is therefore the bearer of memories of bodily experiences and the intentionality, emotions, feelings and thoughts that

accompany them. In order to recall something that we have experienced, we have to reactivate the patterns and reconstruct the memory as a mental image. Of course, most of the body being referred to here is the brain. The activity of the brain, rather than its substance, is the life body into which patterns, pathways and maps as processes are committed to memory in neural networks (Damasio, 2012).

The body of formative, etheric forces is by no means a fixed and constant entity. It is mutable, and for that reason it is essential to the learning process. The human life body is both highly socialized and individualized. Steiner's astonishing idea is that, in my words, the body of structuring processes not only structures the organism it enlivens, but once the process of formation is largely completed and the basic patterns and structures are established in the brain – for example, patterns of behaviour, language, habits and dispositions – then these forces can be directed to shaping the psychological processes of the mind. This includes generating mental images in response to sensory experience, constructing memories which can then be manipulated, sorted, combined and developed when we think about something or try to solve problems. Up until this transition, the body learns through what it experiences, and these experiences are directly woven into the life-body/physical structures, that is, into neural networks that form the basis for behavioural patterns. Sense impressions are more or less embodied without reflection. Language learning is the most obvious and also the most complex example of mimetic learning before the second dentition, which Steiner says marks the conclusion of the first major phase of development. Children learn their native language(s) by participating in social practices accompanied by talk. They do not learn by conscious reflection, learning and applying rules or memorizing lists of words but by observing and participating, whilst unconsciously noticing structures and 'correct' forms. We learn language by using it (Tomasello, 2003).

In terms of Waldorf education, after the second dentition the child begins to develop an embryonic autonomous life of soul/psyche that is later emancipated at the age of puberty in a third 'birth', following the physical birth and the birth or emancipation of the life body. In other words, the formative life processes now become the basis for learning, and they enable the individual to gradually build up an inner life of thoughts, feelings and intentions. The third 'birth' at puberty leads to the individualization of the psyche. The main reason for withholding formal learning and instruction in literacy and numeracy in Waldorf schools until the age of 6 years is to allow the life-processes to complete the initial structuring of the brain. Only then can they healthily be directed to abstraction; to do so prematurely would weaken the child's constitution by deflecting the life-processes away from the physical organs thus leading to ill-health in the long run. Waldorf education has been following this approach for 100 years and the evidence (as I discuss in Part 4) is that Waldorf pupils and alumni shows no negative consequences in their learning and academic progress, despite starting formal learning later than children in other school systems. However, proving causal relationships between early academic learning and long term health issues is unlikely to be possible given the number of factors involved. The relevance of this approach only makes sense within this overall holistic theory of learning. My personal sense is that children need time to gather rich direct and imaginative experience of the world before literacy filters this experience. Orality is the living hand in the glove of literacy, as Barry Sanders (1994) memorably expressed it.

As we have seen Steiner structures the development of the child into three approximately seven-year stages, from birth to the change of teeth and from there to puberty. These seven-year phases have played an important role in traditional presentations of

Waldorf education. Today it is more appropriate to see them as institutional transitions that take account of the ideal-typical maturation of children and young people that manifest in significant changes in the relationship on the one hand between the body, psyche and individuality and on the other, between the person and social world. Each phase requires a different pedagogical approach; early years up to the change of teeth, childhood to puberty and then adolescence. Given what has been said above about the transformation of part of the life-processes into learning processes at the second dentition (Rawson, 2018), respecting the nature early childhood maturation during the first six years is probably the most important of the stages, in terms of supporting healthy development.

During the first phase of development up until the change of teeth and partial emancipation of the life processes, the predominant modes of learning are imitation, emulation and mimetic learning. Steiner writes, "Children imitate what happens in their physical environment, and in this process of imitation their physical organs are cast in forms that thus become permanent" (1996b, p. 18). This refers to everything that can be perceived by their senses and includes "all moral or immoral action, all wise or foolish actions that children see" (ibid., p. 18). The educational implications of this are profound. As Nicol and Taplin (2017, p. 51) put it,

> all adults have a great responsibility when in the presence of children and have the opportunity to contribute to their healthy development. In response to children's unconscious commitment to imitate their surroundings, adults must be fully conscious of all actions in the presence of children because, from the young child's position, the adult will always and inevitably be modelling how human beings behave.

Another consequence that Steiner highlights is that children actively engage with the world with their limbs but also with their imagination. Rather than providing them with toys that are complete in all details, they should have simple objects and complete these with their imagination. In this way the child's brain, according to Steiner, is modelled in a plastic way, not with completed images but with patterns that can be continuously modified. This means that the life forces, that in this age are still shaping and forming the organs and in particular the brain, embody flexible and mutable patterns, capable of adaptation. The child enters into a reciprocal relationship with the world through participation and in so doing, establishes basic structures and relationships.

In *The Education of the Child*, Steiner makes the case that in the first phase of life, children should be shown what to do so they can participate in meaningful practices. Once the life forces have begun to become free, they can be educated through images and the imagination and through experiencing that some things have meaning and value:

> whatever is filled with deep meaning that works through pictures and allegories is proper for these years . . . it is not abstract concepts that work in the right way on the growing etheric body, but rather what is seen and perceived – indeed, not with external senses, but with the mind's eye.
>
> (1996b, p. 23)

The appropriate modes of learning at this age are discipleship and authority, but not "authority compelled by force, but authority that they will accept naturally without question. Through this they will build up their conscience, habits and inclinations" (ibid., p. 24). Just as a lack of moral integrity experienced in young children will be

embodied and the child will remain insecure, so in the second life phase, between the change of teeth and puberty, children need models of human coping, learning, meaning making and being moral. Steiner refers to this as the need for reverence and veneration and for access to people or stories of "outstanding people of history, life stories of great men and women"; role models rather than abstract moral exhortations. Steiner's focus was on experience of real examples that can be retained as habits; not successful people as such but of good people, moral heroes.

In order that these models make sense, it is of course necessary that the pupils also experience the opposite – examples of evil, egotism and darkness that goodness and light can overcome. To this end, the curriculum includes a wealth of legends and myths in which all forms of human weakness and strength are played out. These stories, Steiner suggests, have a very different impact when they are narrated by a good storyteller rather than simply being read out. Most effective are parables, images and metaphors and least effective are intellectual explanations. The story well told is a meeting of beings, mediated by a person for whom the story has meaning. The impact of other media is therefore likely to be less because, in Steiner's terms, the story has to be communicated in an authentic way, in the sense that it is authored by a person, which is what happens when the storyteller chooses, prepares and convincingly tells the tale in her own words. Steiner also emphasizes the necessity of schooling the child's memory, because this enables the life body to be transformed. We can understand this better when we place this thought in the context of the learning process, as I do in Part 2. Here Steiner anticipates work by Bruno Bettelheim and Jerome Bruner and others on the importance of narrative in education.

Although children should have authentic experiences, Steiner notes that the lessons should not be based only on sense experience. Here we encounter one of the difficult terms for translation. In the original, Steiner says *sinnlicher Anschauungsunterricht*. *Sinnlich* means using the senses and *Anschauung* has a range of meanings spanning view, outlook, way of seeing, contemplation and intuition. In this context it means the kind of understanding based on a way of looking at the world. What Steiner is criticizing is the assumption that children, or indeed anyone, can understand the world simply by looking at it and experiencing it. I take up this point in Part 2, when talking about teaching methods.

Regarding the third phase of child development, after puberty Steiner added very little in *The Education of the Child*, except to point out that this phase marks the emergence of the ability to discern and form judgements. He emphasized the importance of the personality of the teacher in guiding the students to become independent thinkers. He also makes the point that is relevant at all stages of education; that it is vital that teachers should develop a clear awareness of how various teaching strategies and methods actually affect young people, and that the teacher always has to develop the right sense of tact to make the right decisions in each individual situation. This tact can be developed through engagement with spiritual scientific ideas.

Key ideas

- The spiritual understanding of the human being offers practical insights into education.

- Human development occurs in four dimensions – in the physical body, in the growth and development of the body through life processes (life body), in the mind and the emergence of psychological aspects (sentient body) and in the existential self or spiritual core of being (the I).
- In the development of the child, early childhood, childhood and youth require quite different modes of learning that Steiner characterized respectively as imitation, guidance through authority, and the development of personal judgement.
- The human I is active in different ways in the learning process throughout the three phases of childhood, transforming the three other dimensions or 'bodies'.

References

Appiah, K. A. (2019). *Lies that bind. creed, country, colour, class, culture*. London: Profile Books.

Barfield, O. (1988). *Saving the apearances. A study in idolatry* (2nd ed.). Middletown, CT: Weslyan University Press.

Barfield, O. (2010). Introduction. In O. Barfield (Ed.), *The case for anthroposophy. Being extracts from Riddles of the Soul by Rudolf Steiner* (pp. 1–18). Oxford: Barfield Press.

Barfield, O. (2011). *Speaker's meaning*. Oxford: Barfield Press.

Biesta, G. J. J. (2009). Good education in an age of measurement: On the need to reconect with the question of purpose in education. *Journal of Educational Assessment, Evaluation and Accountability*, *21*(1), 33–46.

Biesta, G. J. J. (2010). *Good education in the age of measurement: Ethics, politics and democracy*. Herndon, VA: Paradigm Publishers.

Bortoft, H. (1996). *The wholeness of nature: Goethe's way towards a science of conscious participation in nature*. Hudson, NY: Lindisfarne Books.

Bourdieu, P. (1992). *Outline of a Theory of Practice* (R. NIce, Trans.). Cambridge, UK: Cambridge University Press.

Bransby, K., & Rawson, M. (2020). *Waldorf education for the future: A framework for curriculum practice*. Retrieved from https://www.steinerwaldorf.org/wp-content/uploads/2021/02/1.Waldorf-Education-for-the-Future-A-Framework-for-Curriculum-Practice.pdf

Cassirer, E. (1944/1962). *An essay on man*. New Haven, CT and London: Yale University Press.

Childs, G. (2003). *Rudolf Steiner: His life and work*. Edinburgh: Floris Books.

Crotty, M. (1998). *The foundations of social research: Meaning and perspective in the research process*. London: Sage.

da Veiga, M. (2016). Grundmotive im philosophischen Denken Rudolf Steiners und ihr Bezug zu Methoden und Fragestellungen in der Phänomenologie und der analytischen Philosophie. In J. Schieren (Ed.), *Handbuch Waldorfpädagogik und Erziehungswissenschaft: Standort und Entwicklungsperspektiven* (pp. 82–115). Weinheim Basel: Beltz Juventa.

Dahlin, B. (2010). Steiner Waldorf education, social-threefolding and civil society: Education as cultural power. *Research on Steiner Education*, *1*(1), 49–59.

Dahlin, B. (2013). Gloves of ice or free hands? A nomadic reading of Rudolf Steiner and Bergson and Deleuze and others on knowledge as nonrepresentational and the importance of Aesthesis. *Other Education: The Journal of Educational Alternatives*, *2*(2), 67–89.

Dahlin, B. (2017). *Rudolf Steiner. The relevance of Waldorf education*. Berlin: Springer.

Damasio, A. (2012). *Self comes to mind: Constructing the conscious brain*. London and New York: Random House.

Davy, J. (Ed.). (1993). *A man before others. Rudolf Steiner remembered. Personal recollections of those who knew him. A collection of personal memories from the pages of The Golden Blade and other sources*. London: Rudolf Steiner Press.

Dewey, J. (1938). *Experience and education*. New York: Touchstone.

Douven, I. (2017, Summer). Abduction. In E. N. Zalta (Ed.), *The Stanford encyclopedia of philosophy*. Standord, CA: Metaphysics Lab, Standford University.

Gabriel, W. (2014). Erziehungswissenschaft und Waldorfpädagogik: erlkenntnisswissenschaftliche Zugänge zu Rudolf Steiners pädagogischem Impuls. In P. Heusser & J. Weinzirl (Eds.), *Rudolf Steiner: Seine Bedeuting für Wissenschaft und Leben heute (Rudolf Steiner: The significance of his work for science and life today)* (pp. 228–266). Stuttgart: Schattauer.

Gadamer, H.-G. (2013). *Truth and method* (J. W. a. D. G. Marshall, Trans. Rev. 2nd ed.). London: Bloomsbury.

Gamble, C., Gowlett, J., & Dunbar, R. (2014). *Thinking Big: How the evolution of social life shaped the human mind*. London: Thames & Hudson.

Gidley, J. M. (2011). *Steiner, Rudolf (1861–1925) encyclopedia of the sciences of learning* (pp. 3188–3191). New York: Springer.

Gidley, J. M. (2016). *Postformal education. A philosophy for complex futures*. Weinheim Basel: Springer.

Göschel, J. C. (2012). *Der biographische Mythos als pädagogisches Leitbild: Transdisziplinäre Förderplanung auf Grundlage der Kinderkonferenz in der anthroposophischen Heilpädagogik (Biographical myth as pedagogical leading thought: Transdisciplinary learning support on the basis of the child study process in anthroposophical curative eduction)*. Dornach, CH: Verlag am Goetheanum Athena.

Hay, D., & Nye, R. (2006). *The spirit of the child* (Rev. ed.). London and Philadelphia: Jessica Kingsley Publishers.

Hemleben, J. (1963). *Rudolf Steiner*. Reinbek bei Hamburg: Rowohlt.

Heusser, P. (2016). *Anthroposophy and science. An introduction*. Frankfurt am Main: Peer Lang Edition.

Hughes, G. R. (2012). Rudolf Steiner's epistemology and its relation to feminist thought in America. In R. McDermott (Ed.), *American philosophy and Rudolf Steiner: Emerson, Thoreau, Peirce, James, Royce, Dewey, Whitehead, Feminism* (pp. 227–250). Great Barrington, MA: Lindisfarne Books.

Ingold, T. (2000). *The perception of the environment: Essays in livelihood, dwelling and skill*. London and New York: Routledge.

Kaiser, U. (2020). *Der Erzähler Rudolf Steiner. Studien zur Hermeneutik der Anthroposophie*. Frankfurt am Main: Info 3 Verlag.

Kannenberg, M. (2011). *Rudolf Steiner in Berlin: Rudolf Steiner's Impulse, Begegnungen und Wirkenstätten in Berlin 1897–1923*. Dornach, CH: Verlag für Anthroposophie.

Kiersch, J. (1978). *Freie Lehrerbildung. Zum Entwurf Rudolf Steiners* (Vol. 11). Stuttgart: Verlag Freies Geistesleben.

Kiersch, J. (2010). "Painted from a palette entirely different". A new hermeneutic approach to Steiner's esoteric courses for teachers. *Research on Steiner Education*, *1*(2), 64–72.

Kiersch, J. (2012). Spiritualität auf der Suche nach Wirklichkeit- in pädagogischer Hinsicht. In P. Loebell & P. Buck (Eds.), *Spiritualität in Lebensbereichen der Pädagogik. Diskussionsbeiträge zur Bedeuting spiritueller Erfahrungen in den Lebenswelten von Kindern und Jugendlichen* (pp. 325–342). Opladen; Berlin; Toronto: Verlag Barbara Budrich.

Kinzler, S., & Tillmann, D. (2018). *Die Stunde der Matrosen. Kiel und die deutsche Revolution. Ausstellungskatalog*. Darmstadt: Konrad Theiss Verlag.

Kugler, W. (Ed). (1994). *Rudolf Steiner als Hauslehrer und Erzieher Wien 1884–1890. Beiträge zur Rudolf Steiner Gesamtausgabe*. Dornach, CH: Veröffentlichungen aus dem Archiv. Doppelheft Nr. 12/113 Frühjahr 1994.

Lachman, G. (2007). *Rudolf Steiner: An Introduction to his life and work*. New York: Tarcher/Penquin.

Lakoff, G., & Johnson, M. (2003). *Metaphors we live by* (2nd ed.). Chicago: University of Chicago Press.

Lieberman, D. (2013). *The story of the human body*. London and New York: Penguin.

Lindenberg, C. (1992). *Rudolf Steiner. Mit Selbstzeugnissen und Bilddokumentaten*. Reinbek bei Hamburg: Rowohlt.
Lindenberg, C. (2013). *Rudolf Steiner a biography*. Great Barrington, MA: Steiner Books Inc.
Marx, K. (1965/1844). Estranged labour. In T. B. Bottomore (Ed.), *Karl Marx: Early writings* (pp. 121–129). London: George Allen & Unwin.
Maton, K. (2012). Habitus. In M. Grenfell (Ed.), *Bourdieu: Key concepts* (pp. 48–64). Durham: Acumen.
McDermott, R. (2009). *The new essential Steiner: An introduction to Rudolf Steiner for the 21st century*. Herndon: Lindesfarne Books.
McDermott, R. (2012). *American philosophy and Rudolf Steiner*. Great Barrington, MA: Lindisfarne Books.
McMillan, M. (1904). *Education through the imagination*. London: Swann Sonnenschein & Co.
Miller, L. (2015). *The spiritual child*. New York: St. Martin's Press.
Mücke, J., & Rudolf, A. A. (1955). *Erinnerungen an Rudolf Steiner und seine Wirksamkeit an der Arbeiter-Bildungsschule 1899–1904*. Berlin: Verlag R.G. Zbinden.
Nicol, J., & Taplin, J. T. (2017). *Understanding the Steiner Waldorf approach*. London: Taylor & Francis.
Paull, J. (2011). Rudolf Steiner and the Oxford conference: The birth of Waldorf education in Britain. *European Journal of Educational Studies*, *3*(1), 53–66.
Paull, J. (2018). The library of Rudolf Steiner: The books in English. *Journal of Social and Developmental Sciences*, *9*(3), 21–46.
Paull, J., & Hennig, B. (2020). Centenary world maps of the global diffusion of "The School of the Future". *Journal of Social Sciences and Humanities*, *6*(1), 24–33.
Rawson, M. (2003). *The spirit in human evolution*. Fair Oaks, CA: AWSNA Publications.
Rawson, M. (2010). Rudolf Steiner's research methods for teachers. *Research Bulletin for Waldorf Education, 25–40*(1), 33–37.
Rawson, M. (2012). Contemplative practice and intuition in a collegial context: An action research project in a Waldorf school. *Research Bulletin of Waldorf Research Institute*, *17*(1), 47–54.
Rawson, M. (2014). Practices of teacher learning in Waldorf schools: Some recommendations based on qualitative inquiry. *Other Education: The Journal of Educational Alternatives*, *3*(2), 45–68.
Rawson, M. (2018). The case for practitioner research using complementary theory in Waldorf pedagogy. *Research on Steiner Education*, *8*(2), 15–32.
Rawson, M. (2020a). A Theory of Waldorf Education: Part 2 the role of study and artistic exercises. *Research on Steiner Education, 11*(2), 23–36.
Rawson, M. (2020b). A Theory of Waldorf Teacher Education: Part 1. Learning dispositions. *Research on Steiner Education, 11*(2), 1–22.
Rawson, M. (2021). Using artistic, phenomenological and hermeneutic reflective practices in Waldorf (Steiner) teacher education. *Tsing Hua Journal of Educational Research, 440*.
Rose, S. (1998). *Lifelines: Biology, freedom and determinatism*. London: Penguin.
Rosslenbroich, B. (2014). *The origin of autonomy. A new look at major transitions in evolution*. New York and Heidelberg: Springer.
Sanders, B. (1994). *A is for Ox: The collapse of literacy and the rise of violence in an electronic age*. New York: Vintage Books.
Schad, W. (2003). Chronobiologie ist Ätherforschung. *Tycho de Brahe Jahrbuch für Goetheanismus*, *2003*, 20–36.
Schad, W. (2015). *Zeitbindung in Natur, Kultur und Geist*. Stuttgart: Verlag Freies Geistesleben.
Scharmer, O. (2016). *Theory U: Leading from the future as it emerges- The social theory of presencing* (Rev. 2nd ed.). San Francisco, CA: Berrett-Koehler.
Schieren, J. (2011). The scientific credibility of Anthroposophy. *Research on Steiner Education*, *2*(1), 90–98.
Schieren, J. (2012). The spiritual dimension of Waldorf education. *Research Bulletin for Waldorf Educaion*, *19*(1), 19–29.

Schieren, J. (2015). Anthroposophy and Waldorf education: A dynamic relationship. *Research in Steiner Education*, *6*(2), 139–149.
Schieren, J. (2016). Lernen in der Waldorfpädagogik. In J. Schieren (Ed.), *Handbuch: Waldorfpädagogik und Erziehungswissenschaft: Standortbestimmung und Entwicklungsperspektiven (A handbook of Waldorf pedagogy and the science of education: Positions and developmental perspectives)* (pp. 427–446). Weinheim Basel Basel: Beltz Juventa.
Schieren, J. (2019). Waldorf pedagogy and educational science. In C. Willmann & L. Weiss (Eds.), *INASTE. The next generation of Waldorf teachers* (pp. 8–9). Vienna: INASTE.
Schneider, P. (1982). *Einführung in die Waldorfpädagogik*. Stuttgart: Klett-Cotta.
Schwartz, B., & Sharpe, K. (2011). *Practical wisdom. The right way to do the right thing*. London: Penguin Books Riverhead Books.
Selg, P. (2009). *Rudolf Steiner und Felix Koguzki: Der Beitrag des Kräutersammlers zur Anthroposophie*. Dornach, CH: Verlag des Ita Wegman Instituts.
Selg, P. (2011). *The fundamental social law. Rudolf Steiner on the work of the individual and teh spirit of the community* (C. E. Creeger, Trans.). Great Barrington, MA: SteinerBooks.
Selg, P. (2014). *Rudolf steiner. Life and work* (Vol. 1). Great Barrington, MA: SteinerBooks.
Shepherd, A. P. (1961). *A scientist of the invisible*. London: Hodder and Stoughton.
Steiner, R. (1928/1925). *The story of my life* (H. Collinson, Trans.). London: Anthroposophical Publishing Co.
Steiner, R. (1935). *Two essays on Haeckel*. London: The Rudolf Steiner Publishing Co.
Steiner, R. (1963a). *The philosophy of spiritual activity* (R. Stebbing, Trans. and H. S. Bergman, Ed.). West Nyack, NY: Rudolf Steiner Publications.
Steiner, R. (1963b). *Truth and science* (R. Stibbing, Trans. and P. M. Allen, Ed.). West Nyack, NY: Rudolf Steiner Publications.
Steiner, R. (1966) Autobiographical sketch: "The Document of Barr". Notes by R. Steiner on a meeting with E. Schuré 9.9.1907 in *The Golden Blade*, 1–9.
Steiner, R. (1968/1886). *A theory of knowledge implicit in Goethe's world conception*. Spring Valley, NY: Anthroposophic Press.
Steiner, R. (1973). *Anthroposophical leading thoughts* (G. Adams, Trans.). London: Rudolf Steiner Press.
Steiner, R. (1977). *Towards social renewal: Basic issues of the social question*. London: Rudolf Steiner Press.
Steiner, R. (1979). *Erziehungs-und Unterrichtsmethoden auf anthroposophischer Grundlage* (Vol. GA304). Dornach, CH: Rudolf Steiner Verlag.
Steiner, R. (1982/1920). Spiritual knowledge of man as a fount of educational art, lecture 21st September, 1920 in Stuttgart. In T. van Vliet & K. Kiniger (Trans.), *Balance in teacher*. Spring Valley, NY: Mercury Press.
Steiner, R. (1985/1919). The threefold social order and educational freedom (E. Bowen-Wedgewood & R. Mariott, Trans.) *The renewal of the social organism* (pp. 70–76). Spring Valley, NY: Anthroposophic Press.
Steiner, R. (1988/1886). *The science of knowing* (W. Lindemann, Trans.). Spring Valley, NY: Mercury Press.
Steiner, R. (1994). *The social basis for education* (B. Masters, Ed.). Forest Row: Steiner Schools Fellowship.
Steiner, R. (1996a). *Anthroposophy (A fragment)*. Hudson, NY: Anthroposophic Press.
Steiner, R. (1996b/1907). *The education of the child*. Hudson, NY: Anthroposophic Press.
Steiner, R. (1996c). *The foundations of human experience*. Hudson, NY: Anthroposophic Press.
Steiner, R. (2009). *The riddles of philosophy: Presented in an outline of its history*. Hudson, NY: SteinerBooks.
Steiner, R. (2010). *Gegenwärtiges Geistesleben und Erziehung: Ein Vortragzyklus Ilkley (Yorkshire) 5 bis 17 August 1923* (Vol. GA 307a). Retrieved from http://anthroposophie.byu.edu/vortraege/307a.pdf

Steiner, R. (2011a). Sozialethik: für Edith Maryon, persönliche Bemerkung zu. In *"Ausführungen der Dreigliederung des sozialen Organismus" 1919 Wahrspruchworte* (Vol. GA40). Dornach, CH: Rudolf Steiner Verlag.

Steiner, R. (2011b). *Theosophy: An introduction to the supersensible knowledge of teh world and the destination of man* (M. Cotterell & A. P. Shepherd Trans. Rev. 150th Anniversary ed.). Forest Row: Rudolf Steiner Press.

Steiner, R. (2015). Brief reflections on the publication of the new edition of 'The Philosophy of Freeedom' in *From Symptom to Reality in Modern History, lecture VI, 27.10.1918*. A. H. Parker, Trans. from GA 185. London: Rudolf Steiner Press. Retrieved January 17, 2021, from https://wn.rsarchive.org/Lectures/GA185/English/RSP1976/19181027p01

Tomasello, M. (2019). *Becoming Human. A theory of ontogeny*. Cambridge, MA: The Belknap Press of Harvard University Press.

Tomasello, M. (2003). *Constructing language: A usage-based theory of language acquisition*. Cambridge, MA: Harvard University Press.

Uhrmacher, P. B. (1995). Uncommon schooling: A historical look at Rudolf Steiner, anthroposophy and Waldorf education. *Curriculum Inquiry*, *25*(4), 381–406.

van Manen, M. (1991). *The tact of teaching*. Ontario: The Althouse Press.

Walach, H. (2017). Secular spirituality- what it is. Why we need it. How to proceed. *Journal for the Study of Spirituality*, *7*(1), 7–20.

Wagemann, J. (2016). Erkenntnisgrundlagen der Waldorfpädagogik. In J. Schieren (Ed.), *Handbuch Waldorfpädagogik und Erziehungswissenschaft: Standort und Entwicklungsperspektiven* (pp. 31–81). Weinheim Basel: Beltz Juventa.

Welburn, A. (2004). *Rudolf Steiner's philosophy and the crisis of contemporary thought*. Edinburgh: Floris Books.

Wiehl, A. (2015). *Propädeutik der Unterrichtsmethoden in der Waldorfpädagogik (Propadeutic of teaching methods in Waldorf pedagogy)*. Frankfurt am Main: Peter Lang.

Zajonc, A. (2009). *Meditation as contemplative inquiry: When knowing becomes love*. Geat Barrington, MA: Lindesfarne.

Zander, H. (2016). *Rudolf Steiner. Die biographie*. Munich: Piper.

Zander, H. (2019). *Die Anthroposophie. Rudolf Steiners Ideen Zwischen Esoterik, Weleda, Demeter und Waldorfpädagogik*. Paderborn: Ferdinand Schöningh.

Zdrazil, T. (2019). *Freie Waldorfschule in Stuttgart 1919–1925. Rudolf Steiner-das Kollegium-die Pädagogik*. Stuttgart: Verlag Freies Geistesleben.

Part 2 Generative principles of teaching and learning in Waldorf education

Summary: Part 2 gives an overview of the generative principles related to teaching and learning within Waldorf education. After a brief introduction on working with generative principles, I offer five vignettes describing teaching situations spanning the age range and types of subjects to give the reader unfamiliar with Waldorf education a glimpse of its practice. These include some tasks for the reader to reflect on and a brief outline of the generative principles at work in the pedagogical examples shown (e.g. the role of the teacher, the overall curriculum approach). Following this is a sequence of generative principles. Each principle is described and explained. After each principle, I list the kind of skills that teachers need and the possible research questions that arise.

Generative principles and how to work with them

Angelika Wiehl (2015a, 2015b, 2017) has described Steiner's educational approach as a propaedeutic, from the Greek *pro* meaning before and *paideuein*, meaning to teach a child. This word literally means, what you need to know before you act pedagogically, or how to prepare oneself to teach. Steiner frequently referred to the Waldorf School as a method-school. The word method also comes from the Greek, *méthodos* meaning the path to reaching a goal, or metaphorically, the way one travels and the experiences one has on the journey (www.merriam-webster.com).

Wiehl calls the Waldorf method a procedural approach to pedagogy that "it is not a methodology or system of specific practices" (2017, p. 4). The procedure involves starting with the ideas of Steiner's pedagogical anthropology (the Foundations) and including new ideas generated from this, and then following the core aims of the education, teachers choose pathways (*méthodos*) to learning that are appropriate for the learning needs of the pupils in the given context. The Foundations and the generative principles based on these form a meta-level for pedagogy and curriculum. Pedagogy refers to understandings of the relationship between teaching and learning and curriculum is the answer to the question: what learning situations and experiences enable children and young people to develop as persons and to learn the skills and knowledge they need, and when can these best be offered? In other words curriculum is a response to the developmental tasks of the pupils and students. When the generative principles are applied they lead to certain practices over time, including curriculum, that have proven to be effective, though they periodically need to be reviewed and evaluated (see Bransby & Rawson, 2020).

Thus the onus is on teachers to be creative. Instead of relying on existing knowledge, prescribed learning schedules, pre-prepared teaching materials and course books, the teacher is supposed to read the given situation, form judgements and plan the lesson accordingly. Obviously, novice teachers are more dependent on guidance.

As I discuss below in the section on curriculum, this is a new interpretation of curriculum within Waldorf education. Bransby and Rawson (2020) have developed a new framework for developing curriculum out of the Foundations and the generative principles outlined here.

In what follows, I define the main generative principles and describe the typical practices that either Steiner directly introduced or which have developed subsequently. These principles can be used as heuristic ideas or living theories (Whitehead, 2018) to plan and later evaluate practice at the meso- and micro-levels.

The purposes of generative principles are (briefly summarized):

- Planning teaching
- Self-evaluation
- School evaluation
- Teacher development
- Research

I have organized the generative principles into two loose categories; how the teaching and learning is organized and the role of the teacher, though these two categorizes cannot really be separated. The generative principles are:

1 Waldorf education takes the spiritual dimension seriously
2 Sense of coherence is the basis for healthy learning and well-being
3 The quality of learning depends on the quality of the teacher's preparation
4 Children and young people need to be ready to learn, and they need time to learn
5 Learning is a rhythmical process
6 The learning processes are structured over time
7 Block teaching strengthens learning dispositions
8 Direct experience is the basis for good learning
9 Activating the imagination through vivid pictorial descriptions and images is another powerful starting point for learning
10 A phenomenological approach enables the organic growth of knowledge
11 The teaching must be artistic
12 The self-activity of the students is essential to learning
13 Good teaching and learning depends on the development of the senses

Five vignettes of Waldorf practice

The lower school main lesson

The children arrive at the classroom door shortly before 8 a.m. Most have arrived in a fleet of yellow minibuses with the name of the school on the side that have gathered them from all directions across the city. In each bus there is a bus assistant, as well as the driver, to ensure that all the children are present and to confirm this with lists of the children sent by the school office. The school has an efficient system for parents to notify the school if their children cannot come to school for some reason, so the bus only stops where necessary and still reaches school on time, even in the chaos of a large city in India. The children are all about 9 years old and there are about 24 of them. They are in grade 3 in a Waldorf school. They line up at the door with their

school bags on their backs, having left their shoes outside. The teacher greets each one at the door by name: "Good morning, Sanjay, how are you today?" "Good morning, Shivanya-teacher, I am well, thank you", the child replies.

The children enter the classroom and place their bags around the side of the room, which is empty except for a large, patterned carpet on the floor with a teachers' desk at the side and a blackboard at one end. Around the sides of the room are piles of simple, handmade chaukis, low wooden tables about knee high, some cupboards, an open cupboard with racks of wooden boards for drawing and watercolour painting. Next to the blackboard in one corner is a table resembling an altar with fresh flowers, some crystals, a bundle of cereals including rice, wheat, millet and some lentils in thin dried pods, and an incense burner, which is lit at the beginning of the lesson. Because the windows are wide open so that a fresh breeze can cool the room, a candle is lit in a lantern, otherwise it would blow out. The school does not use air conditioning or electric fans, as most schools in India would, but relies on traditional methods of cooling rooms using shade and the circulation of air. The children are wearing a mixture of Western-style and Indian clothes suitable for the heat, whilst the teacher is wearing a colourful traditional salwar-kameez (knee-length tunic, slit on both sides) and leggings.

The teacher rings a small bell, and the children form a circle. There are a few things to clarify – who is absent, who has returned from being absent – the visitor (me) is welcomed and the teacher bids the children a formal good morning, to which they reply in chorus. They then speak the morning verse together. This is spoken in all Waldorf schools around the world and translated into the local language. Recently a small book was published in celebration of Waldorf 100 which contained translations in 39 languages. However, in India the language of instruction is English, so the morning verse is also in English. The children all speak the regional language of their family, of which there are 22 official ones (e.g. Kannada, Marathi or Tamil), and many other local dialects. The parents of the Waldorf school tend to come from a wide variety of backgrounds in India; about half speak Hindi and educated parents all speak English. The school also teaches Hindi and the local language in effect as foreign languages, but these are not foreign to all the children.

Following the morning verse, the children sang some songs (in all three languages) and recited times tables accompanied by rhythmical stepping and clapping. They did some exercises involving throwing several beanbags across the circle in particular patterns, during which they had to be alert and aware of the whole circle. Then the children took out a couple of long ropes, went outside and did some skipping. Three children stayed behind and cut up the fruit that everyone had brought. Each child brings one piece of fresh fruit and this is cut up and shared. After the children came back inside, summonsed by one of the 'cooks', they stood in a circle again and recited and enacted a poem with actions that related to harvesting and threshing that was to be performed shortly in the monthly school festival. They seemed to know all the verses by heart and 'performed' the poem in what can only be described as a serious but natural way. Led by the teacher, they then took out their wooden flutes and played some tunes together. Some children also played various simple Indian percussion instruments. After that, they sat cross-legged on the floor and a group of five children stood up at the front and spoke their 'birthday verses'. Each child receives a report at the end of the school year which includes a verse either composed or selected by the teacher for that child, something appropriate and uplifting. Each

day of the week, the children whose original birthdays fall on that weekday all recite their verses by heart. The 'weekend' children are split between Friday and Monday. During these birthday verses, the class is quiet and attentive. If a child forgets a word, others in the class will gently remind them. It appears that by the end of the year most children know the verses of all the others, though no one has required them to learn them.

Following this, the fruit break was announced and the children each took a metal plate from a rack where they had evidently been put to dry after washing. The 'cooks' – different children each day – have this task, whilst others water the plants in front of the classroom, sweep the veranda, tidy the shoes, shake out the carpet outside, wash the painting boards, glasses and brushes after painting, wash the cups and plates and fill a big jug of fresh water. These tasks are somehow fitted in during breaks, sometimes even in the lessons and after school. A rota drawn up by the teacher hangs near the door, along with a circular calendar marking each child's birthday throughout the year. Before they ate, a grace was spoken:

> For the dark earth that cradles the seed, for the rain that brings forth the green leaves, for the stars that give form to the flowers and the warm sun that ripens the fruit, for all this goodness and beauty we thank the creator spirit.

They do not know that this verse was spoken by the children at the York Steiner School in England. One of the handwork teachers there embroidered the verse and it hung for many years in the dining room, and it may still be there. The final line has been changed. The English version was "our heavenly Father, we thank thee", which is both gender-wise and religiously inappropriate, though it did rhyme better. Like many small traditions in Waldorf schools, this one had its origins in Camphill in Scotland, an anthroposophical community of people, some of whom have mental disabilities. My classroom in York was once decorated by a large Indian bedspread with sewn-in mirrors representing the starry sky at night. Thus, there is a kind of intercultural exchange, as indeed there has always been throughout history, especially when people saw something they valued. During the fruit break the children chat, and some get up and walk to others, but generally the meal is eaten quietly.

Following this event, the children got their chaukis and placed them in a three-quarters circle, sitting or kneeling on the floor. The teacher then asked the children to discuss in pairs what they had experienced the previous day in the main part of the lesson. Then she asked three pairs of children to report. The children told her about their visit to the school garden where they harvested lentils, brought them to a covered area and threshed them onto a large cloth, then gathered them together and placed them in baskets. The teacher then asked what was important to know, and a number of children volunteered answers. The teacher sometimes asked individuals who had not put their hands up. The children related that lentils are ripe in July; the plants need to be cut and left in the field to dry; you shake the plant and see if it rattles to see if the lentils are dried; they ripen quickest at the bottom of the plant and you have to make sure only the ripe ones are shaken out; the plants are quite prickly, so it's best to wear gloves but not everyone had gloves; you have to check the plants to see if all the pods have separated, and the remains go on the compost, or are kept as straw. Then the lentils in the baskets are carefully thrown up and down to let the wind blow away the

chaff. This apparently led to some spillage, for which some criticism was given. It was remarkable to learn how much the children had observed, even for an experienced Waldorf observer!

Then the teacher recalled the various steps of the process verbally and made a list of these on the blackboard; cut, dry, harvest, thresh, winnow, store. The children were then asked to write short texts describing what they had done, using the words on the blackboard. They made notes in their main lesson books. On the blackboard was a drawing of lentil plants. Those who had finished could copy the drawing into their books. Then they were asked to close their books (after previously being told that they had a few more minutes left). Those who had not finished were to finish this in a lesson later that day. The teacher then asked who knew how to cook lentils. Some children knew but most didn't, which reflects the fact that most of them live in families in which cooks generally do the cooking, just as many of these middle-class families have servants who clean, do the washing, and often drive for the family, which is not unusual in India. The names of some dal dishes were written on the blackboard but most of this part of the lesson was verbal, in which the teacher explained the uses of lentils.

While the children were working, the teacher took the opportunity to visit a couple of children and offer them support with their book work. Then she systematically called six children to the front and looked through their main lesson books, pointing out possible improvements and ensuring that everything was up to date and offering a word of praise or advice.

The teacher then put a stool in front of the class – her storytelling stool – and introduced her story by asking the children briefly how Krishna and his brother Balarama were born, and how even as an infant Krishna was able to overcome the wet-nurse with poisoned milk and the demon who came as a gust of wind and another as a loose cartwheel. They answered briefly and the story continued. Krishna's parents Nanda and Yashoda moved their whole family to live next to a forest of Tulsi plants – just like the Tulsi bush in front of our classroom. Krishna grew up with local cowherds and developed a fondness for butter. He was always playing pranks on the milkmaids, stealing the butter and hanging it from the rafters. When he was older, he was given the task of taking the cows to graze. He entertained the other gopas – the cowherders – with his flute and protected the cattle from many enemies and ensured that life was full of affection, love and security. Krishna's brother Balarama watched over the orchards and was a farmer and ploughed the fields. He even made canals to channel the water from the river to the fields. When he was older, he opposed the rituals of the Rishis – the enlightened ones – of pouring ghee into the fire, preferring acts of charity and kindness. Krishna suggested they worship not Indra, the god of the sky, but Govardhan, the local mountain, that trapped the clouds and made it rain. When Indra found out, he punished the people by sending a deluge to wash their crops and fields away. Krishna lifted up the mountain with his finger and held it over the village like an umbrella and protected the whole village. Then Indra saw that he was no ordinary lad (see Pattanaik, 2010, for the whole story).

After finishing this part of the story, the teacher stood up and the children, somewhat awestruck (as indeed was the visitor), stood up too. The teacher gave them some information about later lessons in the day, reminded them not to be late for buses later and bid them farewell. They cheerfully left the classroom and went out to play.

Tasks for the reader

1 Can you identify what aspects of this account are familiar to your own experiences of school and where the differences lie (apart from the lesson being in India)?
2 What do you think the purposes of the different aspects of the lesson are, assuming they are all meaningful?
3 Would you have read the account differently if these questions had been posed before you read the text?

Generative principles in the lower or primary grades

- The class teacher is responsible for teaching the first two hours of each day, ideally from grade 1 to grade 8.
- This two-hour 'main lesson' involves a range of activities and is given usually for four weeks.

Optics in grade 7

Grade 7 are in the middle of a four-week block on optics. They have been exploring various phenomena relating to light, mirrors, shadows and colours using prisms with simple practical experiments. Today was a certain culmination and this had required major preparations. The classroom was to be made totally dark and the question for the lesson was: Can we see light? Following the morning verse, some singing and mental arithmetic, the teachers asked the students to find their notes from the previous day's lesson, in which a series of pairs of mirrors had been used to try to establish how many times an image could be reflected back and forth when the mirrors were held opposite each other and a small object was placed between them. Neighbours shared each other's drawings and compared how many reflections there were. The teacher prompted a short discussion, with the outcomes that most children felt the number was probably unlimited but that you couldn't see this because we are always looking 'round the corner'. Two girls suggested, "We think that if you could look through the mirror directly at the other mirror, there would be no end of reflections until they were too small to see". A boy said, "Yes, like the police have, two-way mirrors". Another added, "Yes, but only two-way from one side". Another pupil said that if you could look through the mirror exactly at the object, you would only see one reflection. Another said that then you wouldn't see any reflection, just the object itself. This was briefly discussed. The teacher moved the conversation on without resolving the question by saying that for today's experiment, they would need to make the room totally dark.

The class then set about fixing thick, black-lined curtains that the teacher took out of a box, using black masking tape to ensure that no light crept in around the edges. This took some time, and several patches of light had to be double-taped. The classroom light was turned off to see if everything was dark, and the class immediately discovered a crack of light under the door. This had to be blocked by laying a towel on the floor in front of the door.

When all the preparations had been completed, the class was quite excited. The teacher then asked if light could be seen. There was quite a discussion, but the majority felt that yes, light could be seen, citing electric lightbulbs that shone when turned

on, or the sun could clearly be seen shining; one pupil even mentioned the moon, but some others said that was due to reflection of sunlight, though they couldn't fully explain this to the satisfaction of the pupil, who defended his position. The teacher wound up this discussion by changing the topic to the situation that was about to occur once the classroom became totally dark. Some pupils asserted that no one would be afraid, and this was supported by a number of others, looking at each other for affirmation. It seemed to be a matter of pride. One girl explained that nothing bad could happen because the door was closed and the only people in the room were the teacher, the class and me as visitor. At this moment, the whole class looked at me as if trying assess my potential danger. A voice spoke up asking, what if someone has to go to the toilet? This was met with some mirth, but the teacher made the suggestion that to be sure, everyone who needed to, should go to the toilet before the door was finally 'sealed'. The teacher then added that she was sure that no one would do anything to spoil the scientific mood of observation, reminding the class that this after all was a scientific experiment, not party fun! There would be no giggling and no provocation or surprises! This seemed to calm the class down and when the door was opened, more than half the class left to go to the toilet but were back in what seemed to me a highly unrealistically short time.

When all the children were back in the class and settled, the towel was laid across the gap at the bottom of the door and it really was very dark and silent. Before the lesson the teacher had put a cardboard box that had been painted matte black on the table. Inside was a powerful torch. A hole had been cut in both sides of the box, one for a hand, covered in a black cloth. The second hole was about 2 centimeters in diameter. The teacher put her hand in the box and switched on the torch, which was pointing away from the class towards the blackboard. A patch of light appeared on the blackboard. The teacher asked if the children could see the light. Most said yes, both on the door and coming out of the hole. Then the teacher asked me to hold a hand mirror opposite the hole in the box. After a bit of searching, I located the beam and reflected the light upwards towards the ceiling. The light on the blackboard disappeared and it took a few moments for the children to find the patch of light, now somewhat wider, on the back wall. The teacher asked again, "Can you see the light?" The children were less certain but said yes, on the back wall. Then the teacher gave a child in the front row another hand mirror and after a bit of searching, the child redirected the light beam in another direction, up to the ceiling. The teacher then turned the torch off, then back on and then off again and put the lights in the classroom back on. It suddenly seemed very bright and a bit strange. The teacher asked again whether we could really see the light. The students were uncertain. One child suggested that we can only see what the light touched, like the edge of the hole in the box or where the beam lit up the wall. "Okay", the teacher said, "let's try that again and see what happens". This time she asked a pupil to hold the torch in the box and turn it on. Everyone settled down, the lights were turned off and the torch came on. It took a while for the two mirrors to get aligned, and now the light patch was in a different place. Then the teacher stood up and clapped two blackboard dusters together, creating a cloud of dust particles. Suddenly everyone could see the beam of light zigzagging across the space! The children were amazed and started talking. The teacher put the light back on, turned the torch off (one could even see the beam in daylight a bit) and took down the curtains. There was a lively buzz of talk.

The teacher then asked the pupils to take out their notebooks and write down everything that had happened. After a few minutes getting organized, they were all busy

thinking and writing. The teacher went round looking at what they were doing and checking that the homework from the previous day had been done, commenting on whether the drawings were accurate. There was some discussion about the sequence of events, which the children clarified amongst each other. About 15 minutes before the end of the lesson, the notebooks were put away. Then a girl stood up and gave a short report on a book she had read, ending with the recommendation, "This is a book for people who like horses". The girl was thanked by the teacher, and the lesson was ended with everyone standing up and the teacher giving some information about part of the playground being out of bounds because the grass was being reseeded. The class and teacher said goodbye and the children went out to break.

Tasks for the reader

- Can you identify the various learning processes involved?
- Can you identify the learning outcomes and the teacher's intentions?
- Would you organize the lesson differently?

Generative principles of science teaching in the class teacher period of the middle school (grades 6 to 8, age 12 to 14)

- Phenomena, in this case light, are experienced and described before definitions or laws are mentioned.
- The pupils experience the phenomena firsthand.

A craft project involving grades 5 and 10

It is a blustery day in June, and a large group of grade 5 students (11 years old) and grade 10 students (16 years old) are assembled outside the tool shed in the school grounds. They have gathered all the wheelbarrows, spades and shovels from the school garden and the teacher is handing out other hand tools from the tool shed to some of the older students, who distribute them to the younger ones to carry.

The backstory is somewhat complicated. Grade 10 have just finished doing a site survey of a corner of the school grounds as part of their surveying block under the guidance of the maths teacher. Using non-digital theodolites, levelling staffs, surveying poles, tape measures, rodometers and calculators, they had done an accurate survey of an area of the grounds including a small stand of trees, two ditches, a road and couple of old huts. The survey had to meet the standards of the local planning authorities because the school was hoping to get planning permission to build an outdoor classroom space there. The school's architect had also been involved. Because the planning had been delayed, it had been decided to build a temporary structure that did not require planning permission in the short term, so that an outdoor clay oven could be built with an outdoor kitchen so that bread and pizzas could be baked.

The previous year, in grade 9, this class had cleared the space around the trees, laid a path and built a wooden footbridge across the ditch, as part of their grade 9 work block. They had also cleared a space for the planned outdoor classroom. The current year's grade 9 had constructed a timber-framed, open-sided building with no walls, consisting mainly of a roof with timber frame and corrugated roofing material, supporting wooden posts on a hard-core base. The weather had been too bad, and they

had been unable to build the clay oven as planned. As grade 10 were finished with their surveying, they had a few spare days, and only three pupils were needed to transfer the surveying data onto an accurate map of the site. The class teacher and craft teacher in grade 5 had the idea of doing a joint project helping to build and equip the outdoor space. The plan was to build an outdoor oven using bricks for the lower part and a domed clay oven.

Grade 5 have been learning whittling craft skills in the woodland area since Easter, learning to carve green wood with sharp knives. They had started by sawing logs with double-handed cross-cut saws, stacking the wood to dry, splitting the logs with wedges and a froe (a blade at the end of a shaft that is hit with a wooden mallet), using a side axe to roughly shape the pieces to be carved into forked pegs, butter knives and spatulas. They had also learned about lighting fires, drying wood, collecting kindling and so on. In grade 6, woodwork using seasoned wood begins and continues through to the high school, covering carpentry, the common joints, learning about the different kinds of wood and their origins and uses, leading up to making furniture and in some cases making musical instruments. Their role in the combined project was to help source and prepare the clay, help with the bricklaying and later make some green wood stools and trestle-tables for the kitchen.

The two classes were split into a number of groups, each with two grade 10 students as team leaders. The previous day, grade 10 had drawn up plans for the work and broken it down into tasks, which were described in short texts and copied onto sheets so that each group knew what their task was and what the overall sequence was to be. One group dug clay from the side of a ditch, which the bridge building had exposed, and cleared brambles. Another group transported (with wheelbarrows) and stacked bricks, sand and cement that had been delivered to the school entrance. The clay was also transported to the site and piled up on a thick plastic sheet. A group of grade 5 pupils (in shorts and bare feet) trod the clay, removing impurities and lumps and adding some water, and covered the clay so it wouldn't dry out. Another group cut and gathered hazel and alder wands of the right length for weaving into a basket shape as a form for the oven. Another group started laying bricks with cement in a square about a metre high. This was carefully filled with rubble.

At the end of the day, all tools were cleaned and put back in the tool shed. The work was 'inspected', and plans were made for the next day. This included cementing a layer of flat paving stones on top of the base of the oven, which would form the floor of the bread oven. A large basket shape was made with hazel wands with an opening at one side that would later be the door. Another hole was cut in the basket for the chimney. This was inverted on the oven platform and several layers of clay were applied by hand to the outside of the oven, and the edges of the openings were rounded off with clay. Shortly before the end of the day, a fire was built with dry kindling and brushwood, which burned quickly at a high temperature inside the oven space, burning off the basket frame inside and hardening the clay skin of the oven. At a later stage, the metalwork teacher fitted a steel chimney and, using fire cement, sealed this at the upper opening, with the help of a couple of students. Some grade 10 students made a door with a handle out of stout oak lined with tin, to lean on the door opening during baking to keep the heat in. A few weeks later, the oven was fired by laying a fire inside with twigs, letting it burn down, raking out the ashes and embers and then placing bread dough on the hot stones. The bread was then shared and enjoyed.

The whole process was documented by the grade 5 pupils with illustrations and texts describing all the various stages. The oven was used regularly for grade 3 baking and making pizzas during school festivals. The oven required periodic repairs but served well for several years.

Tasks for the reader

- What educational aims do you see in such a project, and what do you think the pupils learned?
- What elements of curriculum do you think are addressed?
- What practical arrangements would be needed to conduct such a project?

Generative principles

- Handcrafts educate the will through engaging with the material world in meaningful and meaning-making ways.
- Craftwork is a way of bringing the human being back into a relationship with nature.
- Work is a form of transformation in which the human spirit gives new form to matter and thereby transforms the person.
- Craft activity involves ancient practices, rich in meaning relating to the materials, their origins and nature, tools and social practices that carry deeply embodied cultural understandings.
- Understanding hand technology provides a conceptual basis for digital technologies.
- Craftwork is about doing something well for its own sake and creating aesthetic value.

Art history, grade 9

Grade 9 students are 15 years old. There are 36 students in the class and the main lesson begins at 8 a.m. This lesson is on the Wednesday of the third and last week of three. The topic is art history, and this is the second art history block in the school year. At the beginning of the block, the students received a paper which briefly outlined the contents of the block.

> *In this block we will discuss the nature of art, its meaning and different roles in society and explore its origins in some early cultures. Each student will research and present the art of an early culture or of recent indigenous peoples. The aim is to learn how to study, describe and interpret different art forms in different cultures and to explore what art can mean for us.*

The pedagogical aim is essentially to school the students in accurate observation, description and comparison before coming to any subjective judgements, particularly at an age when they are prone to high levels of subjectivity in their judgement. It is a rigorous method that teaches them to look before they form a judgement. In effect they are learning hermeneutic methodology, though they are not told this. This is a skill that not only applies to art but also is an excellent way of developing powers of judgement.

This block description was followed by a brief outline of the forms of learning. These include looking at works of art singly and in groups, group discussion, doing written and verbal descriptions following a particular sequence, drawing and copying works of art, whole class discussions, individual reflections expressing a personal relationship to the works and completing a portfolio with prescribed categories of content (sketches of parts of the works studied, at least one full copy in drawing, three picture descriptions, notes on the work of art looked at and information about the artist and their times, a formal essay on an aspect of the whole theme). The block description then outlined the criteria for assessment (active and constructive participation in the lessons, understanding of the art discussed expressed orally and in text, picture analysis of at least three works of art, piece of individual research and a verbal, pictorial or written presentation, submission of a portfolio of work by a set deadline).

In the first block, the class discussed what art is and what kinds of art there are. This led to interesting discussions as to whether cooking, ballroom dancing or DJ-ing were art forms and if so, what distinguishes art from craft and simply doing things well. The class identified that one can categorize art in varying ways, essentially as arts involving space (drawing, painting, sculpting, architecture) and those involving time and live performance (music, dance, theatre, poetry, etc.) and combinations of these. By the end of the block, the class had learned that art is one of the oldest known forms of human cultural activity and apparently occurs in all societies. It does not appear to have immediate survival value but probably had economic value from early on, though not exclusively. Art appears to be a response to a human need to create or articulate meaning.

In the first block, the class studied Ice Age art from around the world, including the forms it took and the places it was made. They discovered that the earliest known art was already highly developed (recent discoveries of cave paintings in Borneo, cave paintings in La Grotte Chauvet, the mammoth ivory figures from the Schwabian Alps, Aboriginal rock art) and was probably related to ritual, involved several media including music, dance and storytelling and had symbolic meaning. Practical activities included recreating Ice Age cave art using charcoal, chalk and pastel colours in the range of colours used at the time, painted onto surfaces of plaster, semi-dried clay, papier-mâché or cardboard to imitate cave walls or clay-modelling to imitate sculpture (some students tried carving wood). The students also wrote pieces of creative writing, imagining the production and use of such art. They individually or in pairs researched and presented various indigenous art forms (Aboriginal, Inuit, San, etc.).

In the second block, the class focused on art in early urban cultures. In class, the students looked at ancient Egyptian wall-painting and sculpture and compared this with ancient Greek sculpture and Etruscan and Roman painting and sculpture. This was mainly done using projected photographs. As well as learning about the cultural context and techniques of such works of art, the students made copies in drawings but also enacted the sculptures by imitating the stance of the figures and experiencing the forces of balance and tension, and how these evolved from static archaic forms to dynamic figures that appear to be breathing and in movement. This activity enabled the students to experience what the artists were trying to express far more effectively than mere observation. A visit to a museum was used to look at a few examples of Roman sculpture. At the end of the block, the class looked at examples of sculpture from West Africa and Polynesia.

The main lesson on this day begins with the students arriving and having informal conversations in the classroom. The teacher stands at the front and asks the class to stand up. There follows a brief formal greeting, followed by the class speaking the morning verse together. The students sit down, and one student leads the class in a song accompanied by the guitar. Another student recites two poems and asks the class to comment on them in pair work. This is followed by a brief news report by two students who have chosen an item from the day's news, in this case a report from the civil war in Syria and a story about the condition of refugees on a Greek island near the Turkish coast. This is discussed. Many of the art works we are looking at come from the region, and this is commented on too.

This is followed by a review in groups of the individual project work. Each student explains to the others what stage they are at in their preparations and what their main focus is. One member in each group makes brief notes for the teacher. The class then comes together, and the teacher leads a brief review of the picture observed the previous day. Each student is to mentally recall the picture (a slightly less than life-size sculpture from the excavations at Pompeii depicting the goddess Isis in marble found in the temple of Isis, showing a blend of Egyptian attributes, Hellenistic Greek and Roman portrait style). They reconstruct the image verbally in pair work and then make a few notes in their journals. After this, we discuss in the class what the image portrays and what is particularly interesting about it (the wonderful blend of Egyptian, Greek and Roman style elements, belying strict classical art categories). The teacher summarizes what has been said and one student makes brief notes on the blackboard (for the benefit of students who need this support). Then the teacher introduces the theme of how art is rarely fully isolated either from the past or from other cultural traditions. An image is shown of a painting by Vincent Van Gogh reflecting Japanese woodcut style, a sculpture by Pablo Picasso showing the influence of Benin sculpture from West Africa and James Sterling's Staatsgalerie in Stuttgart, considered a classic example of postmodern architecture blending elements from different cultural styles (Egyptian, Classical Greek and Roman plus modernist glass and steel).

The next task, using the existing picture material the students already have, is to work in pairs to order the sculptures in a sequence showing the increasing mobility and dynamism of the figures, then find a free space in the schoolhouse to take up the stance of the sculptures and photograph each other using their smartphones. The task is to order the photographs and make a picture gallery for their portfolios (at home, or in the computer room).

The last 15 minutes or so are for the students to work quietly on their portfolios and write descriptions of what it felt like to take up the various poses. During this quiet working time, various students went to the teacher for some feedback on texts or to show their portfolio to identity what was missing. The lesson ended at 9:50 a.m. informally.

Tasks for the reader

- This main lesson differs in a number of ways from the first vignette, grade 3 in India. What was similar and what was different in terms of the lesson structure and tasks?
- Can you imagine what competences are addressed in such a lesson?
- Can you identify the various learning activities?

Generative principles for teaching the study of art

- The approach is symptomatic rather than attempting a comprehensive overview of art history. Art expresses how people relate to the world and themselves in different times and places and the consciousness that accompanies this.
- It is important that pupils learn to make accurate observations and interpret hermeneutically what they see in a systematic way.
- Art is experienced as an important field of culture and knowledge through which people in different cultures strive to give expression and meaning to what is important to them.
- Learning about techniques, materials, gestures, embodied experiences and social/cultural contexts is important for the depth of experience and opportunities for identification.
- Study of art is separate from, though obviously related to, practicing fine arts.

A foreign language lesson in grade 12

In this school in Germany, foreign languages are taught in blocks of 90 minutes a day for three weeks (15 double lessons). Traditionally, foreign languages were taught after the main lesson, usually in three 45-minute lessons a week. In this school (one of a growing number who now teach foreign languages in blocks), the blocks are usually located in the second track (the first being the main lesson for 100 minutes at the start of the day), though they can also be in the main lesson track. Grade 12 (18-year-olds) have three blocks of English and three blocks of Spanish alternating throughout the year. Each block has a theme. In the previous year in grade 11, the themes had been making newspapers, analysing and writing short stories and the topic of identity and young native Canadians based on Drew Haydon Taylor's play *Toronto at Dreamer's Rock*, set in the late 1980s. In grade 12, the students have worked on Shakespeare's *The Tempest* (performing scenes in groups and then putting the scenes together in an impromptu internal performance), and now in the second block they are studying the post-colonial experience using short stories and images. The main text being studied is Petina Gappah's story *An Elegy for Easterly*, set in a township in Zimbabwe.

This lesson is in the third and last week of the block. In the first two weeks, the students read several short stories by authors writing in English from Nigeria, India and South Africa. The story that provoked the most discussion was Chinua Achebe's *Dead Men's Path*, which portrays a reforming head teacher in a rural school who ignores local advice that a path running through the school grounds linking the village to the graveyard should be kept open because it is used by the 'in-coming' unborn children. The night before the school inspectors are due to arrive, the school grounds are trashed. The story contrasts traditions with rational modernity, which the modern head teacher sees as the kind of superstitions he is trying to overcome. Opinions in the class were strongly divided, with some pupils tending to side with the head teacher and others taking the villagers' view that the path should be kept open for the unborn children.

At the beginning of the block, the students were given a sheet which outlined the content and aims of the block, along with assessment criteria: constructive participation; speaking the foreign language at all times in the lesson; understanding

of the themes discussed as evidenced by three written texts (in English as foreign language), one factual, one analytical and one piece of creative writing; evidence or corrections, list and short commentary of the texts worked on; and reflection on the block, this last piece not being assessed. The whole lesson is conducted in the foreign language.

The lesson begins with formal greetings and the morning verse in English. There are a number of Muslims in the class and a girl from Togo in West Africa, who belongs to a charismatic Christian faith group, and another couple of students who were unsure if they believed in a God. In some prior discussions (at the beginning of the school year) about the verse, which contains the line "to thee O Spirit of God, I seeking turn . . .", all members of the class agreed and were willing to participate in speaking the verse and that each person can imagine whatever they wish when they speak these lines. The discussion had been fruitful and interesting. Following the verse, the pupils formed pairs for 5–10 minutes of "small talk" in English, covering a range of everyday topics familiar to the students. Then in groups of four, the students shared their experiences of the text (*An Elegy for Easterly*) that had been read over the previous three days and weekend, with the aim of clarifying that they all understood the story. This took around 20 minutes and was followed by classroom discussion about the themes in the story. Most of the discussion turned on two episodes, the rape of the mentally unstable and ostracized Martha and Josephat's wife's search among the Evangelical church communities for help to get pregnant. The teacher winds up the discussion by asking two students to summarize the issues and talking points in the story for those wishing to take notes. The students are given a choice of tasks, which they can now start if they have finished previous tasks (students generally work at their own pace in the quiet working time). The tasks are to write an analysis of the story showing the stylistic devices used by the author (which have been discussed in this and previous blocks and include narrative perspective, evocative but sparse prose, repetition and juxtaposition of surprising elements, brief authorial comments within the narrative, inclusion of non-English words in Shona or other local languages, etc.), to describe the social and economic conditions in the township or to produce a piece of creative writing based on the story (e.g. Martha's earlier life, eyewitness account of one episode, story of a newcomer from the country arriving in Easterly, etc.). The students work until the end of the lesson. In this period, the teacher goes through texts with individual students, making suggestions for improvements or explaining some point of grammar to a couple of students who request this. An outside observer would probably notice the relaxed relationships between students and teachers. The teacher has taught the class for a number of years and is very experienced.

Tasks for the reader

- If this description was your lesson and you had to write a lesson plan with an explanation, how would you formulate the plan for this lesson?
- What elements of self-activity and scaffolded learning can you identify?
- Can you imagine what teaching methods over the years have led to this kind of lesson with 18-year-old students, i.e. what would need to happen before students could have a 90-minute lesson in a foreign language with a challenging theme like post-colonialism?

Generative principles for foreign language learning

- Learning other languages supports the development of the person and intercultural understanding, as well as having practical use.
- Two foreign languages are taught from grade 1 (age 6) onwards, and these are continued throughout the whole school.
- Language teaching acknowledges the holistic nature of language and uses a holistic learning approach, that is, involving the body, feelings and ideas.
- Second-language learning is modelled on natural first-language learning, though modified to take account of the age of the learners and what is afforded by the classroom situation. Languages are learned through immersion in a rich linguistic environment, through interaction in meaningful social practices. This means that all lessons are conducted in the target language and involve age-sensitive activities in the classroom.
- Children therefore need a lively and rich language environment.
- Orality precedes literacy and remains the core activity in language lessons. Both foreign languages are taught orally for the first three years. The aim is for each learner to develop voice in each language. It is important for learners to experience a range of voices in the language.
- The emphasis is not on learning the language *per se* but using the language as a medium for age-sensitive activities (play, conversation, poetry, song and literature, film, issues relating to identity, learning about the cultures that speak these languages, learning about other subjects in the foreign language such as geography or history, etc.).
- Authentic literature and materials are preferred.
- Language learning cultivates aesthetic sensibilities.

Generative principles for teaching and learning

Principle 1: Waldorf education takes the spiritual dimension seriously

Generative principles:

- Waldorf pedagogy does not promote any particular worldview or religious beliefs, though it does actively acknowledge a spiritual dimension to human beings and nature and cultivates this in its practice.
- Spirituality manifests in experiences in which the person feels herself to be part of a greater whole, and when she experiences the disposition or potential to become freer.
- Taking account of the spiritual dimension in understanding the human being and the world changes the relationship of the teacher to the teaching, to the pupils, to the teaching material and to herself.
- Teachers no longer interpret their experiences in materialistic terms only but try to imagine real relationships and connections that can't be directly perceived.
- They work with the idea that this invisible world is a reality that has real effects on people and life. Events are deemed to have a meaning, even if this is not initially apparent.
- They encourage a relationship to spirituality through their pedagogy both implicitly and explicitly, though leaving pupils free to respond as they are capable and willing to.

- Teachers engage in their own spiritual development.
- Teachers are able to find appropriate ways to represent the spiritual background of Waldorf education, that leaves others free yet does not deny this dimension.

Spirituality used to be something seen in the academic world as "intellectually invalid, scientifically unverifiable and ultimately meaningless . . . [however] the advent of a post-modern world, with its rejection of such pseudo-rationalistic dogma, has brought with it a revival of spirituality" (Wright, 2000, p. 2). This has happened to such an extent that many school curricula now include spirituality as a learning goal. As Wright adds,

> at both the popular and academic levels a reaction against the failures of modern rationalism, materialism and capitalism, coupled with a new-found freedom from the dogmas of traditional religious discourse, has helped to create a renewed interest in spiritual questions and issues.
>
> (ibid.)

Recognition of spirituality is not to be confused with the resurgence of religious fundamentalism, but is something more inclusive, less connected to ethnic or nationalistic identities. Wright defines spirituality as "our concern for the ultimate meaning and purpose of life" (2000, p. 7). It is linked both to what is mysterious and elusive and to that which is "vital, effervescent, dynamic and life-giving" (ibid.).

The approach to spirituality that Waldorf education takes is in many ways a more concrete understanding of spirit, without contradicting Wright's definition and characterization. The fact that this Waldorf understanding is so detailed and concrete is what makes it problematic for many people who recognize the spiritual dimension, though in a vague, undefined way. Reading or hearing about Steiner's detailed descriptions is overwhelming and for that reason, perhaps also suspect. Academic accounts of spirituality obviously have a very different tone. I reiterate, however, that Waldorf education does not have to engage with the whole structure of Anthroposophy but only with that part of it that is relevant to understanding learning and development. To take a theological analogy, I do not need to study theology (Orthodox, Catholic, Protestant or Nonconformist) to understand the parable of the good Samaritan. Or to take a scientific analogy, I do not need to understand the complexities of time, space, light, energy and matter to design, build and operate a wind turbine. The bigger picture always helps but is not essential and is certainly not a matter of belief. I acknowledge that not all Waldorf colleagues take this view.

Other than the fact that spirituality is considered by many to be unscientific, or rather inaccessible to conventional scientific methods, there are other reasons why spirituality (often pejoratively linked to spiritualism) is treated with suspicion. As Schieren (2012a) points out, spirituality is often associated with exclusiveness, sectarianism, dogmatism and devotion to a guru. It is also often seen as the privilege of the initiated and therefore seen as undemocratic. This exclusivity is a criticism often levelled at Anthroposophy as a "one-man-science". There are, of course, quite other, more democratic links to spirituality. From the 17th to the 19th century in England, the roots of democracy lay not in ancient Athens but in the Nonconformist movements, in particular in the Quakers, Shakers, Diggers, Levellers and other radical Protestant Dissenters (Hill, 1972), who basically believed that every man *and* woman

has access to spirituality and that the divine can be found everywhere in the world (Hessayon & Finnegan, 2016).

What does taking the spiritual dimension seriously mean in education, if it is not just confined to religion and ethics lessons? The Waldorf approach makes a number of assumptions about the spiritual that informs its practice. Following Steiner's theory of knowledge, there is an epistemological and ontological aspect of how we generate knowledge, which involves the spiritual dimension. The pedagogical anthropology on which Waldorf education is based recognizes the spiritual dimension in the human being in several ways – through the non-material processes of the body of formative forces, through the psychological-spiritual aspects of thinking, feeling and willing and in the recognition of the I as the spiritual core of individuality. Furthermore, there is the cultivation of spirituality in classroom practices through the cultivation of wonder, reverence, awe, a sense for higher spiritual meaning through ritual and the use of verses that have an explicit spiritual content. Through history, art, literature and ethics, the pupils learn about the ways that different cultures experience their relationship to the world and the spiritual dimensions, and how this is expressed in different religions (see Rawson, 2020a, 2020b, 2021).

In some countries, Waldorf schools also offer religious lessons. These are nondenominational and overlap closely with meanings of the term spirituality, whereby spirituality has both an individual level of experience but also a communal dimension. In such lessons, children experience different realms of spirituality through story, biographies, study of religious texts from different religions and their practices. In some Waldorf schools in some countries, there is a tradition of voluntary nondenominational, originally Christian religious rituals for pupils who do not belong to any particular faith community and whose parents wish this. This ritual tradition is very much a minority activity, tolerated on the whole with respect by other teachers who for a variety of reasons do wish to participate. Some would say that the whole curriculum contains perspectives on spirituality, values, ethics, holistic perspectives, reverence, wonder, respect and ritual, which are considered vital for children to experience in their lives and thus there is no need to have extra lessons to provide this. One could call this inclusive spirituality. Others take the view that this is insufficient, that a sacral space needs to be created, free of other educational expectations. I take the inclusive view and suspect that the need for an extracurricular space takes a too-conventional view of the rest of school learning. I align myself personally with the British tradition of nonconformists and dissenters who see the divine potentially everywhere and mistrust the role of the preacher in whatever form they manifest. In other countries in which education and religion are strictly separated, these practices do not occur.

As I have explained in Part 1, Steiner's theory of knowledge assumes the involvement of human consciousness in the construction of reality by bringing sense perception and concept together, and the source of the concept is the spiritual world of ideas. Thinking is a spiritual activity because thought is the nature of the fabric of the world, what holds everything together in a dynamic warp and weft of relationships, and the individual act of thinking is the individualized manifestation of this (Schieren, 2012b). This is the source and nature of individual autonomy; it is not given but it can be achieved through self-development. Consciousness of reality is not an everyday state of mind but one that can be developed through meditative practice, though anyone can have inklings of this state at any time. Steiner's approach was to learn how to do this intentionally and with consciousness. We are embedded in the spiritual world but

we are asleep to it; we are unconscious of this dimension. Thus, spirituality in Waldorf terms is both a process of knowledge and emancipation based on autonomous thinking, because both strengthen the individual's ability to live in the spirit.

The pedagogical application of this epistemology is explained later in most of the generative principles. As Schieren (2012b) notes, this manifests in sense-based direct experience-based teaching, the use of imagination, developing skills and abilities. This approach applies right across the curriculum, and he gives the example of form drawing, in which the child must embody the forms and then internally engage with them before being able to bring them forth through the inner activity of forming, rather than following guidelines. Being able to draw a straight line or curve freehand requires an inner experience of straightness or curvature. Equally, animals and plants must be experienced in their morphology, behavior and context and inwardly visualized and 'felt' in their shapes and movements. Imagining images requires a similar involvement. Above all, the Waldorf approach to spirituality means recognizing the spiritual dimension in each person and creating spaces for this to emerge in the form of personhood.

Holistic experiences of all kinds imply that everything somehow belongs to a whole that is not reducible to parts without loss of meaning, whether this be the ecology of the biosphere or the functions of the brain (which the neurologist Thomas Fuchs, 2017, describes as an ecological system in itself). In Steiner's system, all these processes and holistic entities are non-material and therefore belong to what he called the soul-spiritual dimension, even though of course we need physical organs and especially the brain to be aware of this. A spiritual perspective means not accepting reductionist accounts of this process that say that the sensations, feelings, intentions and thoughts we have are produced by our nervous system and brain.

We are moving in the spiritual dimension when we give things meaning and when we represent this meaning through symbols. I realize that some readers will say, yes, but that is just social construction, and indeed in a sense, it is. But is social suggestion, to borrow Berger and Luckmann's (1966) phrase, or social construction *material*? It is the invisible force, the sum total of activities generated by how people behave, talk and think that comprises the lifeworld (Gallagher & Zahavi, 2012). It is the ideas that both shape and come to expression in culture. People speak figuratively of the spirit or Zeitgeist of 1968, but one can also see complex cultural explosion as a non-material and therefore spiritual phenomenon that manifests, as spirit must, in matter. We know that things have meaning in relation to their origins and context, which are also not visible. In sociological terms, structures are organizational but also mindsets, ideas and moods (think of how fear and anger, or hope, can motivate millions of people). If there is a higher order, a divine dimension or reality, there may also be a whole batch of lesser and lower orders of human origin. Spiritual is not *per se* good, better or best. That is a problem with most definitions of spirituality – they tend to be entirely positive, which also makes them suspect. There are dimensions of the spirit I don't want to go to. The recently identified self-portrait of Vincent Van Gogh from Oslo is probably the only painting he did of himself whilst suffering psychosis (report in *The Guardian*, 20th January 2020). It is a portrait of spirituality that even he could barely face.

People respond to feelings, hunches and intuitions and often act on them. Not everything we do can be rationally explained. Many such feelings and emotions are profound and affect us deeply, such as the feelings we have when we encounter beauty, goodness or truth in people and in the world. We value creativity, imagination, novelty, uniqueness and charisma, and we feel deep satisfaction and elation when we

attain what seemed impossible and when something appears special to us. These are all manifestations of a (positive) spiritual dimension. The subjective experience that I cannot explain such experiences is already spiritual when I sense that my knowledge of them cannot be reduced to contingency. The point is not whether someone else has a rational explanation for the phenomena I experience; spirituality starts in the realm of subjectivity but is often accompanied by a sense of certainty that the experience is real. Steiner's method is to observe this process exactly and then gradually gain certainty in our experiences, which also includes finding explanations for them. The point is the meaning that the experience has for me. Rationalism is arrogant when it declares my experience to be invalid or a mere illusion. It just means that it has another explanation, one which may or may not have meaning for me.

We can have profound religious experiences or feel deep peace when we meditate or do yoga. We can experience deep loss at the death of friends and family, or we can even be touched and moved by events on the other side of the world. All these phenomena are spiritual and so common that we take them for granted, although if we had to explain their nature, we would be challenged to come up with rational accounts. We can, of course, explain such experiences away, denying them and belittling their significance, being embarrassed that we allow these experiences to affect us. Nonetheless, the academic discipline of children's spirituality (and the journals that belong to it) has long documented that spirituality cannot be explained away as psychological affects or illusions but is a form of self-authenticating knowledge based on real experience (Cole, 1992).

Research reported by Hay and Nye (2006) suggests that two-thirds of the adult population are aware of the spiritual dimension to their experiences, despite a socialization that tells them that this is nonsense. Spirituality in childhood is even more common but frequently unrecognized. The problem with spirituality, according to Hay and Nye (2006) is that it has become largely privatized and is overlain with modern Western assumptions of rationality that obscure, suppress and even repress "the natural spirituality of the human species" (2006, p. 32). Induction into adult society may close down children's access to spirituality and turn it "from something explicitly reflected upon, and therefore potent within political and social life, into something implicit and vague, disconnected from the mainstream of human activity" (ibid.). Hay and Nye speak of the social destruction of spirituality and the disillusionment of childhood spirituality. They ask whether the "loss of spiritual coherence is only the surface appearance of a more profound loss or suppression of spirituality which begins in childhood" (ibid., p. 48). No one can estimate the damage caused by the suppression of spirituality, though the symptoms of when people lose their sense of purpose and meaning, when they suffer burnout or seek to hide their emptiness in escapism and fantasy, in entertainment and consumption, in alcohol, drugs and the endless quest for going beyond their limits, of breaking through to their true selves – have been widely documented (e.g. by Paul Verhaeghe, 2015). It is paradoxical that in such a materialistic world, so many people – especially those who can afford to – turn to all manner of spiritual practices.

The idea of taking the spiritual seriously means taking all these experiences seriously, integrating them into our understanding of the world. It is not a question of choosing which is more real; it's simply a question of acknowledging that we already possess faculties that mediate knowledge of the spiritual to us. Imagination and intuition are ways of knowing the world that actually complement the other forms that

we know, such as science. Educational thinkers like Elliott Eisner (1991, 1994) have long made this point. Philosophers like Ernst Cassirer (1944/1961) have pointed out that human beings understand the world using symbolic systems including myth, religion, art, literature and science. Steiner says that we can go beyond the limitations of rational thinking by developing intuitive thinking. Jennifer Gidley (2017) argues that if we want to take long-term responsibility for our actions, we need the powers of imagination and intuition, not as fantasy of wishing but as postformal, integral, holistic forms of knowledge.

Why is it that so much of education, despite the addition of spiritual values to most school curricula, is still basically reproducing materialistic notions of what education is for; filling up the empty vessel with the predigested, prescribed and testable knowledge predicted to be 'necessary and useful' for the job market in a totally unpredictable and highly insecure future. How much history is useful to the future information technology (IT) worker if those history lessons have not even taught us to recognize the ongoing evils of colonialism and xenophobic nationalism? What use is ethics or democratic civics teaching when it is obvious that the world of power has a totally different orientation?

Steiner's concept of the I as spiritual core of being is something that Waldorf teachers work with. We can try to see that the developing, emergent human being manifests the interaction between material and non-material processes, and we can participate in structuring these processes. What Jan Göschel (2012) calls the biographical mythos, the signature of individuality, may be experienced in our thinking when we open ourselves to the child or young person's being through a process of constructive hermeneutics. It makes no sense considering the spiritual if we do not recognize this dimension in the child or indeed in every other person. If we only privilege certain types of knowing, we miss the opportunity to nurture these other ways of knowing. If we value art and the imagination as much as academic subjects, this sends a strong signal to children that this is important (something neurologists such as Gerald Hüther (2008) have been saying for years).

Peter Loebell (2012a) refers to what he calls the totality of a person's capacity, or integrated potential to act responsibly in a given situation, as an indication of individuality and the presence of spirituality. Loebell sees the preconditions for spiritual experience in the person's ability to attend to something, make an inner connection to it and have the experience of meaning in relation to it. The inner strength to open oneself to a new experience is an important precondition of spirituality, and he makes the case that the inability to do this, the inability to let go of existing positions and stances, the inability to tolerate ambiguity and uncertainty, are factors that limit experiences of spirituality. Equally, the willingness of others – in pedagogical situations, the teachers – to accept and welcome the uncertainty and vulnerability of children and young people is a precondition that they can make sense of their experiences.

If we help children to value and respect the achievements of other cultures and their other kinds of spirituality (as Waldorf children in the United States do in relation to Native Americans, as Australian children learn about Aborigines, New Zealanders about Maori culture, Finns and Swedes about the Lapps), then they may be open to other cultures asking questions about their own. If we address the spiritual dimension in the world, particularly through the immense richness of the narrative curriculum, we acknowledge it in ourselves, and this is a strong message to children. At every turn we place the human being in the centre of our focus, in our wisdom and power but

also in our destructiveness. We are a "being darkly wise and rudely great . . . in doubt to deem himself a god or a beast" (to quote Alexander Pope's powerful but not very poetic poem, *The Great Chain of Being*).

Recognizing the spiritual in the world makes a difference to what and how we teach. This is not because we need to offer a religious perspective but rather more fundamentally, we need to offer a perspective of respect, wonder, reverence and intrinsic meaning. Phenomena are taught in ways that they can be experienced for their own sake by the pupils not just because they are useful to us today to know. The world of nature is not just something out there that we can feel sentimental about as we destroy it. Perceiving the spiritual dimension means appreciating the qualities of things, such as colours, forms, living creatures and mathematical properties, and not simply as building blocks to useful, technocratic knowledge. Children and young people need to experience a sense of meaning, which is not the same as naming something and knowing what its uses are.

The spiritual dimension becomes apparent as soon as we start thinking in terms of processes, relationships, underlying connections, rhythms and the meaning these can have. It means approaching the world tentatively with our senses and minds open to the other, listening carefully to what we don't understand, rather than capturing, dissecting, analyzing and controlling it. It is more an attitude towards the world than a set of contents. After all, the Waldorf curriculum covers practically the same topics across the subjects that mainstream schools cover. The difference lies in how and when this is done. Peter Lutzker (2012) has shown how aesthetic experiences through poetry (in English as a foreign language, in particular reading William Wordsworth) can lead to a transformation that brings the student into the presence of the spiritual. The same is true for the chemistry of acids and alkalis (so I am told), or the worms in the Watten Sea mud (which I have seen).

Taking a spiritual perspective of other people and taking this into account means treating pupils with care and respect and being open to what they can become, rather than fixating on what we think they are and have become. Identifying and labelling students as problems, as challenging or even as 'good' may limit us from seeing their potential, what is emergent. The asymmetrical nature of relationships between adults and children and youth means that what we think of them is far more powerful than what they think of us. Taking each individual seriously as they are, yet being open to what they can be, is essentially a spiritual attitude because it acknowledges a possible reality that has not yet materialized. The central thought is that the pupil standing before me is only a limited manifestation of who this person is and can become. This attitude alone, if genuine, is more important than any speculation about a person's 'true self' or nature. Being spiritual is not about saying prayers to a God or gods (though it can be), nor is it about celebrating certain traditions or practicing certain customs. These elements may support a spiritual approach, but they may equally have the opposite effect. Of course, being moral and behaving ethically as a teacher obviously has the effect of creating a trusting atmosphere and setting a good example, and teachers should do this anyway, but this is not necessarily accompanied by recognition of a spiritual dimension.

In Barz and Randoll's (2007) study of former Waldorf pupils, 68.6% of those questioned said that their experiences at Waldorf schools favoured their spiritual experience and capacity for subsequent experiences. The study suggests that the 12 or 13 years in a stable but heterogenous group of differently abled people – the Waldorf

class – helped mutual recognition and fostered a sense of individuality in each. They felt that the equal value placed on academic, practical, artistic and social learning sensitized them to the importance of aesthetic and spiritual values, as did the formative and ipsative assessments. The ethics of care of the teachers and the encouragement of self-activity all contributed to a basic sense of values. Not least, of course, the rituals and the daily sharing of the morning verse, in which higher powers and spirituality are related to school activities – looking into the realms of the world, looking within and the prayer-like element of asking for strength and presence of mind for working and learning.

Finally, taking the spiritual dimension seriously as a teacher means taking one's own self-development seriously – and not only for selfish reasons but as a professional necessity. If I am to really understand the learning process and development, I need to recognize that there are 'hidden' dimensions to this. Nor is it a question of simply believing in the system; one has to come to insights oneself through reflection, meditation and learning to cultivate and trust in pedagogical intuition. My view is that teachers should espouse only as much about spirituality in their educational practice as they can honestly base on their own experience. That applies for the spiritual content of Anthroposophy – I can only base my practice on Anthroposophy in as much as I understand and experience it. As Jennifer Gidley (2016) has argued, a postformal education is essential today; that is, one that does not see the development of intellectual rationality as its highest endeavor, but goes further and seeks to create learning spaces in which individuals can develop a holistic, integral and spiritual consciousness. This assumes that teachers have first got that far.

Teacher skills

The teacher:

- Has a personal relationship to the spiritual dimension and reflects on what this means in terms of her approach to the teaching content and to her understandings of pupils
- Can act pedagogically in ways that enable children and young people to have spiritual experiences in their encounters with the world
- Recognizes and values the expression of spirituality in children and young people

Research questions:

- Spiritual insight often occurs in intuition. Intuitive ideas and insights can be explored retrospectively in terms of their fruitfulness through self-observation. Journaling can be helpful to monitor such experiences over time.
- Spirituality in children and young people as identified by a sense of higher meaning, inwardness, numinous experiences in relation to certain people, places or objects, natural reverence and, particularly in young people, the search for higher meaning and identity, can be observed or can be evidenced in children's talk, writing and artwork. As I describe in the section on research, spiritual experiences in children are very common, though often not recognized.
- Teachers can gain a sense of the spiritual dimension in individual children and young people by contemplating the way they inhabit their bodies, how they relate

to others and the world. In such individual case studies (discussed later in the section on *knowing the child*), the aim is to develop a sense for the signature of individuality as it expresses itself in movement and behaviour.

- Steiner's idea of using boundary ideas to expand our awareness of spiritual realities can be applied systematically in any form of reflective practice, and this can be the subject of autoethnographic research. Likewise, ideas drawn from the Foundations can be used as heuristics in hermeneutic investigations of pedagogical practice.

Principle 2: Sense of coherence is the basis for healthy learning and well-being

Generative principle: Healthy learning depends on maximizing the learning content, using a minimum of means and resources and effective use of time without the pupils losing their sense of coherence, which is the basis for healthy learning. Indicators for sense of coherence are understanding, self-efficacy and meaning making.

Steiner's experience as a private tutor for the Specht family taught him that teaching has to be economical in terms of preparation, efficiency and effectiveness, and he also learned that children learn through strong experiences. Regarding teaching economy, he said:

> The Waldorf idea is that all the teaching will be so organized that the maximum amount of learning experiences or content can be brought to the pupils in the minimum amount of time using the simplest possible means, but in such a way that the individual children never have the feeling that they have lost the overview and that they don't understand.
>
> (Steiner, 2003, lecture 30th December 1921)

Steiner often referred to this idea as soul-economy, or economy of teaching, in relation to the well-being of children and young people. His description of teaching here, of maximum content with minimum means in the shortest possible time, is a startling definition of efficiency. However, the significant qualification is that the children should never have the feeling that they don't understand, because this could have a negative effect on their sense of well-being and would harm their health. Steiner's experience was that if children were given too much information that they could not 'digest' in the sense of understanding and being able to assimilate it, then this would drain their vitality. This is a radically psychosomatic perspective on learning. The way a child is taught can have both short-term and long-term impact on their health and well-being. Other educational theories may criticize overload in cognitive learning as ineffective because this may weaken motivation and attention, but health effects, positive or negative, rarely play a role. Most theories of learning don't refer to the health effects.

Waldorf practitioners have been convinced that what Steiner calls soul-economy and other aspects of healthy education do lead to healthy learning, and gradually they have developed a deeper understanding of how this psychosomatic effect actually occurs. The key, however, to understanding Steiner's notion of health, well-being and learning came with Aaron Antonovsky's (Antonovsky, 1987, 1996) notion of health promotion through salutogenesis. In the last 20 years, Waldorf education has increasingly drawn on salutogenic theory to both research Waldorf practice and theorize its methods (e.g. Zdrazil, 2000; Glöckler, 2002; Marti, 2006; Wiehl, 2015a, 2015b).

Antonovsky (1923–1924) was an Israeli-American medical sociologist and public health researcher who, after many years of studying the general factors that lead to health and ill health, formulated the construction of salutogenesis (Lindström & Lindström, 2006; Lindström & Eriksson, 2011). Instead of focusing on the pathological factors that lead to ill health and disease, he explored why some people who are exposed to the same risks and stressors nevertheless enjoy long-term health and psychological well-being. He called this approach salutogenesis, a neologism based on the Latin word for health and well-being (*salus*) and the Greek word for origins (*genesis*). Over the years, salutogenesis theory has been applied to a wide range of fields, including increasingly in education (Mittelmark et al., 2017). Lindström and Eriksson (2011) see salutogenesis as an umbrella concept that covers a range of factors that enhance health, well-being and resilience that are not limited to a sense of coherence, including concepts such as empowerment, self-efficacy, quality of life, hope and action competence, among others. It is important to note that the construct of salutogenesis was developed in medical sociology and whatever unique individual factors there are, the social context is fundamental. Creating salutogenic environments involves the whole community, the whole school and the whole child, and approaches involve high levels of participation in community activities (Jensen, Dür, & Buijs, 2017). Antonovsky's approach focuses on all the people within the school system and seeks to promote salutary (i.e. health-generating) factors rather than merely avoiding risks and stressors, and looks at the whole person rather than only at the health condition. Antonovsky identified that some people have a salutogenic orientation to life, which he defined as a manifesting sense of coherence. A basic disposition to a sense of coherence is a strong indicator that the person is resilient and able to maintain a salutogenic or health-creating orientation.

People who have a sense of coherence generally have the belief that the challenges and tasks they face are comprehensible. The factor of *comprehensibility* links to Steiner's notion that children should have the feeling that they understand what they are being taught. The second factor, *manageability* or self-efficacy, relates to the person's sense that they have the resources they need to act and that they can manage to do what is required of them. They also have a sense of *meaningfulness*. They generally feel that the tasks they face are interesting, personally motivating and worth their while investing time and effort. These criteria can readily be applied to the situation of a learner, and one only needs to consider what state occurs when the criteria are not fulfilled; the learner has the permanent feeling that what she has to learn is incomprehensible and that she cannot understand it, she can't manage the tasks she is given, she doesn't cope and doesn't know how to get support and the whole exercise feels like a waste of effort and lacks any subjective meaning for her. We can imagine what this constant state of mind does for a learner's motivation.

Applying the principle of soul-economy in teaching means observing if the pupils generally have a sense of comprehensibility (Do they understand?), coping and manageability (How well are they coping?) and meaningfulness (How do I know that they find it meaningful?). This means having indicators for these qualities across the subjects. It also means being able to adjust the teaching, the tasks, the social organization and context of the lesson to enhance sense of coherence. It means gaining a picture of the school culture and whether this encourages learning in this sense and what values it places on the range of activities in which children can gain experience of a sense of

Components of sense of coherence	*Level 1*	*Level 2*	*Level 3*
Comprehensibility Understanding-in-life	Understanding in context through stories, use of primary tools, basic cultural techniques (literacy, numeracy, communicative skills)	Exploring, researching the world and understanding information, understanding through comparison	Developing powers of judgement, being able to analyse and reflect
Manageability Certainty in action, self-efficacy	Participating, communicating and cooperating	Producing and presenting	Creating, solving problems, making conceptual models
Meaningfulness Making meaning	Empathy, identification with objects of learning	Taking responsibility, sharing	Taking one's own position, doing good, helping others

Figure 2.1 The elements of a sense of coherence in developmental terms

coherence, as practical subjects like woodwork, artistic activities and social projects all provide different contexts for experiencing coherence.

We can show in tabular form (see Figure 2.1) the various elements of a dynamic sense of coherence and the activities associated with this in three developmental levels. The levels correspond to some extent to age but in a more general sense to complexity of learning. Understanding, for example, occurs at different levels; in context with support or, to take the other extreme, being able to form judgements and reflect.

Economy of teaching is also a question of teaching strategies. The teacher must be able to choose which teaching method is most likely to enable all the pupils to quickly grasp what they should learn and structure the steps of learning to make this possible with the least possible means and materials. Sometimes this involves organizing things so that the students experience something directly; sometimes they need narrative and images; and sometimes it is easiest to let them read texts or work with other media. The teacher has to be able to observe how each of these methods achieves the goals intended. Economy means maximum learning with minimum input *but* minimum stress.

There is a virtue in using minimum means to achieve this. However, sustainable learning is like healthy nutrition; quality ingredients from sustainable sources, prepared in a way that activates the self-activity of the eater (i.e. not pre-processed food that superficially stimulates through salt or sugar) and has a long-term positive energetic effect. The analogy, according to Steiner, is not a coincidence, as the life-processes that organize our nutrition and regenerative processes when transformed are the same processes in learning. Learning is a process not only of digestion but of building new 'substance' and gaining new energies. It is a process of taking in, assimilating forces and releasing energy to build skills and abilities.

Teacher skills

The teacher can:

- Observe and identify the effect of their teaching on the learning behaviour of their pupils; in particular, to observe if they generally have the feeling that they can understand and manage, and that it is meaningful for them
- Choose learning strategies that are age and task appropriate and are effective in terms of resources

Research tasks:

- One obvious research task is establishing indicators across subjects and learning activities that show a sense of coherence in learning situations in schools.
- There is some evidence among Waldorf alumni that former Waldorf pupils are in some respects healthier both psychologically and physically (see Part 3). This is an area that needs far more research, not merely to demonstrate the efficacy of Waldorf but to inform education at large.

Principle 3: The quality of the teacher's preparation influences the quality of learning

Generative principle: Good learning depends on the quality of the teacher's preparation, her relationship to the material being taught and her understanding of the pupil's learning capabilities.

Another major factor that contributes to a sense of coherence and healthy learning is the quality of the teacher's preparation of the teaching material. Steiner realized that burdening children too early with abstract ideas makes them lose interest, which can only be regained by pressure and stress. It also drains their vitality and regenerative forces, which apart from being unhealthy, is ineffective because they will soon forget what they have learned, as memory is related to the forces of vitality (it is hard to focus and recall details when one is tired or stressed). The best motivation is intrinsic. Steiner never tired of warning about the risks to the individual but also to society and the economy by wasting children's energies learning things that no one really needs to know. Without genuine curiosity and interest, capacities cannot grow. It is likely that this concern was greater in the 1920s when school was largely about rote learning, than it is today, but it remains an important question to ask: Is everything children and young people have to learn today necessary and comprehensible?

Steiner frequently made the point that "the question of pedagogical quality is not only one of the interest, hard work and devotion of the students, but the interest, hard work and devotion of the teacher!" (Steiner, 2004, lecture 14th August 1923). If the pupils are to understand, manage to learn and find the contents and activities meaningful, the teacher must prepare the material thoroughly, identify which examples will illustrate the phenomenon most clearly with the minimum of explanation and present it in a lively, interesting way that engages the pupils' interest. The teacher's own genuine enthusiasm for the subject is an active factor in this process. This enthusiasm comes through intense preparation, often done well in advance.

Referring primarily to education up to the age of 14, Steiner gave the example of a teacher needing three hours of preparation for 30 minutes of intensive learning on the part of the pupils. After the three hours, *all* the teacher needs to prepare, he adds, is the artistic presentation! This is obviously not an exact figure but an indication of the relationship between teacher activity and pupil activity. The teacher has to find and select the material that is just right for this class; have a clear idea what aspects the pupils should learn; and judge what is age appropriate, interesting and relevant to the pupils, taking account of what they already know, what they need to know and which aspects highlight the essential elements. This is, of course, what course books and prescribed curricula do. Steiner's point was that this should be more focused on the specific situation.

The teacher should be able to judge for herself what phenomena and which details, examples, illustrations, images and information are most salient to their pupils' understanding – which means how it can enhance their ability to place experiences in a wider, meaningful context. Furthermore, the teacher's genuine interest and identification with the topics and the enthusiasm this generates will have a positive effect on the motivation of the pupils. The world is full enough of interesting things that, given enough pedagogical tact to sense what the children might find interesting, should be possible to engage them, especially if the teacher can introduce the subject in a way that immediately enables them to identify with it. Furthermore, there is an accumulative effect; if the teacher usually chooses interesting things about the world and is evidently genuinely interested in these, the children will come to expect this and assume that something worthwhile is coming, thus enhancing their anticipation, which Steiner saw as an important quality in education. He sometimes emphasized that telling pupils, for example, "We can't understand this yet, but we will by grade 9", strengthens and sustains their will and interest.

Here we see an important aspect of the role of the Waldorf class teacher. Her authority comes from the children's long-term experience that *this person* is choosing and preparing interesting things *for them* rather than simply delivering a set programme, and that she role-models the enthusiasm that learning about the world can generate. Because many of the topics taught are new to the teacher, there is a freshness of discovery that is communicated to the pupils. This is far more important at this age than fine-tuned, routine presentations, or indeed the delivery of material someone else has prepared.

In the example of our lesson in India, the class teacher has not been called upon to present much. But she has previously prepared the whole situation by arranging the visit to the garden months in advance, organizing with the gardening teacher that the children could sow the lentils in the first place, periodically weed them and then harvest them. It would hardly have required much extra motivating to harvest what they had sown and tended. The teacher has obviously also informed herself about lentils.

That is why Waldorf class teachers don't use textbooks or use standardized lesson plans or scripts that others have created, though these may be of help to inexperienced teachers. This is an attitude rather than an iron rule. There is literature on the various subjects and themes that class teachers can refer to in order to prepare their lessons, but they still have to choose what content they wish to use. It is the inner effort and personal engagement of the teacher and the clarity of her intentions that are important. The work done by the teacher in preparation is in direct relation to the quality of learning of the pupils. In evaluating lessons, one criterion is the quality of the preparation.

If the lesson is well prepared and the presentation clear, there should be no need to explicitly state what is important and should be noticed. This should be obvious and logical from the presentation itself. This means that the teacher doesn't continuously point out what is important and needs to be learned, doesn't draw the conclusions for the children, doesn't give them the definition or the answer to the problem, but allows them time to explore the experience for themselves and offers help when needed.

The art of teaching is in the preparation and then in the presentation of the material. The pupils have their own experiences; they do not need to be told what to experience. You can take a horse to water, but you can't direct it to drink. However, one can make the horse thirsty and ensure that the water is sweet. One way of making it 'sweet' is in the verbal presentation, when the teacher can select and emphasize elements that make the topic accessible to different types of learners and even individual learners. An example of this would be to start with what the students are familiar with, to "gather them up where they are". Another would be to add details that appeal to different interests; for example, appealing to an empathetic response, or dramatic elements, or elements of surprise. In effect, the kind of rhetorical devices which are used by actors or speakers are of use to the teacher.

This brief description shows that the use of standard teaching materials is not going to generate the same level of personal identification for the teacher and perhaps the pupils. Gert Biesta (2012) has made the point that pupils often want to learn something from somebody, though it is not just about learning; it is also about entering into relationships:

> education is never just about learning, but is always about the learning of something for particular purposes. In addition I wish to argue that education is always about learning from someone. Whereas the language of learning is an individualistic language – learning is after all something you can do on your own – the language of education is a relational language, where there is always the idea of someone educating somebody else.
>
> (2012, p. 12)

It is this relational aspect of the teacher giving something of herself, something that she has created and shaped, that adds a further dimension to simply engaging with ideas through other media. It is more personal. We should not underestimate how young children feel that 'our teacher' has done something special for us. That is the basis of the teacher's authority. It is an authority that transforms over the years to respect, as the pupils become more capable of accessing the world with their own sense of inquiry and wish for meeting.

Of course, pupils can understand much from other media when they are motivated to look with 'open' eyes, as opposed to looking with the question, "What are we supposed to see?" It is clearly not a question of either the teacher presenting the material or students doing their own research and making their own discoveries, as both can be meaningful, but there is no doubt that the personal commitment of the teacher makes a lasting impression on students, as alumni research shows (Barz & Randoll, 2007).

As John Hattie and Klaus Zierer (2018, p. 129) point out, the influence of the teacher on the learning of the pupils depends on the mutually influential effects of the teacher's mastery of the subject, her pedagogical skills, didactic competence and the passion that

arises out of this in the encounter between teacher and pupils. The pupils' motivation comes from three sources:

- Their own intrinsic interest in the world, which, following Steiner, is an expression of their will, which in its primary gesture is to engage with and participate in the world
- The teacher's role as enthusiastic learner
- That both teacher and pupils are embedded in a learning community and learning culture

One need only imagine the opposite; the pupil feels alienated from the world because she encounters it only through artificial and indirect media and abstract concepts, in an instrumentalized, institutionalized context, in which students are required to do certain things and in which acquiescence is rewarded and non-compliance leads to further alienation, in which the teacher or teaching medium has in itself no personal interest in the matter that it is delivering, that external instances are in control, surveillance is ever present, targets have to be achieved and this is what constitutes the learning culture. The robot as teacher would lack this dimension of personal effort, enthusiasm and commitment. I can imagine using robots or learning software at the stage at which what has been learned needs to be practiced. If a teaching/learning software could match exercises to each child's individual learning level, correct the answers and give feedback, it could have a role, but presenting a topic in a lively, imaginative way, responding to the specific learning group is something that in the foreseeable future needs a teacher.

In the lesson example detailed earlier, the relationship is already established and the teacher in this lesson does relatively little teaching of new input (the uses of lentils in cooking). This is an example of economy, because in this instance the teacher was able to take the children to an activity outside the classroom. In another case, the experience may have been organized in the classroom – for example, an experiment with light and shadows, or storytelling presented by the teacher in the case of a history lesson. In each case exists the principle of 'saturation' preparation, maximum material in minimum time with the simplest of means, though in a way that takes all pupils in the class along on the journey, and each one has the feeling she is being addressed and involved and that basically what is being presented can be understood.

Teacher skills

The teacher can:

- Identify what is essential in what has to be learned and how this can be mediated effectively to all pupils
- Evaluate the quality of her preparation by the way the pupils respond to the lesson, as indicated by the level of self-activity that the lesson prompts and in what is recalled on the next day, at the end of the block and by the start of the next block in this subject
- Adequately prepare the lesson plan, materials, tasks and activities in relation to the developmental tasks, and these can be assessed in relation to expectations and the intentions of the lesson plan

Research tasks:

- It would be interesting to study how teachers actually prepare their lessons and how this relates to the pupils' response to this.
- A comparison between different modes of presenting pupils with new experiences and the kinds of knowledge generated could offer criteria for effective learning.

Principle 4: Children and young people need to be ready to learn, and they need time to learn

Generative principle: Learning occurs most effectively when the learners are ready and attuned to learning.

A basic principle of Waldorf education is giving children and young people time to learn and introducing them to new learning steps when they are ready. Being ready means having the inner resources to learn in a healthy way. However, the teacher or school has to initiate and scaffold whatever learning processes are necessary to attain the target level of learning. As we have seen, Waldorf education holds off on formal learning until the formative processes within the child's body become free to shape and form the psyche, and in particular to structure cognitive processes such as focused memory, forming mental images and the first steps in grasping abstraction (for example, in literacy and numeracy). It is not a question of waiting until all children are 'ready' to take the next step, which in some cases might involve lengthy delays in the whole learning process. Rather, the Waldorf approach is to work to a normative timetable, which theoretically finds the majority of pupils with the developmental resources and needs to take the next step at a given age. Within this framework there is considerable tolerance for individuals. Some may be ready to move on faster than the others and need extra tasks, whilst others need more scaffolding and support to catch up. Treating the class community as the norm, prioritizing social processes of doing things together at the same time and having class teachers who accompany the process over many years makes it possible to include all students and meet their learning needs. The point is not to test whether children are ready – some countries have elaborate school-readiness procedures – but to identify how ready, or not, each child is and plan the learning accordingly. The question is not, "Is the child ready?"; rather, "Is the school ready for this child?" Diagnostic assessment can help teachers identify how each child needs support.

If we look at the grade 3 class in India described earlier, the lesson begins with a series of rituals and activities that are aimed at enhancing readiness to learn. Before children can really learn and make sense of their experiences and assimilate this into what they already know, they first have to be as ready to learn as possible under the circumstances. Children come to school from a wide variety of home circumstances, and the journey to school may be long and eventful. Children may be tired, worried, unhappy or unable to concentrate for all kinds of reasons, but the first thing that happens to them in school is that they are personally welcomed and greeted by a friendly adult, and of course by their classmates. However, before this, the room is prepared. Just as the teaching material is prepared, so too are the spaces in which the lessons occur. The classroom is a special place and under the given circumstances, the room should be arranged and decorated in a way that welcomes and provides an environment that is conducive to the activities there.

A classroom is a place of work, but it is also a special place where significant things happen on a regular basis. It is a blend of workshop, artist's studio, sanctuary and holy space. One steps over a threshold to enter and leave it, and it is a special space because of the care and activities that occur within it. It reflects the changing seasons in its decorations and seasonal displays, and it has the tools we need – blackboard, chalk, sink, painting boards, desks and so on – that have their places, as in any well-organized workshop. A classroom is space dedicated to special practices. I used to bring in the cloths that I have always gathered on my trips around the world and use them for the nature table or to decorate the teacher's desk. When you enter a Waldorf classroom, you can usually tell quickly what the theme of the moment is, not only from the pictures hanging around the walls but also from the artefacts one can find, i.e. craftsman's tools, bundles of willows, triangles and protractors, maps, mirrors and Native American headdresses.

The children gather around, and the lesson begins with a formal greeting and the morning verse. There are two verses, one for grades 1 to 4 and one for grades 5 to 12. Steiner referred to the verses, which he composed, as a kind of prayer intended to evoke a mood of reverence and gratitude and an encouragement to engage with the world. Images in the verse open to the different realms of the world – mineral, plant and animal – in interest and gratitude. Then it directs our attention inwardly, appealing to a (nondenominational) divine being for strength to work and learn. The effect of speaking this verse (which of course, with older pupils, needs to be discussed so that it doesn't simply become routine) every day over many years no doubt has the effect of giving a thoughtful, reflective, meditative quality to the daily work if it is 'performed' with the appropriate seriousness and matter-of-factness of ritual. The same is true of the birthday verses. In the class teacher period, each child has her own verse that has been chosen individually. It is an intimate and at the same public ritual shared within the class community.

What follows is a phase of the lesson in which a series of activities are performed as a kind of routine with the explicit aim of focusing the children as a group (singing or reciting together) and awakening individuals in their bodies and senses through coordination, repetition, exercising the memory (e.g. times tables), and of bringing the group as far as possible into a state of relaxed attentiveness that makes learning possible. Many children today have a lifestyle that denies them the amount of healthy movement, meaningful activity and rich, real sensory experience that they need to enable their senses to develop in a healthy way. As a result, they often feel unconsciously uncomfortable in their body, can neither be still nor really physically active and breathe deeply. The symptoms of lack of concentration and focus, the inability to listen and open themselves to the world are complex. I do not want to go into the causes here. The task of the teacher is to do whatever she can to help the children get into a receptive and active state necessary for the kind of learning that Waldorf schools offer. I call this phase of the lesson *warming up and tuning in*. Just as musicians have to tune their instruments and play their way in, or athletes have to limber up and stretch and become mentally attuned to the tasks ahead, children need something similar before the process of learning can really start.

The time this requires and the nature of the exercises will vary from class to class, day to day and year to year. The important thing is to observe what effect the activities actually have on the children, whether it livens them up, calms them down, activates them or makes them more passive, makes them aware of others in the room (singing or playing flutes together, reciting poems) or helps them focus on their fine

coordination (finger exercises, body geography, e.g. put your left thumb on your right knee). Some exercises can train the child's balance and coordination of feet (skipping), others help coordinate the hands and eyes (juggling), or they can activate listening (identify objects by their sound alone, e.g. shaking different objects in a match box, tapping different materials together), memory (recalling objects we see briefly), recognizing patterns, moving quickly or very slowly in a controlled way, lip-reading, hand signs, drawing shapes on people's backs and letting them guess what letters or numbers were drawn. Mental arithmetic tasks, constructing geometrical shapes in the imagination (such as imagining a green point about an arm's length in front of your closed eyes, move this point upwards and to the right, then down to the right and stop level with the starting point; join up the two points to form a triangle and fill this in with red colour, then tip the triangle backwards until all you see is the baseline, and so on). This exercise can have a very concentrating effect on a class of 11-year-olds.

If the children are not awake in their bodily senses of touch, life and vitality, movement and balance, they may lack the basic skills of movement and coordination, attention, ability to adapt to others in confined spaces – all of which are preconditions for learning in school – they are at a disadvantage and their social behaviour often makes it difficult for other children. In earlier times, children developed most of these skills playing in nature or playing games on the streets. This children's culture has been largely lost today, not least because of the amount of time children sit using computers or watching television.

To this end, the model of the *moving classroom* for the first two school grades was developed to enable children to catch up on this stage of development. The lesson may begin with movement games involving balance, mirroring movements, coordination skills and sense of touch. Like the Indian school described previously, the children do not sit on chairs at desks but use specially made benches that allow space for two children to open up their writing and drawing books side by side. The benches can be turned over, and like gymnastic benches they can be used for balancing or climbing over or under. The children either sit on the floor or on firm cushions filled with grain seed. Rather than being supported by a chair, the child has to inwardly maintain balance and focus. The idea is that the classroom is a flexible space for movement and activities that are limited if the room is full of desks. It is also felt that too much sitting on chairs weakens the back muscles and prompts a lazy posture. Sitting upright using their own muscle power focuses and strengthens attention. The research that has been done in schools using this model certainly suggests that the children are on the whole more relaxed and focused (Auer, 2019).

It may be that the children need a longer phase of warming up and attuning on a Monday because of the aftereffects of the weekend but a shorter one on Thursday, when the weekly rhythm has settled. A maths lesson with a lot of cognitive calculating and thinking may need to be balanced with activities that require movement or modelling beeswax into shapes. The teacher has to judge when is the right moment to move on, given all the other things that have to happen. New teachers may follow a strict schedule of timed activities, but the more experienced teacher can judge how long each activity should last.

Some Waldorf teachers refer to this part of the lesson as the *rhythmical part*. In reality, the whole lesson and the whole school day, in fact the whole school week, is rhythmical, so calling one part rhythmical is a bit of a misnomer. The lesson has phases in which certain activities predominate and these depend on the pedagogical

aims one has. A very important part of rhythm within a single lesson is the balance of activities that involve being active and being quiet, speaking and listening, working in pairs or groups and working alone, paying close attention and being more relaxed and perhaps most importantly, the emotional states that accompany learning, such as seriousness, respect, reverence and gratitude but also lightheartedness and humour. It is sometimes said that each lesson should involve laughter and tears. Laughter is certainly a regular feature; however, it doesn't have to come to tears, but there should be elements of feeling touched and moved. The sign of a good lesson is that the children have worked hard and are energized by the end – not overenergized, because that will be a problem for the next teacher – but let us say, fulfilled and ready for a short break. They should not be exhausted, drained, tensed up – as they often are after a test. They should be in a fit state for the next lesson.

The reader can recognize the rhythmical phase of the lesson in the Indian class example described previously. In the grade 7 example, this takes less time but the practical activities of darkening the classroom are also a preparation for the pupils. In the high school lessons, attunement is also important through singing or recitation but alert interest through self-activity is more important than routine.

However, it should be very hard to say what a typical lesson looks like, though in many schools there is a more standardized pattern, either because the teachers have learned that things have to follow a certain pattern or perhaps because similar circumstances pertain. What is important is that the teacher knows exactly what she wants to achieve and then chooses the means and a structure that enable her to realize those aims. Some schools decide to start the day in a communal way. One school I know starts each day with 30 to 45 minutes of community activities in mixed-age groups. These include things like preparing food, cleaning, gardening, sweeping leaves or snow, chopping wood for the classroom stoves or seasonal activities around the schoolhouse and gardens. The older students supervise the younger ones. After this work, the children go back to their classes already attuned and ready to start their schoolwork. A similar thing happens in afterschool care, where the pupils are supervised in mixed-age groups, usually sharing a meal together, resting, playing, working and doing homework, before then going home.

Once the pupils are as warmed up and attuned as possible – the ideal state is rarely attained – a phase of recall occurs and then new experiences are introduced. Following this is a phase of individual or group working. In the younger classes, the end of the lesson may involve listening to a story. Then the lesson is formally ended.

One of the aspects of the lesson described earlier that I felt was most effective was the fruit break. The fact that the children share their food and take turns at preparing it and that all participate in clearing up, was not only a strong social element in the learning group – it also enabled a good break to structure the main activities. Refreshed both in terms of the break but also in terms of some healthy food and drink, the students were 'fit' for the next stage of the lesson. This is a regular feature of the lessons in Waldorf schools in India.

Teacher skills

The teacher can:

- Judge when children are ready to learn
- Use a range of methods that efficiently help the pupils to be attuned to learning

- Conduct the lesson effectively, as indicated by the levels of positive participation of all the pupils in learning activities and the condition they are in at the end of the lesson (e.g. exhausted or refreshed)

Research questions:

- What factors enable children to pay attention in school, and what weakens their ability to focus?
- What methods enable children to be both focused and relaxed in classroom learning situations?

Principle 5: Learning is a rhythmical process

Generative principles:

- Transformative learning leads to change in the whole person body, soul and spirit and is therefore the motor of development.
- Informal learning happens in and through life, including life in school. Formal learning is structured by the pedagogy.
- Waldorf education works with a number of micro and macro rhythms of learning, involving daily, weekly, monthly and annual rhythms that ensure healthy learning.
- The primary rhythm of learning involves attunement, rich experience, forgetting, recalling, sharing and forming concepts, practicing and growing abilities leading to transformation.
- Knowledge is formed in the learning process through experience, through engaging with the material and forming judgements and through constructing living concepts.
- Learning leads to dispositions to skills, attitudes and knowledge in the form of competences, which in appropriate situations can be enacted as performance.
- The way that dispositions are learned strongly influences how the skills and knowledge are learned. Competent action depends on three factors: the way a competence is learned, the reason and occasion for the performance of the skill and the way in which the actor applies the skills, especially the degree of self-confidence and certainty with which this is done.

From the perspective of Steiner's spiritual anthropology (which I have described in detail elsewhere, Rawson, 2018), the process of learning starts with experience in the world. In his account of learning in the book *Theosophy*, Steiner (2011b) points out that the 'treasures' of experience do not remain unchanged in our soul; they undergo change. This comes about through at least two processes; each time we recall an experience, this is akin to a new perception, only not in the external world but internally, we change this memory. Each memory is modified in the new situation, because we have new, related experiences or simply because we are in some way changed, and thus we retain this modified version when we stop attending to it. This happens naturally as we grow and develop. Secondly, the mind is continuously relating and sorting experiences unconsciously. This process occurs even when the focus of our attention is elsewhere; that is, when the mind is unconscious but active. This happens particularly when we are asleep and when the flow of external experiences is reduced considerably.

Steiner goes on to describe how memories eventually fade, leaving, however, their fruits behind. The fruits of experience are taken up by the spirit in the form of abilities. He writes,

> This is how the spirit transforms its treasure trove of memories. It abandons to fate anything that can lead only to images of individual experiences, keeping only the power to heighten its own abilities. We can be sure that not a single experience goes to waste, since the soul preserves each one as a memory, and the spirit extracts from each one whatever it can use to enhance its abilities and enrich its life. The human spirit grows as these experiences are worked over and assimilated. Thus, although we do not find our past experiences preserved in the spirit as if in a treasure vault, we do find their effects in the abilities we have acquired.
>
> (2011b, pp. 67–68)

The I extracts a 'force' to develop its abilities. The example he gives is learning to read. Once we can read, we don't need to recall all the preparatory stages of learning to read, we can simply read, and reading changes our whole relationship to the world. We are fundamentally transformed by becoming literate. How can we understand this process of extraction? Steiner uses the analogy that when we digest food, we do not literally use the broken-down substance from, say, a cheese sandwich to regenerate, rebuild and energize our bodies. Rather, in breaking down the food substance, we release forces that our body can use as energy to activate our processes of regeneration and nourish our muscles or brains in activity.

Abilities such as being able to read, ride a bicycle, play the violin or speak a foreign language change who we are. They become in effect part of the I – a kind of metaphorical 'I-substance'. The I thus becomes more experienced and capable and can do things it couldn't do or didn't know before. In other words, the person has been transformed through learning. Thus, abilities become dispositions that change the way we experience the world and ourselves. We can be disposed to see things in new ways and think about them in particular ways and this informs what we do, including how we relate to the world and what and how we learn.

In the final chapter of *The Riddles of Philosophy*, Steiner describes this process as follows:

> One can observe how the human soul is transformed by experiences that represent, in a certain sense, repetitions of earlier experiences. If somebody has read an important book in his (sic) twentieth year and reads it again in his fortieth, he experiences it as if he were a different person. If he asks without bias for the reason for this fact, he will find that what he learned from his reading twenty years previously has continued to live in him and has become a part of his nature. He has within him the forces that live in the book, and he finds them again when he rereads the book at the age of forty. The same holds true with our life experiences. They become part of man himself. They live in his I.
>
> (2009, Part 2 Chapter VIII)

This transformative idea of life-long learning lies at the heart of Waldorf learning theory. Waldorf education approaches learning from just this position. It takes a strongly developmental approach by suggesting an ideal type of learning based on

the relationship of the child to the world and how this changes through maturation. The pedagogical approach deliberately shapes and modifies the mode of learning. Hence the mode of learning in the first phase up until the change of teeth is typified by mimetic learning (Wulf, 2014), emulation and imitation. The anthroposophical reason for this is that the I of the child is de-centred and distributed in her environment and therefore participates in all she perceives. Steiner's (1996) description is even more precise; he says the spiritual part of the head 'sleeps' in the environment. That means unconscious cognition is occurring. Because of this, the child's unconscious experiences of her environment stream through the senses directly into the brain and body – he refers to the young child as wholly sense organ – and the forces, thus activated, structure the brain. This accounts for Steiner's frequent references to the effect of early literacy learning; the little black marks (i.e. letters in a book) are embodied without the child's ability to generate meaningful mental images associated with them. Thus the letters remain metaphorically like indigestible stones in the stomach. That's how children learn language orally, develop their senses, construct body maps and learn to coordinate their limbs. The body in a sense is the child's first environment that needs to be explored. The body's interaction with the world, mediated by the 'world' senses (i.e. those directed to the outside world), but also the body senses of movement, vitality, balance and touch, model and structure the neural networks that build what Damasio (2012) calls 'the body in the mind', which is the basis for the core-self that constructs a dynamic but coherent pattern of relations between body and world retained over time in memory (knowing how to tie my shoes, cook a meal, knit). The brain, according to Damasio (2012), has inherited dispositions from its long evolution that determine 'standard' responses to events. These, he tells us, blend with the mind maps we generate through experience. This is the beginnings of what Steiner (2011b) called the process by which the mind retains the fruits of experience as memory.

In this spiritual anthropological account of learning the I is the agentic core, the answer to the question, "Who learns?". The body and the animated body of life processes are involved, preserving the experiences mediated by the senses and 'selected' by the psyche on the basis of inherent learning structures, learned behaviours and biographical interests. The outcome of learning is the growth of skills, abilities, habits, dispositions and, of course, both procedural and propositional knowledge. When these serve the biographical interests of the person and when they facilitate a deeper relationship to self and world, the learning process can be said to be transformative. I will now look at the specific phases in the process of learning (see Figure 2.2).

	Learning process
1	Taking in rich experience
2	Forgetting
3	Recalling, sharing, clarifying
4	Making meaning, constructing constructs
5	Practicing and applying
6	Growing abilities
7	Transformative learning

Figure 2.2 The sequence of learning processes

Taking in rich experience – experiencing the world

The first stage in the learning process is noticing, directing one's attention intentionally towards the object, opening one's self to it and taking it in as a percept. If this is accompanied by a keen sense of anticipation, it can also call forth a response of curiosity, wonder, and of encountering something significant. The encounter includes initial lived experiences, intensified by observations and, in some cases, through experimentation, all of which are intended to provide a rich experience that leads to a process of *becoming aware and the forming of percepts*. An initial encounter with the phenomenon is deepened to the point of becoming an experience. This involves an initial recognition and verbalization of what has been perceived, even if this does not yet involve a systematic relating to other experiences. We see this in the grade 7 optics lesson. Associations arise, other memories are activated, and the experience is naturally embedded in emotions and feelings. An important aspect of this initial encounter is that the new experience may in some way interrupt and discontinue the learner's existing experience and view of things that cannot be easily assimilated into that existing picture. This encounter may be surprising, puzzling or even shocking and cannot easily be ignored or glossed over. It needs to be dealt with. This can be done by rejecting and forgetting it as soon as possible, by treating it as a temporary perturbance before 'normal service is resumed', or it can lead to a questioning of existing assumptions, a reordering of the ways things are or indeed to transformation. The interruption may be cathartic.

Forgetting – forming an unconscious connection

Rich experience is followed, inevitably, by a process of forgetting as the learner's attention moves on to the next experiences. In a more profound sense, it means leaving at least 24 hours between the initial experience and recalling. For this reason, wherever possible, the initial experience is not further discussed and certainly not analysed until the next lesson, the following day, or whenever the next lesson is. From the perspective of Waldorf theory, the richer the primary experience, the more we can form a relationship to a phenomenon. As complex beings ourselves, we are disposed to identify with complexity and thus we expect complexity in other living beings. If our seeing is richly imbued with meanings, we will experience the world in greater depth. In forgetting, we connect unconsciously with what we have experienced and is now embodied, and we relate this to our existing body of knowledge. The difference, metaphorically speaking, is that we experience our experiences from the inside out, rather than from the outside in, because what we retain is our bodily and emotional response to the original percept – including aspects we have unconsciously perceived, which are often considerable. If we have seen an autumn forest of red- and yellow-coloured leaves, what we recall is the effect this experience had on us. The image of the leaves is bound up with our response to them in that situation. In the unconscious dimension of the psyche, the new experience connects with our existing, associated embodied experiences.

In the Foundations lectures, Steiner (1996) identifies three modes of consciousness: full waking consciousness in thinking, semi-consciousness in our feelings and unconsciousness, which is not absence of consciousness, in our will. Consciousness, according to Steiner, is always non-physical. However, in waking consciousness under

normal circumstances it is reflected by the neural processes like a mirror. It does not penetrate the physical, chemical processes of the brain. Our daytime awareness cannot penetrate the unconscious processes in the rest of our organism, which is a good thing because, Steiner says, we would not be able to cope with this. When our consciousness enters this domain, we experience pain, and we would experience the metabolic processes only in the form of images – the kind of images that occur in pathological states, in dreams, nightmares, visions like those portrayed in mythology or in the paintings of Hieronymus Bosch.

In the unconscious spiritual dimension, there are no boundaries between subject and object, and therefore intentionality, shared awareness and relationships, of which we are often initially unaware, are integrated. Undistracted by the continuous flood of sensory input, the mind can sort the experiences of the day using value systems and dispositions that prioritize experiences, in effect choosing what is more relevant and what is less relevant. If the teacher has made a strong inner connection to a phenomenon and structures her presentation to draw attention to what she believes is important, the child perceives these intentions – quite literally through all the senses, but especially the senses of language, thought and the sense of the other as an I – though unconsciously. Thus, the child's unconscious shared 'attention' is drawn especially to these experiences that have been structured by the teacher and bear her individual style, which the sense of the other as I would reveal. Then these experiences are likely to be given priority in the sorting and relating process.

The neurologist Manfred Spitzer (2007) has made the point that sleep consolidates what has been experienced and learned during the day and that lack of sleep or interrupted sleep has a measurable and negative effect on learning. Spitzer summarizes the results of many years of research into this phenomenon that has even shown which parts of the brain are involved in this complex process. Particularly important is the unconscious but deliberate process of assessing the importance of experiences. Here emotions and motivation have a positive influence, whilst stress has a negative effect. Experiences that are prefigured or anticipated are assimilated and prioritized more effectively than other experiences. This highlights the function of the teacher's preparation in selecting and emphasizing what is important already during the presentation. In the grade 7 optics lesson, the intentions of the teacher are very clear in my account, though I doubt the students present were as conscious of these. Experiences that are accompanied by a positive feeling of social belonging will be recalled more thoroughly.

The process of forgetting is a form of inner connecting to the experience. The young child, according to Steiner (2007, p. 141, lecture 12), actually 'sleeps' when she is awake with regard to her cognition and does not especially 'reflect' on the events of the day during sleep. After the age of about 10 years, Steiner tells us, the sense impressions of the day are particularly important in sleep. After puberty, this changes, and what the young person experiences in sleep is a kind of evaluation of the day's experiences in terms of their relevance to her biographical intentions. With older students, the biographical questions that preoccupy them may resonate in an inner connection as the echo of an experience in the spiritual or unconscious realm. The psychologist Ole Dreier (2008) speaks of young people pursuing their concerns and interests, sometimes unconsciously, as they navigate social situations. They notice and are drawn to things relevant to them and their interests. Their feet, as it were, take them towards or away from situations in an intuitive manner, rather than through rational and reflected

decisions. Something similar happens in the unconscious process we are calling sleeping consciousness. This interest can manifest as a half-conscious inkling during a lesson. During the phase of unconscious 'processing', more complex inner connections may be formed. Angelika Wiehl and Michael Zech (2017) express the adolescent's biographical interests as follows,

> For young people it is important that school learning processes or everyday experiences offer them opportunities to identify with their own individuality because only then can they engage with the world through their will. Successful teaching is based on learning processes, projects, initiatives and opportunities for creative activity that are experienced as fulfilling. If that is the case then even developmental hindrances, obsessive behaviour, fixations on bodily development or narcissism, typical of youths at a certain age, can be overcome. In an education based on relationships, a bridge can be built between the I and the world in the youthful psyche that forms a basis for transforming egotism into love and interest in humanity, the world and the environment and thus to a social form of individualism.
>
> (Wiehl & Zech, 2017, pp. 294–295)

Language plays an important role in this process. If a young person experiences that language mediates figurative, symbolic or metaphorical meaning beyond its surface meaning, these 'hidden' levels of meaning can be experienced as a kind of emancipation and as potential that counters the experience of meaningless and randomness in language. Poetry, literature and drama contain dimensions of meaning that mere information lacks. The same is true of art generally, which brings more to expression than what we see on the surface. This is one of the reasons that art from different cultures is taught in grade 9, as in the vignette detailed earlier. Poetry and art work in the unconscious, weaving connections together that the surface does not reveal, particularly if it is not merely looked at intellectually but in a way that embodies the experience. Holistic science offers glimpses of a bigger, more complex or more fundamental context, and inklings of this can have a similar effect. The inspiration that such experiences prompt is often indirect; making a connection in the unconscious is like opening a window, through which other new impulses and resonances can enter. Inspiration is not rational, nor bound by causality.

Anke Grotlüschen (2004) has elaborated the notion of expansive learning, which describes the motivation young people have to learn when they experience that a given learning situation affords them opportunities to be agentic, to be active in pursuit of their biographical intentions and ideals. This interest is not necessarily a self-interest, in the sense of "What's in it for me?", though it can be, but much more an openness to the other in the world. Ideals run the risk of remaining fantasies if they are not based on a deep connection to the world. This deep connection belongs to the will of a human being, which is embedded in the world and woven into the relationships. In the will the human being is least conscious, which is why Steiner treats will, unconscious activity and sleep more or less synonymously. But pure will is purposeless and in a real sense, mindless; it needs to be directed by the conscious mind. Literature, art, scientific ideas, natural phenomena and encounters with people can all prompt an inner quest for meaning and give life a direction. Being creative and productive likewise releases new energies. Their source is the unconscious spiritual dimension, and such experiences can lead into a spiral involving waking, dreaming and sleeping consciousness.

The opposite of expansive learning (Grotlüschen, 2004) is defensive learning, in which the motivation to learn is reduced to avoiding stress, to relinquishing agency and submitting to external forces, such as the need to gain grades, pass tests and get qualifications. Much school learning prompts defensive rather than expansive learning. Indeed, when students apply their skills, knowledge and energy to fulfilling tasks set by the 'system', rather than expressing their own creativity and interests, it can lead to alienation, the condition that occurs when the human being feels that she is unable to identify with the products of her own spirit. When school assignments are done because they have to be done to gain grades and avoid problems with teachers and parents, then they become meaningless chores, things you just do. A pupil may feel a degree of pride, even satisfaction in her work, but this is not the same as setting oneself a task and fulfilling it. The problem with awarding prizes (best essay, best sporting achievement, best student, etc.), is that the system only works for a few 'winners' and creates a lot of 'losers' and losers quickly get alienated, sometimes with disastrous social consequences. If education values expansive learning, then each learner can have the feeling that she is achieving something of value to her. I don't have to be the best; I just want to do *my* best. Closure of intrinsic learning motivation can occur early if the pupil's will is directed exclusively to external learning outcomes.

Recalling, sharing and valuing

Behind Steiner's idea of remembering or recalling is the idea that the past experience is not simply retrieved like a saved document in its latest version but is actually a new perception, only not of something in the world but of something in the mind. We reconstruct a mental image from what has been retained in the unconscious mind (in Waldorf terms, in the life-body). This is a view of remembering that is supported by neurological science. Remembering is actually:

> the composite memory of the sensory and motor activities related to interaction between the organism and the object during a certain period of time. The range of the senso-motor activities varies with the value of the object and the circumstances. . . . Our memories of certain objects are governed by our past knowledge of comparable objects or situations. . . . Our memories are prejudiced, in the full sense of the term, by our past history and beliefs. Perfectly faithful memory is a myth. . . . The brain holds a memory of what went on during an interaction, and the interaction importantly includes our own past, and often the past of our biological species and of our own culture.
>
> (Damasio, 2012, p. 133)

That means that memory is not just a record of an object or event but of our *interaction* with it. Damasio notes that the nature of the question posed to the memory alters the course of the recollection process. The research on mirror neurons (Woodruff, 2018) shows that the sense impressions we have of other people activate imitative processes in our own body, which our own mirror neurons register. Recalling an experience 're-fires' these neural patterns, which are then perceived anew, thus generating a new mental image. Though these facts were unknown to Steiner, they underpin the Waldorf approach to the recall component of learning.

Recalling is an important part of the learning process in Waldorf education and relates to a number of learning dispositions. In every Waldorf classroom, the experiences in the previous lesson are reconstructed by the pupils in age-appropriate ways. Prompted by the teacher, each pupil is asked to make the effort to recall the relevant experiences. It is important that this does not take the form of a test but instead takes the form of open questions, that is, without there being a clear and limited 'right answer'. Ideally, this should involve each pupil in an active inner process of reconstruction. The purpose of the exercise is to prompt inner activity on the part of each student, not simply to establish 'where we left off' in order to connect and move on. The example of the grade 7 optics lesson shows the children in pairs reconstructing the experiment with mirrors from the previous lesson. Clearly, verbal questions cannot be asked individually of an entire class, and so the teacher can either trust that all pupils are thinking about their experiences, or the students can make notes or drawings about what they remember. For younger children, recall takes the form of narrative – retelling the story of our experiences and making sense of this.

Having recalled, recollected and reconstructed the past experiences, the next stage is a sharing of these in the class. Each individual will have noticed different aspects of the experience, though there are considerable overlaps. This round of exchange involves noticing what others recall and comparing that with one's own memories. Thus, a rounded, multi-perspective picture is formed. Gaps can be filled in and aspects can be clarified. The experience is placed in a wider context, related to existing knowledge and to other experiences in a process of deepening knowledge.

Making sense, making constructs

Gradually a shared recollection is formed, and this then leads to the third stage of recall, the co-construction of a concept or the characterization of the original phenomenon. Here the movement goes from what *I* recall, to what *we* recall and what *we* identify dialogically as the most relevant and important features of the experience. These can be named or defined and documented. If children and young people form their own collective concepts, and each has individually contributed to this in a more or less active way, they are more likely to remember it. Thus, general concepts are individualized. It doesn't matter if these concepts are also known to millions of other people; the point is, in this process each individual reconstructs it herself anew in individual ways. At a later stage, the teacher can point out that scientists refer to this using technical terms, definitions and formulas.

The process of active, targeted recall is also a process of forming a conscious connection to the experience. At some stage in the whole cycle, the individual begins to make sense of the experience. It begins to appear comprehensible and significant and the learner takes up a position in relation to it. Peter Loebell (2000) calls this an experience of self-evidence, a self-evident form of understanding, a moment of individual recognition which is an experience of certainty. This is the last of three stages in learning new knowledge. The first step was the initial activity of *noticing* the phenomenon. This is followed by *forming a relationship* to it and now the third step, that of *self-evidence*, means that the learner has entered into a relationship to the object at several levels, the meaning of which has been assimilated into the learner's world view. Peter Lutzker (2007, p. 425) describes this process as "a decisive basis for the learner's autonomy", though this is not necessarily permanent. It is a momentary sense

of having grasped something. It is, however, likely to be consolidated through application and practice, the stages that follow in our analysis.

Thus, through these three learning processes – encountering the world, forming a relationship and coming to an understanding in the sense that a living concept has been formed – a significant part of the learning process has been described. However, further processes are necessary to complete the transformative process.

Reflection is another aspect of recalling and is an important aspect of learning. This must be done in age-sensitive ways that do not push the children into intellectualizing and reifying their experiences and thus losing connection to their will to do better. Reflection is a broad spectrum of activities that require a reconstruction of a situation and its review using relevant criteria in relation to the task. If children and young people have criteria that they understand that relate to the nature of the task, they can judge to what extent they and others have fulfilled the criteria, and what they might need to do differently in the future. The criteria in the art and foreign language lessons are examples of this. Given the range of individual and shared tasks that pupils in a Waldorf school have across practical, artistic, social and academic tasks, they learn what criteria are relevant and appropriate. They can also reflect on the process, the challenges and difficulties, the satisfaction and pleasure they experienced.

Reviewing and reflecting on collective activities helps children learn how to make judgements. The first stage of simple reflection is to note what happened in relation to what should have happened. The second stage is to note differences, unexpected situations and how the pupils reacted. The third stage is a personal reflection about what each person enjoyed most, didn't enjoy and finally what could be done better next time. Children can be encouraged to assess whether they did what was asked of them adequately. Whatever form of reflection is used, it has the aim of stimulating the child or young person's will to do better. By giving pupils goals – later they can set their own goals as well – that can be achieved that are both individually and collectively meaningful, they learn how to do better.

Sharing, perhaps exhibiting and valuing the outcomes of schoolwork, strengthens both the sense of collective achievement (that is what we have done) and self-efficacy (that is what I can do). At the end of any lengthier, complex process, it is always helpful to display all the outcomes. The things knitted or crocheted in handwork, the vegetables harvested in the garden, the wooden bowls carved in woodwork, the geometrical drawings, the stories and poems written in creative writing, should be read out or printed in a collection. Each class can show their neighbouring classes what they have been working on in a sharing event at the end of the week, and main lesson books can be displayed at parents' evening. The original idea of the monthly festival was that each class would show something of the block they had just finished to the rest of the school. The Waldorf view is that sharing is a more socially conducive form of gratification than prizes for achievements.

In their model of learning, John Hattie and Gregory Donoghue (2018) explain the relationship between surface learning and deep learning in relation to conceptual understanding. Surface learning is about knowing things and what they are called, whilst deep learning means knowing how things relate and what they mean. A shallow learning strategy is one in which the learner stays at the level of learning reproducible facts and ideas. Deep learning involves relational learning (relating ideas to other ideas) and extended or abstract learning (Hattie, 2012, p. 60). In the learning phases discussed thus far, the first stage that I have described as direct experience is shallow.

It involves getting to know something, including the vocabulary related to it. In the phase of recall, sharing and forming concepts, the learning moves to a deeper level involving connections and relationships.

Surface experiences and shallow knowledge need to be consolidated through memory and recall, the latter requiring effort and will, before it can become deep learning. Deep learning starts with the ability to relate experiences and conceptualize them and involves an iterative process of application and practice.

Practicing and applying what has been learned

Learning is a skill that needs practicing. Learning also benefits from a working memory, belief in learning, encouragement and expectations, and the Waldorf method supports each of these. Learning also requires what Hattie and Donoghue (2018) call will, comprising learning dispositions, habits of mind and what they call a growing mindset, which might be translated as transformative learning that changes the more one actively learns and living concepts that grow as one learns. The third factor that Hattie and Donoghue mention is what they call thrill. Learning that is exciting and enjoyable is basically more likely to be effective.

Wiehl (2017) considers repeated practice central to the Waldorf method because it is only through continuous practice that skills become embodied and become dispositions. Applying what has been learned by practicing is necessary for the growth and consolidation of abilities and skills. Practice involves repetition of something one can already do. If the conditions remain the same, it becomes a kind of training that establishes habits and routines, and this conditions the body to certain activities and responses. Repetition with variation in similar and gradually more unfamiliar settings not only strengthens and extends the skill; it also contributes to a deepening of embodied understandings. If we take a craft example, we can imagine that through applying certain tools or techniques to different tasks or using different types of materials, a basic disposition becomes skilled artistry. Spitzer (2007) has pointed out that in order for learning to become a disposition and ultimately an ability, the learner must be immersed in continuous practice in authentic situations. What is implicit is not necessarily consciously understood, but it becomes an ability through practice, and this means having many examples. Children should experience many good examples of the skill or activity. If they have enough examples in real situations, they will discover the rules for themselves. This is a principle in foreign language learning.

Practicing also means improving what you can do. Abilities grow when students take a piece of work that they have done, receive feedback and make corrections and then do the same task again with the aim of getting a better result. This also means learning consciously from mistakes and developing resilience in learning. Just as writers often write many drafts of a text until they are satisfied, so students can take pride in their work, have a sense of self-efficacy that they have achieved something and develop real skills. Musicians and athletes know very well that practice improves performance, but students often need encouragement to do the same in school lessons. Another learning disposition acquired through practice and applying is what Claxton (2009) calls playfulness, which involves trying out and experimenting creatively by applying what has been learned creatively in unexpected and unfamiliar ways. This too is an important aspect of practicing.

Growing abilities

Waldorf education takes a long-term perspective on learning, an interpretation of life-long learning that takes both an embodied and biographical perspective. Dispositions and abilities learned in childhood and in youth predispose a person throughout their lifetime. Three further factors need to be taken into account. As the human being ages, the relationship between her body, psyche and spirit changes and this has corresponding influence on her learning capacity. Living concepts can continue to grow and expand in their relationship to reality, leading ideally to greater understanding and practical wisdom. Secondly, learning increasingly becomes a matter of self-development and *Bildung*, in the sense of becoming more experienced. In German educational philosophy, *Bildung* means the self-formation of the person through engaging with the culture she is embedded in. Peter Faulstich (2013) describes Bildung as an ongoing biographical process through which individuals engage bodily and mentally with the world, embodying culture as they do so and thus weaving together self and world, body and spirit, individuality and sociality in search of possible and secure identities, and thereby developing their personality within the opportunities and restraints of the social situations they are embedded in. Throughout this process, the person's experience of competence grows through performance, and performance is always limited or afforded by the available opportunities in life and society.

Thus, the shortest time frame of learning is from one lesson to the next, ideally with a night of 'forgetting' between the initial experience and recalling, forming a relationship to the experience and understanding its meaning. The next learning process, then, is that of cultivating abilities through practice and application. During this phase of learning, which is short-, medium- and long-term, dispositions and abilities gradually develop through practice and through repeated phases of new input through the block-teaching followed by periods of 'forgetting' and inner consolidation. Growing abilities often need more support in the form of scaffolding or in the form of support, guidance and feedback or assessment for learning.

Transformative learning

Holistic learning culminates in the transformation of the whole person. This can be observed in biographical case studies (Wiechert, 2012a, 2012b; Göschel, 2012), and of course young adults can observe this in themselves through reflection. Biographical learning (Alheit, 2018), which is a form of transformative learning, occurs when individuals can recognize and utilize opportunities in their lives, or in their educational contexts, to further their biographical interests. Transformative learning is a process of change in which the person enacts a fundamental change in her understanding of the world, self and relationship to others (Illeris, 2014). It is more than the acquisition of knowledge and skills, though these may be part of the transformation; it is a productive and iterative process of becoming a person-in-society (Jarvis, 2018). It is not so much self-discovery as self-formation. Nor is it likely to be restricted to a single event but is an ongoing process of establishing stable and coherent identities across social practices and over time. It is one of the aims of the curriculum to provide a sequence of formal and non-formal learning opportunities for individuation and what Gert Biesta (2013) calls subjectification, the process of coming into being as an agentic subject. Waldorf education aims to contribute to socialization, qualification

and subjectification. Teachers also support this process through their pastoral care and long-term commitment to monitoring and understanding developmental and biographical growth. This occurs through long-term relationships between teachers and pupils, through pupil case studies and detailed text reports (rather than grades) that aim to characterize the child or young person's development as a whole person over the past year, setting developmental tasks for the coming year.

In transformative learning, the I as spiritual core of individuality comes to expression in the ongoing activity of constructing and remodelling identities (Loebell, 2012b). The central aim of Waldorf education is to support this process of individuation (Rawson & Richter, 2000; Götte, Loebell, & Maurer, 2016). Following Michael Zech (2018), Waldorf pedagogy in the upper school (ages 15–19) particularly involves teachers enabling transformative learning and the formation of the person through the realization of their biographical intentions, interests and potential. In modified form, this applies to the education of children and young people of all ages. Topics are chosen and presented in ways that offer opportunities for transformative learning, whatever other curriculum aspects have to be considered. This is a process of self-becoming in which the young person generates meaning and identity as an outcome of the learning, which at the same time generates the necessary conditions for further learning.

Teacher skills

The teacher can:

- Plan and structure lessons on the basis of a sound understanding of the learning process
- Choose teaching strategies and learning activities that follow the daily, weekly, monthly and annual rhythms of learning
- Identify skills, attitudes and knowledge in the performance of authentic tasks
- Adjust their teaching on the basis of analysis of the observed effects on the pupils' learning behavior
- Support pupils in each stage of the learning process in appropriate ways
- Observe and recognize children's and young people's biographical development over the long term

Research questions:

- As this presentation of learning stages is fairly unique in terms of learning theory, a detailed theory of learning needs to be developed and applied empirically. Waldorf learning theory has only recently been formulated (Loebell, 2000, 2017; Schieren, 2012a, 2012b; Rawson, 2018, 2019a). Considerably more research needs to be done to establish an empirical basis for it.

Principle 6: The learning processes are structured over time

Generative principle: The teaching follows the natural learning processes, and this provides a structure to the lessons and a range of different activities engaging the whole person.

As we have seen, the core learning activities involve new experiences that are recalled the next day, discussed, contextualized and understood as a concept, then applied and practiced. One could describe this as a natural sequence of learning steps that occurs even without formal learning. However, in formal learning situations they can be structured in a way that can be planned, monitored deliberately varied and incorporated into an ongoing process over longer periods of time – which is what school is. If we put these learning processes into a structure that can be applied over a series of lessons, we discover that we have a complex series of overlaps because most days, something new is added to what has already been experienced and the various processes of recalling, understanding and practicing each have their own time requirements, which means that teachers must have both a systematic plan and an intuitive grasp of where the students are in the learning process.

If we reduce this complexity to a series of simple diagrams, the challenge becomes clearer. Let us start with the introduction of new experiences over the course of a typical five-day school week in the main lesson. Let us assume that Monday is the first day of a new main lesson block, so there is nothing from the previous day to recall (though a discussion connecting to what was done in the previous block is essential). The new experience is labelled E1. On the next day this is recalled (recall 1). The following day, this is taken up and understood as a concept, which I label C1 (i.e. concept related to phenomenon E1). This concept or rule can now be practiced and applied (which can mean a variety of activities, from applying a rule to documenting the whole experience or practicing something that has been learned). This can be called A1. Thus, we get the theoretical sequence of learning stages across the week, as seen in Figure 2.3.

Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
Introduction of new topic: first experiences E1	Recall of E1	Understanding concept E1 =C1	Applying C1	(Perhaps applying C1 again to a wider context)

Because a new experience or phenomenon is usually introduced every day, day 2 – Tuesday – already has a new experience, E2, which logically leads to its own sequence of activities that run parallel to the E1 process. Thus, the weekly lesson now looks like this:

Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
Introduction of new topic: first experiences E1	Recall of E1	Understanding concept E1 =C1	Applying 1	
	New experience E2	Recall of E2	Understanding concept E2 =C2	Applying C2

On Wednesday, a further new experience is introduced, E3. This fits into the daily and weekly lesson plan as follows:

Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
Introduction of new topic: first experiences E1	Recall of E1	Understanding concept E1 =C1	Applying 1	
	New experience E2	Recall of E2	Understanding concept E2 =C2	Applying C2
		New experience E3	Recall of E3	Understanding concept E3 =C3

If we add a further new experience on Thursday, the sequence has to be continued over into the next week. Thus, Week 1 looks this:

Week 1				
Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
Introduction of new topic: first experiences E1	Recall of E1	Understanding concept E1 =C1	Applying 1	
	New experience E2	Recall of E2	Understanding concept E2 =C2	Applying C2
		New experience E3	Recall of E3	Understanding concept E3 =C3
			New experience E4	Recall of E4
				new experience E5

If we continue the sequence into the following week and arrange the activities in the kind of sequence that always starts with recall, then we get the following sequence:

Week 2				
Monday	*Tuesday*	*Wednesday*	*Thursday*	*Friday*
Recalling E5	Recalling E6	Recalling E7	Recalling E8	Recalling E9
Understanding E4	Understanding E5	Understanding E6	Understanding E7	Understanding E8
New experience E6	New experience E7	New experience E8	New experience E9	New experience E10
Applying C3	Applying E4	Applying E5	Applying E6	Applying E7

Figure 2.3 Tables showing the arrangements for structuring the learning stages

In the lower school grades, we may imagine this process concretely in terms of the teacher introducing a particular tree on day one – for example, an oak – describing its forms (trunk, bark, roots, leaves, fruits, profile) and ecological location, perhaps with the help of a picture and branches and leaves. The next day, this is recalled from memory. To this picture may be added further information about the animals that are closely associated with oak trees, the special qualities of the wood and its uses (e.g. the way that Viking shipwrights used the naturally grown shape of the timber at the 'elbow', where a branch grows out of the trunk, to construct the boat's frame because it requires no further shaping and follows the grain of the timber, making it incredibly strong, or to fashion the rowlocks utilizing the natural hole in the wood – facts that are meaningful to the students who have recently studied the Vikings and their boat-building skills). The following day, all the information about oaks is gathered together to characterize the oak tree. The application can take the form of applying the same categories to other tree species or drawing a detailed illustration of a typical oak tree and its various uses in boat building and house building.

In the higher grades, it may be more relevant to follow the recall with a discussion and to integrate the process of forming a concept. Indeed, Michael Zech (2018), in summarising learning in the upper school, identified three processes:

1 Taking in new material;
2 A work process involving opening up, interrogating and developing the material, analysing and structuring it; and
3 Forming concepts – taking an individual position in relation to the new material and then integrating this into the overall general understanding of the theme.

The second and third stages can be integrated into a 40- to 50-minute process in which individuals engage with the subject, with the aim of enabling individual insight and self-positioning rather than reducing the material to a certain specific or predetermined concept as outcome. Zech quotes Wilfred Sommer, saying that Waldorf education trusts in the potential of each individual to be able to identify what is relevant and what is not relevant, to be able to reduce complex experiences to their basic and essential elements, to be able to make exact observations and to be able to approach the phenomena themselves. Sommer also claims that Waldorf students are capable and should be trusted to make their own judgements, rather than simply accepting rules and laws as universal or absolute. Forming their own judgements prevents them from premature closure, shutting off valuable avenues of thinking and opportunities for dialogue. Making and justifying their own judgements means cultivating criticality and reflexivity. Characterizing instead of defining has always been the Waldorf approach, a dialogic approach so that the students become capable of dialogue judgement (Sommer, 2010).

Teacher skills

The teacher can:

- Plan the elements in the learning process on the basis of a sound understanding of the learning process in a flexible and multidimensional way

Research questions:

- What has been described here is a very generalized view. It would be helpful to take a fine-grained look at how the learning process varies in terms of how individual themes and subjects need to be taught in the different grades, so that guidelines can be formed.

Principle 7: Block teaching strengthens learning dispositions

Generative principle: block teaching enables the children and young people to immerse themselves in a field of knowledge and learn dispositions and skills related to that field for a period of three to four weeks, two hours each day.

Another aspect of having 'time to learn' is the so-called main lesson and block teaching method. One of the innovative features of the first Waldorf school was that each school day started with a 90- to 100-minute lesson, usually referred to as the main lesson, which is actually a translation of the German word *Hauptunterricht*. The term 'main lesson' is actually not very helpful for the rest of the teaching, because if this is the *main* lesson, the rest of the day must be in a sense subsidiary or secondary, which would be a misunderstanding. To get around this, some schools use the same abbreviation, ML, on the timetable but call it 'morning lesson'.

During this time, the class teachers work with the class on a theme or subject for three or four weeks. This was important for Steiner to avoid the fragmentation of the day into a random sequence of 45-minute lessons. A main lesson block is devoted to a theme for three to four weeks. The idea of blocks – 90 minutes a day over four weeks – allows the students to immerse themselves in a theme or subject and enables the teacher to develop a theme while attuning to optimize the learning process. Steiner's idea was

> that you cannot do anything more destructive to the human heart and mind than arranging [the timetable in random short lessons] in this way for young people's powers of concentration to be so thoroughly undermined. What we must begin upon when reforming the sphere of education socially is pre-eminently the timetable, that arch-enemy of everything to do with genuine education.
>
> (Steiner, 1994, p. 38)

The 90- or 100-minute morning lesson allows for attuning, recalling previous experiences, discussion, presentation of new material, group work and other classroom activities and offers a strong rhythm for the day.

The idea of block teaching has major consequences, particularly in the middle and upper school, as more and more subjects are introduced, such as the sciences. The block system means that there may only be one geography, chemistry, biology or physics block per school year. Even subjects such as maths, language and literature may have only three blocks per year (12 weeks). As I explain in detail in the next section, the idea of allowing experiences to sink down into the unconscious is a central part of learning in a Waldorf school. Conventional teaching may be concerned that the children will forget everything in the long gaps between the blocks. The Waldorf approach, however, aims to establish habits, dispositions and abilities rather than short-term memory. One does not forget how to ride a bicycle once one has learned. The point of block teaching is not to produce short-term memorizing of facts but

long-term learning. Prolonged immersion followed by a period of 'fallow' or forgetting during which subconscious assimilation occurs, followed by recall and reactivation of skills, has proven to be highly successful. All Waldorf schools use block teaching, and there is no evidence that the children know less. A recent study compared state school high school students with Waldorf students in history. The Waldorf students, despite having block teaching and significantly fewer lessons in total, knew as much as their peers and even showed somewhat higher levels of historical understanding and narrative competence (van Norden & Schürenberg, 2019).

If one immerses oneself in a field – say, the geography of North America – for a month in grade 7 (age 13), what is likely to be recalled after a year? They probably won't know many facts, perhaps a few surprising or noteworthy ones. If the block has been successful, the children will, however, have learned how to look at a continent geographically, what questions to ask and why. One looks for the main features, mountains, river systems, coastlines and latitudes and their impact on climate and vegetation and what kind of environments this produces, as well as what this means for plants, animals and humans. They learn to look at how landscape and climate influence economic activity and the main routes for transportation. In other words, they have learned to look at the earth *geographically*, which is different than looking at it *culturally*, though the perspectives are related and are examined as such in the upper school in historical and cultural studies.

The students have in effect become disposed to the way of looking at the world that the science of geography offers us. The same is true of looking at the world from a chemical perspective of substances and transformations of states – solid, liquid, gaseous, etc. In the class teacher period, the aim is not to train future chemists or geographers but to learn how the world can be approached from, for example, a geographical, chemical or historical perspective. In other words, block teaching encourages the development of dispositions that influence ways of seeing the world. With my understanding of the relationship between the geography of a landscape and human activity or types of vegetation, I have a model for looking at any continent. This disposition has grown since grade 1, when the children did nature walks and marked the turning of the seasons. They have also had rich narrative sources in tales and legends in which nature is characterized often in archetypal ways. They have learned about farming, animals, plants, rocks, soils and landscapes. They have explored their local geographical surroundings. They have learned about their local region and country, following its rivers and mountains, coastlines and plains, and they bring this embodied knowledge to studying a whole continent. Their geographical knowledge has grown organically, the parts relating to the whole, expanding outwards from the familiar to the unfamiliar, using their imagination. Because they have walked the local urban and natural terrain, mapped it, formed mental images of it, imagined it changing over time (geography is always about time, often deep time), they have rich embodied understandings to feed their exact imagination of things and places they can't visit.

At some point they will have heard the story of the Chinese emperor who wanted a map of his empire. Each time his cartographers returned, he found some detail known to him personally which was not on the map – the cherry tree at the bend in the river where he liked to rest whilst hunting, with its blossom and sometimes delicately decorated with snow. And so each year, the mapmakers were sent away to make an ever more detailed map, and gradually the map got larger until it was so large that it had to be unrolled in a large room and the Emperor had to walk across it in his silken socks

to study the various parts. Soon the map took up large sections of the Forbidden City. Each time, the Emperor was unsatisfied because some detail or other was missing. At some point, his advisors plucked up the courage to advise him that soon he would need a map as large as the kingdom itself to portray all the details. The downside of this would be that it would be bad for the economy, because it would hinder trade and block out the sunlight to the farmers' fields and thus reduce his tax income. The story, like all good stories, teaches us much; the need to be selective, the need for reduction and abstraction, the need to have a legend of symbols to explain the map, the need to be able to expand the symbols in our mind with our imagination to bring us back to the reality of the place being represented, to supplement the representation with exact imagination based on rich sensomotor experience in the world. We need to simplify and contract reality so we can grasp it but then expand it again in our imagination. This is an important lesson in understanding the role of symbols and formula, and the categories we use to reduce the complexity of the world. Perhaps it also teaches the students how to deal with megalomaniacs in power without losing your head!

We could unpack the history curriculum in much the same way, when it has to do with the development of historical consciousness (Zech, 2012). In developing this disposition, I can look at different periods of history because I know that we need to look at social structures, cultural forms and economics factors including agriculture, industry and trade, and for each of these categories I have a frame of reference and look for symptomatic moments in history that reveal, for example, changing consciousness.

The block system also benefits from a preview at the beginning of the block showing the pupils where the journey will be going and a summary of key ideas and concepts at the end (as in the examples in the vignettes). It was also Steiner's assumption that at the end of each year there would be a lengthy phase of recapitulating the main aspects in each block by way of consolidating the dispositions and identifying the links between different topics and themes.

The high school still offers a general and non-specialized education right up to grade 12 at the age of 18, in which all students essentially have the same experiences, regardless of whether they will later study chemistry or law or become carpenters. Nevertheless, in the high school the subjects are taught by specialists who have studied these subjects. If during the class teacher period, the primary task is to learn in an inclusive social group, after puberty the pupils have the opportunity to also become acquainted with the practice of geographers, chemists, artists, craftspeople in their specialist workshops, laboratories, art studios and music rooms. Here they have the opportunity to sojourn in hybrid practices (Pratt & Kelly, 2007), that is, practices that are not fully authentic places where these practices are conducted in the world – such as in a university, a company or studio – but where this practice is located in school. The teacher is also a practitioner of that field. Learning physics in a lab with a physicist is a hybrid place of the practice of doing physics.

Given the marked variation in maturation and especially the trend of puberty coming earlier (Largo, 2017), it can be argued that the shift to a high school mode of learning should come earlier today. Steiner usually spoke in terms of the period from the change of teeth to puberty and the significance of the changes that puberty brings for the teaching and learning approach. Indeed, many schools do reduce the class teacher period for grade 6 or 7, and the high school teachers take over the main lessons, creating a kind of junior high school model. In a recent compilation of articles on the class teacher period, a number of prominent Middle European authors, all class teachers,

argued against this (Röh & Zdrazil, 2016). However, I feel that the point here is that the teaching style significantly changes with and after puberty. Whether the class teacher is qualified in the widest sense to make this transition is a matter that can only be determined locally by the college of teachers.

Other subjects such as arts and crafts are also taught in blocks. Here, the reason is more obvious – crafts and arts require time for real skills to develop though immersion in the activity. The school day is structured in most Waldorf schools in such a way that after the main morning lesson there are a series of subject lessons. The theory is that the school day should be rhythmical with a balance of activities. Some simplistic accounts speak of head (main lesson), heart (music, languages) and hands (crafts), but this ignores the fact that all subjects require all three parts of the human being – or they should, if the learning is to be healthy. Certain Waldorf traditions have emerged, often based on bogus justifications (Steiner said . . . which may or not be the case, but which ignores the context and the reality). There are four primary factors that need to determine a school timetable; the actual learning needs of the pupils in question, the actual effect of the teaching, the need for a balance of activities and the pragmatics of organizing a timetable taking the availability of people and rooms into account. Having done the school timetable over many years, I can state that we usually *reduced* the factors that we tried to take into account to about 25! The most effective timetable model, in my view, is to have all subjects in blocks of 45 or 90 minutes (depending on the nature of the activity). This certainly is the case in high school. It makes timetabling a much easier activity because there are fewer changes in each day, and it is also much less stressful for pupils and teachers to not be constantly shifting from one lesson to another.

In recent years, there has been a trend to introduce block teaching into second-language teaching. Waldorf schools ideally teach two second languages from grade 1 onwards. These lessons traditionally fall outside the main lesson structure, usually in the middle part of the day. The original reason for this was that the foreign language teachers were all class teachers, and this had to be timetabled outside of the main lesson time. Thus, a tradition within the schools developed that seemed to promote that language learning only works with regular practice and therefore needs regular lessons. Similar arguments were made for music. However, the reality is that two or three 45-minute lessons per week in each language inevitably means that some days the children have two languages, one after the other, for timetable reasons, and they may also be later in the day when the children's capacity for this kind of learning is not ideal. Language learning is very much about immersion in the language and developing long-term dispositions. Many second-language teachers now prefer to teach the languages in alternating blocks, usually of four, sometimes six weeks. That means that the younger children have a single lesson every day in one of the languages. Then the language changes, and they learn the other language for four or six weeks. In high school, they get double the amount of time. All the arguments for block teaching remain and the gap is relatively short. Most schools that make this change do not change back.

Teacher skills

The teacher can:

- Structure the main lesson in a healthy way that optimizes the learning process, engages the students and creates a mutually supportive learning community

Research questions:

- This is an area that deserves more research attention, as it is fundamental to the Waldorf approach.
- There are many aspects of block teaching that could be studied, including the learning effect at different ages, with end-of-year reviews for consolidation and without, what effect blending blocks with regular lessons has and the quality of dispositions, knowledge and skills.

Principle 8: Direct experience is the basis for learning

Generative principle: Learning through direct experience is powerful.

As we have seen in the vignettes, there is a strong emphasis on learning through direct, rich sensory and hands-on experience. In grade 3, the children went to the garden and harvested and threshed the lentils, and this process was set within a longer term-context because they also planted and tended the crop over months. The general rule is to start as concretely as circumstances permit with tangible things. In the lower grades, numbers start off as quantities of conkers, acorns, stones, beans and beads and then the abacus before becoming abstract numbers. In a first-grade class in India, I saw that each child had a hand-sewn cotton bag containing 12 small polished stones, which were used for making patterns and sets with numbers on the ground. This is another form of direct learning. At another level, writing poems, short stories and dramatic scenes is a practical activity that makes the subsequent learning of analysis more 'hands-on'.

Letters or written symbols are introduced through tactile, haptic means (drawing in sand, being moved bodily in space, modelling in wax or plasticine, etc.), then drawn and copied into lines and finally turned into handwriting. In language lessons, the various word types are characterized: verbs are activities, nouns collect attributes, prepositions point to places, verb tenses are organized in spatial relationships of time, past, past in the present, ongoing present and so on. The children walk in the woods; take buses and trains; take boat trips (in cities like Hamburg, Amsterdam, St. Petersburg or Bangkok, it's the best way to see the city); explore their surroundings on foot; visit farms, mills, nature parks, outdoor museums and later, of course, art galleries; and visit factories and container terminals – wherever one can go with a class to experience things directly. The activities have to be meaningful, not simply running through location, filling out checklists – seen this, done that!

Not all subjects can be taught in such a practical way but many can. Block teaching provides time for project-style teaching not only in the outdoor classroom but also in the school classroom. Many science experiments and observations can be done in the classroom. Indeed, the Pedagogical Research Centre in Kassel, Germany, produces a whole catalogue of science equipment for the classroom and lab.

However, it is not sufficient to simply witness things or do things. An expectation needs to be created, and the things being witnessed or done need a context that has been provided beforehand. It goes without saying that trying to explain what we are looking out for while standing on a windy cliff in the rain with half the pupils out of earshot is not effective. They need to know why we are going to the cliff (and naturally, also, how far from the cliff edge we should stand – here, vivid imagination of the consequences may help) and what we are looking for through the explanation. The aim is to direct attention, not say what it means.

Teacher skills

The teacher can:

- Create lively learning situations in which the pupils can participate through hands-on doing and close observation that involves many senses

Research questions:

- How do teachers create direct learning experiences across the curriculum?
- How best can hands-on experiences be managed and structured?

Principle 9: Activating the imagination through vivid pictorial descriptions and images is another powerful starting point for learning

Generative principles:

- Vivid descriptions activate the child's imagination, feelings and bodily sensations, as do images that suggest situations that can be completed in the child's imagination.
- Exact imagining requires an act of will.

There are some places we can't go, and so the children are encouraged to develop their imagination. Imagination needs to be based on real experience. The more direct sensory experiences children have had, the more easily they can imagine things vividly. Much depends on the teacher's direct presentation. The primary mode is narrative, and this is supported by other media – pictures, maps, charts, film and so on. Through using narrative, the teacher has a flexible and more immediate way of stimulating the child's response through identification and inner images – the child can vividly imagine what is being described and is therefore inwardly active in a form of self-activity in response to what is taken in. The teacher, too, needs to call on embodied experience in order to be able to evoke realistic experiences.

There are two kinds of vivid pictorial description, each with different functions. The first is lively, detailed description, which enables the child to visualize the object being described. The second type is metaphorical and uses analogy to hint at other meanings. Steiner gives the example that if a teacher wants to show the child by analogy that the human psyche can undergo transformations, then she can describe the life cycle of the butterfly, the metamorphosis from egg to larva (caterpillar), to pupa and adult butterfly. However, if the teacher then draws attention to the analogy and points out the equivalence in the transformations, which are of course not exact, or draws out the moral, then the meaning remains abstract. If the teacher trusts in the sleeping, unconscious wisdom of the child, then the child will make the connection in her own way, in her own time. Later, perhaps years later, this topic will come up and the analogy will be drawn, and the child will be able to inwardly connect to a living image of transformation that informs her concept of change. If children raise questions about the meaning of a story or want to discuss it, it should be encouraged. I believe Steiner's point was that we should let the stories 'speak' for themselves and avoid adult explanations.

This is what is referred to as using *living imagery* to engage the child's thoughts, feelings and will. Waldorf education works a lot with analogy, metaphor and simile, often with moral purpose. This can take the form of a pedagogical story for younger children, in which a moral message is contained in a story, often thought up by the teacher to match the given situation without spelling it out. A class of children who are squabbling among each other and are being distracted may be told the story of the crew of a boat who were squabbling over the way the profits on the cargo should be shared out, failed to heed the warnings of the lookout and so hit the rocks. Though all manage to rescue themselves, they lose their cargo. The idea is that the message of the story does not need to be explained but works subliminally, as it were.

In both types of pictorial image, the idea is to enable the children and young people to construct age-appropriate and individual concepts that, unlike definitions, can grow and expand over time with experience. Steiner (1996) referred to these as living concepts, as they can grow and evolve. Constructing images in the mind requires imagination, which is a powerful force in the psyche and plays a vital role in Waldorf education. With our imagination, we are able to use our will to generate images from within the mind which are not merely reflections of the world we have perceived. Of course, being able to imagine something real – as opposed to fantasy, which by definition is unreal – requires a foundation of embodied experience in the real world. As Tim Ingold puts it, "we do not have to think the world in order to live in it, but we do have to live in the world in order to think it" (2000, p. 418). From the perspective that we dwell in the world, Ingold argues that there is one kind of activity that does not involve a palpable engagement with the world, and that is

> activity of a special kind we call imagining . . . [which] carries forward an intentionality, a quality of attention that is embodied in the activity itself. . . . Where this process of imagination differs from other forms of activity, and what makes it so special, is that attention is turned inwards on the self: in other words, it becomes reflexive. I dwell, in my imagination, in a virtual world populated by products of my own imagining.
>
> (ibid., pp. 417–418)

In the imagination, we rehearse and go over real activity. The activity of the imagination is real because it is based on embodied prior sensory experience in the real world but also because, following Steiner, its origins lie in the world. Our will is in the world, fully engaged, thus we stand in our will in a relational connection to the world. The will flows through the activity of the limbs in the world into our organism and on meeting our physicality, is 'thrown back' into consciousness, taking on image form. Imagination differs from sense perception, according to Steiner, though they are closely aligned, in that imagination always has the tendency to transform, whereas the mental images we form through sense perception have the tendency to fade. It is difficult to overestimate how important it is to develop the powers of imagination as a self-renewing, sustainable source of ideas and possibilities. Waldorf education makes use of the imagination in a number of different ways on a daily basis – exact recall, anticipating outcomes, realizing experiences artistically, problem solving and in all manner of practical work.

Steiner frequently referred to an approach that he *didn't want* in the Waldorf school. He criticized what in German is called *Anschauungsunterricht* – literally,

teaching based on looking at something (*anschauen*). The term was widely used in various progressive approaches including John Amos Comenius's *Orbis sensualium pictus* (1658/1979) and is best known in Johann Pestalozzi's (1801) use of pictures with the aim of making the learning of concepts concrete – a kind of show-and-tell method. Steiner's objection was (e.g. see Steiner's lecture in Aarau on 11th November,1921; 1995) that what the child cannot receive through this method is the relational, moral and spiritual meaning of the object. This cannot be grasped by the child simply through looking at the picture and hearing the word.

However, in the teacher's pictorial description, these dimensions are implicit – assuming the teacher understands the meaning – and therefore intuitively apprehended by the child. Indeed, Steiner goes so far as to suggest that simply showing pictures (and presumably naming the objects) encourages a materialistic understanding. Rather, the young child should have the following experience: that is my teacher, she knows and understands what is important. The child 'reads' this in the teacher's gestures and the gesture of her words.

This is not an argument for not using visual material but simply the recognition that the verbal description permeated with meaning and relevant detail and context gives a more holistic experience. Looking at pictures actually requires a hermeneutics of image to understand how the image was constructed and the effect it was intended to have. Someone who is used to regularly illustrating stories, as Waldorf pupils are, will be familiar with these aspects. Likewise, looking at maps makes the most sense when the person herself has made maps and knows what they show and what they don't show.

Teacher skills

The teacher can:

- Evoke vivid inner pictures of events, people and situations through narration using verbal descriptions and images of phenomena and be able to express complex ideas in pictures, analogies and metaphors
- Provide text or other materials from other media that activate the pupils' imaginations

Research questions:

- What are the preconditions for effective and vivid oral presentations?
- How can teachers best learn storytelling and presentational skills?

Principle 10: A phenomenological approach enables the organic growth of knowledge

Generative principles:

- The learners encounter the phenomena being studied first, engage with it, thus forming a relationship to it and understanding it in context. Finally, they construct a concept and meaning for the phenomenon.
- The teachers also develop a relationship to the phenomenon in a similar way.

- Encountering and meeting the other – as human being or world – involves an interruption to existing knowledge and relationships but is done in a spirit of respectful, open interest.
- The learning process is performative and aesthetic and thus transformative.

Steiner's theory of knowledge, described in Part 1, translated to the pedagogical field of teaching and learning, starts with the phenomenon as given and then brings in concepts that make the phenomenon comprehensible in a larger context. On the one hand it shapes the pedagogical approach, and on the other hand it directs the teacher's attention to certain understandings. Steiner's approach to generating knowledge through learning starts with being open to the phenomenon, helped by a certain mood of wonder and curiosity, then moves to an initial reflection and proceeds to a perceptual knowing, or a knowing-in-experiencing through which the phenomenon initially becomes a personal experience. As we saw, building on Goethe's scientific approach to gaining knowledge of nature, Steiner (2011a, 2011b) developed three dimensions of the I-world relationship: firstly, the world as we find it, given to us through the senses; secondly, the world that the person forms a personal and meaningful relationship to that leads to a modification of her existing understanding of the world; and thirdly, the individual knowledge one gains of more general concepts, principles and laws. This more conceptual knowledge, however, retains a quality of striving towards a more complete, more generalized understanding.

This epistemology has led to what is generally referred to as a phenomenological approach in the natural sciences and a symptomatological approach in all forms of culture (Zech, 2018; Wiehl, 2015a, 2015b, 2017). Symptomatology refers, in a subject such as history, for example, to the choice of significant moments in history that are representative or symptomatic of particular historical processes, such as the emergence of a new consciousness or trends towards the emancipation of the individual, or totalitarianism in the modern world. In studying art, it means first describing the work and how we experience it, then putting the work into context – for example, accessing the iconography of the picture's language or the medium used (e.g. sculpture) and asking in what relationship the work stands to cultural values at the time. The question in symptomatology is: what comes to expression through this object, both in terms of its specific message and for a historical way of seeing or being? In phenomenology, it is an approach that starts from the phenomenon – for example, in physics or biology – and proceeds to the principle, the definition, the law or the category.

In the Foundations, Steiner outlines three stages of knowing that are relevant in education – experiencing, forming a judgement and coming to a living concept (in German, *Schluss*, *Urteil*, *Begriff*). This quintessential Waldorf method of coming to knowledge, or rather to the experience of knowing, has been described in the following terms: "the encounter becomes experience; and out of the experience the concept crystallizes. Encounter, experience, concept-perception, feeling, idea: these are the three steps in every genuine learning process" (Barnes & Lyons, 1979, p. 7). Michael Zech (2018, pp. 41–42) describes the same process as follows:

1 Taking in new experiences
2 Opening up the phenomenon, analysing, structuring, ordering and relating
3 Taking an individual position in relation to it, forming a concept and connecting this to a wider field of knowledge

The first stage of the phenomenological approach involves having a bodily, sensory experience; the second is a personal, dialogical and relational engagement with the phenomenon; and the third step involves forming a living, often provisional, concept, which from the Waldorf perspective is a spiritual activity. The second stage involves a personal subjective and also inter-subjective, social and cultural process through language, dialogue and socio-cultural expectation. Thus body, psyche and spirit are involved.

The learning process is thus structured by the teachers to follow this general sequence and applies to both science and humanities subjects and, actually, to all other educational activities involving learning. As we have seen, these steps of forming knowledge map onto the first three phases of learning, as discussed earlier.

In order to take in rich experience, the learner has to be able to attend to the objects being studied with some degree of interest and openness. The interest of the pupil begins with and grows with the development of a personal relationship to the part of the world being studied, and this relationship arises not only out of a desire to get to know the world but also out of a need for self-efficacy. Very young children tend to take hold of what they encounter in order to get to know it but also to see what they can do with it. The same gesture is true of older children and young people, though they have learned to engage with the world in different, often more circumspect ways. In science and in practical craft activities, the phenomena are predominantly located in the world. In the humanities, the phenomena arise more inwardly in relation to the thoughts and feelings that arise through engagement with a piece of literature, music, art or philosophical ideas. However, following Steiner's epistemology, the phenomenon always includes the world outside, the 'inside' of that world (that is, the inherent qualities of the object) and our response to it. From the perspective of the learner, the phenomenon arises through our engagement with it.

The phenomenological approach used in Waldorf schools encourages an encounter with the phenomenon before learning the concepts associated with it. From an epistemological perspective, the pupil's attention is directed by the teacher towards the object in question, thus establishing a shared and collective intentionality. This directs each individual's will, attention and interest towards the phenomenon and engages her in an activity with or towards it. In young children, this is often generated by imitation and the cognitive capacity to create and follow joint agency (Tomasello, 2018). Joint agency means *we* do this and that, *we* attend to this or that. Joint agency and shared intentionality are part of our social makeup as human beings. Children learn what to attend to by participating in social practices that give these things significance and meaning. This is bound up with being embedded in a learning community of practice with peers within a specific learning culture, in which more expert members – i.e. the teachers – have authority and coordinate and lead the processes. 'Instruction' in the lower school in a Waldorf setting invariably means following what the teacher is doing. With children in the middle school (ages 11 to 14), this is done wherever possible through hands-on experimentation, accurate observation and description. The process *does not* start with the law or definition ("metamorphic rocks are formed by extreme pressure deep in the earth's crust . . .") but with an invitation to engage with the phenomenon ("feel the weight and describe the colour, texture and structure of this piece of rock . . ."; or, "strike the tuning fork and put it in a glass of water, and then on different objects including the tip of your nose, and observe and describe what happens"). In the high school, phenomena are explored, where possible measured,

and data is collected, analysed and interpreted. In all three encounters, there is the possibility of transformation through new discoveries.

In the previous section in which I outlined the learning phases, I pointed out that the encounter with the world through rich experience can involve an interruption – an encounter with something unexpected that prompts an experience of otherness. This belongs to what Biesta (2013) calls subjectification, or the coming into being of the person as subject. Biesta's notion of subjectification draws on a phenomenological philosophical tradition and specifically on Hannah Arendt's (Arendt, 1958) notion of our 'natality' or beginnings. This refers to the situation of our 'coming into presence' or 'coming into the world'; that is, the process or series of events in which something new comes into being through the action of the subject (Biesta, 2013, pp. 141–143). This coming into presence, however, is always a coming into the presence of others. Thus, our subjectivity is dependent on how others respond to our natality, thereby creating a space for our action, for our becoming. Biesta (2013) points out that the notion of subjectivity as coming into presence needs to be complemented by the notion of uniqueness. Here he draws on Emmanuel Levinas's (Levinas, 1991, 1998) notion that we need others in order to experience that we are different. Biesta (2013) explains the significance of Levinas's idea and takes it a stage further by pointing out that it is not crucial to know what makes us unique but to know when it matters. It matters when we are responsible for the other (Biesta, 2013, p. 144). Responsibility is something that only we ourselves can take responsibility for. We cannot be morally coerced to take it, otherwise it is not responsibility in this sense. Taking responsibility is an act of a subject who recognizes the intentionality of her relationship to the object. Thus, subjectification as an educational function involves enabling people to come into presence through the (personal) other, for whom they take responsibility.

Biesta highlights the pedagogical value of "situations we can find ourselves in, situations in which we are literally singled out and in which our uniqueness matters" (Biesta, 2013, p. 145). As he puts it, this uniqueness is an event, rather than a quality one can have or claim to know. It occurs in encounters between people that cannot be planned or predicted. These are situations in which the subject is called upon to answer the question posed by life, "So what are you going to do about this – remain who you are or change?" This leads Biesta to argue that subjectivity-as-an-event cannot be produced by education, but teachers can create conditions in which it is more likely to occur.

Through pedagogy, teachers can create situations that call for subjectivity, but they cannot guarantee that this will occur. Biesta (2013, p. 23) argues that it is important to ensure that our educational arrangements, "our curricula, our pedagogies, our lesson plans, the way we run and build our schools, and the ways in which we organize schooling in our societies – do not keep our students away from such experiences". Linked to his notion of coming into being through the other, Biesta has introduced the idea of a 'pedagogy of interruption' (Biesta, 2019). In educational contexts, the pupil may be confronted with something (or somebody) that causes a disruption of habitual patterns that awakens the person to the other (i.e. that which interrupts). This intrusion can occur because what the student encounters is something unknown, something strange and different or something that offers resistance. However, a pedagogy of interruption is not just a question of adaptation and the assimilation of something that comes from outside, nor the unfolding of what is within but about an ongoing dialogue between self and other. In this encounter, there are three possible responses.

The first is to overcome, master and defeat what resists us. Thus, we affirm and retain our existing status, and no change occurs. The second is to avoid the other. In either case, nothing is learned in the sense of learning as becoming something that is not yet there. Biesta calls the third option the educational 'space'. This is where education as self-formation begins, which is a process that cannot be totally controlled and with uncertain outcomes. Resistance enables the person to see herself and her relationships to the world from another perspective. These 'interruptions' are by their nature events, rather than a systematic or habitual process. Interruptions may awaken the subject reflexively to a particular situation and its structures. Thus, becoming a subject is not an outcome in the usual sense of educational aims and outcomes but rather a series of events through which the subject comes into being, in and through encounter with the other. Rawson (2019b) has shown in a study with Waldorf students that many projects, practicals and internships offer opportunities for interruption that may prompt subjectification events, especially if these are supported by scaffolded reflection that can help reify and verbalize these experiences and their meaning.

Käthe Meyer-Drawe (2012) gives a similar account. She takes a phenomenological approach to learning as experience in which the existing world that a person has internalized is interrupted and a new one is constructed. Thus, learning bridges the gap between what is no more and what has yet to become. Learning begins at the point where what is known is no longer adequate and what is not yet known is not yet available. Learning is characterized by fractures or ruptures (or interruptions). These are experiences we have when we choose to or are forced to look at what we know with new ways of seeing, listening, feeling, knowing, understanding and doing (2012, p. 15). Meyer-Drawe uses the old German word *Widerfahrnis*, which refers to something that comes towards us in life unexpectedly and generates an experience. She links *Widerfahrnis* with the classical Greek notion of pathos – meaning to suffer, as used in connection with Greek drama, in particular in the expression *pathei mathei* – learning through suffering (2012, p. 150). Learning, especially in the sense of transformation of the person, is 'suffering' through experience, and the precondition for this is the willingness to be open to what comes towards us.

Learning, according to Meyer-Drawe (2012), is both a new 're-membering' or a new reconstruction of what was tacitly known and proceeds from what has been bodily learned that is raised to consciousness through reflection. In both cases, the productive difference between before and after is brought into movement. In both cases, existing horizons and structures of knowing are reconstituted. In this sense, learning is not based on experience; learning *is* experience, and therefore learning is always, in some way, a learning about oneself. Meyer-Drawe contrasts this phenomenological account of learning with the predominant learning discourse that sees learning ideally as a machine-like optimization of smooth processes that solve problems with a minimum of disruption that aims for reproduction without variation and other inadequacies, as free from stress and friction as possible. Phenomenological learning, she says, is messy, awkward, subversive and anachronistic and involves the person getting tangled up with the world. Changes brought about through interruption can open up new opportunities for responding to the world and for agency.

Rather than being 'delivered' in 'pre-packed' form, the concepts are brought forth through the pupils' own agency, though guided by a teacher. In this sense, the pupil (re) connects her being with the being of the world in this process of knowledge-generation, thus bringing both the reality of the phenomenon and her own being

iteratively into being. Thus, the concepts that are produced by the process epistemologically also have an ontological dimension, as they bring the reality of the phenomenon into being as well being a process of becoming for the learner. In bringing forth a new concept, the learner brings herself as a more integrated person into being. The mind is embodied, and the body learns, and thus the whole develops.

The notion that the human being has a body, psyche *and* spirit and that the I as spiritual core of the individual integrates these over the course of growth and development has major implications for teaching and learning. As the core of individual being, the I participates in the sense-perceptible world through the body. Through social interaction, it participates in the lifeworld of cultural and social relations and through thinking, it accesses the spiritual world. The I has the function of integrating these bodily and psychological experiences with the spiritual into what we would call today an embodied mind. In the first stage of the knowledge-process, the I experiences itself when it takes an interest in the object of the lesson, explores it and 'plays' with it in forming a perception. At the same time, the process allows the free play of thoughts, associations and memories to be prompted by the experience. This process of engaging with the world is one in which the whole body is involved in movement. In engaging with the world in a direct and concrete way, we literally position our feet and legs in relation to the forces of gravity. This, according to Steiner, ensures that thoughts are grounded, centred in the person's being and orientated towards a fixed point, namely the centre of the earth. The pupil experiences herself in her body primarily through the experience of gravity and the continuous play of forces needed to retain balance in movement. The body gives us a first-person, direct and centred experience of the world.

If the phenomenon being studied is not physically present, then the pupil's imagination has to be activated in calling on embodied knowledge of sensory experience to activate embodied patterns of movement and sensation, so that she can vividly imagine the phenomenon. Through vivid narrative, the pupil's body maps can be activated to approximate the relevant experiences (e.g. of standing on a windy deck with cold spray hitting the face, of the deep sense of loss felt by the beloved, of the moral dilemma of choosing the right path, of the prisoner facing torture to reveal his comrades' hiding place, etc.).

These cognitive processes play into the sensation of the perceived object. Thus, we can see how the first step in the knowledge process is intimately bound up with the experience of the body. The I forms the perception, which has a physical and relational correspondence in the movement of the feet and legs, literally taking a stance, which is why Steiner used the term conclusion (in the sense of deciding on a position) for this first part of the knowledge process. Edith Stein (1917/1989), in her phenomenological exploration of empathy, noted that the perception of another sentient body in space enables the observer to both experience the stance of the other but also to realize her own corresponding stance because of the intentional nature of the relationship of subject and object. Empathy means being able to take the stance of the other (or walk a mile in her shoes). The relationship between body and mind in this respect alters with maturity and in particular with the onset of puberty, when a much greater consciousness of gravity emerges.

In the second phase of the knowledge process, the subjective effect of the object on the observer-participant becomes relevant. Conventional epistemologies ask us to distinguish between subjective and objective; however, there is a median position, a

movement between linking these polarities. Between suppressing the personal in the (usually unrealistic) attempt to be purely objective and giving oneself over totally to sympathy or antipathy, the learner engages in a process of balancing both aspects to come to a judgement based on a feeling for what is relevant and right. This is a process of 'weighing up' the evidence, often accompanied by the body language of the hands and arms 'weighing' the relative merits of the arguments and data. This is a process of dialogue and negotiation that precedes a judgement. By the third stage, that of forming a concept, sufficient distance from the physical and psychological dimensions must be achieved in order to grasp the nature, context and meaning of the object being studied – or, in anthroposophical terms, to engage in a spiritual process. The pupils now try to relate what they have learned to the bigger picture and the wider context, physically symbolized by the 'cosmic' all-embracing spherical form of the head. Thus, the threefold structure of the body, feet, legs and will, arms and feelings and the head and thinking, are integrated, and the whole process occurs within a dynamic oscillating between unconscious participation and conscious reification.

This understanding adds a new dimension to the question as to what learning and development mean, and what distinguishes the Waldorf approach from most others. However, there are close affinities to some strands of current thinking, in particular to more recent phenomenological anthropology. Thomas Fuchs (2017) makes the important distinction between two fundamental aspects of body, between the body as "an observable and materially composed object among other objects" and the body as "living centre and medium for the enactment of life" (ibid., p. 211) – in German, distinguished respectively as *Körper* and *Leib*, the body and the lived-in body. The dualism implied by the distinct categories of mind and body is overcome by the concept of embodied subjectivity,

> which constitutes all conscious processes as permanently enacted on the basis of the lived body's overall state, and thus being always embedded in this subjective state. Given that this bodily background experience participates in the basic co-extensivity of subjective body and physical body, this also applies to all apparently 'mental' conditions of human consciousness. The subject of all mental activities is always a *bodily subject*, 'incarnated' in the whole physical body, and it is thus a *spatial* subject as well.
>
> (ibid., p. 211)

Fuchs draws attention to the dual aspect of the person as living being and the process of knowing; the bodily side of the organism provides us with a basis for a naturalistic, objectifying relationship to the world through our sense organs and the experience of being a body in space under the force of gravity and the animated bodily aspect that mediates us an integral, personal, subjective, lifeworld-orientated and conscious lived experience.

Wilfred Sommer (2016, pp. 518–519) maps the three phases of Waldorf teaching/learning onto Fuchs' phenomenological dual aspect of the human being. The subjective and integral aspect of experience comes to expression in the initial encounter with the phenomenon, driven by interest and attention that derives from a central perspective of the learner by virtue of the centrality of bodily experience. Because we 'take' our body with us, our initial stance is always the perspective from our own body as zero-point. At this stage, the encounter is experiential but not yet explanatory.

First phase	*Second phase*	*Third phase*
Lived body (*Leib*) in time	Oscillates between lived body and physical body	Physical body (*Körper*) in space
First-person perspective		Second-person perspective
Body as subject		Body as object
Immediate lived lifeworld experience		Reflected reified experience
Centralized perspective		Eccentric perspective
Integral expressions of life		Physiological processes
Subjective participatory stance		Naturalistic, objectifying, reifying stance

Figure 2.4 A phenomenological perspective on the Waldorf teaching/learning phases (after Fuchs, 2017; Sommer, 2016)

The third stage of conceptualizing, in which the object can be apprehended in its inherent lawfulness, requires a degree of distance from the original experience, and thus from an 'eccentric' or non-central, non-personal perspective, maps onto our experience as physical body. The second phase of teaching/learning oscillates between these two poles. We can see this in Figure 2.4.

As Sommer expresses it,

> the phenomenological approach of Waldorf teaching requires a performative approach because it is not only seeking to understand a phenomenon or the meaning lying *behind*, *under* or *over* the latent world-structure. Rather it seeks to grasp the reality-constituting moment of coming-into-being itself.
>
> (2016, p. 521)

It applies Steiner's theory of knowledge in a way that takes account of his spiritual anthropology – the threefold nature of the human being as body, psyche and spirit through the activity of the I as core of being. The ideas and concepts that can be experienced by the pupils through this process arise through performative acts of meaning.

The term performativity has a number of meanings (see Burnard & White, 2008). Here, performativity (also *performative* and the verb *to perform*) refers to the notion that action and reflection flow together in doing, which of course engages the whole human being, body, mind and spirit, and this implies a bringing-into-being of something latent or potential. As an educational concept, it places more emphasis on process than outcome and suggests that education should be more concerned with the construction, application and performing of knowledge than on merely testing performance after the event.

The Waldorf approach aligns with the research field known as the culture of performativity in education, which as Wulf and Zirfas (2007) explain, refers to the notion that knowledge and education refutes the traditional distinction between doing and reflecting on experience. Educational processes can be increasingly understood as

both reflexive and performative, particularly when the learners themselves are active and productive in generating experience and knowledge. Such processes are often also understood as transformative and therefore as the possibility of bringing about emancipation and system change, particularly when this includes extending the action radius of the individual and the group. A performative perspective is not just a reflexive cultural practice; in the sense of Judith Butler (1995), it can be understood as an act of resistance. As Claudia Ruitenberg (2007) argues, Butler's conception of performativity in education sees agency as beginning where sovereignty of the self wanes. Ruitenberg writes,

> educators must conceive of students, and students of themselves, not as autonomous agents, nor as passive recipients of tradition, but rather as subjects whose actions and identities both depend on, and can make changes to, discourses that precede and exceed them.
>
> (2007, p. 266)

Knowing and identity depend on the performing of acts of knowing and constructing identity as the counter-practice, as it were, both to essentialism and to passive acceptance of standardized knowledge and socially desired identities. If the discourse is one of phenomenological learning, then the student's agency can both contribute to and change this discourse, as it is not a method set in stone but rather a fluid movement.

The performative perspective focusses on bodily co-presence, the relationships in space, physicality, phonetic qualities, eventfulness and rhythm in the generation of meaning. Steiner's notion of art, not as a means to an educational end but as a process in itself, incorporates many of these aspects. The bodily co-presence refers to the experience of shared intentionality and attention that form the basis for human communication in the first place (Tomasello, 2018). This means that the experience of the individual in a performative process is enhanced by the intensity of the resonance of shared experiences in a shared social space. The artistic construction of the learning situation and aesthetic documentation are further aspects that enhance the performative aspect of Waldorf teaching and learning. Embedded in a learning community with shared practices, such enacted, and in the theatrical sense, dramatic presentations and learning situations reduce the asymmetry of teacher–pupil relations by virtue of their participatory quality. Furthermore, such learning situations frequently involve an uncertain act of discovery for the teacher as much as the pupils, as outcomes are often unpredictable. The teacher can plan the learning situation but not what everyone will experience nor what each will contribute to the ensuing discussion. Likewise, not all outcomes can be secured. Such a learning situation can make boundaries of knowledge more fluid for all concerned, teachers and pupils – something that mainstream education often goes to some length to avoid.

The rhythms of participation and distance, direct experience and recall, expectation and actual experiences are, following Wulf and Zirfas (2007), themselves part of the performative process. Finally, the third phase of knowledge generation has the performative quality of combining centric and eccentric perspectives, because generating meaning through a new concept is an individual act of recognition of something that has general validity. The individual, from her centric perspective, constructs a concept that has generalized and therefore non-centric validity and perspective. The process is both participatory and reifying through conscious meaning making.

Sommer also draws on the work of Erika Fischer-Lichte (2008), a theorist of performativity in the theatre, to account for the Waldorf lesson. Fischer-Lichte has appropriated the term autopoiesis, originally from biology (see Stephen Rose, 1998), to characterize the unified and self-regulating system of interaction and feedback between audience and performer. In the autopoietic system of performance, transformation can occur. Sommer (2016) suggests that something similar can happen in the third stage of knowledge generation in the Waldorf classroom, in which both pupils and teacher may experience transformative learning. Regular experience of this process would lead to a habitus or disposition.

> Bildung [formative educational] processes become transformative processes. The event-character of aesthetic educational experiences would be borne by such transformative processes, leading to boundary experiences of knowledge that enable the person to recognize that she too is capable of transformation. Along with the distinct effect of strengthening sense of self-efficacy, this process could lead to and stimulate the development of the person.
>
> (Sommer, 2016, p. 527)

The performative and aesthetic perspective is explicitly associated by Fischer-Lichte (2008) with human development and also with a post-Enlightenment, post-rationalist process of boundary crossing, in which the embodied mind replaces the dichotomy of mind/spirit and matter and the activities of describing and mastering the world, including self-mastery, are replaced with the attempt to construct a new integral world through performing in life, as in performance in art. Sommer argues that Waldorf education's commitment to the development of the person capable of engaging freely with the world is brought to expression through an artistic and phenomenological teaching and learning process that is performative.

One can also identify a correspondence between the Waldorf phenomenological process and Hartmut Rosa's (2019) notion of resonance in the relationship between subject and world that this generates. Resonance is described as a relational and fruitful intersubjective relationship between subject and object, a state which is the opposite of alienation and therefore, from a salutogenic perspective, enhances well-being and health through a sense of coherence (though Rosa does not make this connection). In an educational context, Rosa speaks of a resonance triangle connecting teacher, pupil and the object of the lesson. The function of the theoretical triangle is to analyse whether and how those involved come to a resonant relationship with some aspect of the world. In a successful lesson, the triangle arises when the pupil is open and 'captivated' by the theme and feels involved and at home in the process. The teacher's enthusiasm reaches and inspires the pupils, and the content feels relevant and meaningful to all concerned. In an unsuccessful lesson, the content feels irrelevant and boring, the teacher feels threatened by the pupils, cannot reach them and feels that they are uninterested and can only be motivated by external pressure (e.g. bad grades and their consequences). The pupils are bored or overtaxed by the tasks and experience school as a location of alienation. I would add, following what has been discussed earlier in the generative principles, that this is a context that is likely to provoke defensive learning and a weak sense of coherence, whilst the conditions of the successful triangle are

likely to lead to a sustainable sense of coherence and are motivated by expansive learning.

In the successful model lesson, the teacher's role is that of the tuning fork in the metaphor of resonance; that is, the person who performs the decisive act to set the process in motion. Rosa points out that this ideal closely resembles the optimal conditions for learning identified in John Hattie's (2012) famous meta-study of learning. The teacher is the catalyst for the responsive emotional involvement of the pupils and for the necessary focussing of attention. The ability to experience resonance is the precondition, according to Rosa, for education as a transformative process of learning.

Waldorf education aims to support the free development of the person as a resource for society. It does so with a curriculum and a method of teaching and learning that takes the developmental tasks into account in ways that engage the self and enable it to become embodied. This aim is supported by a spiritual anthropology that portrays the spiritual core of the human being engaging with the body, the animated body and, in engagement with the social and cultural lifeworld, bringing forth its own psyche with the capacities of thinking, feeling and willing. The Waldorf method of block teaching using the phases of learning aligns in many aspects with the models of phenomenological anthropology outlined here. Furthermore, all the subjects in the Waldorf curriculum have the potential to be taught in this phenomenological way, though each requires its own particular approach. The humanities, for example, require a more hermeneutic approach and what Steiner calls a symptomatological approach. Fischer-Lichte's (2008) aesthetic performative approach offers a model for understanding the processes of the Waldorf teaching-learning approach in the main lesson as one of transformative learning.

This is a significantly new way of interpreting Waldorf education, one that builds on Steiner's spiritual anthropology and uses contemporary theory to not only explain the Waldorf approach but to provide a framework for researching and developing it.

The significance of this approach in a world in which the Fridays-for-future movement of young people is trying to draw the political (adult) world's attention to the consequences of a detached, exploitative, profit-driven relationship to the world, can hardly be overestimated. Given many of the problems we are wrestling with, such as climate change and its implications for our behaviour and lifestyle, climate change denial and the general trend of ignoring scientific expertise, the willingness to believe fake news, the increase of xenophobic nationalism, intolerance and the increased willingness to accept violence, the risks of burnout and loss of meaning for the individual, the Waldorf approach may be the basis for a new, open relationship to the other, to the world and to ourselves. Practiced daily – even inexpertly – in age-sensitive ways over a 12-year school career, dispositions may be learned that form the basis for a deep and meaningful connection to the world and to meaningful and meaning-making identities. The phenomenological approach that Waldorf education practices certainly deserves to be considered as an alternative to existing mainstream approaches that do not appear to be very successful in talking about the issues just listed. Most people educated over the past 30 years cannot fail to have noticed that we have an ecological and social imperative to change, yet there is little discernible change in behaviours that make matters worse. Of course, the Waldorf approach makes some major assumptions about the nature of the human being and the nature of knowledge and learning that many educationalists find might find hard to accept.

Teacher skills

The teacher can:

- Structure lessons in such a way that enables a phenomenological or symptomatological approach
- Have a sense themselves for understanding phenomena and the knowledge processes connected to them
- Enable pupils to generate knowledge, rather than being merely recipients of existing propositional knowledge. This means starting from the child's position and existing mode of understanding
- Ensure that the curriculum expands in age-sensitive and developmental ways, so that the knowledge of pupils may grow with them

Research questions:

- This is an area of Waldorf education that has rarely, if ever, been researched except in the wider sense of alumni studies that have identified attitudes. Even though the phenomenological method has been practiced for the best part of 100 years in Waldorf schools and has been described here and there, the link that Sommer and others make to contemporary phenomenological theory opens up new possibilities.
- Among the research questions, one would be to explore how the phenomenological method is practiced in actual classrooms, what kind of performative knowledge is generated and how this perhaps differs from conventional approaches. It would seem obvious that hermeneutical phenomenological methods (e.g. Kvale & Brinkman, 2008) would lend themselves to such studies.

Principle 11: The teaching must be artistic

Generative principles:

- The teaching uses skilled artistry to shape and structure the lessons to optimize the learning experience of the pupils.
- Skilled artistry means being aware of all the possibilities in a given situation and working with them in such a way to realize potential in all involved.
- Aesthetic sensibilities enhance the learning environment.

Teaching is not only an art; it is a craft and a science, and the craft of teaching is skilled artistry. The notion of education as an art has a long tradition. What does it mean in practice? This is another complex concept that risks being all things to all people but, like other such notions that Steiner continuously referred to, it is best thought of as a landscape one can move around in. Sometimes it helps to characterize it by what it is not. Being artistic in pedagogy *is not* following a recipe or a predetermined plan or learning programme but involves micromanagement and choices made in situ. It means assembling the means and methods at one's disposal to create learning situations for particular groups of children at a particular time and place and with specific pedagogical intentions, and then making the whole into a composition – this is an artistic activity. As Heinz Zimmermann noted,

> without this artistic element, Waldorf education would deteriorate into dogmatism and sectarianism. At every turn the teacher must be able to see what has come about thus far and what can potentially come about in future. The teacher's artistry depends on a sensitive perceptiveness, a richness of ideas and a sense for what is unique. Respect for the being who is trying to develop towards freedom.
>
> (1991, p. 23)

Christof Wiechert describes the art of education as,

> a form of teaching that breathes, pulsing between concentration and relaxation, between intensive listening and independent activity; that passes through intensifying phases, approaching a theme, presenting it, then deepening it. Related themes are added, then moments too of reflection. Independent work alternates with group activity.
>
> (Wiechert, 2012c, p. 94)

He goes on to compare the structure of a main lesson with a sonata form; "the theme is presented (exposition), then developed and led into an intensified repetition (reprise) before its summation and conclusion (coda)" (ibid). The coda can follow on the next day after a period of recall. This artistic approach heightens the pupils' sense that what they are learning is significant and that they are involved in something that has an inner development and momentum. As Wiechert writes, this is an ideal but one that shows that teaching has to be thought of as an art form rather than as a routine or functional event. It also involves the teacher taking what Wiechert calls a 'stance' that conveys commitment and enthusiasm. This is one of a number of aspects of aesthetics, including the care and design of the environment, the choice of teaching material and literally the materials used and the aspect of doing something well for its own sake.

The notion of skilled artistry or skilled practice that I want to suggest here is not only intuition and inspiration but also application. I confess to being strongly influenced in my understanding of artistry by Richard Sennett (2008) and by Tim Ingold's (2000) work on skill. Like Sennett (2008, p. 290), I try to avoid the terms 'creativity' and even 'art' because of their Romantic associations with genius. Skilled artistry is learned. It is trained practice applied in situ. Art and craft belong together, "because all techniques contain expressive implications. This is true of making a pot; it is equally true of raising a child" (Sennett, 2008, p. 290). I follow Ingold in seeing skill as an ecological property of a person embedded in an environment, in this case the teacher, the class and the pedagogical situation. Skilled practice calls for qualities of care, judgement and dexterity as the bodily response to the overall situation. How is skilled artistry learned? Craftspeople and children do not learn by being given a body of representations or information "but by introducing novices into contexts which afford selected opportunities for perception and action, and by providing scaffolding that enables them to make use of these affordances" (Ingold, 2000, p. 354). The skilled practice or artistry of teaching means knowing the 'material' and its intrinsic qualities and relationships. This means knowing the children and young people. It means knowing the 'tools' and how and when to apply them and to what effect. As Ingold points out, skilled artistry means working from the activity of imagining, which carries within it an intention projected into the future, but that is not the mere implementation of a model that determines the outcome. Imagining, says Ingold (2000, p. 418), is also

reflexive; its activity is not only directed to the external world but is also turned within and can relate to notions of what could be, to an anticipation of what can become. As Otto Scharmer describes, we have within us the potential to experience what is emerging, and this requires generative listening. The preconditions of this opening are that we "get our (old) self out of the way in order to open a space, a clearing, that allows for a different sense of presence to manifest" (2009, p. 13). However, as Ingold also points out, skilled artistry means dwelling in the present in the actual world, situated and embedded in a context.

Of course, skilled artistry in teaching also has an aesthetic dimension and purpose, and everything the teacher does is an aesthetic statement that signals to the children that they have been conscious in their choice of colours, forms and materials. If things are worth doing, they are worth doing well. This moral notion is not to be understood pedantically in the sense of standards but that we try to do things in a way that brings out the best in the situation, whether it is neat, readable handwriting, carefully illustrated main lesson books, the way the chaukis are stacked up carefully, the nature table that is ordered with care or the way attention is given to detail. The satisfaction and mutual recognition come from the learning community when we do our best, not that *we are* the best. Aesthetics are an expression of higher values, of creating a higher meaning, revealing hidden meanings and processes.

In my many experiences doing land art with school pupils, student teachers and teachers – and in many different cultures – I am always amazed how little explanation this activity requires. Land art is an artistic activity that involves using or playing with natural materials such as leaves, stones, branches, earth, sand, ice and so on, in their environment (usually outdoors) to construct arrangements that bring certain experiences connected with the material and the setting to expression. With the minimum of introduction, the participants disappear into the environment (and I do this in school playgrounds and city streets, as well as in nature) and come back deeply satisfied. When we examine the products, there is no need to analyse, we just admire. The artist Andy Goldsworthy, who has done more than most to show us the possibilities of land art, writes, "fixed ideas prevent me from seeing clearly. My art makes me see again what is there, and in this respect I am also rediscovering the child within me. . . ." (Goldsworthy, 1990, p. 6). The human need to not simply have artefacts but give them shape, and therefore enhance their meaning, is as old as human culture. To give shape and form and colour to our working and living environment is to care and to recognize that things have their place in the world and that we can respect that. This aspect is an important part of being artistic in Waldorf education.

How do we recognize artistic teaching? Probably the first thing to say is that it is unobtrusive, almost invisible because the lesson appears to have flow and the processes appear natural and relatively effortless. Sometimes, teachers who are natural educational artists are themselves unaware analytically of how what they do works. Novices will not always recognize this when they observe expert teachers. The transitions from one activity to the next are smooth, or at least do not unduly break the flow. However, artistic does not necessarily mean calm, harmonious or quiet. Quiet lessons may be a result of a lack of interest and initiative among the pupils, or it may be based on fear (especially when being observed).

Artistic means there is rhythm but also contrast. There is a balance of activities that awaken and energize and those that calm, between active doing and talking and listening. There should be mood changes from excitement and laughter to seriousness and

introspection, between thoughtfulness or reverence and lively engagement. The point is that all these activities have a purpose directed towards realizing the overall learning aims, which include aspects such as participation and not just measurable outcomes. Therefore, a certain structure must be apparent but not a rigid time plan. The teacher must have alternatives if things don't work or turn out to be inappropriate. The 'work of art' that is a lesson is, of course, a performing art in which all players participate. It has both a choreography and script and also allows for improvisation. It is not a work of art if it is standardized, though of course it may make use of standardized elements, composed into a whole that is relevant to the situation. Waldorf teaching uses a wide range of teaching methods familiar to other school types; the choice of which is based on the judgement- which activities enhance the learning experience of this group at this moment.

Teacher skills

The teacher can:

- Understand the developmental needs of their pupils and the educational and developmental tasks in their context
- Develop mastery of their pedagogical 'tools'; that is, the various teaching methods and their effects when implemented in certain ways
- Blend this knowledge into a meaningful whole set of activities unique to the situation in order to enable learning
- Have an aesthetic sense for creating a cared-for space and learning environment

Research tasks:

- Build up a repertoire of possible teaching methods ('tools') and assess their function and usefulness through observation.
- Reviewing and analysing lessons from the perspective of their artistry. This can be done through participation and observation but perhaps also through video analysis.

Principle 12: The self-activity of the students is essential to learning

Generative principles:

- All learning is self-formation and requires self-activity.
- This needs to be recognized and supported by teachers.
- Self-learning is enabled by participation in practices.
- Students document their own learning in self-made main lesson books, folders or portfolios and other artefacts.

Steiner frequently made the point that all education is self-education or self-formation. One of the meanings of this is that education has to activate the learner in her self-activity. As will have become clear so far, much of the method of Waldorf teaching involves enabling the children to have direct experiences. The aim of scaffolding in learning is to enable and support the self-activity of the learner. Another aspect is that

children are encouraged to do things and to make things themselves or together with others. The self-activity of the pupils also needs to come to expression through the way they document their learning, so that the recording of learning becomes an integral part of the learning process. Instead of using textbooks, the children make their own 'text' books of the main lesson themes. In a large book, which they decorate and design a cover for that reflects the content, the students record what they have learned in their own words and pictures. When the children are older, in some schools, they literally make their own books and paper. They may copy drawings from the blackboard and even important short texts, but the basic idea is that they document their own learning in their own words as soon as they can write. Not only do the children take pride in their work, but they also learn about the importance of presenting information and about presentation, which has become an important aspect in many fields of work.

This self-activity is of course prominent in the arts and crafts, in which aesthetic and useful (usually also aesthetic) things are designed and made. Another aspect involves the many opportunities for project work, self-study and teamwork. Self-activity is guided by reflection on aesthetics and on the effectiveness of what the students produce. Is the message clear and appropriate? Does the form reflect and enhance the content? Does the presentation contain all the important aspects? In the upper school, a portfolio is increasingly used as a medium for pupils to select, present and reflect on their experiences and work (Koch, 2010) The essence of a portfolio is that the owner chooses what material or work should be included, and this choice should be explained and justified and on the whole be individualized and yet self-explanatory. As in many creative processes, it is helpful to have models and exemplars to learn from and orientate oneself to, so self-activity in the learning process often has to be scaffolded by teachers and more competent others.

Differentiating tasks allows students of all interests and aptitudes to participate. The Waldorf model is basically inclusive and non-selective. One of Steiner's key ideas from the threefold social idea was that there should ideally not be a social divide between people who would later have a professional and academic job or a practical profession. It is a challenge, of course, for the teacher to structure the lessons in such a way that students of all interests and abilities can have a fruitful experience. Even in the youngest classes, there is always a wide range of interests and aptitudes. The key to the inclusion of differently abled individuals is to enable participation. In a Waldorf class, children and young people of all abilities learn side by side. There is no streaming or setting and all children basically do everything, though they will do it in different ways.

However, the clue to this method is twofold; firstly, all learners are embedded in a supportive learning community and secondly, the learning steps are differentiated. Pupils share the same initial experience and its recall, but when it comes to applying what has been learned, different tasks are given according to individual need. There are three primary levels of application that can be varied as required. The first level is simple reproduction. The learner is asked to repeat or reproduce what has been presented. The second level is comparative. The learners apply what has been learned to similar or comparative situations. The third level is the open question for which there is no set question or answer, but rather the learner determines what would be appropriate questions and how these would need to be answered. Or indeed, the learner can apply what has been presented to entirely new contexts. This requires a higher level of comprehension. To give an example from geography: the first level would be to reproduce information that has been given, the second task level may be to apply the approach learned in one

context (e.g. understanding one geographical region) to another and the third level of question would either be to identify relevant questions and possible answers or indeed to explore some entirely new aspect of the theme not raised by the teacher.

In project work, different learning types can be guided to take different approaches. Some can choose practical projects, making something, whilst others may choose an artistic challenge or a scientific study (to name just a few options). Each pupil plays to her strengths and faces her challenges in a supportive environment. Waldorf teachers differentiate in four different areas, which can be briefly summarized as follows.
Content: Students can:

- Be set individual learning goals
- Be allowed to choose their themes for individual work
- Be given different tasks
- Set individual emphasis within given topics

Learning pathways: Students can:

- Be given more or less time for tasks
- Be given the same time for open tasks (i.e. each student does what she can in the given time)
- Use different modes of presenting their work, such as orally, in writing, enacting a scene, drawing a picture or making a graphic

Material: Students can:

- Be given material with different levels of difficulty
- Be given material in different media (text, image, film, graphic, etc.)
- Be given different kinds of learning support

Social form: Students can:

- Work alone, in pairs or in groups
- Use discussion in lessons
- Work in teams with specified roles

Formative feedback is often given to students in differentiated and ipsative form; that is, in relation to that individual's learning progress rather than in a standardized way, because individuals can often respond to feedback that is tailored to their needs. I will discuss this later in the section on assessment for learning.

Teacher skills

The teacher can:

- Understand the importance of self-learning and work to enable it
- Are capable of supporting self-activity
- Are able to differentiate tasks for different student groups
- Understand that their own self-activity is a role model for their pupils

Research questions:

- What kind of scaffolding supports self-activity, and what limits it?

Principle 13: Good teaching and learning depends on the development of the senses

Generative principle:

- The quality of learning depends on the development of the child's senses because this mediates her relationship to the world and to her own body.

Waldorf education assumes that healthy sensory processes are fundamental to learning and development, which in itself is not a particularly radical idea. What is unique to Waldorf education, however, is Steiner's expansion of the categories of sensory experiences. Therefore, it is worth unpacking this in somewhat more detail, particularly by putting it in the context of more recent theories of perception. In the Foundations (1996, lecture 8), Steiner introduced the 12 senses and grouped them into three categories:

1 Will-orientated senses: senses of touch, life, movement and balance
2 Feeling-orientated senses: smell, taste, sight and warmth/temperature
3 The cognitive senses: hearing, speech, thought and perceiving the I of another person

Steiner explains that most experiences are multi-sensory and that the I reintegrates the various sensory modalities into mental images that form our initial percepts. This involves a process of unconscious subjective judgement as to what we consider relevant. There have been a number of explanations of the 12 senses in a pedagogical context (König, 1999; Soesman, 2006; Aeplli, 2013), but few of these accounts have taken account of current theories of sensory perception. Very briefly, I can outline what the senses mediate:

1 The sense of touch mediates the body's boundaries, thus providing a basic sense of the distinction of bodily self and the other.
2 The life sense, or sense of vitality, mediates the body's experience of homeostasis – the state of equilibrium between bodily processes and our experience of our interior state of well-being or disease. It also mediates sensations of pain, hunger, thirst, exhaustion, nausea, vulnerability, sexual arousal, energy, relaxation, vitality and excitement. This sense is largely responsible for our bodily sense of self, what Damasio (2012) calls the proto-self, which is the basis for the core self. Over time, this provides us with the foundation for an embodied sense of autobiographical continuity, or in other words, the history of our body as core identity. The sensors for this sense are located in all our inner organs.
3 The sense of movement is known by science as the sense of proprioception and provides us with the experience of our own bodily movements, muscle tension and position in space. It is also the basis for perceiving movement of other things or people and indeed the effect that the statics of buildings and large structures

(including large objects in nature, like trees and mountains) have on us. It gives us a sense of the space we are in. We have sensors of movement in all our muscles.

4 The senses of balance, smell, taste, hearing and sight are as commonly understood by medical science.

5 The sense of warmth or temperature mediates the difference between the ambient temperature and the temperature of objects around us in relation to the body's own warmth. It also senses the bodily processes involved in establishing our body temperature, which are very individual. Bodily warmth is closely related to our sense of comfort, security and emotional warmth.

6 The sense of word, speech or language enables us to distinguish speech from other sounds, though not the meaning. It enables us to perceive gestures, body language and facial expressions (and sign language) and forms; that is, it indicates that they have meaning but not what that meaning is. We follow the melody and intonation of speech and the emotionality within it, and it enables us to identify distinctions between languages. The organs of language perception include the entire muscle system but particularly the movements of the chest and larynx that produce and imitate speech (Lutzker, 2017, 2018).

7 The sense of thought mediates the thoughts of others as coherent structures – for example, the structure of a sentence or a piece of music. It conveys the experience that something has meaning and relationships but does not convey to us the content of the thoughts. It enables us to perceive patterns, sequences and structures spatially but also temporally. The organs of perception are our entire cognitive processes.

8 The sense of the other person as a sentient being and an individual. It enables us to recognize their unique signature, style, voice in both literal and figurative senses and their biographical mythos.

Thus, together the senses mediate the world to us, though not its meaning. Meaning is provided by the concepts we intuit; or in the case of children, they are intuited from the behavior and speech of those around them, which is internalized. The point about the development of the senses is that the quality of what is 'given' and 'received' varies depending on how open we are and how quickly we identify what we have experienced and what its meaning for us is.

Steiner arranged the 12 senses in various ways, but Auer (2019) has regrouped these for educational purposes into, firstly, the senses directed towards the states of the body, including the senses of touch, life (or sense of vitality/vital sense), movement and balance and secondly, the senses directed towards the world. Amongst these, he distinguishes between the communication senses, including the senses of sight, hearing and touch, the sense of language or form, the sense of thought or meaning and the sense of the other as I. He distinguishes these from what he calls the senses that have the function of providing warning signals, such as the senses of taste, smell and warmth. Schmelzer and Deschepper (2019) refer to the senses of hearing, language, thought and the sense of the I and of the other as the social senses, which I think is helpful.

Currently, science recognizes nine senses – sight, hearing, touch, taste, smell, balance, temperature, pain and proprioception or self-movement. At present, it is not known for certain how sensations are translated into qualia; that is, into subjective conscious experience (Lutzker, 2017, p. 188). Steiner's account does not solve the

problem, but he does provide an exact phenomenology of experience in his book *Theosophy* (2011b). He distinguishes between the *sentient body*, which comprises the organization of physical, chemical and electrical processes that mediate sensation, and the *sentient soul* that experiences them. In the boundary between these two domains, or rather in the translation from the body to soul/psyche, is where the qualia problem lies. It is also important to note how Steiner defines sense experience. He defines a sense organ as the bodily processes that mediate the ability to recognize the nature of an object, a process or another living being as it manifests in the physical world, i.e. not any psychological or spiritual quality.

If we think of word or language perception, this can be an incredibly complex process involving thousands of 'bits' of information in the form of movements, gestures and sounds. This makes it clear that all 12 senses and their neurological processes have a physiological basis. Each sense organ, which in the case of touch or warmth may cover most of the body's surface, and each organ function is part of the body and mediates sensations based on physical phenomena. The senses mediate no spiritual qualities, only what is perceptible to the physical senses and their neurological processes. Secondly, a sense impression is not the same as recognizing the object, naming it and forming a judgement about it. A pure perception is a conglomeration of sense data without meaning. This is hard to observe because we habitually organize our sense impressions into percepts built on previous experience. Furthermore, most impressions we have are a blend of different sense impressions, vision, movement, balance, sound, smell, etc., which the I then integrates into a lived experience.

The body senses provide us with the experience of being in our body, of well-being and a sense for boundaries. Auer (2007, 2019) highlights the pathological conditions that can occur when children do not feel secure in their body and lack a sense for boundaries. The sense of touch mediates the boundary between our body and the surrounding environment and what physical qualities the things we encounter have. At the same time, this provides us with a self-perception and the basis for a body image. The sense of vitality or life sense provides a basis for the sense of well-being, or signals to us and the world when we are unwell. Rhythm is an important way of establishing well-being. Balance is not only a question of equilibrium whilst moving but also provides an embodied basis for balance in a wider psychological sense. Children who have a poor sense of balance have a permanent underlying sense of anxiety because they feel physically unbalanced and precarious. The sense of balance develops, like all senses, through use.

The sense of our own movement and the positions of our limbs is a highly differentiated sense ranging from upright walking and posture, running, dancing, climbing and so on, to fine motor control, coordination and gesture, mimicry and imitation. It is also active in enabling us to move together with others in space in play or sport but also in the dance of social life, with its cultural expectations of proximity and intimacy. It provides us with perceptions that enable us to orientate ourselves in space and to follow lines, whether of writing or music notation, to read maps or even to orientate ourselves whilst looking at images. In all forms of movement, balance is involved and enables us to focus our eyes on things whilst moving, estimate distance and determine spatial depth, particularly in hand–eye coordination.

Steiner's theory of the senses has opened up very important new perspectives on child development, particularly how the senses develop in young children and what fosters and hinders this. Traditional children's play provides ample opportunity for all the body-orientated senses to develop. Social interaction schools the social senses.

What is unknown, because it is not generally recognized as a factor and therefore is not researched, is the effect of prolonged inactivity in front of digital media instead of engagement with the real world.

One perspective of the senses that I think is very valuable is that from the anthropologist Tim Ingold, who takes an ecological perspective on perception. He writes,

> Perception is not an achievement of a mind in a body, but of the organism as a whole in its environment, and is tantamount to the organism's own exploratory movement through the world. If mind is anywhere, then, it is not 'inside the head' rather than 'out there' in the world. To the contrary, it is immanent in the network of sensory pathways that are set up by virtue of the perceiver's immersion in his or her environment.
>
> (Ingold, 2000, p. 3)

Perception is therefore a relationship between an environment and an organism. It is a skill that is learned and which develops over the course of human ontological development because, as the psychologist James Gibson, who developed the theory of affordances, puts it, "learning is an education of attention" (1979, p. 254). Ingold adds, "as such, it is inseparable from a person's life in the world, indeed continues as long as he or she lives" (2000, p. 167). For Gibson, perception is the activity of the whole organism moving through and engaging with its environment. What the organism perceives are not things but the possibilities the situation affords the subject for activity. In other words, perception is the process of attending to and responding to the affordances in the environment, which then manifest as skill when we engage with the world. Meaning arises through such productive engagement (Ingold, 2011, p. 11). This engagement has always been mediated in human beings by cultural understandings. Here lies one of the central tasks of education – to mediate the way children encounter the real world, and to some extent help them to distinguish between real and virtual worlds.

Teacher skills

The teacher can:

- Understand the different senses and sensory processes and recognize the need to cultivate the various senses in the pedagogical environment
- Support children and their parents to recognize how important cultivating the senses is and help safeguard them from activities that numb the senses to rich and subtle experience (e.g. in language, through colour, form and texture, through movement and coordination, through directly encountering nature and natural materials, which are far richer than artificial or virtual things)
- Develop their own sensory processes through artistic activity, experiences in nature and observation

Research questions:

- Many aspects of the senses are medical and cannot be researched by teachers, but all the claims and assumptions referred to previously need to be observed

in practice and researched, such as cultivating the lower, bodily senses through movement and coordination.

- It would be good to research the effects of activities such as the moving classroom, sense-based early music experiences, eurythmy and craft activities. Though such effects are difficult to observe, one can use qualitative methods to describe the activities and how teachers and pupils experience them. It is not always necessary to demonstrate through comparison that one approach is more effective than another; though this can be done, sometimes it is helpful to have accurate descriptions of lived experiences.

References

Aeplli, W. (2013). *The care and development of the human senses: Rudolf Steiner's work on the significance of the senses in education* (Rev. 4th ed.). Edinburgh: Floris Books.

Alheit, P. (2018). Biographical learning- within the new lifelong learning discourse. In K. Illeris (Ed.), *Contemporary theories of learning. Learning theorists . . . in their own voice* (2nd ed., pp. 153–165). London and New York: Routledge.

Antonovsky, A. (1987). *Unraveling the mystery: How people manage stress and stay well*. San Francisco, CA: Jossey-Bass.

Antonovsky, A. (1996). The salutogenic model as a theory to guide health promotion. *Health Promotion International*, *11*(1), 11–18.

Arendt, H. (1958). *The human condition*. Chicago and London: University of Chicago Press.

Auer, W.-M. (2007). *Sinnes-Welten. Die Sinne entwicklen. Wahrnehmung schulen. Mit Freude leben*. Munich: Kösel-Verlag.

Auer, W.-M. (2019). Entwicklung und Pädagogik der Sinne. In A. Wiehl & W.-M. Auer (Eds.), *Kindheit in der Waldorfpädagogik*. Weinheim Basel: Beltz Juventa.

Avison, K., Rawson, M., & Richter, T. (2014). *The tasks and content of the Steiner-Waldorf curriculum* (2nd ed.). Edinburgh: Floris Books.

Barnes, H., & Lyons, N. (1979). Education as an art. The Rudolf Steiner method. In E. Piening & N. Lyons (Eds.), *Educating as an art* (pp. 7–15). New York: Rudolf Steiner School Press.

Barz, H., & Randoll, D. (2007). *Absolventen von Waldorfschulen. Eine empirische Studie zu Bildung und Lebensgestaltung (Alumni/Gradutaes from Waldorf schools. An empirical study of education and life stories)*. Wiesbaden: VS Verlag für Sozialwissenschaften.

Berger, P., & Luckmann, T. (1966). *The social construction of reality. A treatise in the sociology of knowledge*. London: Allen Lane and Penguin.

Biesta, G. J. J. (2012). The future of teacher education. Evidence, competence or wisdom? *Research on Steiner Education*, *3*(1), 8–21.

Biesta, G. J. J. (2013). *The beautiful risk of education*. Boulder, CO: Paradigm Publishers.

Biesta, G. J. J. (2019). What is the educational task? Arousing desire for wanting to exist in the world in a grown-up way. *Pedagogia y Saberes*, *50*, 51–61.

Boland, N., & Rawson, M. (in press). Waldorf curricula and post-colonialism. *Research on Steiner Educaion*.

Bransby, K., & Rawson, M. (2020). *Waldorf education for the future: A framework for curriculum practice*. Retrieved from https://www.steinerwaldorf.org/steiner-resources/academic-articles/

Burnard, P., & White, J. (2008). Creativity and performativity: Counterpoints in British and Australian education. *British Educational Research Journal*, *34*(5), 667–682.

Butler, J. (1995). *Excitable speech: A politics of the performative*. New York: Routledge.

Cassirer, E. (1944/1962). *An essay on man*. New Haven, CT and London: Yale University Press.

Claxton, G. (2009). Cultivating positive learning dispositions In H. Daniels, H. Lauder, & J. Porter (Eds.), *Educational theories, cultures and learning* (pp. 177–187). Abingdon: Routledge.

Cole, R. (1992). *The spiritual life of children*. Boston, MA: Houghton Mifflin Company.

Damasio, A. (2012). *Self comes to mind: Constructing the conscious brain*. London and New York: Random House.
Dreier, O. (2008). *Psychotherapy in everyday life*. New York: Cambridge University Press.
Eisner, E. W. (1991). Forms of understanding and the future of education. *Educational Researcher*, *22*, 5–11.
Eisner, E. W. (1994). *The educational imagination: On the design and evaluation of school programs* (3rd ed.). New York: Macmillan.
Faulstich, P. (2013). *Menschliches Lernen. Eine kritisch-pragmatische Lerntheorie*. Bielefeld: Transcript Verlag.
Fischer-Lichte, E. (2008). *The transformative power of performance* (S. I. Jain, Trans.). Abingdon: Routledge.
Fuchs, T. (2017). *The ecology of the brain*. Oxford: Oxford University Press.
Gallagher, S., & Zahavi, D. (2012). *The phenomenological mind* (2nd ed.). London and New York: Routledge.
Gibson, J. J. (1979). *The ecological approach to visual perception*. Boston, MA: Houghton Mifflin Company.
Gidley, J. M. (2016). *Postformal education. A philosophy for complex futures*. Weinheim Basel: Springer.
Gidley, J. M. (2017). *The future. A very short history*. Oxford: Oxford University Press.
Glöckler, M. (2002). *Education as preventative medicine: A salutogenic approach* (Rev. ed.). Sacramento, CA: Rudolf Steiner College Press.
Goldsworthy, A. (1990). *Andy goldsworthy*. London: Cameron Books.
Göschel, J. C. (2012). *Der biographische Mythos als pädagogisches Leitbild: Transdisziplinäre Förderplanung auf Grundlage der Kinderkonferenz in der anthroposophischen Heilpädagogik (Biographical myth as pedagogical leading thought: Transdisciplinary learning support on the basis of the child study process in anthroposophical curative eduction)*. Dornach: Verlag am Goetheanum Athena.
Götte, W. M., Loebell, P., & Maurer, K.-M. (2016). *Entwicklungsaufgaben und Kompetenzen: Zum Bildungsplan der Waldorfschule (Developmental tasks and competencies: Towards an educational plan for Waldorfschools)* (2nd ed.). Stuttgart: Verlag Freies Geistesleben.
Grotlüschen, A. (2004). Expansive learning: Benefits and limitations of subject-scientific learning theory. *Vocational Training: European Journal*, *36*. Retrieved from http://www2.cedefop.europa.eu/etv/Upload/Information_resources/Bookshop/423/36_en_grotlueschen.pdf
Hattie, J. A. C. (2012). *Visible learning for teachers: Maximizing impact on learning*. London: Routledge.
Hattie, J. A. C., & Donoghue, G. M. (2018). A model of learning: Optimizing the effectiveness of learning. In K. Illeris (Ed.), *Contemporary theories of learning. Learning theorists . . . in their own words* (2nd ed., pp. 97–113). London and New Yoek: Routledge.
Hattie, J. A. C., & Zierer, K. (2018). *Visible learning. Auf dem Punkt gebracht*. Baltmannsweiler: Schneider Verlag Hohengehren.
Hay, D., & Nye, R. (2006). *The spirit of the child* (Rev. ed.). London and Philadelphia: Jessica Kingsley Publishers.
Hessayon, A., & Finnegan, D. (2016). *Varieties of seventeenth- and early eighteenth century English radicalism in context*. Abingdon: Routledge.
Hill, C. (1972). *The world turned upside down: Radical ideas during the English revolution*. London: Penguin.
Hurrelmann, K., & Quenzel, G. (2013). Lost in transition: Status insecurity and inconsistency as hallmarks of modern adolescence. *International Journal of Adolescence and Youth*. doi: 10.1080/02673843.2013.785440
Hüther, G. (2008). *Die Macht der inneren Bilder. Wie Visionen das Gehirn, den Menschen und die Welt verändern*. Göttingen: Vandenhoeck & Ruprecht.
Illeris, K. (2014). *Transformative learning and identity*. London and New York: Routledge.

Ingold, T. (2000). *The perception of the environment: Essays in livelihood, dwelling and skill*. London and New York: Routledge.

Ingold, T. (2011). *Being alive. Essays in movement, knowledge and description*. London and New York: Routledge.

Jarvis, P. (2018). Learning to be a person in society: Learning to be me. In K. Illeris (Ed.), *Contemporary theories of learning: Learning theorists . . . in their own voice* (pp. 15–28). London and New York: Routledge.

Jensen, B. B., Dür, W., & Buijs, G. (2017). The application of salutogenesis in schools. In M. B. Mittelmark, S. Sagy, M. Eriksson, G. F. Bauer, J. M. Pelikan, B. Lindström, & G. A. Espnes (Eds.), *The handbook of salutogenesis*. Cham: Springer.

Koch, T. (2010). *European portfolio certificate handbook*. Brussels: European Council for Waldorf Education and Comenius Lifelong Learning Programme.

König, K. (1999). *A living physiology: Lectures and essays*. Botton: Camphill Village Press.

Kvale, S., & Brinkman, S. (2008). *InterViews: Learning the craft of qualitative research interviewing* (2nd ed.). Thousand Oaks, CA: SAGE.

Largo, R. (2017). *Das Passende Leben. Was unsere Individualität ausmacht und wie wir sie leben können*. Frankfurt am Main: Fischer.

Levinas, E. (1991). *Totality and infinity*. Dordrecht, NL: Kluwer.

Levinas, E. (1998). *Entre-nous. On thinking-of-the-other*. New York: Columbia University Press.

Lindström, B., & Eriksson, M. (2011). From health education to healthy learning: Implementing salutogenesis in educational science. *Scandinavian Journal of Public Health, 39*(Suppl. 6), 85–92.

Lindström, B., & LIndström, M. (2006). Conceptualizing salutogenesis adn Antonovsky in public health development. *Health Promotion International*, *21*(238–244).

Loebell, P. (2000). *Lernen und Individualität: Elemente eines individualisierenden Unterrichts (Learning and individuality: Elements in individualised teaching)*. Weinheim Basel: Beltz Juventa.

Loebell, P. (2012a). Identität, Individualität udn Selbstgestaltung. In M. Basfeld & W. Hutter (Eds.), *Identitätsbildung im pädagogischen Prozess* (pp. 21–40). Baltmannsweiler: Schneider Verlag Hohengehren.

Loebell, P. (2012b). Spiritualität und (Waldorf-)Schulpädagogik. In P. Loebell & P. Buck (Eds.), *Spiritualität in Lebensbereichen der Pädagogik. Diskussionsbeiträge zur Bedeuting spiritueller Erfahrungen in den Lebenswelten von Kindern und Jugendlichen* (pp. 21–42). Opladen, Berlin and Toronto: Verlag Barbara Budrich.

Loebell, P. (2017, May). Lernen im Kontext der Waldorfpädagogik. *Lehrerrundbrief*, *102*, 31–45.

Lutzker, P. (2007). *The art of foreign language teaching: Improvisation and drama in teacher development and language learning*. Tübingen and Basel: Franke Verlag.

Lutzker, P. (2012). Poetry and illumination- aesthetic education in the presence of the spiritual. In P. Loebell & P. Buck (Eds.), *Spiritualität in Lebenbereichen der Pädagogik. Diskussionsbeiträge zur Bedeutung spiritueller Erfahrungen in den Lebenswelten von Kindern und Jugendlichen* (pp. 243–260). Oplade, Berlin and Toronto: Verlag Barbara Budrich.

Lutzker, P. (2017). *Der Sprachsinn: Sprachwahrnehmung als Sinnesvorgang* (2nd. ed.). Stuttgart: Verlag Freies Geistesleben.

Lutzker, P. (2018). Rudolf Steiners Hypothese eines Sprachsinns und ihre Bedeutung für den Fremdsprachenlehrerausbildung der Waldorfschulen. In H. Kern, T. Zdrazil, & W. M. Götte (Eds.), *Lehrerbildung für Waldorfschulen* (Vol. Beltz Juventa, pp. 249–273): Weinheim Basel: Beltz Juventa.

Marti, T. (2006). *Wie kann Schule die Gesundheit fördern? (How can school nurture health?)*. Stuttgart: Verlag Freies Geistesleben.

Meyer-Drawe, K. (2012). *Diskurse des Lernens (Discourses of learning)* (2nd ed.). Munich: Wilhelm Fink.

Mittelmark, M. B., Sagy, S., Eriksson, M., Bauer, G. F., Pelikan, J. M., Lindström, B., & Espnes, G. A. (2017). *The handbook of salutogenesis* Retrieved from www.ncbi.nlm.nih.gov/books/NBK435831/. doi:10.1007/978-3-319-04600-6

Pattanaik, D. (2010). *Jaya. An illustrated Retelling of the Mahabharata*. New Delhi: Penquin India.

Pratt, N., & Kelly, P. (2007). Mapping mathematical communities: Classrooms, research communities and masterclass hybrids. *Journal for the Learning of Mathematics*, *27*(2), 34–39.

Rawson, M. (2018). Life processes and learning in Waldorf pedagogy. *Research Bulletin for Waldorf Educaion*, *XV111*(2).

Rawson, M. (2019a). A complementary theory of learning in Waldorf pedagogical practice. *Research on Steiner Educaion*, *9*(2), 1–23.

Rawson, M. (2019b). The development of the person: Subjectification and possible links to non-formal learning situations and expansive learning in Waldorf (Steiner) education. *Other Education Journal*, *8*(2), 2–25.

Rawson, M. (2020a). De-colonizing your curriculum – Some tips for auditing. *Waldorf Resources*. Retrieved from http://www.waldorf-resources.org/single-view?tx_news_pi1%5Baction%5D=detail&tx_news_pi1%5Bcontroller%5D=News&tx_news_pi1%5Bnews%5D=8582&cHash=2dbc788a2edf780c3204815255941972

Rawson, M. (2020b). Waldorf education and postmodern spirituality. In U. Stoltz & A. Wiehl (Eds.), *Education-spirituality-creativity. Reflections on Waldorf education*. Wiesbaden: Springer.

Rawson, M. (2021). Spirituality and subjectivity in Waldorf (Steiner) education: A postmodern Bildung perspective. *International Journal of Children's Spirituality*, *26*(1/2).

Rawson, M., & Richter, T. (2000). *The edcational tasks and content of the Steiner Waldorf curriculum*. Forest Row, UK: Steiner Schools Fellowship Publications.

Röh, C.-P., & Zdrazil, T. (2016). *Themenheft klassenlehrer*. Dornach: Pädagogische Sektion am Goetheanum.

Rosa, H. (2019). *Resonance. A sociology of our relationship to the world* (J. C. Wagner, Trans.). Cambridge: Polity.

Rose, M. (1998). Towards a living grammar. *Paideia. A research journal for Waldorf Education*, *17*.

Ruitenberg, C. W. (2007). Discourse, theatrical performance, agency: The analytical force of 'performativity' in education. *Philosophy of Education*, 260–268.

Scharmer, O. (2009). *Theory U: Leading from the future as it emerges- the social theory of presencing*. San Francisco, CA: Berrett-Koehler.

Schieren, J. (2012a). The concept of learning in Waldorf education. *Research on Steiner Education*, *3*(1), 63–74.

Schieren, J. (2012b). The spiritual dimension of Waldorf education. *Research Bulletin for Waldorf Educaion*, *19*(1), 19–29.

Schmelzer, A., & Descepper, J. (2019). *Menschenkunde Verstehen. Vierzehn Essays zu Rudolf Steiners Voträgen zur "Allgemeinen Menschenkunde"*. Stuttgart: Edition Waldorf.

Sennett, R. (2008). *The craftsman*. London: Penguin Books.

Soesman, A. (2006). *The twelve senses: Wellsprings of the soul*. Stroud: Hawthorn Press.

Sommer, W. (2010). Upper school teaching at Steiner Waldorf schools: Cognitive challenges for the embodied self. *Research on Steiner Educaion*, *1*(1), 19–32.

Sommer, W. (2016). Zur Rolle der Allgemeinen Didaktik in der Waldorfpädagogik (Concerning the role of general didactics in Waldorf pedagogy). In J. Schieren (Ed.), *Handbuch: Waldorfpädagogik und Erziehungswissenschaft: Standortbestimmung und Entwicklungsperspektiven (A handbook of Waldorf pedagogy and the science of education: Positions and developmental perspectives)* (pp. 479–532). Weinheim Basel: Beltz Juventa.

Spitzer, M. (2007). *Lernen. Gehirnforschung und die Schule des Lebens (Learning. Brain research and the school of life)*. Heidelberg: Spektrum akademischer Verlag.

Stein, E. (1917/1989). *On the problem of empathy*. New York: Doubleday.

Steiner, R. (1994). *The social basis for education* (B. Masters, Ed.). Forest Row: Steiner Schools Fellowship.

Steiner, R. (1995). *Waldorf education and anthroposophy: Nine public lectures, Februray 23, 1921–16. September, 1922*. Hudson, NY: Anthroposophic Press.
Steiner, R. (1996). *The foundations of human experience*. Hudson, NY: Anthroposophic Press.
Steiner, R. (2003). *Soul economy. Body, soul and spirit in Waldorf education (lectures presented in Dornach 23.12.1921–5.1.1922*. Hudson, NY: Anthroposophic Press.
Steiner, R. (2004). *A modern art of education (lectures presented in Ilkley, Yorkshire August 5–17, 1923*) (J. Darrell, Trans.). Great Barrington, MA: Anthroposophic Press.
Steiner, R. (2007). *A modern art of education (Lectures in Illkley, 1923*). Hudson, NY: Anthroposophic Press.
Steiner, R. (2009). *The riddles of philosophy: Presented in an outline of its history*. Hudson, NY: SteinerBooks.
Steiner, R. (2011a). *Sozialethik: für Edith Maryon, persönliche Bemerkung zu "In Ausführungen der Dreigliederung des sozialen Organismus" 1919 Wahrspruchworte* (Vol. GA40). Dornach, CH: Rudolf Steiner Verlag.
Steiner, R. (2011b). *Theosophy: An introduction to the supersensible knowledge of the world and the destination of man* (M. Cotterell & A. P. Shepherd, Trans. Rev. 150th Anniversary ed.). Forest Row: Rudolf Steiner Press.
Stockmeyer, E. A. (2015/1965). *Rudolf Steiner's curriculum for Steiner-Waldorf schools. An attempt to summarize his indications* (R. Everett, Trans., 5th ed.). Edinburgh: Floris Books.
Tomasello, M. (2018). *Becoming human. A theory of ontogeny*. Cambridge, MA: The Belknap Press of Harvard University Press.
Van Norden, J., & Schürenberg, W. (Eds.). (2019). *Lernprogression narrativer Kompetenz im Geschichtsunterricht. Ein Vergleich von Waldorf- und Regelschulen*. Frankfurt-am-Main: Wochenschau Verlag.
Verhaeghe, P. (2015). *What about me? The struggle for identity in a market-based society*. Melbourne and London: Scribe Publications.
Whitehead, J. (2018). *Living theory methodologies*. Retrieved July 14, 2019, from www.actionresearch.net/writings/arsup/livingtheorymethodologies.pdf
Wiechert, C. (2012a). *Solving the riddle of the child*. Dornach, CH: Verlag am Goetheanum.
Wiechert, C. (2012b). *Teaching. The joy of profession. An invitation to enhance your (Waldorf) interest*. Dornach, CH: Verlag am Goetheanum.
Wiechert, C. (2012c). The art of observing children. In D. Mitchell (Ed.), *Observing the class. Observing the children* (pp. 13–16). Chatham, NY: AWSNA Publications.
Wiehl, A. (2015a, December). Individuation, Sinnerlenem und Erkenntnis. Prodädeutik der Waldorfpädagogischen Unterrichtsmethodik (Individuation, meaning making and knowledge: A propadeutic account of teaching methods in Waldorf education). *Research on Steiner Education*, *6*, 117–122.
Wiehl, A. (2015b). *Propädeutik der Unterrichtsmethoden in der Waldorfpädagogik (Propadeutic of teaching methods in Waldorf pedagogy)*. Frankfurt am Main: Peter Lang.
Wiehl, A. (2017). Das propädeutische Methodenkonzepz der Waldorfpädagogik. In J. Schieren (Ed.), *Enzyklopädie Erziehungswissenschaft Online* (Vol. Waldorfpädagogik). Weinheim Basel: Beltz Juventa.
Wiehl, A., & Zech, M. M. (2017). *Jugendpädagogik in der Waldorfschule. Studienbuch*. Kassel: Pädagogische Forschungsstelle.
Woodruff, C. C. (2018). Reflections of others and of self: The mirror neuron system's relationship to empathy. In L. Stevens & C. C. Woodruff (Eds.), *The neuroscience of empathy, compassion and self-compassion* (pp. 157–187). London, New York and San Diego: Academic Press.
Wright, A. (2000). *Spirituality and education*. London and New York: Routledge Falmer.
Wulf, C. (2014). Mimetisches lernen. In M. Göhlich, C. Wulf, & J. Zirfas (Eds.), *Pädagogische Theorien des Lernens* (pp. 91–101). Weinheim Basel: Beltz Juventa.
Wulf, C., & Zirfas, J. (2007). *Pädagogik des Performativen. Theorien. Methoden. Perspektiven*. Weinheim Basel: Beltz Juventa.

Zdrazil, T. (2000). *Gesundheitsförderung udn Waldorfpädagogik* (PhD Dissertation), Bielefeld. Retrieved from http://webdoc.gwdg.de/ebook/fk/2003/uni-bielefeld/disshabi/2002/0020.pdf

Zech, M. M. (2012). *Der Geschichtsunterricht an Waldorfschulen (History teaching in Waldorf schools)*. Frankfurt am Main: Peter Lang: Internationaler Verlag der Wissenschaft.

Zech, M. M. (2017). Geschichtsunterricht in der Pubertätsphase – Ankommen im und Ausgehen vom hier und jetzt. *Lehrerrundbrief*, *106*(May), 67–85.

Zech, M. M. (2018). Bildung und Waldorfpädagogik. In S. Sigler, W. Sommer, & M. M. Zech (Eds.), *Handbuch. Oberstufenunterricht an Waldorfschulen (Handbook of upper school teaching in waldorf schools)* (pp. 12–55). Weinheim Basel: Beltz Juventa.

Zech, M. M. (2019). "Seven-year periods" as heuristic tools or: Why Waldorf education works. *Waldorf Resources*. Retrieved from https://www.waldorf-resources.org/single-view/seven-year-periods-as-heuristic-tools-or-why-waldorf-education-works

Zimmermann, H. (1991). The foundations of Waldorf education. In N. Göbel (Ed.), *Waldorf education worldwide* (pp. 18–23). Berlin: Freunde der Erziehungskunst.

Part 3 Communities of learning

Generative principles

Summary: Part 3 continues with descriptions of the generative principles that relate to the context of learning, how the teaching is arranged, the role of the teachers' college and the question of teacher education. The following generative principles are discussed (the numbering is continued from Part 2):

14 The Waldorf class is a learning community
15 The teachers are a collegial learning community with responsibility for the educational leadership of the school
16 The curriculum maps out possible learning situations and learning pathways in relation to the developmental tasks
17 Becoming a Waldorf teacher is a process of transformative learning

Principle 14: The Waldorf class is a learning community

Generative principles:

- Children and young people learn best in a learning community, which provides a context for learning, identity, sense of belonging, growth of knowledge and meaning.
- Most issues of inclusion and differentiation are related to questions of participation.
- One of the primary tasks of the teacher is to ensure that all members of the class/learning community can participate to the fullest extent.

A Waldorf class is a classic example of a learning community, with many shared experiences, assumptions, expectations, ways of talking, its own narratives, rituals and ways of being and becoming. Etienne Wenger (1998), who has elaborated the notion of learning in communities of practice into a major and influential theory, weaves together the processes of learning, generating knowledge, identity and community within the construct of a community of practice or a learning community. Membership of a learning community involves both participation, which is largely unreflected because it is what one does, and reification, which means that individual members reflect on their roles and positions and those of the other members and reposition themselves. The process of reification means fixing an idea of the community and the roles one has, but this is continuously 'forgotten' in participation and thus undergoes a change, which only becomes apparent at a later moment of reflection.

The learning community is an ongoing, shifting constellation of members, with new members joining or existing members leaving, in both cases bringing about a realignment

of the whole structure. Simultaneous membership of several communities of practice is usual, and school students belong to families and other communities (sports teams, clubs and organizations like the Scouts). Where communities of practice overlap, there are often tensions at the boundaries, which can be sites of intensified learning (Wenger, 1998). Teachers are also members of the learning community of the class (and obviously other communities too). New teachers first have to adjust themselves to the class, and the class ultimately adjusts itself to each new teacher. Wenger's construct offers Waldorf teachers a very important heuristic for analysing pedagogical situations.

Each class, and each school, has its learning culture. A learning culture comprises the actions, interactions, talk, dispositions, expectations and interpretations of all the participants in that culture. It is not simply the location and environment in which learning occurs. As Lave and Wenger argue (1991, p. 35), learning is "not merely situated *in* practice . . . it is an integral part of generative social practice in the lived-in-world" (1991, p. 35). Social practices are not simply reproduced; they are brought about by people learning, though some cultures may resist change. Because learning cultures themselves are culturally situated, expectations and norms from outside the school may influence the way people in the learning culture may think, feel and act in ways that are more persuasive than either unconsidered values or espoused internal ideals. The learning culture in a Waldorf school may, for example, conflict with societal expectations or with traditional school cultures.

Barbara Rogoff (1995) is another scholar who has explored how children learn in learning communities. She describes the process of learning using the metaphor of apprenticeship, though not with the classic understanding of apprenticeship with a master but rather in which the community of practice, including the experienced members (i.e. teachers), is 'master'. A Waldorf class, over 12 years, actually passes through different phases of apprenticeship. In the lower school, the class teacher is the 'master' instructor and role model; in the middle school. this role is supplemented by more specialist subject teachers (gardening, woodwork, music, sport, etc.) in their own 'workshops', thus resembling more a traditional apprenticeship; and in the high school, the students learn with subject specialists in their own communities of practice (scientists, artists, craftspeople, academics).

Rogoff (1995) identified three levels or modes of learning through participation as apprentice. The first is cultural apprenticeship, which occurs through observing and joining in. The second involves guided participation, which involves scaffolding and support. The third is participatory appropriation, which refers to how individuals change once they can participate fully in the practice. Rogoff describes this as a process of becoming rather than of acquisition. This perspective on learning helps us to focus less on the outcome of learning and more on the process and what it means for the development of the person. Cultural apprenticeship is being part of a class community within a school culture. Guided appropriation is what teachers do much of the time when they give pupils tasks, and participatory appropriation covers many forms of self-learning both in the classroom and outside it.

Rogoff (2014) also identifies seven criteria for effective learning in learning communities. These, somewhat modified to apply to a Waldorf setting, are:

1 The learners are incorporated in and contribute to community endeavours
2 The learners are eager to contribute, collaborate and belong and each member is willing to help others

3 The activities are organized so that all can participate
4 The goal of the learning situation is participation so that all can contribute, allowing others to participate, taking responsibility and contributing to belonging
5 Learning occurs through keen attention to and contribution to events, guidance is provided across the community and there are expectations that people will learn in the learning culture
6 Communication is based on shared reference in collective activities through verbal and nonverbal communication, narratives and dramatizations are used to convey values, and explanation is always in the context of the shared activities or in anticipation of such activities
7 Assessment for learning is used in relation to the collective endeavor

We can translate these conditions into the situation in a Waldorf classroom, in which there are activities with explicitly and implicitly shared assumptions, expectations and values with a shared and evolving language and often enhanced through ritual. An atmosphere of collaboration and inclusive participation is encouraged. Participation is expected, enabled and valued. Teachers offer guidance, but as can be observed in any classroom, children frequently help each other. Formative feedback is given to encourage and enable better participation. Whilst this form of learning is most apparent in younger classes, it remains a core aspect of learning throughout the school. The question of inclusion has to be seen in this context, the question being, what do we have to do to enable this person with her special needs to participate in our learning community? The question of differentiating tasks relating to a shared experience has been mentioned earlier.

The ideal, frequently realized, in a Waldorf school is that the class community stays together from grade 1 to when they leave school, usually after grade 12 (sometimes grade 13). Few teachers stay with the class that long, though some foreign language and eurythmy teachers may manage this feat. The class community is experienced by students as an ongoing profound field of learning. It is different from being in a family, though relationships are formed that may sustain years after school, and it is different from the workplace, though it has some overlaps. Parents sometimes ask what happens if their child is unhappy in her class or with her class teacher, given that the relationship is going to be over the long term. The answer is that of course, this can and does occasionally happen. Working through such crises can be a valuable learning experience for all concerned. The onus is on the college of teachers, as the responsible body, to ensure that the teachers involved are aware of this, and when the children have reached the age of about 11 or 12 and can also begin to take active responsibility for social processes in the class (often through class representatives elected by the other pupils), a process is begun with the aim of recognizing and ameliorating the problem. The point about such problems is that if handled intelligently by the pupils and/or teachers, significant learning opportunities can be had by all concerned.

Thus, one can see that the class community, which is the basic organizational unit within a Waldorf school culture, has a pedagogical function of its own.

Teacher skills

The teacher can:

- Deliberately contribute to the formation of a learning community by creating an atmosphere of collaboration, inclusion, respect for others and willingness to help each other

- Create a narrative for the class that establishes a sense of belonging to a learning community
- Organize learning activities in which all pupils can participate at their own level
- Offer pupils appropriate support to enable participation and inclusion
- Assess to what extent all learners are included

Research questions:

- Does this theoretical perspective help understand processes in Waldorf classes between pupils, between pupils and teachers and in the development of both pupils and teachers?
- Does this model help to understand the processes of changing teachers and new teachers entering the learning community?
- Does this model help us to research questions of inclusion?

Principle 15: Teachers support their pupils' learning and development by generating active knowledge using assessment for learning

Generative principles:

- Assessment for learning is a vital support for pupils' learning and development.
- The learning being assessed takes the whole person into account, in ways described in this book, and includes assessment of socialization processes, qualification outcomes and appreciation of the development of the person.
- Assessment evaluates the things that Waldorf education values and is comprehensive.
- Waldorf practice uses formative, ipsative and summative assessment for different purposes.
- Because learning should be experiential and social, performance assessments are usually appropriate (e.g. naturally occurring evidence).
- Assessment gives teachers important feedback on their teaching as part of their planning and reviewing process.
- Assessment of pupils should be supportive, specific and helpful.
- Assessment should be effective, unobtrusive, embedded in classroom practice and unbureaucratic, yet also well documented.

Assessment for learning

Waldorf education recognizes that assessment for learning, or learning-centred assessment, is a vital support for learning and development and to enable pedagogical quality development and high levels of teaching effectiveness. The phrase *assessment for learning* interprets learning as defined in the section on learning and takes the whole human being into account. It is not just about measuring the delivery of cognitive knowledge and subject-based knowledge and skills but has to do with making judgements about pupils' learning processes and overall development and appreciating their development as persons, their achievements and interests. It is also about recognizing emergent qualities in the person.

Knowing how pupils are learning, what difficulties they may have and what they can do well enables teachers to offer the right kind of support at the right time and provides teachers with essential information about the effects of their teaching. Therefore, formative modes of assessment are used to accompany learning processes and

to give individual pupils ongoing specific feedback. Feedback is a powerful tool for supporting learning (Hattie & Clarke, 2019). It should be kind, specific and helpful. It is important that pupils understand in age-sensitive and language-appropriate ways, which may be implicit (i.e. reinforced by teacher behavior and classroom practice) or explicit (i.e. cognitive), what the learning aims are and what criteria will be applied to assess a particular assignment or a whole learning block (what Americans call rubrics), how to achieve those aims and how they are currently doing. My experience is that simply informing pupils cognitively (e.g. a text on the blackboard, or verbally) what the criteria for success are often does not help them to grasp what is important. And we also wish to avoid merely learning for the test ("learn this because this is what you will be tested on"). Waldorf education works with the understanding that young children assume that what their teachers teach is important. Teachers also prepare their lessons in such a way that they understand what is important. Helping children understand what is important *is pedagogy* and takes many forms, including the richness of the child's classroom experiences, the nature of the presentation, summarizing key points, reviewing and recalling – each retrieval reinforcing and enriching knowing. Older pupils can be provided with criteria in writing so they can refer to it, but this applies to teaching at all levels. This is done according to individual school practices. Teachers use informal and formal formative assessment in their ongoing lesson reviewing and planning processes.

Ipsative methods are used in one-to-one formal and informal pedagogical conversations, in which the learner is helped to recognize her own progress measured against her own previous achievements and levels of participation. Thus, the pupil has the feeling that she is not competing with others or external standards but is trying to do better or simply maintaining her own high standards. Children and young people undergo all manner of developmental crises prompted from outside or through inner changes, and this often impacts on school learning. They need to be helped to see what the symptoms mean and what the causes are so they can recognize the effect they are having on them, rather than having the feeling of "I am too stupid" or "it's the teacher's fault that I don't get it". Then they can find ways of moving on with the help of the teachers.

Waldorf schools practice individual case studies in which a group of teachers who teach a particular pupil meet, share their experiences of the pupil, look at her work across all subjects and activities, discuss her situation in school and take into account her overall development throughout childhood (as much as is known). The parents are involved and can contribute their perspective and when the pupils are old enough, they too can contribute their own views in age-sensitive ways (though they usually don't participate in the case study itself). The teachers try to form a picture of the pupil's learning and developmental pathway, how it has been and how it is now in the current situation, and they try to sense the nature and character of the emergent person and her biographical intentions. The purpose of such case studies is to grasp the 'meaning' or signature of the pupil's biographic trajectory through her life course so the teachers can understand her situation better. Of course, the process may also lead to possible support and interventions. It is also a significant pedagogical learning process for the teachers involved.

Summative methods are used at the end of learning blocks and in year-end assessments to show performance of groups or individuals over time and for purposes of quality assurance.

The emphasis is on assessment being useful for pupils and teachers, which means that it has to be uncomplicated, unbureaucratic, intermeshed with practice and effective and not a burden for all concerned.

Modes of assessment include observation and monitoring of performance or activities; projects; group work using checklists with criteria, including naturally occurring evidence; feedback on main lesson books, presentations and portfolios (in which pupils select the work they want assessed); end of written block tests; oral exams (often in small groups); and self-evaluation (see Rawson, 2015).

Values

Assessment and evaluation are expressions of the values we hold and enable us to identify what we value and make judgements about this. This means formulating what we value, finding appropriate indicators for this, observing closely and interpreting intelligently and insightfully. Judgements about quality are always related to purpose – why we are teaching what we teach – and that in turn reflects the functions of education that we hold as an educational philosophy. Each school has to find the right balance, so that the domains of socialization (being), qualification (skills and knowledge) and becoming a subject (dispositions, motivation, interests) are respected and do not crowd each other out. Therefore, Waldorf schools formulate their own values as well as recognizing external evaluation criteria.

Typical values that manifest in learning situations and can be assessed include:

1 Constructive participation in the social practices of the class and allowing and enabling others to participate
2 Ability to work with others
3 Listening when others (and not just teachers) speak
4 Being able to articulate one's thoughts, feelings and intentions in ways that others can understand
5 Understanding of the topics being taught as expressed in words, text and images
6 Application of relevant subject-related skills
7 Ability to learn from mistakes and improve their work
8 Ability to work independently and with others in projects
9 Self-assessment and setting own goals
10 Aesthetic presentation of work
11 Motivation as manifest in behavior

In forms of formal assessment, these can be grouped into fewer items.

Self-assessment

Pupils are encouraged to assess their own activities in age-sensitive ways. Younger children are invited to express their feelings and thoughts about their activities. Older pupils can participate in formulating appropriate criteria (what makes a good presentation?) and self-assess. Students in the upper school can apply the same criteria used by the teachers.

Individual annual reports

Each year, the teachers write a brief summary of what was taught in the year and an appreciative evaluation of each child's performance, participation and progress in text form, characterizing the child's strengths and weaknesses. Typical items referred to include participation in classroom practices, progress across the subjects, particular achievements, social behavior and personal development. The report is both summative, in that it summarizes the pupil's attainments in relation to normative expectations, and ipsative, in that it addresses the individual in relation to her overall development in comparison to the previous year. Comparing annual reports should give a picture of the overall development of the person. It also offers practical advice for the coming year, perhaps in one area that the pupil herself can tackle.

Lesson planning and reviewing as assessment for learning

Good practice in Waldorf schools involves teachers drawing up, actioning and assessing:

- Annual class plans: Making an annual plan for each class covering all subjects (class teacher and subjects share this in class planning meetings), also to ensure synergies and avoid overlaps.
- Block plans: Each main lesson block (four weeks) or block of subject lesson is planned and outlined, showing the main contents, skills framework, new knowledge (retrieval and revision of embodied knowledge), capacities framework and dispositions being practiced (these are long-term).
- Lesson plans: Outlines structure of the lesson, sequence of activities, content, tasks and activities.

These plans are all reviewed, and outcomes are briefly documented (e.g. including some copies/photographs of examples of student work from weak, typical and very strong students showing the span). From grade 6 (age 12) onwards, pupils are given a brief text description of the content of the forthcoming block and the assessment criteria, showing possible levels of outcome. At the end of the block, each student receives a brief summary of attainment based on the criteria that the pupils were given at the beginning and feedback about the next learning step.

Principle 16: The teachers are a collegial learning community with responsibility for the educational leadership of the school

Generative principles:

- The teachers are collegially responsible for the educational quality and development.
- The primary function of the college of teachers is to be an ongoing 'university' for the development, evaluation and research of the education and the continuing professional development of the teachers.
- Whatever leadership and management forms are chosen, they have the core function of sustaining the integrity of the school (without which the education has no

> meaningful application) and enabling and optimizing the learning and development of each individual, starting with the pupils but including the teachers and other staff, and even the parents – though the pupils clearly take priority.

Steiner's (1977) model of social renewal saw the functioning principle of education, as part of the cultural domain, as freedom. The same applies to the arts and cultural life generally, and to science and religion. These activities should be free of all determination by the state or the economy. Only those who practice it and take responsibility for it should determine what is taught and how schools should be run. As a result, the teachers in a school should have the full responsibility for the curriculum and pedagogy, appointment of teaching staff and admission of children. They do this in a Waldorf school by constructing and maintaining a common vision based on a common understanding of the generative principles.

At the founding of the Waldorf School, Steiner made it clear that the school should be run as a "true teacher republic" (1996, p. 30), which in the historical context of 1919 in Germany meant run by the workers, with each individual fully responsible for the whole. There should be no school director or external authority specifying what should be taught, when and how (including, we can add today, Waldorf authorities and official curricula). Readers who want to gain an overview of everything Steiner said about school leadership and management should read Gladstone (1997), who compiled and collated all the relevant references.

In 1923 in Ilkley, England, Steiner explained what this meant in practice:

> The meetings are really a living university for the college of teachers – a permanent training academy, as it were. They are so indeed, and for the reason that every practical experience gained by the teacher in school becomes, in turn, part of his (sic) own education. And he who derives such self-education for himself from his teaching work, gaining on the one hand a profound psychological insight into the practical side of education and on the other side into the different qualities, characters and temperaments of the children, will always be finding something new, for himself and for the whole college of teachers. All the experience acquired from the teaching should be 'put into the pool' at these meetings. In this way the College of Teachers in spirit and soul becomes a whole where each member knows what the other is doing, what experience has taught him and what progress he has made as the result of his work with the children in the classroom. The college of teachers becomes, in effect, a central organ whence the whole life-blood of the practical teaching flows and helps the teacher to maintain his freshness and vitality. Perhaps the best effect of all is that these meetings enable the teachers to maintain their inner vitality instead of growing old in soul and spirit.
>
> (Steiner, 2007, p. 208)

In modern terms, this means that the teachers are to practice illuminative practitioner research (Rawson, 2018). The school leadership in a Waldorf school rests with the college of teachers, which is a non-hierarchical collective of colleagues or peers. The school leadership (see Bush, 2011, for a distinction between leadership and management) is responsible for the vision of the school, its policies and the quality of the education. The management – that is, the implementation of policy and school administration – can be organized in whatever way the leadership sees

appropriate, but the education clearly remains in the hands of the educators. In practice, both leadership and management can be and are regularly delegated to individuals and small groups by the college of teachers within procedures agreed to and regulated by the teachers' body. This form of collegial, non-hierarchical leadership is by no means easy but is worth striving for, as it contributes significantly to the school culture. This is of great value as an example of counter-practice in an age of hierarchical, standards-driven managerialism and the application of economic principles to education, which ultimately is not a commodity or service that can be bought and sold but is a basic human right and has the core function of enabling the education of the person as individual, whatever else society demands in terms of qualification.

Parents of course have a right to participate in the community life of a Waldorf school and may even take positions of responsibility, and in most schools this is in fact the case. This needs to be done in such a way that it does not restrict the primary principle that teachers are primarily responsible for the pedagogy.

There is much more that could be written about leadership and management in relation to Steiner's intentions, the historical development of schools and their highly varied practice internationally. It is a highly contested areas of Waldorf education, though I think the generative principles noted earlier would find common ground across all versions of Waldorf practice. How this principle is applied in specific places at certain times is bound to vary. As in most things Waldorf, there are those who think the way things are done in their school is the historically correct way. As I have pointed out more than once, the local is often mistaken for the general, especially when it claims proximity to the true path.

What characterizes the quality of leadership in a Waldorf school, and was clearly intended by Steiner, is the commitment to a spiritual understanding of its role. Steiner specifically located the professional ethic of leadership in spiritual context, offering an esoteric image for how this can be realized. Steiner referred to the need for each teacher to undertake the responsibility for the education and that in order to be able to do this, each individual needs to be active in her spiritual development, cultivating what could be called her higher self. All forms of personal ambition, egotism or self-interest are destructive of the basis on which collegiality is based. This is the precondition for recognizing the spiritual intentions in one's colleagues. Trusting the other calls for a perception of the other as a person with a spiritual core. Collegiality is based on genuine mutual recognition and respect. Only then can the whole be more than its parts and metaphorically be an organ for what is emerging as positive potential for the school.

Glenys Woods and Philip Woods (2008) have specifically linked holistic or developmental democracy and spiritual awareness as aims of school leadership and identified these qualities in their study of Steiner (Waldorf) schools in the UK. They identify six aims of developmental democracy in creating

> an environment which enables the capacity for human potential to be realized (for intellectual reasoning, aesthetic sensibilities and spiritual awareness). The primary point of democratic leadership is to strive towards a way of living – in and through relationships – which is orientated towards cultivating the fullest development of this human potential.
>
> (2008, p. 102)

The other aims of democratic and spiritual leadership that they list are ethical rationality, discursive rationality, decisional rationality, therapeutic rationality and social justice.

A key to the Woods and Woods understanding of spiritual experience includes a sensing of transcendent power, which manifests as inner strength, sense of purpose and a sense of what is real. Noetical quality refers to the experience of knowing that is not based on the intellect alone. It is a self-evident experience of reality. An ethical quality of experience is related to conscience, the sense of doing good, being honest and being altruistic. Spirituality is experienced as life enhancing and affirmative. Together these experiences can "steer navigational feelings and has the potential to orientate social action" (2008, p. 104), particularly in the field of school leadership but also, obviously, in pedagogical action. Woods and Woods prefer a trialectic view of social dynamics, meaning a threefold relationship between structure, person and agency, in which the school culture and organizational structures and values provide a context for the person to act with others out of spiritual insight. "Democratic agency involves profound participation, which is authentic involvement guided by spiritual awareness and shared symbols of democratic ideas and ideals" (2008, p. 104). In some ways Woods and Woods, as outsiders, have identified more clearly than many insiders what school leadership in a Waldorf school is about.

Apprehending or presencing the future as it emerges, as Otto Scharmer (2009) puts it in his analysis of organizational development, is the central task of leadership in a Waldorf school. The future has not yet manifest; it is always in a state of becoming. Neither is it predetermined. What happens and what can happen is also shaped by who we are and whether we recognize the opportunities afforded by life, and whether we are capable of taking these opportunities makes a huge difference. If we leave that to chance, there are many other ways that things can go. By collectively opening ourselves up to the future, by ensuring we are capable and willing to grasp what comes towards us, we can steer it in ways that further rather than hinder development. What hinders development? What prevents presencing? Mostly we do, because of our lack of vision and what Buddhists call grasping and aversion – hence the importance of self-development. Scharmer's analysis of what gets in the way (see his chapter on *The grammar of the social field*) highlights the problems of old and limiting patterns of judgement, the voice of cynicism and the voice of fear, all of which can lead to "absencing" – that is, shutting down our "capacity to relate to the future that wants to emerge" (2009, p. 248), involving not seeing, de-sencing, absencing and destroying.

In order for teachers to take responsibility for the leadership of their school, they need to base this on evidence provided by reviewing, reflection and practitioner research and continuing professional development. Some studies (Rawson, 2010, 2014) have shown that most Waldorf teachers feel ill equipped to research their practice, partly because they have not learned how during their initial teacher education and partly because the demands on them in school are such that they barely have time to deal with ongoing issues, let alone carry out research. These studies also make the case for incorporating contemplative methods in practitioner research. The way forward, I believe, is in partnerships between teachers, schools and universities. In Germany, the Freie Hochschule Stuttgart currently has a research project establishing just this – a partnership between schools and researchers to explore using the methods of practitioner research and issues in schools and to introduce and monitor innovative projects. Part of the project involves student teachers, which is a win–win situation for all involved.

Teacher skills

The teacher can:

- Take responsibility for their own actions and for the whole school
- Have the social skills necessary to contribute to collegial working
- Develop the leadership skills necessary to contribute to running a school
- Learn to be a critical and reflective practitioner and practitioner researcher
- Have the inner strength to overcome their egotism to be able to serve the education and the pupils in creative and responsible ways
- Develop the powers of presencing
- Learn and use reflection and practitioner research to evaluate and develop their practice

Research questions:

- Using models such as that of Woods and Woods (2008) for holistic school leadership, the lived experiences of teachers in actual collegial processes can be explored.
- The sheer diversity of leadership and management models internationally could be described and analysed.
- Not only can practice be researched using methods of small-scale case studies or action research, but the impact of such research in terms of knowledge generation and enskillment can be evaluated.
- The use of contemplative methods can be researched to build on the few studies that exist.

Principle 17: The curriculum maps out possible learning situations and learning pathways in relation to the developmental tasks

Generative principles:

- In Waldorf education, curriculum refers to the entire educational approach that influences the learning and development of the children and students, including what is taught, how, when and why it is taught, and the practices these are embedded in.
- Curriculum is the answer to the question as to how pedagogy can respond to the developmental tasks.
- The primary unit of pedagogy in a Waldorf school is the class of children and young people of the same age (defined by the school year).
- Everything taught in a given school year is linked by certain guiding ideas based on the developmental tasks facing the students in a given school year. This is known as the horizontal curriculum.
- Subjects (e.g. maths, geography, music, etc.) follow a vertical development from grade 1 to grade 12 (known as the vertical curriculum).
- Certain key concepts (e.g. wholeness, interrelatedness, principles of form starting with straight lines and curves, transformation, emancipation, etc.) grow as living concepts that are revisited and expanded throughout the curriculum. This is the spiral curriculum.
- There is no definitive or canonical Waldorf; curriculum is emergent.

The primary aims of curriculum in Waldorf schools are:

- Fostering healthy development in each child
- Enabling children to realize their potential
- Helping children to develop the skills they need to contribute to society (Rawson & Richter, 2000, p. 7)

In a more recent presentation, the aims are described as seeking to educate the whole person in the three domains of educational purpose described by Biesta (2013) by enabling:

- Qualification: how students acquire the dispositions, skills, knowledge and cultural capital to participate in and contribute to the world of work and civil society
- Socialization: how students learn the dispositions, skills, knowledge and abilities to actively participate in their community and within a multicultural society and be able to establish and maintain coherent identities
- Individuation: how students develop agency, judgement, sense of coherence, a feeling for responsibility, ethical dispositions, the ability to orientate their lives, be creative and develop a relationship to spirituality (Bransby & Rawson, 2020)

Biesta refers to becoming a subject and in the Waldorf discourse, this is usually referred to as individuation or the development of the person.

Though it is one of the most distinctive features of Waldorf education, the Waldorf curriculum is in fact a highly diffuse object of research (Zech, 2016). It is considered by some practitioners to be a complex, multilayered work of art that brings the developmental nature of the human being to expression and that cannot be altered in anything more than superficial ways. However, it is also this aspect that highlights the paradox between the need to adapt curriculum to changing circumstances and different cultural contexts and yet offering a consistent, coherent and normative model. The definite article and capital letters in the phrase *The Waldorf Curriculum* highlight the problem. Is there a single, definitive, official and authentic Waldorf curriculum or are there Waldorf curricula, plural? Is *Waldorf* an attribute that can be given to different curricula? Can a curriculum be Waldorf-inspired but not exclusively follow a recognized Waldorf curriculum? This is an issue for public, charter or academy Waldorf schools that are usually required to fulfil state requirements in the curriculum.

First we need to define what the term curriculum refers to because in its Waldorf usage, it is a translation of the German *Lehrplan* and the meanings of both terms are not synonymous (see Horlacher, 2018, Boland and Rawson, in press). In short, I follow Dahlin's (2017) characterization of curriculum in Waldorf education: this includes:

> everything that children or students in a Waldorf school or preschool may experience or are supposed to experience, consciously or subconsciously, during their school day. Thus, not only the contents of teaching and learning, but also the way

> the teachers teach, and the teachers themselves as persons are included in "the curriculum". Even the aesthetics of the internal and external architecture of the school house is part of it, because in an ideal Waldorf school, this aspect of the external environment is consciously designed to support children's development. All aspects of the curriculum are related to potentials for the learning that children have in different ages and phases of development.
>
> (Dahlin, 2017, pp. 87–88)

This comprehensive understanding of curriculum shows that it is more than just content and outcomes. Furthermore, the function of the curriculum is to provide a sequence of learning experiences that prompt development in specific ways at specific times – a model that Waldorf education deems to be healthy in the sense of fostering sense of coherence. Curriculum structured in heterogeneous year groups (the Waldorf class) can lead to an alignment of developmental processes across the group.

Götte, Loebell and Maurer's (2016) book on learning outcomes (full title translated: *Developmental tasks and competences: towards the educational plan of Waldorf schools*) introduced a new concept of curriculum: developmental tasks. The construct has been adopted across a number of disciplines to "characterize the conflict between the physical, mental and ecological demands, with which individuals are confronted during various stages of life" (Hurrelmann & Quenzel, 2015, p. 261). Götte et al. (2016) define developmental tasks from a Waldorf perspective in terms of the dynamic between the child's individual potential and the educational and social requirements of socialization and qualification.

Bransby and Rawson (2020) redefine the link between developmental tasks, curriculum and institutional transitions. They suggest that curriculum takes the following factors into account:

- The generative principles of Waldorf education based as they are on the Foundational understanding of the nature of developing human being
- Generalized maturational pathways that take certain key developmental processes into account (e.g. trajectories of physical, emotional, social and cognitive development)
- the local social and statutory expectations regarding skills and knowledge
- individual learning needs

This leads to a layered curriculum comprising;

- a macro level of general and ideal-typical developmental themes that probably apply in any Waldorf school
- a meso level of idea-typical topics and skills and knowledge progressions that take local conditions, culture and requirements into account
- a micro level of actual teaching and learning plans.

This approach provides a framework to include some aspects of purist and traditional adaptionist stances to curriculum (Oberman, 1997); the 'universality' of the curriculum is retained at the maco-level, whilst the meso level offers scope for cultural adaptation. However it does challenge some traditional readings of Waldorf curricula that tend to suggest a close match between the content in each class and the developmental stage of the pupils at that age, because

> from the perspective of the heterogeneous developmental situation of the individuals in a [typical Waldorf] class, we cannot assume that age-specific stages are the norm. Rather they are set by the teaching. General developmental phases such as seven-year stages or changes around the age of 10 or 12 that can be observed, do not simply occur, but are brought about by the pedagogical approach and orientate themselves on the one hand on ideal-types, and on the other, on the actual situation of the pupils.
>
> (Zech, 2017, p. 70)

Studies such as those reported by Largo (2017) highlight that developmental variation is the norm, and therefore it makes no sense to refer to typical grade 3 or grade 9 pupil. However, using such ideal types enables teachers to compare the reality of a particular developmental situation in a class, or indeed of individual pupils, with an ideal type, which is not a statistical norm but a construct deemed to embrace most relevant factors. This means that the ideal type is not prescriptive, it is not what has to be taught, but rather provides an orientation for the teacher. Thus, the curriculum and pedagogy in a Waldorf school uses the cultural construct of curriculum as an orientation that take the development tasks into account. This leads to a common content for the whole class, though the individual learning tasks within this are differentiated.

Clearly, novice teachers should adopt the practices within their school, but more experienced teachers can modify their approach. This assumes that teachers can judge when the traditional ideal type needs to be modified and that this is based on some kind of research on practice. This is, of course, not easy because teachers who are used to teaching a particular curriculum content are likely to be convinced that this is developmentally right. Nevertheless, in a culture of critical reflection and discourse, and particularly when new teachers enter practice, questions will be asked about curriculum relevance. Thus, the ideal types can be modified over time and cultural space.

Traditionally, Waldorf curricula have not described specific learning outcomes. Under pressure to be comparable with state school curricula, Rawson and Richter (2000) described minimum learning outcomes for maths, English and second language, and in Germany, Götte, Loebell, and Maurer (2016) used a competence model (with subject, methodological, social and self competences) to describe typical outcomes. Bransby and Rawson (2020) have described a school-based learning process in which a range of potentialities (modes of relating to self and world) transform into skills, defined as knowledgeable action with purpose, which can become more general abilities and later, life-long capacities. These processes can and should monitored so that learning can be supported. The interdisciplinary nature of the thematic block-teaching approach means that developing skills (e.g. in literacy) need to be tracked across subjects.

The historical perspective is helpful in understanding what curriculum means in Waldorf discourse. The first compilation of the school's curriculum practice was published by one of the founder teachers, Caroline von Heydebrand, in 1925, written shortly after Steiner's death with a dedication to the school's founder, and published in October. This appeared initially as a special edition of the newsletter of the Waldorf School Association (*Mitteilungsblatt Waldorfschulverein Sonderheft*, October 1925) and had 40 pages, including a note to the members that the booklet should not be made accessible to the children. In 2016 the German Richter curriculum had over 600 pages, by 2019, it had over 900. Heydebrand's booklet was the main source of curriculum for many years. In the introduction, she briefly explains Steiner's method in relation to curriculum. This was to present his understanding of the nature of the developing human

being from the perspective of his anthroposophical anthropology, on the basis of which the methodological and didactic details necessary for practicing of the art of education could be developed. At the end of this process, there are some suggestions for the actual teaching content. She explains that on the basis of this, each teacher was expected to develop and extend the curriculum out of their own insight and experience. She wrote,

> what is presented here as the sequence of teaching material for the various classes should not be taken dogmatically as a rigid law. The ideal curriculum must follow the changing picture of human nature in the different age-groups, but like every ideal it must deal with the reality of life and adapt itself to this.
>
> (von Heydebrand, 1925, p. 3)

Since 1921 Steiner had intended to write down a more formal curriculum, since this was required by the state authorities (Zdrazil, 2019). Indeed, when the first school inspector, Herr Dr. Eisele, visited the school in early 1922, he complained about the lack of a curriculum in which he could recognize what the learning goals for each class were (Zdrazil, 2019, p. 479). The teachers had given him a sketch outlining what had been done in each class, but this told him little about the goals. He felt it was inappropriate to judge the Waldorf School by the standards and expectations of the state schools, a problem that remained with Waldorf curricula until the beginning of the 21st century.

Von Heydebrand structured her account of the curriculum of the Waldorf School with a short characterization of "the being of each individual developmental stage". This started with the age span of 6 to 9 years (age 6 or 7 being the age of school entry). This was then followed by a brief summary of what was taught in each subject in each class. The curriculum starts with painting and drawing in grade 1, followed by writing, reading, speech, local surroundings, arithmetic, English and French, eurythmy, music and handwork, and so on, up to grade 12 (age 18). The last entry was technology in grade 12. Jennifer Militzer-Kopperl (personal communication) has offered the view that what von Heydebrand collected in her *Lehrplan* was actually curriculum-as-taught in the Stuttgart Waldorf School, in other words, micro-level curriculum, rather than Steiner's original macro level ideas and the meso-level examples he suggested for local use. However, this micro-level *Lehrplan* has been taken as the blueprint ever since. This interesting perspective has been made possible by Bransby and Rawson's (2020) new layered framework for curriculum analysis and development. A version of this booklet was translated by the English Waldorf teacher Eileen Hutchins and published in 1966 with additional notes on the teaching of English in England, and this was later reworked by Roland Everett, a Waldorf teacher, and published in a printed edition in 1989. Stockmeyer (2015) later collated all the mainly meso-level curriculum references from Steiner's discussions with teachers and the lectures in a single volume organized by subject. This was also published in German in 1951 and later in English in 1966.

These two sources were the basis for curriculum in Waldorf schools up until the 1990s, along with secondary literature in which teachers outlined the curriculum in their subject field. Such presentations elaborated on the core curriculum, though they often described the practice tradition in their own schools. There was little debate about curriculum issues except in science teaching, where there were controversies in the German literature about the exact way in which the phenomenological approach should be practiced. For most subjects, a few books established the canon. Frequently, only one of these was translated, thus narrowing the canon even further. Some of these translated texts became international standards, though they in fact represented

a local tradition that was often soon outdated. In many ways, it is fortunate that Waldorf teachers are generally resourceful and creative and generate their own material.

Following the collapse of the Soviet Union and the reunification of Germany in 1989, Waldorf schools were founded across the former Eastern Bloc countries, many of which were able to gain state recognition and funding during a period of liberal and progressive governments. This also led to a demand for more formal curricula showing the content, aims and outcomes of Waldorf education in ways that could be compared to state curricula. Tobias Richter in Vienna produced a curriculum for Austria, which was taken up by the International Conference (at that time known as the Hague Circle) and expanded in manuscript form by the German Waldorf schools' movement in 1995. This version was still very much like an expanded von Heydebrand. It provided some developmental descriptions and extensive lists of content in all subjects. This work was first made available in 1995 as a manuscript, and was then published in 2003, with revisions in 2006, 2010, 2016 and 2019. Recently, the editorial commission has inaugurated a fundamental discussion about the role and nature of curriculum.

In the late 1990s in the UK, interest from Tony Blair's New Labour in 'bringing Steiner into the maintained sector' led to a comparison between the then English National Curriculum and the Waldorf curriculum (Mepham & Rawson, 1997). In 2000, an international version of this was published in the UK (Rawson & Richter, 2000). This text included a number of innovations including an explanation of the early years of Waldorf, an account of school organization and issues such as assessment and student monitoring. It also contained minimum attainment goals for grades 1 to 8 (ages 6 to 14) for maths, English and first foreign language. These aspects all appeared for the first time in a Waldorf curriculum (and are still missing from all German versions). A revised edition appeared in 2014 (Avison, Rawson, & Richter, 2014).

Subsequently, Waldorf schools and their representative bodies in many countries have had to produce similar systematic curricula to justify their education to state authorities. The Rawson and Richter (2000) book has often served as a basis for this. At least 18 translations are known to me, the latest being Mandarin, which appeared in 2018.

Curriculum has become problematic for two basic reasons. The first has to do with what counts as the norm, and the second has to do with rapidly changing developmental tasks. Firstly, the curriculum has been 'translated' into other languages and educational cultures almost from the beginning, as schools were founded outside of Germany (in the UK in Kings Langley in 1922 and London in 1925; in Hungary, Budapest and Norway, in Oslo, in 1926; in the Netherlands, in The Hague in 1927; in the United States, in New York, and in Czechoslovakia, in Prague, in 1928; etc.). Published versions of curricula came much later. Gradually local literature (poems, songs, texts) was added to or replaced the German versions, and local historical and geographical perspectives were taken. As Waldorf education evolved everywhere, including in the German-speaking world, ways of celebrating festivals and shaping school cultures and practices were developed and many variations emerged. Along with many local innovations, a number of what Steve Sagarin (2009) calls Waldorf myths emerged; that is, practices that were claimed to be initiated by Steiner but in fact were later, local inventions.

However, until recently, the basic assumption has been that the curriculum used in Waldorf schools is based on an assumed historical 'original' core curriculum and where necessary, equivalents are sought in the local culture where this is felt necessary. Equivalence implies that something has the same or similar value in relation to a standard or norm. In these translations and adaptations, an assumed norm relating to content and a body of practice is taken for granted and is referred to as *The Waldorf*

Curriculum, with a definite article, indicating uniqueness. That means anyone developing curriculum in say, China, uses *The Waldorf Curriculum* as a norm and seeks where necessary, equivalent contents that match the original as closely as possible.

The problem here is that *The Waldorf Curriculum* means different things to different people in different places. For most of the world that does not speak or read German, *The Waldorf Curriculum* has in practice over the past 20 years, during which the number of Waldorf schools worldwide has increased significantly, been Rawson and Richter (2000), or in its latest form, Avison et al. (2014). This book has been translated (bearing in mind that every translation is an interpretation), and in some but not all cases, locally modified. Historically speaking, though, Rawson and Richter was itself a heavily revised version of Richter (1995, which itself drew on existing German traditions and texts). It also drew on Mepham and Rawson (1997, a comparison of the English National Curriculum and Waldorf curriculum as practiced in the UK) and a US version that was in circulation in manuscript form. The revision of the subject contents was done by teams of UK subject specialists (all listed in the original 2000 version) and included an early-years curriculum and a series of guidelines to themes such as pedagogical assessment, school leadership and management, which were not part of previous curriculum presentations. This curriculum also contained recommended learning outcomes for maths, mother tongue and first foreign language, which was a novelty in Waldorf literature. Rawson and Richter also included a number of idiosyncrasies belonging to the editorial team, which were local, even personal innovations.

To my knowledge, none of the recent German versions of the curriculum have taken account of curriculum developments in other parts of the world. The direction of travel has been one way. The question of what *The* (canonical) *Waldorf Curriculum* actually is can best be described as ambivalent, as it depends on your source and your perspective.

Gert Biesta's (2002) story about how London-based Greenwich Mean Time (GMT) replaced local Plymouth and Exeter time when the railways were being built in England in the 19th century is an instructive metaphor. London time had precedence because the political power was concentrated in London, not because London time was any more real or true. As Biesta points out, the general and standard is often only another local version but one that has gained historical hegemony. The difference in the Waldorf context is that no central authority is insisting on what the canon is. There is, however, a widespread assumption that there is a definitive, authentic and original curriculum. For some this is the German version, for many others it is the English adaptation and for others, such as the Australian Waldorf school who have their own state recognized Australian Steiner Curriculum Framework, it is their local variation. That there is such a thing as *The Waldorf Curriculum* is taken for granted as a reified fact, as an Archimedean point of orientation, but it actually exists simultaneously in several locations both as artefact (published books, online publications) and as idea-in-practice in all its multiple local, even personal variations. The fact is, there are various serious, well-researched versions of the Waldorf curriculum and there are the ideas of curriculum that live in many individual minds. None of these can be declared to be truer or more real than the others, though many people take it for granted that one of them is *the* one! As Boland and Rawson (in press) suggest, the Waldorf curriculum has always been emergent and the spread of Waldorf education has been rhizomic, with new schools and variations on curriculum emerging from a common, though hidden, network of ideas.

The second problem with the notion of Waldorf curriculum is the rapidly changing conditions under which children and young people develop, the impact of mainly neoliberal, mainstream education policies and major socioeconomic change. Governments and

transnational bodies (e.g. the Organisation for Economic Co-operation and Development [OECD], the European Union, the United Nations Educational, Scientific and Cultural Organization [UNESCO]) continuously require education to add further important topics to the curriculum, such as ecological consciousness and sustainability, intercultural capabilities, the changing nature of work and need for life-long learning, the disappearance of job security and many traditional professions, the impact of digital technology, inclusion and most recently, social media and fake news. The Richter curriculum is over 900 pages long because it has to be to cover all the things that could/should be taught, and it still doesn't cover everything. The sections on economics and money, civil society, civil rights and law, politics, democratic education, sociology, psychology, sex education, gender and media studies are thin or nonexistent (though Til Ungefug, 2017, has written an excellent book in German on the subject, and the post-2019 Richter will eventually include these themes). Also, political and economic interests have changed the way they value and measure educational outcomes and this has an-often hidden-impact on Waldorf practice. All these factors mean that understandings of education and curriculum have changed in the public sphere, in the academic world and undoubtedly also for Waldorf teachers and parents. How does Waldorf education deal with these demands in terms of curriculum? The answer is that some work is being done, but this is largely uncoordinated and struggles to overcome the barrier between the German-language Waldorf movement, which is the largest and best resourced, and the rest of the world.

So what, at any given moment, is the definitive curriculum for Waldorf schools in 70 or so countries? If we apply Bransby and Rawson's (2020) layered model of curriculum, we can show the relationship graphically as follows:

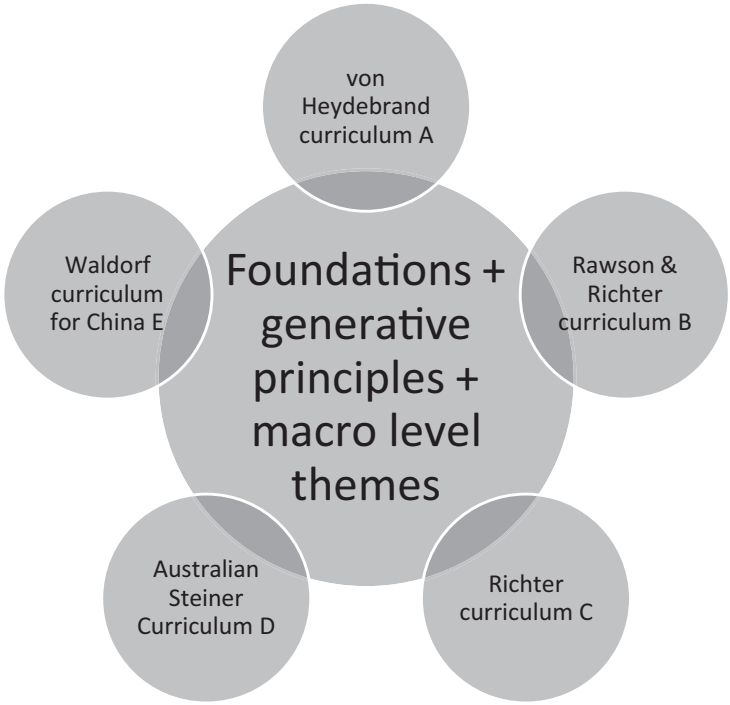


Figure 3.1 The symbolic relationships of various Waldorf curricula to the core ideas of the education

Looked at closely, there are many issues related to translating Waldorf curriculum ideas either from German or from English into different cultures, including methods of introducing literacy in countries without alphabets, narrative material and literature, history, arts and crafts, gender education and celebrating festivals.

It has also become increasingly apparent in recent years how Eurocentric traditional presentations of Waldorf curricula are (Boland, 2015, 2017a, 2017b; Rawson, 2017, 2020). This applies particularly in areas such as story material, literature and history teaching. I suspect that many Waldorf class teachers are unfamiliar with postcolonial critiques of Eurocentric perspectives on history and the rise of civilization, emphasizing the virtues of theocratic states, writing, laws and monumental architecture, rather than the fact that these urban societies were based on slavery and brought with them the burdens of disease, warfare, oppression of nomadic peoples and ecological fragility (see Scott, 2017). The later contributions of Asia and Islam are often glossed over and the European Age of Discovery is valorized rather than being seen as The Age of Colonization. The aim of history teaching in Waldorf schools is to enable students to develop what Zech (2016) calls historical consciousness, that is, to understand how social and cultural change occur and to appreciate that other people at different times and in different places have a different way of relating to self and world. This requires multi-cultural and ecological perspectives. Many Waldorf teachers outside of Europe still feel it necessary and desirable (perhaps because Western history is deemed more important) to teach the classic European periods of history such as the Greeks, Romans, Medieval history and the Renaissance. As these periods of history obviously influenced the European cultures that colonized much of the world, there is a certain validity in looking at European cultural traditions. Far more important, in my view, would be to teach how cultures across the world have always influenced each other, both positively and negatively. From each geographical location and culture perspective, history begins with us and then relates to them, and ultimately 'they' hold up a mirror to us.

The problems of translating curriculum also apply to the terms that seem to be common to both languages. Some German words used by Steiner belong to quite a different cultural context. Kwame Appiah (2019) has shown that the terms 'culture' and 'civilization' are used in German in quite different ways from the English or French equivalents. Also, the German concept of *Abendland*, literally the land of the evening, where the sun goes down, is not merely a Eurocentric (and illogical) geographical perspective but a matter of cultural identity laden with cultural assumptions. Abendland referring to the cultural West, is a term that is used today by right wing nationalists who want to save German Christian culture from what they see as an Islamic invasion. Thus phrases such as 'cultural epochs', or 'evolution of consciousness' associated with certain cultures may be problematical if naively translated into English or other languages without insight into their associations. In their German usage, words such as culture and civilization tend to be essentialized, leading to fixed notions of cultural identities-the Romans, the Greeks, the East. The work of Edward Said (1979) on orientalism does not seem to have had much impact on the Waldorf discourse. The whole notion of cultural epochs within the Waldorf curriculum needs re-thinking (and is not even an original suggestion from Steiner but belongs to one of Sagarin's (2009) Waldorf myths). This critique does not generally apply to high school history teachers, though they often have to wrestle with the state's requirements for a national and sometimes nationalistic focus on history. Even history teachers who

have read history at university often have to struggle to emancipate themselves from national narratives, as we have discovered in our annual international meetings of Waldorf history teachers in Kassel in Germany. British, Irish and Indian teachers, or Russians, Ukrainians, Georgians, Armenians and Hungarians often have quite different understandings of the 'facts' of history, as do South Korean, Japanese and American colleagues. There is much to do in this field-but at least these history teachers meet regularly and talk to each other!

This lack of consciousness unfortunately sometimes gets transported through Western mentors to developing Waldorf schools in Asia. When advising teachers in India or China about the history curriculum, I always emphasize the generative principle that history is taught symptomatologically to illustrate and exemplify important moments in history that show the cultural consciousness of the times, often at moments of crisis and change, and I highlight all the points of intercultural contact between China and other regions, including the West, throughout history. One of the most important topics in history is the significance of the ancient Silk Roads and its modern equivalent in the Belt and Road policy of China today. History always has to be global history, perhaps looked at from a local perspective, but there have never been times when humanity was not in contact with its neighbours and ideas travelled far. People have always made contact and made babies with each other, which accounts for our universal and shared genetic structures!

Neil Boland has also called for a more place-based pedagogy and has introduced the idea that we need a Waldorf 2.0. He writes,

> As Waldorf education heads into its second century, its sustained growth speaks to a lasting resilience of the pedagogy and a lasting ability to meet the needs of families and children in a wide range of social, cultural and geographic situations. To continue to do this effectively, it is imperative that the education is contemporary, relevant and respectful to all and that it represents a path towards a more socially just future. Just as the world has moved on since 1919, it behoves Steiner education to take strenuous steps to ensure that it has moved on at least as far. Only if it can do this will it be able to reach its potential to help its graduates cultivate innovative and integral ways of knowing and being so they can meet the complex and systemic challenges they face.
>
> (Boland, 2017a, p. 73)

The Waldorf curriculum discourse (limited though it is) also needs to take comparative pedagogy into account. This field of study into educational practices in different cultures reflects on the explicit and implicit understandings about the nature of knowledge, learning, schooling, classroom cultures and the socio-cultural context that the education is located in (Kelly, 2013). The different understandings of all these aspects also play into Waldorf education in all the different cultures it exists in. To say, "yes, but they are all Waldorf" is naïve and overlooks how much taken-for-granted expectations and assumptions colour the way Waldorf is received. The generative principles may be applicable in many places, but educational practice is always located and embedded in existing cultural norms. Having spent half of my career in the UK and half in German Waldorf schools, I am well placed to attest to the differences!

Teacher skills

The teacher can:

- Research their own curriculum practice in collaboration with others
- Use the generative principles to develop curriculum
- Show interest in curriculum developments elsewhere

Research questions:
As already indicated, there has been very little research into curriculum. Some obvious research questions could be:

- Is the curriculum's response to the digital revolution adequate?
- Does the current curriculum do enough to support interculturalism?
- How does story material respond to changed social and cultural values regarding gender, sexual orientation, ethnicity and family roles?
- Can the efficacy of existing Waldorf curriculum practice be evidenced and thus provide support for arguments against the imposition of mainstream educational methods?

Principle 18: Becoming a Waldorf teacher is a process of transformative learning

Generative principles:

- Becoming a teacher is a process of transformative learning involving learning and developing dispositions.
- Teacher learning is located in communities of practice and involves self-development.
- Teacher learning is a continuum from initial teacher education, induction in practice and continuing professional development.
- Teacher learning involves reflection for, in and on practice and practice-based research.
- Teacher education and learning require partnerships between schools and teacher education institutions.

Wiehl's (2017) description of the Waldorf method that I referred to in Part 2 assumes that teachers have a solid understanding of Steiner's Foundations and the ability or disposition to read pedagogical situations. It also assumes that teachers practice the methods of developing pedagogical intuition or tact, as well as various exercises designed to develop one's powers of observation and recall, focus and flexibility in thinking as well as skills in diagnostic assessment of pupils. In other words, Wiehl assumes that teacher education establishes high levels of skill and habits of practice in self-development and that ongoing teacher learning continues to develop these. The bar is set high, though one can't expect teachers with this level of skills to come fully fledged out of teacher education – even when this is full-time.

The question is, how do new teachers learn these skills? The classical model of Waldorf teacher education was developed in Stuttgart with the founding of the teacher education seminar in 1945. The core activities at the Stuttgart seminar were (and still are):

- Engaging with Steiner's pedagogical anthropology (e.g. the Foundations), leading to pedagogical dispositions
- Aesthetic schooling of an experiential understanding and learning certain observational dispositions through artistic activity (usually painting, clay modelling, drawing, music, eurythmy, speech formation and drama)
- Subject-specific didactics and preparation for teaching
- Self-development through meditative and contemplative practice (as summarized in Schiller, 2000)

Today, these activities are embedded in full-time Bachelor's and Master's degree programmes in Stuttgart and six other teacher education centres in Germany. Matthias Jeuken and Peter Lutzker (2019) emphasize that the shared experiences and relationships involved in being a student in a course system that encourages the group to consciously learn together, strongly support this process of transformation and enable the development of social perception that teachers require. Being a teacher also means being a kind of performer, not in the sense of entertaining their pupils but as in the analogy of an actor stepping onto a stage and into a relationship to their audience/pupils on the one hand, and into their role/subject on the other. Jeuken and Lutzker quote Seymour Sarason making this comparison,

> like it or not . . . the teacher as a performing artist is faced with a terribly complex and difficult task that all those in the conventional performing arts confront: How do you put yourself in a role and then enact it in a way that instructs and moves an audience.
>
> (Sarason, 1999, p. 54, cited in Jeuken & Lutzker, 2019, p. 299)

Many teacher education centres in Germany have included theatre clowning and improvisation, which an empirical study by Lutzker (2007) showed can be highly effective.

Jörg Soetebeer (2018), a teacher educator from Kiel, theorizes the importance of transformative learning through artistic and aesthetic experience in teacher education. In anthroposophical terms, it is about learning to experience living processes and understand phenomena holistically. This includes experiences of boundaries, transitions, expansion, growth and contraction, crystalizing and dissolving, sympathy and antipathy, opening and enclosing, linear and curving, balance, stillness and movement and so on. The idea is that such exercises not only make such invisible qualities visible but also help the student teachers to develop a vocabulary to describe such processes. Using arts-based methods of schooling perceptive abilities is a vital and indirect form of teacher learning that can enable students to develop ways of seeing and dispositions to engaging with complexity.

Ernst-Michael Kranich, for many years the leader of the Stuttgart seminar, spoke of the future teacher as someone learning to be a researcher with regard to the pupils and being an active shaper of the lessons (1969, in *Erziehungskunst*, cited in Schiller, 2000, p. 71). He added that the ability to teach would first be developed in contact with the pupils. Most Waldorf teacher education programmes include some kind of school experience, observation and internships, ranging from a few weeks to a year full time (as in my own institution). However, what is lacking here is a theory of how this process is supposed to work. There is a tacit assumption that master-apprentice/

expert-novice models explain teacher learning, though in the absence of research this is hard to affirm. Rawson (Rawson 2020 b & c, 2021) has proposed a communities of practice–based model that includes sojourning in authentic practices, a concept described by Fenton-O'Creevy, Brigham, Jones and Smith (2015) and supports this with data from student teachers.

In many countries, there is no full-time Waldorf teacher education but rather a patchwork of part-time and in-service training or induction opportunities. In other countries, the Waldorf teacher education has in effect merged with mainstream provision. Neither of these statements should be read as implied critique. There is so little data on any Waldorf teacher education, even in Germany, that no judgements can be formed at all about the quality of teachers emerging from these systems. My experience, which is quite extensive and international, suggests that initial teacher education, induction of new teachers and continuing professional development through evaluative and illuminative practitioner research is the best model – though it has rarely been practiced. I would add to this the growth of up to date, practical literature based on genuine recognition of the generative principles, such as Sievers (2017), Allanson and Teensma (2018), Langely & Militzer-Kopperl (2018), Jarman and Saragin (2019) and Sigler (2020).

Teacher educator skills

The teacher educator can:

- Facilitate transformative learning through artistic exercises, study of the Foundations and guided practice
- Support the process of induction as peripheral participation in a school as community of practice
- Understand teacher learning as a continuum of initial teacher learning, induction and continuing professional practice
- Remain in contact with classroom practice

References

Allanson, A., & Teensma, N. (2018). *Writing to reading the Steiner way. Foundations of creative literacy in classes 1 and 2*. Stroud: Hawthorn.

Appiah, K. A. (2019). *Lies that bind. Creed, country, colour, class, culture*. London: Profile Books.

Avison, K., Rawson, M., & Richter, T. (Eds.). (2014). *The educational tasks and content of the Steiner-Waldorf curriculum*. Edinburgh: Floris Books.

Biesta, G. J. J. (2002). How general can Bildung be? Reflections on the future of a modern educational ideal. *Journal of Philosophy of Education*, *36*(3), 377–390.

Boland, N. (2015, December). The globalisation of Steiner education: Some considerations. *Research in Steiner Education*, *6*, 192–202.

Boland, N. (2017a). Travels in education: Towrads Waldorf 2.0. *Educational Journal of Living Theories*, *10*(2), 51–81.

Boland, N. (2017b). Waldorf: An education of its time. *Pacific Journal*, *51*(1), 26–30.

Boland, N., & Rawson, M. (in press). Emergent curriculum based on ideal-types: The example of Steiner/Waldorf education. *Educational Philosophy and Theory*.

Bush, T. (2011). *Theories of educational leadership and management* (4th ed.). Los Angeles and London: Sage.

Bransby, K., & Rawson, M. (2020). *Waldorf Education for the Future: A framework for curriculum practice* Retrieved from https://www.steinerwaldorf.org/steiner-resources/academic-articles/

Dahlin, B. (2017). *Rudolf Steiner. The relevance of Waldorf education*. Berlin.

Fenton-O'Creevy, M., Brigham, L., Jones, S., & Smith, A. (2015). Students at the academic-workplace boundary: Tourists and sojourners in practice-based education. In E. Wenger-Trayner, M. Fenton-O'Creevy, S. Hutchinson, C. Kunbiak, & B. Wenger-Trayner (Eds.), *Learning in landscapes of practice: Boundaries, identity, and knowledgeability in practice-based learning* (pp. 43–63). Abingdon: Routledge.

Gladstone, F. (1997). *Republican academies: Rudolf Steiner on self-management, experiential study and self-education in the life of a college of teachers*. Forest Row: Steiner Schools Fellowship Publications.

Götte, W. M., Loebell, P., & Maurer, K.-M. (2016). *Entwicklungsaufgaben und Kompetenzen: Zum Bildungsplan der Waldorfschule (Developmental tasks and competencies: Towards an educational plan for Waldorfschools)* (2nd ed.). Stuttgart: Verlag Freies Geistesleben.

Hattie, J. A. C., & Clarke, S. (2019). *Visible learning: Feedback*. Abingdon: Routledge.

Heydebrand, V. (1925/1994). *Vom Lehrplan der Freien Waldorfschule*. Stuttgart: Verlag Freies Geistesleben.

Horlacher, R. (2018). The same but different: the German Lehrplan and curriculum. *Journal of Curriculum Studies, 50*(1), 1–16. doi:https://doi.org/10.1080/00220272.2017.1307458

Hurrelmann, K., & Quenzel, G. (2015). Lost in transition: Status insecurity and inconsistency as hallmarks of modern adolescence. *International Journal of Adolescence and Youth*, *18*(3), 261–270. doi:10.1080/02673843.2013.785440

Jarman, R., & Sagarin, S. (2020). *Teaching Waldorf Mathematics in Grades 1–8. Engaging the maths genius in every child*. Stroud: Hawthorn Books.

Jeuken, M., & Lutzker, P. (2019). Lehrerbildung für Waldorfschulen: Das waldorfpädagogisches Studium. In A. Wiehl (Ed.), *Studienbuch Waldorfschulpädagogik* (pp. 292–306). Bad Heilbrunn: Verlag Julius Kinkhardt.

Kelly, P. (2013). Comparative pedagogy: Making sense of cultural complexity. *Research in Comparative and International Education*, *8*(4), 415–427.

Langley, J., & Militzer-Kopperl, J. (2018). *The Roadmap to Literacy*. Maitland, Fl.: MillCity Press.

Largo, R. (2017). *Das Passende Leben. Was unsere Individualität ausmacht und wie wir sie leben können*. Frankfurt am Main: Fischer.

Lave, J., & Wenger, E. (1991). *Situated learning. Legitimate peripheral participation*. Cambridge: Cambridge University Press.

Lutzker, P. (2007). *The art of foreign language teaching: Improvisation and drama in teacher development and language learning*. Tübingen and Basel: Franke Verlag.

Mepham, T., & Rawson, M. (1997). *Steiner Waldorf education in the UK: Aims, methods and curriculum*. Forest Row, UK: Steiner Waldorf Fellowship Publications.

Oberman, I. (1997). *Waldorf history: A case study of institutional memory*. Paper presented at the American Educational Research Association Annual Meeting Chicago, Chicago.

Rawson, M. (2010). Sustainable teacher learning in Waldorf education: A socio-cultural perspective. *Research on Steiner Education*, *1*(2), 26–42.

Rawson, M. (2014). Practices of teacher learning in Waldorf schools: Some recommendations based on qualitative inquiry. *Other Education: The Journal of Educational Alternatives*, *3*(2), 45–68.

Rawson, M. (2015). Assessment: A Waldorf perspective. *Research Bulletin for Waldorf Education*, *XX*(2), 30–42.

Rawson, M. (2017). Some guidelines for developing a global Waldorf curriculum locally: A generative approach *learningcommunitypartners* (Vol. 2017). Horst Germany: Leaning Community Partners.

Rawson, M. (2018). The case for illuminative practitioner research in Steiner education. *Research on Steiner Education*, *8*(2), 15–32.
Rawson, M. (2019). International Waldorf 100 curriculum project: Some guidelines for choosing story material. *Pacifica Journal*, *56*(2), 10–17.
Rawson, M. (2020a). De-colonizing your curriculum – Some tips for auditing. *Waldorf Resources, 2020*. Retrieved from https://www.waldorf-resources.org/single-view/de-colonizing-your-curriculum-some-tips-for-auditing
Rawson, M. (2020b). A Theory of Waldorf Teacher Education: Part 1. Learning dispositions. *Research on Steiner Education, 11*(2), 1–22.
Rawson, M. (2020c) A Theory of Waldorf Education: Part 2 the role of study and artistic exercises. *Research on Steiner Education, 11*(2), 23–36.
Rawson, M. (2021). Using artistic, phenomenological and hermeneutic reflective practices in Waldorf (Steiner) teacher education. *Tsing Hua Journal of Educational Research, 440*.
Rawson, M., & Richter, T. (2000). *The edcational tasks and content of the Steiner Waldorf curriculum*. Forest Row, UK: Steiner Schools Fellowship Publications.
Richter, T. (1995). *Pädagogische Auftrag und Unterrichtsziele-vom Lehrplan der Waldorfschulen. Manuskriptdruck*. Stuttgart: Pädagogische Forschungsstelle beim Bund der Freien Waldorfschulen.
Richter, T. (Ed.). (2016). *Pädagogische Auftrag und Unterrichtsziele- vom Lehrplan der Waldorfschule* (2nd ed.). Stuttgart: Verlag Freies Geistesleben.
Rogoff, B. (1995). Observing sociocultural activity in three planes: Participatory appropriation, guided participation and apprenticeship. In J. V. Wertsch, P. del Rio, & A. Alvarez (Eds.), *Sociocultural studies of mind* (pp. 139–163). Cambridge: Cambridge University Press.
Rogoff, B. (2014). Learning by observing and pitching in to family and community endeavours: An orientation. *Human Development*, *57*, 69–81.
Sagarin, S. (2009) What is Waldorf Education? Weblog: What is Waldorf Education. 30.11.2009. http://ssagarin.blogspot.com/2009/11/what-is-waldorf-education.html. Accessed 2.2.21
Said, E. W. (1979). *Orientalism*. New York: Random House.
Scharmer, O. (2009). *Theory U: Leading from the future as it emerges- The social theory of presencing*. San Francisco, CA: Berrett-Koehler.
Schiller, H. (2000). Die Ausbildung zum Waldorflehrer. Geschichte, Inhalte und Strukturen. In F. Bohnsack & S. Leber (Eds.), *Alternative Konzepte für die Lehrerbildung* (pp. 45–124). Bad Heilbrunn: Klinkhardt.
Scott, J. C. (2017). *Against the grain. A deep history of the earliest states*. New Haven, CT, and London: Yale University Press.
Sievers, U. (2017). *Creative teaching- sustainable learning*. Norderstedt: BOD.
Sigler, S. (2020). *Mathematics in Waldorf Schools. Teaching mathematics in the upper grades*. Kassel: Pädagogische Forschungstelle Kassel.
Soetebeer, J. (2018). Bildende Erfahrungen als Methodenkonzeption der Lehrerbildung. In H. Kern, T. Zdazil, & W. M. Götte (Eds.), *Lehrerbildung für Waldorfschulen* (pp. 116–130). Weinheim Basel: Beltz Juventa.
Steiner, R. (1977). *Towards social renewal: Basic issues of the social question*. London: Rudolf Steiner Press.
Steiner, R. (1996). *The foundations of human experience*. Hudson, NY: Anthroposophic Press.
Steiner, R. (2007). *A modern art of education (Lectures in Illkley, 1923)*. Hudson, NY: Anthroposophic Press.
Stockmeyer, E. A. (2015/1965). *Rudolf Steiner's Curriculum for Steiner-Waldorf Schools. An attempt to summarize his indications* (R. Everett, Trans. 5th edition ed.). Edinburgh: Floris Books.
Ungefug, T. (2017). *Perspektiven der Sozialkunde: Plädoyer für ein unentdecktes Kernfach der Waldorfpädagogik*. Kassel: Edition Waldorf.

Wenger, E. (1998). *Communities of practice. Learning, meaning and identity*. Cambrudge: Cambridge University Press.

Wiehl, A. (2017). Das propädeutische Methodenkonzepz der Waldorfpädagogik. In J. Schieren (Ed.), *Enzyklopädie Erziehungswissenschaft Online* (Vol. Waldorfpädagogik). Weinheim Basel: Beltz Juventa.

Woods, G. J., & Woods, P. A. (2008). Democracy and spiritual awareness: Interconnections and implications for educational leadership. *International Journal of Children's Spirituality*, *13*(2), 101–116.

Zdrazil, T. (2019). *Freie Waldorfschule in Stuttgart 1919–1925. Rudolf Steiner-das Kollegium-die Pädagogik*. Stuttgart: Verlag Freies Geistesleben.

Zech, M. M. (2016). Der Waldorflehrplan- curriculum, lehrplan oder rahmenrichtlinie? (The Waldorf curriculum- as curriculum, syllabus or framework of guidelines?). In J. Schieren (Ed.), *Handbuch: Waldorfpädagogik und Erziehungswissenschaft: Standortbestimmung und Entwicklungsperspektiven (A handbook of Waldorf pedagogy and the science of education: Positions and developmental perspectives)* (pp. 572–597). Weinheim Basel: Beltz Juventa.

Zech, M. M. (2017, May). Geschichtsunterricht in der Pubertätsphase- Ankommen im und Ausgehen vom hier und jetzt. *Lehrerrundbrief*, *106*, 67–85.

Part 4 Waldorf education and the academy

Positions, research and outlook

Summary: In Part 4, I address the reception of Waldorf education in the academic world and the challenges that this brings and then summarize the outcomes of a number of studies that I think shed light on the generative principles of Waldorf education that will be useful for future research. This includes summarizing the outcomes of a number of studies on former Waldorf students (also known as alumni). As most of the research has been published in German, I offer here an overview of the more significant outcomes.

Introduction

The number of academic studies on Waldorf education has grown significantly since the late 1980s, most of it in the last 15 years. Much of this has been published in German and is therefore relatively unknown in the wider world, where English is the language of academic discourse. There are several reasons for this growth.

In Germany, the accreditation of two universities (the Alanus University of Arts and Social Sciences and the Freie Hochschule Stuttgart) offering BAs, MAs and even PhDs (Alanus) in Waldorf education, and the inauguration of a number of professorships, has prompted a significant increase in academic publications – since 1990, there have been about 300. There have been summaries of research in German (Böhle & Peters, 2010, 2011; Randoll, 2010; Randoll & Peters, 2015). This reflects the growth in the number of schools but also the specific intentions of the Waldorf movement to gain recognition for its teacher education, more effectively engage with academic critics and generally contribute to the wider educational discourse.

In the English-speaking world, the growth of Waldorf public and charter schools in the United States (the first, in Milwaukee, Wisconsin, was founded in 1991) led to a modest increase in studies. Likewise, the arrival of publicly funded Waldorf schools (known as academies) in the United Kingdom has also led to more research and to a review of literature relating to the curriculum in English (Pountney, 2019). In Australia, which has by far the most research on Waldorf education in English (Gidley, 2020), growth has perhaps been stimulated by the government recognition of the Australian Steiner Curriculum Framework (https://www.steinereducation.edu.au/curriculum/steiner-curriculum/) and the general status of Waldorf education.

The International Network of Academic Steiner Teacher Education (INASTE; www.inaste-network.com) lists some 15 universities around the world currently offering Bachelor's and Master's degrees in Waldorf education, with some even offering PhDs. The aims of INASTE are to facilitate an international network by providing a "scientific, research-based and artistic approach to the study, development, and teaching of Waldorf pedagogy . . . [through] conferences, congresses,

publications and cooperation" (Willmann & Weiss, 2019, p. 7). It is committed to exploring Anthroposophy as the theoretical foundation of Waldorf education in a "discursive, critical and productive manner, taking account of relevant thinking and research" (ibid). Interestingly, there is very little mutual referencing between the English-language and German-language publications, though the inaccessibility of German is more understandable than the other way around. Practically all of the English studies referenced here are available through any university library's digital resources.

Reception problems

In 1987 in Stuttgart, an academic working group (*wissenschaftliches Kolloquium*) was founded with ten university professors and ten representatives of Waldorf education, as a response to what was seen as the increasingly unfavourable reception of Waldorf in the academic world. This group, with some changes in personnel, has been meeting regularly ever since and has led to a series of scholarly books on Waldorf education (e.g. Bohnsack & Kranich, 1990; Buck & Kranich, 1995; Bohnsack & Leber, 1996; Rumpf & Kranich, 2000; Paschen, 2010; Loebell & Schuberth, 2012; Loebell & Buck, 2015; Loebell & Martzog, 2016; Randoll, 2013). This flow of scholarly literature (in Germany, the tendency is to publish edited books rather than articles in peer-reviewed journals) has prompted a whole series of recent publications by academic publishers – Peter Lang, Beltz-Juventa, various imprints of Springer Science such as VS Verlag für Sozialwissenschaften, Klinkhardt and Schneider – a regular who's who of German academic educational publishing. Among these are important recent publications such as Bauer and Schneider (2006), Krautz and Schieren (2012), Bauer and Schieren (2015), Schieren (2016a), Sigler, Sommer, and Zech (2018), Hüttig (2019), Wiehl (2019) and Wiehl and Bunge (2019), which specifically set out to relate Waldorf to mainstream educational science. A number of these titles address issues of educational significance such as the personality of the teacher or spirituality in education – offering, in the context of other theories, a Waldorf perspective. Schieren (2016a) is a major book of over 1,000 pages with articles by 18 professors (a fact that counts in Germany).

In his recent survey of the academic reception of Waldorf education in Germany, Jost Schieren (2016b) notes that Heiner Ullrich's (2015) judgement on Waldorf appears to be widespread. As I referred to in the Introduction of this book, Ullrich claims that many academics judge Waldorf practice as interesting and worthy of study but that its theory does not merit academic attention. Indeed, most of the growing empirical evidence of Waldorf practice is indeed positive, measured by its own standards and in comparison with mainstream education. However, the second part of Ullrich's judgement can be contested. Whilst Steiner's spiritual anthropology and the Waldorf pedagogical theory built on it may seem inappropriate to some academics, there are others who do take it seriously, not least those teaching on Waldorf Master's and PhD programmes. Here Ullrich is being somewhat disingenuous; either such authors and their books and articles are academic, or they are not. Being associated with Waldorf education does not reduce the status or validity of one's academic qualifications or the quality of one's academic work. Nevertheless, Ullrich's judgement resonates in the public sphere. It was a soundbite frequently used by journalists reporting on the Waldorf 100 celebrations.

There is no doubt that for many academics, Steiner's texts, particularly the more esoteric ones, are challenging. As McDermott comments aptly, "the difficulty with Steiner's disclosures from the perspective of a more conventional philosophical position is simply that he bequeathed too much information and particularly too much that runs against, or falls outside, our usual ways of knowing" (2012, p. xxviii). So how can one deal with Steiner and Waldorf theory in a scholarly way? The answer depends on what one takes as Waldorf theory and to what extent one engages with Steiner's esotericism. Kiersch (2016) argues that academic attitudes to esoteric knowledge are changing. He cites the work of Hanegraaff (2014) and Faivre (2010) to show that esoteric knowledge is being increasingly recognized as a valid but marginalized form of knowledge. Kilcher refers to esotericism as the "uncanny of modernity: as the return of what has been forgotten and believed defeated" (2009, p. 148). He goes on to express the view that esoteric knowledge can be

> adequately grasped by means of a *praxiological* concept of knowledge, which understands knowledge as performance, as culture. Knowledge then refers to the complex of the collective work of interpretation, by means of which sciences are constituted as well as esotericism.
>
> (ibid., p. 144)

There is a growing body of research on Anthroposophy and Steiner's philosophy, which I will not attempt to summarize here (e.g. Welburn, 2004; Sijmons, 2008, 2014; Ullrich, 2008; McDermott, 2009, 2012; Heusser, 2016; Zander, 2008, 2016, 2019), that takes it seriously though not always positively (e.g. Zander). There is a new journal, *Steiner Studies: International Journal for Academic Steiner Research*, which addresses Steiner's works on philosophy, spirituality, aesthetics, religion and history.

However, the double bridge linking the study of Western esotericism to Anthroposophy and from there to Waldorf education has not yet been built. The question is whether it *should* be built. Schieren (2015) takes the view, which I think will gradually prevail, that Waldorf is right to keep the main body of Anthroposophy at arms' length. He cites Steiner in England and Wales in 1923 making the point that the Waldorf School was not an anthroposophical school; "for we do not bring anthroposophical dogmatism into the life of the school, rather we seek to develop methods that are in keeping with general human principles" (Steiner, 1991, p. 172, cited in Schieren, 2015, p. 141). Furthermore, Schieren suggests that many, perhaps most, Waldorf teachers either do not have a grounding in the anthroposophical foundations nor practice the meditative self-development that goes with it but rather adopt the Waldorf approach as a set of functioning practices, without engaging seriously with Anthroposophy. Whilst there is no meaningful practice without theory, because practice is always based on some kind of ideas and intentions (Wenger, 1998), Waldorf theory does not have to be explicitly anthroposophical, which is Ullrich's point.

Schieren's (2015) discussion of the relationship between Waldorf and Anthroposophy marks an important turn in the Waldorf discourse. He makes a case for seeing Anthroposophy as a form of phenomenology and refers to Husserl's methodology of époche, which means bracketing out and critically reflecting on one's preconceived ideas. Taking this stance, he would 'suspend' Anthroposophy in Waldorf education as époche, at least as a set of fixed ideas. What remains are the "requirements and laws of each particular area of practice" (ibid., p. 145), in this case education, a stance

which aligns to some extent with what I have tried to do in this book with the notion of generative principles. Schieren draws on Steiner's ideas on aesthetics to support his argument. For Steiner, art was not "about the transmission of a particular subject matter . . . not a vehicle for any kind of ideal, but must be effective and convincing in itself . . ." (ibid., p. 145). Thus, education is not the realization of anthroposophical ideals but rather the application of anthroposophical ideas in the service of good pedagogy.

Thus, the value of Anthroposophy to Waldorf education is as a set of heuristic ideas that direct the teacher's attention to the pedagogical phenomena, as Rittelmeyer (1990, 2011) has argued. It can illuminate certain phenomena, as indeed can ideas from other sources. In either case, such ideas have to be subjected to critical reflection. I take Schieren's point and support his intentions, though I prefer to take a hermeneutic phenomenological perspective such as that of Gadamer (2013) to Husserl's, because as Gadamer argues, one cannot actually bracket out pre-knowledge or prejudices but one can lay these out and see them as the starting horizon for interpretative inquiry into practice. As a Waldorf teacher, my starting horizon for reflective or illuminative inquiry is Steiner's anthroposophical anthropology, not as a taken-for-granted foundation but as a heuristic possibility. I may ask, "Does seeing this phenomenon, as a manifestation of the function of what Steiner calls the life-body, add to my understanding? Can I triangulate this perspective with others?" Pedagogical phenomena lend themselves to qualitative and interpretative/hermeneutic methods of research (Hitchcock & Hughes, 1995; Eisner, 2017).

Hermeneutics of suspicion or empathy?

Lorenzo Ravagli (2014) has outlined a phenomenology of academic positions, including polemic positions, in relation to Waldorf education and Anthroposophy. He distinguishes between inclusive and exclusive discourses. A scholar taking an inclusive stance is basically willing to engage with a text with an open mind, whereas a scholar who takes an exclusive (perhaps more accurately, an *excluding*) stance judges Waldorf education, and in particular its theory, to be academically invalid because it lies outside the parameters of the researcher's frames of reference. By claiming that it cannot be understood within its own terms, this obviates any attempt to understand it. It is thus possible to conduct research in Waldorf schools by bracketing out the ideas behind the practice. There is still a sense that young academics researching Waldorf education would be taking a significant career risk if they did take an inclusive hermeneutic perspective. This self-imposed restriction applies both to 'insiders' and 'outsiders' in the sense of Herr and Anderson's (2015, p. 37ff) continuum of researcher positionality.

However, suggesting that Waldorf practitioners or indeed external researchers would be taken more seriously if they bracket out or suspend the anthroposophical roots of Waldorf education is fundamentally un-ecological, non-holistic, reductionist or lazy thinking. It is like admiring cut flowers and ignoring the social, economic and ecological conditions of their production (or praising biodynamic wine from France without inquiring into the ideas behind its production). One solution is to provide an account of Waldorf theory that does not deny its spiritual dimension but reinterprets it in contemporary terms, thus making an inclusive hermeneutic and phenomenological approach possible. The risk, however, in this approach is that theory outreaches practice. The complex arguments, elegant theory and thick books being produced on Waldorf theory may have minimal impact on classroom practice. Though some parts

of the academy still have trouble accepting that practitioners can research their own practice, as Herr and Anderson (2015) show, action-research–university partnerships in particular are playing an increasingly vital role in research that makes a difference. This ought to be the future in Waldorf research.

Critique as polemic

Waldorf and Anthroposophy are also subjected to a kind of critique, akin to what Rita Felski (2012) characterizes as a hermeneutics of suspicion, a phrase originally coined by Paul Ricoeur (1970) that was respectful of the need to expose false consciousness and illusions in the ways that Marx, Nietzsche and Freud did. In its contemporary form, however, a hermeneutics of suspicion (or suspicious critique) is a fundamentally adversarial and negative stance. It is secondary, meaning that it does not have to offer a coherent alternative and raises more questions than it has answers to, and it lacks self-reflexivity. It is iconoclastic and derives its identity from railing against assumed established authorities, though Waldorf hardly has this status. Nevertheless, perhaps the more successful Waldorf practice becomes, the more extreme the critique is. This kind of critique does not tolerate rivals and claims a unique perspective. Indeed, following Felski, it is fundamentally arrogant, being intolerant of and denying other perspectives. It purports to be rational and objective, whilst the object of its critique is portrayed as irrational and non-intellectual but seductive. Felski characterizes the qualities of such suspicious critique as constituting "a muted affective state – a curiously non-emotional emotion of morally inflected mistrust – that overlaps with, and builds upon the stance of detachment . . ."(2012, p. 3). This characterizes the tone of many critiques of Waldorf that one can discover on the internet.

Outside of the academic frame, there is a wide spectrum of critiques of Waldorf education that range from well-intentioned but ill-informed to outright defamatory. Some are based on highly selective readings of Steiner that take isolated statements out of context. There is also a critique that denounces Waldorf education, along with homeopathy, vaccination sceptics and biodynamic farming and other holistic practices as belonging to an unreconstructed, pre-Enlightenment, irrational thinking that needs to be exposed for the protection of the gullible and innocent. This position is taken, for example, by the Humanists, UK (the German branch is also very active). This organization has published the following criticism of Waldorf education:

> For several years we have raised serious concerns about safeguarding and poor education standards in Steiner schools which have included evidence of entrenched racism and systemic bullying along with problems with academic rigour and secrecy.
>
> (https://humanism.org.uk/2019/01/31/humanists-uk-welcomes-ofsteds-calls-to-close-failing-steiner-schools/)

This critique may be justified in some cases, but the assumption is that this is not just due to local incompetence but is systemic in Waldorf education and a fundamental weakness of the approach. Such critiques are ideologically based and lump together long-standing issues with homeopathy, which is not a Waldorf theme, failing Steiner academies and accusations of racism. The failure of management in some schools does not mean that Steiner education has failed. Such failures may have more

to do with inadequate training of teachers, because these schools have to blend a Waldorf approach with frequently changing state curriculum requirements and a testing regime that no longer takes the special features of Waldorf education into account, as it once did when Waldorf schools came under the inspection regime of Ofsted and when the first Waldorf academy was established. Steiner schools are not the only schools that have struggled with the redefining of education to include areas that were once part of social services, such as identifying children at risk outside of school. Since late 2019 the UK Waldorf schools, under the guidance of a new leadership, have been working flat out to ensure that schools are able to take the new regulations on board. Readers interested are recommended to look at recent statements on the website of the Steiner Waldorf Schools Fellowship (www.steinerwaldorf.org/statement-from-the-steiner-waldorf-schools-fellowship/).

The Humanists are amongst the main critics of Anthroposophy and thus of Waldorf. Ironically, Waldorf schools in Germany explicitly align themselves with the neo-humanist tradition of Wilhelm von Humboldt, Johann von Goethe and Friedrich Schiller, as Steiner himself did. The Prussian diplomat, Privy Councillor, founder of Berlin University and philosopher Wilhelm von Humboldt (1769–1835) articulated an ideal of a general education aimed at developing individual cosmopolitan, civic values based on rich experiences in the arts, culture and the sciences. He conceived this as a public education for all social classes, an idea that was variously supported by such figures as Georg Hegel, Karl Marx, Friedrich Nietzsche, Sigmund Freud and Theodor Adorno. Some academic critics of current competence-based education policies that see the primary function of schools as serving the needs of the economy (e.g. Nida-Rümelin, 2013, 2014), cite Humboldt as a positive model of an education system that encourages self-formation and democratic self-determination. As Erik Ode (2015) puts it, whilst the humanistic model of Bildung is criticized as anti-economic, pessimistic of technocratic progress, idealistic and unrealistic, he argues that Humboldt's humanistic Bildung in its contemporary, reflexive form (e.g. Klafki, 2007) should actually be seen as the basis for resistance to instrumentalization, dehumanization and the suppression of democratic initiative. Ode is a Professor at the University of the German Army, which in itself is an interesting notion.

The people who call themselves Humanists frequently mention Steiner's racism as if this was an established fact. As a Waldorf practitioner, it is depressing and demeaning to read journalists and bloggers claiming that Steiner was a white supremacist and therefore all Waldorf teachers are racists, and that Waldorf education teaches this ideology and indoctrinates children and parents with these ideas (for which there is no evidence except anecdotal accounts by some individuals who feel let down in some way by their Waldorf education). I hope I am not a racist, nor have I ever met one in the Waldorf world – though I don't deny there may be some, just as there seem to be in most walks of life. For the record, Robert Rose (2013) and Bo Dahlin (2014, 2017) in English and Lorenzo Ravagli (2004, 2014; Bader, Leist, & Ravagli, 2002) in German, among others, have contextualized and explained Steiner's anachronistic and inappropriate language on race. Dahlin (2014) argues that Steiner's use of the term 'race' applies to prehistoric human ancestors and is synonymous with the term 'hominid species' and that following the end of the last Ice Age (Steiner spoke of pre- and post-Atlantean epochs), the term race means early civilizations, what today we would call early complex cultures. Steiner is quoted in Bader et al. (2002) saying that anyone who promotes racial ideals in connection with nationalism and ethnic

identity is promoting the decline and fall of humanity. Steiner actively promoted the idea of a democratic, socially just, multicultural state and spoke out against all forms of political, social, gender and ethnic discrimination and antisemitism. This sadly did not prevent history from taking its course.

It is a fact that all national and international Waldorf associations have issued statements distancing themselves from inappropriate texts by Steiner and pledging themselves to work against any form of discrimination and absolutely reject any form of racism. With Waldorf schools in some 70 countries on all continents, in Jewish, Christian, Muslim, Hindu, Buddhist, Confucian and Shinto cultures, intercultural cosmopolitanism rather than racism is the predominant tenor of social interactions in Waldorf schools. Even if one could show that Steiner had inappropriate views on race, this does not mean that Waldorf education is based on this system. Education is only a small part of Steiner's collected works. Most of the rest has little to do with the education practiced today. However, the more widespread Waldorf education becomes, the greater the intensity with which it is attacked by a few committed critics.

Some key perspectives on Waldorf education

The body of academic knowledge on Waldorf education has its origins early on but only really takes shape in the 1980s. The earliest recorded academic text on Waldorf education was the PhD thesis of Ilse Staedtke in 1929 on Friedrich Schiller and Waldorf education, though I have been unable to access it. The websites Waldorf Answers (www.waldorfanswers.org) and Wikipedia offer a fairly complete list of theses, academic journal articles and books, but many of the earlier ones are not readily available.

In 1980, four articles in the *Teachers College Record*, published by Columbia University by John Barnes, Alan Howard, Hope Leichter and John Davy, offered a comprehensive introduction to Waldorf education. Barnes (1980) introduces the background and central ideas of the education in a way that can form the basis for any study. Howard (1980) argues that Waldorf education offers a moral and ethical basis for social life. Davy (1980) looks at the way Waldorf education cultivates a disposition to liberty, fraternity and equality by enabling capacities for creativity, social tolerance and the perception of the needs of others. Leichter (1980) explores the temporal organization of the Rudolf Steiner School in New York, introducing the notion of the importance of rhythms in education and distinguishing between *microtime*, what we might call timing in teaching; *calendric time*, which is chronological and marks external progressions; and finally *developmental time*, which is individual, over the long term and involves emergent qualities. The article shows how Waldorf education works with these different dimensions of time in the pedagogy.

The earliest academic book on Waldorf education in English that I have found is Mary Caroline Richards's book *Towards Wholeness: Rudolf Steiner Education in America* (1980), which is a personal reflection on the journey of an academic to Waldorf education. Richards was a university teacher and potter. Though she claims not to offer a systematic account of the education, she does in fact do this, though in an idiomatic, biographical way. She speaks of Anthroposophy as a 'tree of life' expanding, rooting and reintegrating ideas that she feels have been held apart by modernity. Her interest in Waldorf is as an integrating pedagogy that does not separate the subjective from the objective, spirit and matter, self and world but seeks to weave these into a new synthesis. The book was positively reviewed by Owen Barfield (1981).

Douglas Sloan's book *Insight-Imagination: The Emancipation of Thought and the Modern World* (1983) made positive reference to Waldorf education and to Rudolf Steiner's ideas on the imagination as a way of knowing. Sloan is an academic and was a professor at Teachers College, Columbia University. As well as creating an intellectual space for the recognition of an education that valued the imagination and the arts, Sloan also made the point that Waldorf education was not born fully made and that it is an emergent, evolving approach based on reflection and adaptation of its core principles to circumstances. In 1984, Robert McDermott's (1984, 2009) *The Essential Steiner* appeared, with commentaries on key Steiner texts, thus making Steiner's extensive works accessible to an academic readership. Richard Blunt's (1996) *Waldorf Education: Theory and Practice* was an academic book published in South Africa but not by a mainstream academic publisher and is rarely cited, though it provides a very systematic account of many of the key aspects of Waldorf education. McDermott (2012) has recently published a collection of essays by academics relating Steiner to American philosophy, which offers a range of perspectives on Steiner that are less well known and are quite distinct from those based on Middle European traditions.

Another article that explains Waldorf education in a concise and clear way from historical and philosophical perspectives is by William Reinsmith (1989). This article emphasizes how the Waldorf approach relates the whole to the parts and the integrated nature of the relationship between mind and body. Henry Barnes (1991) followed up his original article with an exploration of the learning processes in Waldorf education and offers an account of the spiral curriculum used in Waldorf schools, in which living concepts are frequently revisited and expanded. He refers to this process of growing knowledge and capability as an ascending spiral of knowledge:

> As the students mature, they engage themselves at new levels of experience with each subject. It is as though, each year, they come to a window on the ascending spiral that looks out into the world through the lens of a particular subject. Through the main-lesson spiral curriculum, teachers lay the ground for a gradual vertical integration that deepens and widens each subject experience and, at the same time, keeps it moving with the other aspects of knowledge.
>
> (1991, p. 53)

Bruce Uhrmacher of the University of Denver has written a number of papers on Waldorf education, the most frequently cited being in 1995, with the title *Uncommon Schooling*, which gives a historical account of Steiner, his key ideas and the central ideas of Waldorf education. He offers a number of possible reasons why Steiner is not as well-known and accepted in America as Uhrmacher thinks he should be. The main reasons are that his ideas do not fit neatly into any one sphere of knowledge and

> his pantheistic and angelological outlook rattles many contemporary scholars and makes him suspect among scientifically orientated communities. His understanding of human nature, which consists of physical, etheric, astral, and ego bodies demands a great deal of open-mindedness from students trained in mainstream academia.
>
> (1995, p. 382)

Uhrmacher also offers an interesting account, drawing on Max Weber's analysis of charismatic leaders, of the subsequent development of the Waldorf movement.

The reason he recommends engagement with Steiner is the cultural anthropological virtue of "making the strange familiar and the familiar strange". Serious study of alternatives in education offers new perspectives on the mainstream and vice versa. Assuming that alternative schools exist because parents and teachers are dissatisfied with mainstream schooling, these forms of 'other education' can offer new perspectives on "educational issues currently forgotten, dismissed, or simply unexamined" (1995, p. 382). Uhrmacher makes the important point that Waldorf's holistic stance is not just one of curriculum but of a fundamentally participatory epistemology that integrates the human being into world processes in ways that materialistic approaches do not. The arts-based approach, he suggests, offers a balance to technocratic approaches that were dominant at the time (and in many respects still are). Uhrmacher's account of Waldorf education is, in my view as an insider and practitioner, detailed, accurate and insightful.

In a subsequent paper, Uhrmacher (1997) addresses what mainstream education might learn from Waldorf. He starts by making the crucial point that the whole ecological context of an educational practice is vital. This is frequently disregarded in practice not only in integrating Waldorf and state education but also in translating Waldorf to different cultures. He suggests that both are possible but require several criteria that he refers to as strategies for borrowing. These include:

> (a) knowing the contextual framework from which an idea initiates, (b) knowing the contextual framework (or ecology of schooling) where the idea is going, and (c) becoming aware of aesthetic preferences through (d) aesthetic criticism that would result in an informed (e) aesthetic evaluation. These ideas do not tell you what to do, but they point out if an idea is likely to succeed in a different setting.
>
> (1997, p. 77)

One could call these generative principles for translating Waldorf ideas into other contexts, such as public schools (charter schools or academies), and also for translating Waldorf into schools in other cultures with different educational traditions, expectations and understandings of education.

Freda Easton (1997) conducted a study involving successive interviewing of Waldorf students and an evaluation by external researchers of the Urban Waldorf School of Milwaukee, the first public Waldorf school, founded in 1991. As she puts it, Waldorf education

> resonates with increasing numbers of educators and parents because it provides a framework that informs and inspires educators to think about ways to create a learning community that nurtures children's capacities to become whole human beings in a world that is becoming increasingly mechanized.
>
> (Easton, 1997, p. 88)

She identified six core elements of Waldorf education that can be described as generative principles for Waldorf-inspired education in public schools. These core elements can be listed as:

1 A theory of child development
2 A theory of teacher self-development
3 A core curriculum that integrates artistic and academic work

4 A method of teaching as an art that pays careful attention to synchronizing teaching methods with the rhythm of a child's unfolding
5 Integration of teaching and administration
6 Building the school and the greater Waldorf community as networks of support for students, teachers and parents

Easton identifies that

> Waldorf provides a framework for envisioning a renewal of thinking that integrates imagination, inspiration and intuition into our ways of knowing. . . . Artistic work provides opportunities to become more conscious of our inner and outer worlds. It helps children learn to concentrate, pay attention to detail, and envision the whole. It encourages the free expression of the human spirit in more disciplined ways and strives to balance freedom and discipline. By educating "head, heart and hands", Waldorf education seeks to nurture a self-esteem that encompasses aesthetic and moral sensibilities as well as intellectual competence.
>
> (1997, p. 94)

She completes her study with a warning that Waldorf teachers need to evaluate their practice on an ongoing basis, much in fact as Steiner had. These authors provide a solid theoretical basis for research on Waldorf education.

Elliott Eisner (1991, 1994a, 1994b) has highlighted the significance of multi-sensory experience and the importance of the body in learning. His work on pluralistic modes of cognition and Howard Gardiner's (1983, 1997) notion of multiple intelligences have provided a theoretical frame to explore the ways in which content is presented in a Waldorf classroom and the ways symbolic knowledge can be generated in different modes. Eisner's experience was that:

> there is one institution that I know that does pay serious attention to the use of multiple aptitudes and the development of diverse forms of knowing, I speak of Waldorf schools. . . . Waldorf education is based upon a view of human development that acknowledges the importance of the sensory system and the role of the body in learning. It attaches a special significance to the importance of myth and folk tale, and to eurythmy, a form of bodily movement designed to harmonize the various aspects of the child's experience. It emphasizes the importance of image making in children's learning. Its teachers utilize rhythm and other sensory forms to help students understand ideas in arithmetic. In short, Waldorf schools . . . provide, or attempt to provide, a program that affords children a balanced educational diet, one that not only fosters conventional forms of academic achievement but puts a premium on the development of imagination and the refinement of the sensibilities.
>
> (Eisner, 1994a, p. 83)

In his recent book, *Philosophy of Education in Action*, David Nicholson (2016, pp. 101–105) gives prominent space to Waldorf education, based on his earlier empirical studies (2000). He summarizes the role of the teacher as nurturer of human development, as artist, as guiding authority based on mutually respectful relationships.

He claims that Waldorf schools "straddle a teacher-directed and student-centred education, especially in the stages of middle childhood" (2016, p. 105). On the *Continuum of Educational Philosophy*, which the author offers as a conceptual frame, Waldorf practices range from a fixed curriculum and teacher-directed methods to student-centred practices based on experience. Waldorf schools offer a wide range of academic learning opportunities as well as practical skills and artistic experience. Nicholson suggests that the curriculum has remained unchanged since Steiner's time. This may be the case in the schools he has studied, but it is far from the case if we take the whole Waldorf movement into account. I would also disagree with the characterization of the curriculum as fixed.

Nicholson suggests that Steiner's belief in the spiritual nature of the human being "tethers his educational practices to a divine nature while at the same time reaching beyond a purely rational and intellectual approach to meet the needs of children in a practical and developmentally appropriate way" (ibid., p. 105). This is no doubt a fair judgement of Steiner's educational philosophy. However, Waldorf practice does not flow directly from that source but is interpreted in extremely varied and culturally situated ways. That is the problem with identifying Waldorf education with Steiner; it fails to recognize that it is very rare that an educational philosophy, assuming this is coherently articulated by the educational philosopher in question, is implemented as intended, or indeed whether this is even possible. There is always a gap between espoused, intended and lived practice.

In his book *Rudolf Steiner's Pedagogy of Imagination: A Case Study of Holistic Education* (2004, previously published as a conference paper, 2003), Thomas Nielsen describes seven teaching methods based on an interpretation of Steiner's ideas on imaginative teaching. As he points out, many state curricula call for imagination in teaching, but this notion lacks a solid theoretical basis or indeed concrete ideas about how imagination is used in pedagogical practice. Imaginative teaching bridges the gap between thinking and action, and is thus a contribution to postmodern thinking. He writes, "Imaginative teaching may, therefore, be seen as a generic method of dealing with the sense of separation and fragmentation that is prevalent in today's Western society" (2003, p. 18). Based on a phenomenological study of three Waldorf schools in Australia, Nielsen identified the following methods, briefly summarized:

1 Drama: The imaginative enacting of experiences and situations
2 Exploration: The teacher creates opportunities for hands-on constructing of experiences, pupils making their own discoveries through direct and authentic experience
3 Story: Prompting suspense, identification and empathy and the creation of lively inner pictures and the opportunity to link these to their own inner experiences
4 Ritual and routine: Unconscious learning through habits and dispositions, through which images are constructed over time
5 Form drawing: Unconscious learning about space and movement, symmetry and balance and active imagination to visualize the forms
6 Arts of all kinds: Creating images in different media
7 Pictorial discussion: Through carefully guided discussion, pupils come to understand and accept that others have other images, that multiple answers are possible and can be imagined

Nielsen concludes that imagination is not simply added on as an extra ingredient to make learning enjoyable;

> it is the 'crossroads' at which other dimensions of being human meet and find meaning. Moreover, as imagination appears 'secular' and inclusive of both ends of the spectrum of human experience, thinking and doing, spirit and matter, imaginative teaching provides common ground between past and present preferences for either scientific or religious schooling, vocational or liberal education, materialism or spiritualism.
>
> (Nielsen, 2004, p. 263)

Furthermore, the aspects of imagination are embedded in the pedagogical methods "used in imaginative transaction and are part of the transaction itself, while whole-of-school routines, buildings, and the general management of children constitute the forum in which the transaction takes place" (2004, p. 261).

Philip Wright has shown how imagination can be an important method in geography teaching. He points out that "For Steiner schools descriptive narrative (to build a sense of place or region, in its geographical usage) is central to a knowledge-building method that works closely with pupils' biological rhythms" (2013, p. 13). He explains the rhythms of learning used in the Waldorf approach and emphasizes the epistemological aspect of knowledge production and the role of imagination in this, and contrasts academically prescribed knowledge with the journey children make in transforming sense perception and imaginative experience to concrete idea or concept, in ways that adapt "geographical knowledge to suit pupils' evolving consciousness and relationship to the world" (ibid., p. 14).

One area of research has been spirituality in Waldorf education. A series of studies by Henry (1992), Woods, O'Neill and Woods (1997), Woods, Ashley and Woods (2005) Oberski, (2006, 2011), Dahlin (2009), Okumoto (2019), Goldschmidt (2013, 2017), and Pearce (2019) and Rawson (2021 a & b) all show in various ways that spirituality is cultivated not only in religion lesson but also in the cultivation of thinking and through the whole school ethos, and even permeates teacher education. Art is another feature of Waldorf education that has been researched (Ogletree, 2000), often in comparison with state schools (Cox & Rowlands, 2000; Kirkham & Kidd, 2015; Rose & Jolley, 2016). These studies all emphasize not only the quality of artistic production but also the contribution to overall aesthetic education.

A number of studies have looked at Waldorf education in public schools, starting with McDermott et al. (1996), who conducted an evaluation of the Milwaukee public Waldorf School and set benchmarks for subsequent studies of Waldorf in public schools. Woods et al. (2005) followed up their earlier small-scale research (1997) with a comprehensive study of Waldorf education in 21 schools in the UK on behalf of the then Department of Education and Skills to explore the possibilities of including Waldorf education or Waldorf schools into the maintained sector. They concluded that the Waldorf schools covered all the topics in the English National Curriculum – though not always in the same sequence or using the same methods – and they found alignment in a number of other key areas such as seeing pupils as individuals, the value for assessment and learning support and the role of parents in supporting the education.

The main differences being the collegial and distributed leadership in Waldorf schools (which Woods and Woods (2008) subsequently explored in detail) and the

unique educational philosophy, the study concluded that there was much scope for cooperation and mutual learning but warned that transferring practices from one culture to another is fraught with difficulties and may not be appropriate. If Waldorf schools were to be state funded, it would only make sense if they retained their distinctive educational philosophy and approach. They also highlighted a number of specific areas of Waldorf pedagogy that should be valued, such as the approach to foreign languages, the artistic and imaginative approach, the class teacher system for younger children, the emphasis on narrative and speaking, the teacher's reflective activity, spiritual values and distributed leadership. There were also areas that Waldorf could improve, such as having more transparent admissions procedures and organizational and management structures. A number of recommendations were made relating to the inclusion of Waldorf schools into the maintained sector and the disapplication of certain aspects of the national curriculum to enable Waldorf to retain its character and approach. They argued that there should be more research. This has not really happened in any systematic way, partly because of a lack of interest among academics and partly because no funding has been made available, which in itself reflects the lack of political or academic interest.

A report by the Stanford Center for Opportunity Policy in Education (Friedlaender, Beckham, Zheng & Darling-Hammond, 2015) compared the educational outcomes of a Waldorf-inspired public school in Sacramento, California, with other schools in the district and also looked at how the district supports alternative education models. It identified a number of features of the Waldorf-inspired school that it thought were relevant, including Steiner's theory of development, teachers engaged in sustained relationships with pupils and various pedagogical aspects of this, such as the idea that the teacher teaches the child rather than the subject, each child is allowed to develop at her own pace, the learning environment changes with the age of the pupil, learning uses many modalities (art, movement, handwork, speech, speech, reading, storytelling, hands-on experimentation, practical life skills, etc.) in interdisciplinary contexts, formative assessment, long-term relationships between teachers and pupils), well-prepared, thoughtful teachers and powerful parents. Both qualitative and quantitative data showed evidence of successful learning and positive student behavioural outcomes among all social and ethnic groups. The report concludes that this success is in no small measure the outcome of the district's support for achieving a fidelity to the Waldorf approach. In other words, the success was because of the adherence to the Waldorf approach.

There have been other studies, notably by Oberman (2007), comparing public charter schools in California. The most comprehensive report (Larrison & Van Vooren, 2012) compared public Waldorf schools with each other as well as with non-Waldorf charters. The schools surveyed included a variety of urban, suburban, middle-income and low-income settings. Taken together, the studies show that the pupils in these public Waldorf schools are by and large doing well socially, personally and academically and are being supported in the learning and development by committed and capable teachers, well supported by effective administration. Given this success, it is surprising that mainstream academic research has largely ignored this small but significant field. Lilly Siu's (2011) educational doctoral (EdD) thesis explored the development of six formerly disengaged high-risk youth at a Waldorf-inspired community school in California. These students had a history of poverty, educational failure and involvement in drugs, violence, poverty and crime. The study showed evidence of the

students' engagement and progression, which they themselves subscribe to the effect of the school, and were able to achieve academic milestones that "belie their previous educational trajectories" (2011, p. viii).

Richard Pountney's (2019) appraisal of the Waldorf (referred to as SW) curriculum in the UK context offers an important perspective on curriculum with the changing and polarizing curriculum discourse, between the notion of curriculum as prescription of content and outcomes and the notion of curriculum as portrayal of the substance of education and what it means to be an educated person (ibid., p. 2). He notes that the underlying principles of the Waldorf curriculum provide teachers with orientation, values and identity. His study identified possible alignments of the SW curriculum with the national framework, though this needs clearer articulation. It recommends a more transparent and coherent process of curriculum making, having made the point that review and reassessment is essential. The report concludes that the SW movement needs to be bolder in its commitment to this process and to monitoring and evaluating its outcomes.

Summing up, one can say that these studies offer a theoretical, and to some extent empirical, basis for the following generative principles:

- There is a strong emphasis on the imagination as an important form of knowing that balances more cognitive learning.
- An artistic approach is central to cultivating imaginative knowledge.
- Waldorf education works with different dimensions of time, learning rhythms and developmental time.
- There is a clear link between Steiner philosophy and education, notably in the fact that the pedagogy works to integrate body and mind to bring about the embodied mind, which is a new and important idea in education.
- The spiral curriculum enables the organic growth of concepts.
- Waldorf offers a significantly different educational approach to the mainstream and thereby provides a significant comparison for understanding different educational approaches and asking challenging questions of the mainstream.
- Spirituality is cultivated throughout the curriculum (Rawson, 2021a, 2021b).
- There is evidence that Waldorf pupils are more artistically literate and creative than their peers in other schools. This may not just be an outcome of art teaching but rather a reflection of the whole artistic approach.
- The self-development of the teacher and the network of collegial support within and between schools are important factors in the quality of Waldorf education.
- Adopting Waldorf methods into state schools can be done, but this is most effective when the theoretical and practical context of the original ideas is borne in mind.
- Waldorf needs to explicitly review and reassess its curriculum in an ongoing research process.

Alumni research

There have been a number of empirical studies on Waldorf alumni (former pupils), some going back three generations, carried out internationally (Ogletree, 1998), in Germany (Barz & Randoll, 2007a, 2007b; Barz, Liebenwein, & Randoll, 2012), Sweden (Dahlin, 2007), Switzerland (Randoll & Barz, 2007), Denmark (Jensen, Boding, & Kjeldsen, 2012) and United States private Waldorf schools (Gerwin & Mitchell, 2007;

Safit & Gerwin, 2019); and in public US Waldorf schools (Oberman, 2007). It would take up too much space to report on these studies in detail, but I will summarize the most important outcomes:

- A decisive majority of those former Waldorf pupils surveyed said that Waldorf education had 'prepared them for life', and that they had learned the skills of collaboration, creative and critical thinking and concern for the community as a whole.
- They show a self-confessed high level of life satisfaction. The recent US survey (Safit & Gerwin, 2019) showed that around 90% of Waldorf alumni are satisfied with their life, in a way that does not necessarily correspond to their income given that 30% earn less than their parents. They do not generally associate success with affluence.
- Though Waldorf schools are non-selective comprehensive schools (and were historically the first of this kind in most of the countries studied), the number of pupils who graduate from school with a qualification that enables access to higher education (e.g. in Germany, the Abitur) is almost twice as high as the national average (in Germany, 67%). This is almost certainly helped by the fact that on the whole, Waldorf parents are well educated and motivated.
- Pupil–teacher ratios in Waldorf schools are about 1–13.5, which is significantly below the ratio in state schools. Teacher remuneration in all countries is significantly lower than in state schools.
- The majority of former Waldorf students are indifferent, sceptical of, or reject Anthroposophy (which clearly refutes the claim that Waldorf schools seek to 'convert' their pupils to Anthroposophy). Indeed, former Waldorf pupils show very high levels of tolerance towards people of other religious or ethnic backgrounds and high levels of 'world-openness'. Only between 12% and 18% (the higher figure is from Switzerland) of parents have an anthroposophical connection. A lower percentage of Waldorf students are affiliated with any faith community than the national average.
- Data from Waldorf school students who participated in Programme for International Student Assessment (PISA) tests is sparse but shows higher than average achievements, particularly in scientific understanding, in spite of the fact that 50% of former pupils thought their science education was inadequate.
- Waldorf pupils identify with their schools and what is taught in school significantly more than students at state schools with comparable intake. Waldorf students are motivated to work at school less by the aim of achieving grades and more out of intrinsic interest. They also feel that the education is more well-rounded and addresses a wider range of their interests. Most were glad that they were spared the pressure of working to achieve grades for most of their time at school.
- 38% of Waldorf pupils (in Germany) have extra learning help at home, as opposed to only 5% in state schools, which may more reflect parental nervousness than actual need. Trust in the Waldorf approach is not always given among newer parents when approaching the exams. However, most of the survey participants said that they felt that what they had learned in school was relevant to their subsequent lives, including academic studies.
- Waldorf alumni were critical of the quality of second-language learning, and they felt that they got too little feedback on their learning achievements and had inadequate instruction in learning techniques. They felt that there was too little

reference to current affairs and politics. They felt that there was too little science, sport and politics. Most were unclear what the purpose of eurythmy is.

- Most alumni felt that their education enables them to develop social competence, tolerance, fairness, emotional intelligence and the ability to avoid and resolve conflict. They had strong self-efficacy, self-confidence and self-reliance and were creative, flexible and capable of dealing with unexpected and challenging situations. There was a strong sense of all-around capability and inner mobility. Waldorf pupils are significantly less aggressive and prepared to use violence and would rather avoid conflict. They are definitely not racist or prejudiced towards other ethnic or social groups or people who think and behave differently.
- They sometimes felt less well prepared for situations in which competing with others and getting your way was important. Some felt that they lacked ambition and may have been too naïve about the intentions of others.
- There is even evidence that Waldorf pupils are healthier than their peers in other schools during their school time, with fewer days off for sickness and fewer headaches, nervousness and other stress-related symptoms.
- Waldorf alumni tend to enter educational, medical, social, media and artistic professions. Their choice of career is more related to self-fulfillment than money. In their leisure time they tend more to reading, art, playing music, crafts and participation in further training than their peers.
- The most common profession of their parents were teachers and academics (21.1% compared to the national average of 4%), business and salespeople (17.7% compared with 12.2%), doctors and pharmacists (9.4% versus 1.2%), engineers (8% versus 2.6%) and artists (6.8% compared with 0.9%).
- The range of students from other cultural backgrounds than the majority attending Waldorf schools reflects strongly the location of the school. Schools in multicultural communities or parts of a city have much higher proportions of students from different cultural backgrounds than schools in more culturally homogenous environments. In countries in which school fees are high, the social differences are marked. In schools that have a specifically multicultural profile (e.g. in Mannheim or Berlin in Germany), 86% of the parents were satisfied that their children were doing well.
- The conditions in Waldorf schools for children with special needs are generally good, though only where two teachers are in the classroom.
- A study of pupils in urban public Waldorf schools in California (Oberman, 2007) identified three qualities of Waldorf alumni – rigour, relevance and relationship, a framework adopted by the Bill Gates Foundation, which funded the first urban Waldorf school in Sacramento. Rigour refers to the focus on meaningful instruction that engages all students in all subjects, relevance speaks for itself and relationship refers to the high quality of teacher–pupil relations. The study accounted for the above-average academic achievements of the Waldorf pupils by suggesting that this is a result of the focus on the three 'Rs' instead of the usual focus on academic performance alone. The study also points out that one of the problems of integrating Waldorf into public schools is the 'Germanic' origin of the education and its obscure theory, Anthroposophy.

Studies of pupils in school

A study of pupils still in school (Liebenwein, Barz, & Randoll, 2012) concludes that in comparison with pupils from similar social backgrounds at state schools, Waldorf

pupils show greater enjoyment in learning, feel that they get better support from their teachers, have high self-awareness, have less stress in school and experience fewer health problems such as sleeplessness or school anxiety. Teacher–student relationships were judged to be better.

A study by Dahlin (2010) in Sweden showed that Waldorf high school students demonstrated significantly higher levels of moral and civic engagement than peers in state schools at grade 12 (age 18) and that there is a steady marked increase from grade 9 to grade 12. Dahlin concludes that citizenship is strongly cultivated in Swedish Waldorf schools. This trend was also marked in the Woods et al. (2005) study of Waldorf schools in the UK. Through his previous work, Ashley (2005) had noted a tendency among state school students towards 'early closure' in environmental issues, meaning that young people stop developing their views and understandings. The Waldorf study showed this tendency to be less marked. Dahlin poses the question as to whether this comes from the fact that citizenship issues are dealt with too early in the maintained sector, leading to a 'paralysis of analysis', and that the Waldorf approach comes at a time when students can gain a better grasp of the complex issues involved. He draws a link to Gidley's (1998) study in Australia, which showed that young people on leaving a Waldorf school "demonstrated a strong activist will to create positive futures" (Dahlin, 2010, p. 168) and concluded that the strong focus on imagination and the arts disposed them to human and social notions rather than technological aspects of tackling the challenges of the future. Dahlin sees in this a tendency in Waldorf students to be more open to transformative citizenship education that promotes cosmopolitan perspectives and social justice. Dahlin also cites a study in Norway (Solhaug, 2007, in Norwegian), which compared Waldorf with state school students and suggested that the former

> scored significantly higher on tolerance and social engagement, as well as on interest in social issues and participation in non-parliamentary political activity. The state school students, on the other hand, scored higher on factual knowledge and on participation in future parliamentary elections.
>
> (Dahlin, 2010, p. 178)

Dahlin's study shows clearly the higher levels of transformative citizenship than state school peers but admits that parental background may be a significant factor. He concludes, however, that in either case – parental background or school pedagogy – the teaching approach draws this phenomenon out and helps to focus individual agency.

Some of the studies (e.g. Idel, 2007, 2013, 2014) into the relationship between pupils and class teachers over the course of eight years identified that the class teachers often have an implicit 'romantic' understanding of childhood (i.e. that children are formable, open, naïve, obedient, seek authority and are willing to follow when led). If individual pupils do not conform to this image, they are likely to feel misunderstood, not recognized and perhaps inadequate by virtue of the teacher's judgement of their behaviour. One in three former pupils felt this was the case. As this study was done in Germany, it may be a cultural phenomenon related to idealistic notions of childhood. The author of the study poses the questions as to what Waldorf class teachers do to ameliorate this problem or indeed become conscious of it. I am unaware of this being addressed in any of the teacher education institutions, though some seminars have recently been working to develop new forms of reflective practice and reflexivity. This

issue relates to any embodied and unconsidered images that teachers may have about pupils (including gender, character, behaviour, appearance, ethnicity or intelligence) that influences their expectations and their relationships to them. This is an issue that should be looked at in all cultures.

Summary:

- The Waldorf approach encourages pupils to remain open to complex social and ecological issues longer, so that pupils are motivated to do something about them.
- There is a need for research into the unintended, tacit notions of childhood or youth that dispose teachers to privilege some pupils and discriminate against others.
- The culture in a Waldorf school is usually very different from that in many public or state schools. It is important to understand to what extent this culture is intentional and how school cultures influence long-term learning.
- A positive school culture requires high levels of reflexivity in teachers.

This survey by no means discusses all the research on Waldorf education. Its aim was to show the range and diversity and highlight important themes. I have identified no generative principles for this section, though of course the generative principles are themselves also designed to provide possible research questions.

References

Ashley, M. (2005). *Can one teacher know enough to teach year six everything? Lessons from Steiner-Waldorf Pedagogy*. Paper presented at the British Educational Research Association, Annual Conference, University of Glamorgan.

Bader, H.-J., Leist, M., & Ravagli, L. (2002). *Rassenideale sind der Niedergang der Menschheit. Anthroposophie und der Antisemitismusvorwurf*. Stuttgart: Verlag Freies Geistesleben.

Barfield, O. (1981, April). Three paths to education. Towards wholeness, by Mary Caroline Richards. *Commonweal*, 244–248.

Barnes, H. (1980, Spring). An introduction to Waldorf education. *Teachers College Record*, *81*, 323–336.

Barnes, H. (1991). Learning that grows with the learner: An introduction to Waldorf education. *Educational Leadership*, *49*(2), 52–54.

Barz, H., Liebenwein, S., & Randoll, D. (2012). *Bildungserfahrungen an Waldorfschulen (Educational experiences in Waldorf schools)*. Wiesbaden: Springer.

Barz, H., & Randoll, D. (2007a). *Absolventen von Waldorfschulen. Eine empirische Studie zu Bildung und Lebensgestaltung (Alumni/Gradutaes from Waldorf schools. An empirical study of education and life stories)*. Wiesbaden: VS Verlag für Sozialwissenschaften.

Barz, H., & Randoll, D. (2007b). *Bildung und Lebensgestaltung ehemaliger Schüler von Rudolf Steiner Schulen in der Schweiz- eine Absolventenbefragung (Bildung and lifecourse of former pupils at Steiner/Waldorf schools in Switzerland)*. Frankfurt am Main: Peter Lang Verlag.

Bauer, H. P., & Schieren, J. (Eds.). (2015). *Menschenbild und Pädagogik (Image of the human being and pedagogy)*. Weinheim Basel: Beltz Juventa.

Bauer, H. P., & Schneider, P. (2006). *Waldorfpädagogik. Perspektiven eines wissenschaftlichen Dialogs (Waldorf pedagogy: Perspectives of a scientific dialogue)*. Frankfurt am Main: Peter Lang.

Blunt, R. J. (1996). *Waldorf education. Theory and practice*. Johannesburg: Novalis.

Böhle, P., & Peters, J. (2010). Empirische Forschungen an Waldorfschulen ind deutschsprachigen Raum Teil 1 Themenorientierte Übersicht mit Kurzbeschreibungen (Empirical research in

Waldorf schools in German speaking countries, Part 1 Thematic overview with short description of the studies). *Reserach in Sreiner Education*, *1*(2), 165–198.
Böhle, P., & Peters, J. (2011). Empirische Forschungen an Waldorfschulen ind deutschsprachigen Raum Teil 1 Themenorientierte Übersicht mit Kurzbeschreibungen (Empirical research in Waldorf schools in German speaking countries, Part 2 Thematic overview with short description of the studies). *Research in Steiner Education*, *2*(1), 70–105.
Bohnsack, F., & Kranich, E.-M. (Eds.). (1990). *Erziehungswissenschaft und Waldorfpädagogik*. Weinheim Basel: Beltz Juventa.
Bohnsack, F., & Leber, S. (Eds.). (1996). *Sozial-Erziehung im Sozail-Verfall. Grundlagen, Kontroversen, Wege*. Weinheim Basel: Beltz Juventa.
Buck, P., & Kranich, E.-M. (1995). *Auf der Suche nach dem erlebbaren Zusammenhang*. Weinheim Basel: Beltz Juventa.
Cox, M. V., & Rowlands, A. (2000). The effect of three different educational approaches on children's drawing ability: Steiner, Montessori and traditional. *British Journal of Educational Psychology*, *70*, 485–503.
Dahlin, B. (2007). *The Waldorf school- cultivating humanity? A report from an evaluation of Waldorf schools in Sweden*. Karlsbad, Sweden. Retrieved from www.ecswe.org/wren/documents/Waldorf_School_Evaluation.pdf
Dahlin, B. (2009). On the path towards thinking: Learning from Martin Heidegger and Rudolf Steiner. *Studies in Philosophy and Education*, *28*(6), 537–554. doi:10.1007/s11217-009-9147-1
Dahlin, B. (2010). A state-independent education for citizenship? Comparing beliefs and values related to civic and moral values among students in Swedish mainstream and Ateiner Waldorf school. *Journal of Beliefs and Values*, *31*(2), 165–180.
Dahlin, B. (2014). Review of Robert Rose transforming criticism of anthroposophy and Waldorf education- evolution, race and the quest for a global ethics. *Research on Steiner Education*, *5*(1), 156–159.
Dahlin, B. (2017). *Rudolf Steiner. The relevance of Waldorf education*. Berlin: Springer.
Davy, J. (1980). The social meaning of education. *Teachers College Record*, *81*(3), 345–359.
Easton, F. (1997). Educating the whole child, 'head, heart and hands': Learning from the Waldorf experience. *Theory into Practice*, *36*(12), 87–94.
Eisner, E. W. (1991). Forms of understanding and the future of education. *Educational Researcher*, *22*, 5–11.
Eisner, E. W. (1994a). *Cognition and curriculum reconsidered* (2nd ed.). New York: Teachers' College Press.
Eisner, E. W. (1994b). *The educational imagination: On the design and evaluation of school programs* (3rd ed.). New York: Macmillan.
Eisner, E. W. (2017). *The enlightened eye. Qualitative inquiry and the enhancement of educational practice*. New York: Teachesr College Press.
Faivre, A. (2010). *Western esotericism. A concise history* (C. Rhone, Trans.). Albany, NY: State University of New York Press.
Felski, R. (2012). Critique and the hermeneutics of suspicion. *M/C A Journal of Media and Culture*, *15*(1).
Friedlaender, D., Beckham, K., Zheng, X., & Darling-Hammond, L. (2015). *Growing a Waldorf-inspired approach in a public school district*. Standford, CA. Retrieved from https://edpolicy.stanford.edu/sites/default/files/publications/scope-report-waldorf-inspired-school.pdf
Gadamer, H.-G. (2013). *Truth and method* (J. W. A. D. G. Marshall, Trans. Rev. 2nd ed.). London: Bloomsbury.
Gardiner, H. (1983). *Frames of mind: The theory of multiple intelligences*. New York: Basic Books.
Gardiner, H. (1997). Extraordinary cognitive achievements: A symbol systems approach. In W. Damon & R. M. Lerner (Eds.), *Handbook of child psychology* (5th ed., Vol. 1, pp. 415–466). New York: John Wiley & Sons.

Gerwin, D., & Mitchell, D. (2007). *Survey of Waldorf graduates: Phase II*. Wilton, NH: Research Institute for Waldorf Education.

Gidley, J. M. (1998). Prospective youth visions through imaginative education. *Futures*, *30*, 395–408.

Gidley, J. M. (2020). *The secret to growing brilliant children: Volume 1 Steiner education for the 21st century*? Ballina, New South Wales: Bear Books.

Goldschmidt, G. (2013). How waldorf school graduates cope with the challenges they face during military service: A ten-year overview. *Research on Steiner Education*, *3*(2), 99–110.

Goldschmidt, G. (2017). Waldorf education as spiritual education. *Spiritual Education, Religion and Education*, *44*(3), 346–363. doi:10.1080/15507394.2017.1294400

Hanegraaff, W. J. (2014). *Esotericism and the academy. Rejected knowledge & Western esotericism*. Cambridge: Cambridge University Press.

Henry, M. E. (1992). School rituals as educational contexts: Symbolizing the world, others, and self in Waldorf and college prep schools. *International Journal of Qualitative Studies in Education*, *5*(4), 295–309. doi:10.1080/0951839920050402

Herr, K., & Anderson, G. L. (2015). *The action research dissertation* (2nd ed.). Thousand Oaks, CA: Sage.

Heusser, P. (2016). *Anthroposophy and science. An introduction*. Frankfurt am Main: Peer Lang Edition.

Hitchcock, G., & Hughes, D. (1995). *Research and the teacher*. London: Routledge.

Howard, A. (1980). Education and our human future. *Teachers College Record*, *81*(3), 337–344.

Hüttig, A. (Ed.). (2019). *Wissenschaft im wandel*. Berlin: Berliner Wissenschafts-Verlag.

Idel, T.-S. (2007). *Waldorfschule und Schülerbiografien: Fallrekonstruktion zur lebengeschichtlichen Relevanz in anthroposophischer Schulkultur (Waldorf school and pupil biographies: Reconstructive case studies in life story relevance in anthrosophical school cultures)*. Wiesbaden: VS Verlag.

Idel, T.-S. (2013). Der Waldorfschülerhabitus- kulturelle Passungen im Feld reformpädagogischer Privatschulen amd Beispiel der anthroposophischen Schulkultur. In W. Helsper, R.-T. Kramer, & S. Thiersch (Eds.), *Schülerhabitus. Theoretiosche und empirische Analysen zum Bourdieuschen Theorem der kulturellen Passung* (pp. 293–306). Wiesbaden: VS Verlag für Sozialwissenschaften.

Idel, T.-S. (2014). Identitäts- und Personlichkeitsentwicklung an Reform -und Alternativschulen: Schulbiographische Passungen und Subjectivierungserfahrungen (Identity and personality development in reform and alternative schools: School biographical structures and subjectivities). In J. Hagedorn (Ed.), *Jugend, Schuel und Identität* (pp. 687–706). Wiesbaden: Springer.

Jensen, N. R., Boding, J., & Kjeldsen, C. C. (2012). *Didaktisk analyse af Rudolf Steiner skolers laeringspraksis i. o. til 12. klasse (Didactic analysis of learning experiences of pupils in Steiner schools up the 12 class)*. Oslo, Norway: Institut for Uddannelse og Paedagogik.

Kiersch, J. (2016). Wie wird Steiners pädagogische Esoterik besprechbar? In J. Schieren (Ed.), *Handbuch: Waldorfpädagogik und Erziehungswissenschaft: Standortbestimmung und Entwicklungsperspektiven* (pp. 781–794). Weinheim Basel: Beltz Juventa.

Kilcher, A. B. (2009). Seven epistemological theses on esotericism: Upon the occasion of teh 10th. Anniversary of the Amsterdam chair. In W. J. Hanegraaff & J. Pijnenburg (Eds.), *Hermes in the Academy: Ten years' study of Western Esotericism at the University of Amsterdam* (pp. 143–148). Amsterdam: Amsterdam University Press.

Kirkham, J. A., & Kidd, E. (2015). The effect of Steiner, Montessori and national curriculum upon children's pretence and creativity. *The Journal of Creative Behaviour*, *51*(1), 20–34.

Klafki, W. (2007/1985). *Neue Studien zur Bildungstheorie und Didaktik: Zeitgemäße Allgemeinbildung und kritisch-konstruktiv Didaktik (New studies to the theory of bildung and didactics: Contemporary general education and critical-constructive didactics)* (Rev. 6th ed.). Weinheim Basel: Beltz Verlag.

Krautz, J., & Schieren, J. (Eds.). (2012). *Persönlichkeit und Beziehung als Grundlage der Pädagogik*. Weinheim Basel: Beltz Juventa.

Larrison, A. L., & Van Vooren, C. (2012). *Examining public Waldorf: Holistic education in the public sector. Examining parent's perceptions of Waldorf Charters*. Paper presented at the American Educational Research Association, Vancouver, BC. Retrieved from https://www.academia.edu/1584938/Holistic_Education_in_the_Public_Sector_Examining_Parent_s_Perceptions_of_Waldorf_Charters

Leichter, H. (1980). A note on time and education. *Teachers College Record*, *81*(3), 360–370.

Liebenwein, S., Barz, H., & Randoll, D. (2012). *Bildungserfahrungen an Waldorfschulen. Empirische Studie zu Schulqualität und Lernerfahrungen (Experiences of education in Waldorf schools: Empirical study on school quality and learning experiences)*. Wiesbaden: VS Verlag.

Loebell, P., & Buck, P. (Eds.). (2015). *Spiritualität in Lebensbereichen der Pädagogik. Diskussionsbeiträge zur Bedeutung spiritueller Erfahrungen in den Lebenswelten von Kindern und Jugendlichen*. Opladen, Berlin and Toronto: Verlag Barbara Budrich.

Loebell, P., & Martzog, P. (Eds.). (2016). *Wege zur Lehrerpersönlichkeit- Kompetenzerwerb, Persönlichkeitsentwicklung und aktuelle Herausforderungen in der Lehrerbildung*. Opladen, Berlin and Toronto: Verlag Barbara Budrich.

Loebell, P., & Schuberth, E. (2012). *Menschlichkeit in Pädagogik und Erziehungswissenschaft*. Bad Heilbrunn: Verlag Julius Klinkhardt.

McDermott, R. A. (1984). *The essential Steiner*. San Francisco, CA: Harper & Row.

McDermott, R. A. (2009). *The new essential Steiner: An introduction to Rudolf Steiner for the 21st century*. Herndon: Lindesfarne Books.

McDermott, R. A. (2012). *American philosophy and Rudolf Steiner*. Great Barrington, MA: Lindisfarne Books.

McDermott, R. A., Henry, M. E., Dillard, C., Byers, P., Easton, F., Oberman, I., & Uhrmacher, P. B. (1996). Waldorf education in an inner city public school. *The Urban Review*, *28*(2), 119–140.

Nicholson, D. W. (2000). Layers of experience: Forms of representation in a Waldorf school classroom. *Journal of Curriculum Studies*, *32*(4), 575–587.

Nicholson, D. W. (2016). *Philosophy of education in action: An inquiry based approach*. New York: Routledge.

Nida-Rümelin, J. (2013). *Philosophie einer humanen Bildung (The philosophy of a human education)*. Hamburg: Edition Körber-Stiftung.

Nida-Rümelin, J. (2014). *Der Akademisierungswahn: Zur Krise beruflicher und akademischer Bildung (The academicization madness: Concerning the crisis in professional and academic education)*. Hamburg: Edition Körber-Stiftung.

Nielsen, T. W. (2003). *Rudolf Steiner's pedagogy of imagination: A phenomenological case study*. Paper presented at the First International Conference on Imagination in Education, Vancouver.

Nielsen, T. W. (2004). *Rudolf Steiner's Pedagogy of imagination. A case study of holistic education*. Bern, CH: Peter Lang.

Oberman, I. (2007). *Learning from Rudolf Steiner: The relevance of Waldorf education for urban public school reform*.paper presented at Annual Meeting of American Educational Research Association. April, 2007. Chicago. https://eric.ed.gov/?id=ED498362

Oberski, I. (2006). Learning to think in Steiner Waldorf schools. *Journal of Cognitive Education and Psychology*, *5*, 336–349.

Oberski, I. (2011). Rudolf Steiner's philosophy of freedom as a basis for spiritual education. *International Journal of Children's Spirituality*, *16*(1), 5–17. doi:10.1080/13644 36X.2010.540751.

Ode, E. (2015). Bildung als Widerstand- zur Aktualität eines humanistischen Menschenbildes. In H. P. Bauer & J. Schieren (Eds.), *Menschenbild und Pädagogik* (pp. 188–200). Weinheim Basel: Beltz Juventa.

Ogletree, E. J. (1998). *International survey of the status of Waldorf schools*. Chicago. Retrieved from https://files.eric.ed.gov/fulltext/ED438075.pdf

Ogletree, E. J. (2000). Creative thinking development of Waldorf students: A study. *Trans Intelligence Magazine*, *7*.

Okumoto, Y. (2019). Enlivening thinking and speech in search of spiritual identity: The role of 'speech formation' in Steiner's Waldorf education. *International Journal of Children's Spirituality*, *24*(1), 83–96.

Paschen, H. (2010). Wissenschaftliche Zugänge zur Waldorfpädagogik ((Social) Scientific approaches to Waldorf education). In H. Paschen (Ed.), *Erziehungswissenschaftliche Zugänge zur Waldorfpädagogik* (pp. 11–34). Wiesbaden: VS Verlag für Sozialwissenschaften.

Pearce, J. (2019). From anthroposophy to non-confessional preparationfor spirituality? Could common schools learn from spiritual education in Steiner schools? *British Journal of Religious Education*, *41*(3), 299–314. doi:10.1080/01416200.2017.1361382

Pountney, R. (2019). *Towards a review of the Steiner Waldorf curriculum: Report of the rapid appraisal consultation exercise undertaken in March 2019*. Sheffield. Retrieved from http://shura.shu.ac.uk/25321/

Randoll, D. (2010). Empirische Forschung und Waldorfpädagogik (Empirical research and Waldorf pedagogy). In H. Paschen (Ed.), *Erziehungswissenschaftliche Zugänge zur Waldorfpädagogik (Educational scientific approaches to Waldorf education* (pp. 127–156). Wiesbaden: VS Verlag.

Randoll, D. (2013). *Ich bin Waldorflehrer: Einstellungen, Erfahrungen, Diskussionspunkt- Eine Befragungsstudie (I am a waldorf teacher: Attitudes, experiences and points of discussion- a survey)*. Wiesbaden: VS Verlag.

Randoll, D., & Barz, H. (2007). *Bildung und Lebensgestaltung ehemaliger schüler von Rudolf Steiner Schulen in der Schweiz. Eine Absolventenbefragung*. Frankfurt-am Main: Peter Lang.

Randoll, D., & Peters, J. (2015). Empirical reserach on Waldorf education. *Educar em Revista*, *56*, 33–47. doi:10.1590/0104-4060.41416

Ravagli, L. (2004). *Unter Hammer und Hakenkreuz. Der völkisch-nationalistische Kampft gegen die Anthroposophie*. Stuttgart: Verlag Freies Geistesleben.

Ravagli, L. (2014). Polemischer Diskurs: die Anthroposophie und ihre Kritiker. In P. Heusser & J. Weinzirl (Eds.), *Rudolf Steiner. Seine Bedeutung für Wissenschaft und Leben heute* (pp. 332–352). Stuttgart: Schattauer.

Rawson, M. (2021a). Spirituality and subjectivity in Waldorf (Steiner) education: A postmodern Bildung perspective. *International Journal of Children's Spirituality*, *26*(1/2).

Rawson, M. (2021b). Waldorf education and postmodern spirituality. In A. Wiehl & T. Stoltz (Eds.), *Reflections on Waldorf education*. Wiesbaden: VS Verlag für Sozialwissenschaft Springer.

Reinsmith, W. A. (1989). The whole in every part: Steiner and Waldorf schooling. *Educational Forum*, *54*(1), 79–91.

Richards, M. C. (1980). *Towards wholeness: Rudolf Steiner education in America*. Middletown, CT: Wesleyan University Press.

Ricoeur, P. (1970). *Freud and philosophy: An essay in interpretation* (D. Savage, Trans.). New Haven, CT: Yale University Press.

Rittelmeyer, C. (1990). Der fremde Blick.Über den Umgang mit Rudolf Steiners Vorträgen und Schriften. In F. Bohnsack & E.-M. Kranich (Eds.), *Erziehungswissenschaft und Waldorfpädagogik* (pp. 64–74). Weinheim Basel: Beltz Verlag.

Rittelmeyer, C. (2011). Gute Pädagogik-fragwürdige Ideologie? Zur Diskussion um die anthroposophischen Grundlagen der Waldorfpädagogik. In P. Loebell (Ed.), *Waldorfschule heute. Eine Einführung*. Stuttgart: Verlag Freies Geistesleben.

Rose, E. R. (2013). *Transforming criticisms of anthroposophy and Waldorf education- evolution, race and the quest for a global ethics*. Retrieved from www.anthroweb.info/fileadmin/pdfs/RR_Transforming_Criticisms.pdf

Rose, E. R., & Jolley, R. P. (2016). Drawing development in mainstream and Waldorf schools revisited. *Psychology of Aethetics, Creativity and the Arts*, *10*(4), 447–457.

Rumpf, H., & Kranich, E.-M. (Eds.). (2000). *Welche Art vom Wissen braucht der Lehrer? Ein Enspruch gegen landläufige Praxis*. Stuttgart: Klett Cotta.

Safit, I., & Gerwin, D. (2019). *Into the world. How Waldorf graduates fare after high school*. Hudson, NY: Waldorf Publications at Research Institute for Waldorf Education.

Schieren, J. (2015). Anthroposophy and Waldorf Education: A dynamic relationship. *Research in Steiner Education*, *6*(2), 139–149.

Schieren, J. (2016a). Einleitung (introduction). In J. Schieren (Ed.), *Handbuch: Waldorfpädagogik und Erziehungswissenschaft: Standortbestimmung und Entwicklungsperspektiven (A handbook of Waldorf pedagogy and the science of education: Positions and developmental perspectives)* (pp. 9–25). Weinheim Basel: Beltz Juventa.

Schieren, J. (Ed.) (2016b). *Handbuch: Waldorfpädagogik und Erziehungswissenschaft: Standortbestimmung und Entwicklungsperspektiven (A handbook of Waldorf pedagogy and the science of education: Positions and developmental perspectives)*. Weinheim Basel: Beltz Juventa.

Sigler, S., Sommer, W., & Zech, M. M. (Eds.). (2018). *Handbuch. Oberstufeunterricht an Waldorfschulen*. Weinheim Basel: Beltz Juventa.

Sijmons, J. (2008). *Phänomenologie und Idealismus: Struktur und Methode der Philosophie Rudolf Steiners (Phenomenology and idealism: Structure and method in Rudolf Steiner's philosophy)*. Weinheim Basel: Schwabe Verlag.

Sijmons, J. (2014). Rudolf Steiners Philosophie und die Frage nach der Freiheit (Steiner's philosophy and the question of freedom). In P. Heusser & J. Weinzirl (Eds.), *Rudolf Steiner: Seine Bedeutung für Wissenschaft und Leben heute (Steiner: His significance for science and life today)* (pp. 76–98). Stuttgart: Schattauer.

Siu, L. (2011). *"Finding my good side": A case study of student engagement in a Waldorf-inspired community school* (Doctor of Education), Harvard University, ProQuest. Retrieved from https://search.proquest.com/openview/29efb882784c1849f34a5c484300d1cd/1?pq-origsite=gscholar&cbl=18750&diss=y

Sloan, D. (1983). *Insight-Imagination. The emancipation of thought and the modern world*. Westport, CT: Greenwood Publishing.

Solhaug, T. (2007). Steinerskoler i et demokratisk perspektiv: En sammenlikning med offentlige skoler i Norge (Steiner schools in a democratic perspective – A comparisson with public schools in Norway). *Nordisk Pedagogik*, *27*, 150–171.

Uhrmacher, P. B. (1995). Uncommon schooling: A historical look at Rudolf Steiner, anthroposophy and Waldorf education. *Curriculum Inquiry*, *25*(4), 381–406.

Uhrmacher, P. B. (1997). Evaluating change: Strategies for borrowing from alternative education. *Theory into Practice*, *36*(2), 71–78.

Ullrich, H. (2008). *Rudolf Steiner*. London: Bloomsbury.

Ullrich, H. (2015). *Waldorfpädagogik: Eine kritische Einführung*. Weinheim Basel: Beltz Juventa.

Welburn, A. (2004). *Rudolf Steiner's philosophy and the crisis of contemporary thought*. Edinburgh: Floris Books.

Wenger, E. (1998). *Communities of practice. Learning, meaning and identity*. Cambridge: Cambridge University Press.

Wiehl, A. (2019). *Studienbuch Waldorfschulpädagogik*. Bad Heilbrunn: Klinkhardt.

Wiehl, A., & Bunge, M. (Eds.). (2019). *Bilderfahrungen im Zwischenraum von Kunst, Philosophie und Pädagogik*. Salzburg: Edition Kunstschrift im Residenz Verlag.

Willmann, C., & Weiss, L. (2019). *INASTE. The next generation of Waldorf teachers*. Vienna: INASTE.

Woods, G. J., O'Neill, M., & Woods, P. A. (1997). Spiritual values in education: Lessons from Steiner? *International Journal of Children's Spirituality*, *2*(2), 25–40.

Woods, G. J., & Woods, P. A. (2008). Democracy and spiritual awareness: Interconnections and implications for educational leadership. *International Journal of Children's Spirituality*, *13*(2), 101–116.

Woods, P. A., Ashley, M., & Woods, G. J. (2005). *Steiner schools in England* (Research Report No. 645). London: University of West of England.

Wright, P. (2013). Theory of knowledge or knowledge of the child? Challenging the epistemological assumptions of the curriculum debate on geography from an alternative viewpoint. *Oxford Review of Education*. doi:10.1080/03054985.2013.783796

Zander, H. (2008). *Anthroposophie in Deutschland*. Göttingen: Vandenhoeck & Ruprecht.

Zander, H. (2016). *Rudolf Steiner. Die Biographie*. Munich: Piper.

Zander, H. (2019). *Die Anthroposophie. Rudolf Steiners Ideen Zwischen Esoterik, Weleda, Demeter und Waldorfpädagogik*. Paderborn: Ferdinand Schöningh.

Part 5 Waldorf schools around the world

Summary: The Waldorf movement has been international from the 1920s onwards but grew slowly as a school movement. A number of social and political factors changed the educational climate and made it possible for more schools to be founded. There is no central organizational structure to regulate this expansion, though a number of national and international bodies represent the schools or offer them support. We can recognize five overlapping phases: from the founding of the Waldorf School in Stuttgart in 1919 until 1945, from 1945 until 1968, from 1968 until 1989, from 1989 to the Millennium and from there to the present.

The literature

In 2019 Nana Göbel, cofounder of the nongovernmental organization (NGO) *Friends of Waldorf Education*, published a three-volume history of the Waldorf movement (2019a, 2020). She is probably the only person in the world who has actually visited the majority of these schools. No one knows the movement more intimately than she does and she has probably done more than anyone to facilitate Waldorf projects in many poor African, Middle Eastern, Asian and South American countries through her endless fundraising activities and her enabling of support. The *Friends* also organizes year-long voluntary community service programmes for young people in many Waldorf-related projects (www.freunde-waldorf.de/en/voluntary-services). This work is partly funded by the German Ministry of International Development and Co-operation (BMD) and partly through donations from Waldorf parents and pupils (who organize a Waldorf One World day of activities each year). Göbel has also coproduced several volumes of photographs of Waldorf education around the world (Göbel, 2019a, 2019b, 2019c, 2019d; Göbel & Reinthal, 2019). A scholarly study of the German Waldorf movement has also been written (Frielingsdorf, 2019), and a history of the founding of the Waldorf School in Stuttgart has been researched by Zdrazil (2019). Along with two studies looking at the origin and development of Waldorf schools in the United States (Oberman, 1997; Sagarin, 2011), we now have a well-researched overview of much of the Waldorf movement. Most of what I summarize in the next section comes from these sources.

Overview of the phases

The Waldorf school movement grew slowly at first, though it was international almost from the beginning, with schools in England, the Netherlands, New York, and elsewhere being founded only a few years after the Stuttgart school. In Stuttgart, the Waldorf School faced many pedagogical, organizational and financial difficulties from the

start, which is hardly surprising given that everything was being done differently and for the first time. After Steiner's death in 1925, the school had to reorientate itself. It continued to grow, though the financial problems continued. Once the Nazis came to power, the school had to face the loss of teachers with Jewish backgrounds who went into exile, attacks by the authorities, the requirement to include Nazi propaganda into the curriculum and even some Nazi support from within the parent body. The school chose to close itself in 1938, before the authorities took the step. All the other German Waldorf schools and those in the areas occupied by the Nazis were also closed.

After 1945, the German Waldorf schools rebuilt, literally from the ruins, and were soon developing a solid teacher education programme in Stuttgart, which later became a full-time seminar. Through excellent lobbying work, the schools were gradually able to gain state funding, the right to which having been anchored in the new German constitution. By 1949, there were 22 schools in Germany and 23 schools in eight other European countries (four in Switzerland, seven in the Netherlands, seven in the United Kingdom, two in Norway, one in Sweden, one in France and one in Italy). In East Germany, the Waldorf School in Dresden was forced to close in 1949 and it took another 40 years before schools were founded there and in other parts of the former Soviet Union. In none of the countries in which Waldorf education took roots, was it an easy process. Even at the first international Waldorf conference in early spring of 1948 in Stuttgart, attended by teachers from Switzerland, the Netherlands, the United Kingdom (UK) and the United States alongside the German schools, it was noted that the tide in public education was moving towards centralization, standardization, selection along intellectual grounds and testing, a trend which increased with the establishment of the European Community and later, the European Union. In the years immediately after World War II, education had to deal with the aftereffects of war, the traumatized population, destroyed cities, the gaps in education and the many displaced persons and refugees.

In Germany and many other countries, it required considerable courage and idealism for parents to choose Waldorf schools for their children against the pressure from the state and from the churches, as well as from family members unconvinced of the risks that would be taken if the children (and grandchildren) were to attend a Waldorf school. It took a certain kind of person to make this decision – in many cases, people who themselves were well educated, reflective, concerned about the future and alert to the need for social change and renewal. Perhaps because of this parental background – well educated, socially conscious, committed and critical – Waldorf students' attainments in Germany were even then well above national averages (Hofmann, Prümmer, Weidner & Vier, 1981). Anecdotal evidence from the UK shows a similar trend. This is significant because Waldorf schools in the UK were very dependent on their reputation for academic achievement in an educational landscape with many competing private schools.

When the Waldorf movement celebrated its 50th anniversary in 1969, there were 80 schools in total in the world (Argentina 2, Australia 1, Austria 1, Belgium 1, Brazil 1, Canada 1, Denmark 2, Germany 29, Italy 1, Finland 1, France 4, the Netherlands 7, New Zealand 1, Norway 2, South Africa 3, Sweden 2, Switzerland 6, United Kingdom 6, United States 9). In 1945, no Waldorf schools received any form of state funding, and thus they were all dependent on parental funding (school fees and donations from well-wishers) and sacrifices by teachers. By 1948 in the Netherlands, from 1949 in Germany, from 1950 in Denmark, from 1960 in Sweden and from 1970 in

Norway, Waldorf schools received state funding in some form or other. With state funding, however, the state influence on the curriculum, on the age of starting school, on teacher qualifications and on other regulations began to grow. In those countries in which Waldorf education was privately funded, there was relatively little state influence on the education but also limited access.

The overall educational climate changed in 1968 with the student revolts and the upsurge in civil rights, peace and ecological movements. This opening also led to an increased interest in Waldorf education because of its commitment to freedom and creativity and its promotion of school autonomy and self-administration. Many young people who were dissatisfied with the existing system looked for alternatives of all kinds, and a considerable number found an answer in Waldorf education and the possibilities for creating one's own schools and forming school communities. The 1970s saw a rapid increase in newly founded Waldorf schools, notably in Germany, in the UK, and in northwestern Europe generally. In 1970, there were still only 6 schools in the UK; by 1980, there were 17, many of which identified with the so-called 'New Schools Movement', which rejected the traditional fee-paying structures of the 'old' schools and implemented new forms of school communities, influenced by the growing Camphill movement and its open commitment to working in a threefold way, inspired by Steiner's threefold social order and the notion of mutuality and separating subsistence and income from work. My own school, the York Steiner School, was founded in 1979 very much in a spirit of social renewal and the wish to provide an education for children from all social backgrounds, which was only possible through strong community support and the generosity of benefactors. In its early days, the York school had strong Quaker, Buddhist, alternative lifestyle and gay/lesbian roots. No child was turned away because of the parents' financial status and teachers lived on a pittance (and unemployment benefit in those days).

By 1972, there were 73 Waldorf schools in Europe; by 1980, the number had increased to 178. The overall growth in the economies of northwestern Europe, the United States, Australia and New Zealand enabled this growth, even where the state did not support the education. It also meant that Waldorf schools, once established, began to be attractive to a much wider range of parents than those enthusiasts and anthroposophists who could afford it. The expansion of the movement to Southern Hemisphere countries was mainly led by people who attended the European Waldorf training centres such as Emerson College in England, and to some extent Järna in Sweden, Helikon in the Netherlands and Stuttgart in Germany. From the 1950s to the 1980s, Waldorf education was available in many countries in South America to a relatively small group of well-off people, with teachers coming from those centres in Europe. With the growth of the Waldorf schools' movement in the United States, for example, came the growth of local teacher education centres to meet the growing need.

In 1970, the International Forum (originally known as the Hague Circle) was founded to bring representatives of all the countries with Waldorf schools together and from 1983 onwards, World Teachers' Conferences were held in Dornach, Switzerland every four years with around 1,000 participants. Having attended all of these conferences (and having been a keynote speaker and workshop leader and translator at many of them), I can testify to their impact. The quality of the presentations, the experience of such diversity and the wealth of personal encounters boosted the insight into Waldorf education and the confidence of many teachers struggling in marginalized situations

'back home'. Often the lectures were published in translation afterwards, adding significantly to the literature available, especially in the non-German-speaking world. Such meetings (often funded by the Germany Waldorf movement and various foundations) gave a strong international identity to the movement. By 1980, there were 178 Waldorf schools in Europe and 52 in the rest of the world. In February 1989, there were officially 342 schools in Europe and 107 worldwide. Then came the collapse of the Soviet Union and the fall of the Berlin Wall, and this number grew rapidly.

After the Wall came down

1989 marked a major expansion of the Waldorf movement. In early 1989, I travelled with some colleagues from Western Germany to give talks in Leipzig and Dresden in East Germany to large crowds of people wanting to know about alternatives. Being a Brit made me far more acceptable than my colleagues, because many 'Ossies' did not want to hear from 'Wessies' how great their Waldorf education system was. This was part of the whole ambivalence of what then became the reunification of Germany. No one wanted the old German Democratic Republic (GDR) system but at the same time, many did not want Western capitalism or Westerners who 'knew it all'. Before the collapse of the Soviet Union there had been pockets of dissidents, including many people interested in Anthroposophy across Eastern European countries who, with great courage, operated secretly. These were often ignored once the Westerners came in, leading to resentments and to a loss of local authenticity.

Within months of the opening of the Berlin Wall, Waldorf schools were being founded across the old Soviet Bloc countries, including Russia. Schools were founded in Lithuania, Latvia, Estonia, Poland, the Czech Republic, Hungary, Slovakia, Romania, Croatia, Slovenia and Georgia, among others. In the initial, politically chaotic period during the early 1990s, there was a huge flurry of activity founding schools and other educational ventures, before the old elites gradually returned to power and the processes of centralized government control were reestablished in some of those countries. Those were exciting years, and many enthusiastic young people joined the programmes, not always because they wanted Waldorf but often because Waldorf seemed to be a coherent alternative. There was an incredible sense of a new dawn with endless new possibilities. A decade or so later, much of that optimism had been ground down by bureaucratic demands on schools and the rise of national curricula, standardization and control.

In Russia, for example, up to 40 Waldorf schools were founded in the early 1990s during the brief period of liberalization after the collapse of the Soviet Union. From 1991 to 1999, I visited St. Petersburg every year for three weeks as visiting lecturer at the Waldorf Teacher Training Seminar there. At the time, there were four schools; now, there is one small school. The same happened all over Russia as the economic situation got worse, the opposition of the Orthodox Church grew stronger, the birth rate fell drastically, the government introduced new restrictive education policies and the Waldorf curriculum had to be radically adapted. The existence of schools often depended on very local conditions. Some schools survived as state schools, often embedded in huge school campuses under state control. In 2016, there were 20 Waldorf schools in Russia – 12 large schools and eight smaller schools. Of the larger schools, five are state schools (through still requiring parents to pay fees), and seven are independent schools that receive about 15% of their funds from the state. The Russian Waldorf movement

has consolidated and, with considerable support from parents, is actually reestablishing itself. The Waldorf movement is gradually gaining formal state recognition and a state-recognized teacher training programme is planned.

A similar process was happening in Western Europe, though with better longer-term success. Schools were founded in Ireland, Portugal, Italy, France and Spain. However, increasing restraints on Waldorf schools were making life difficult for teachers. By 2000, there were 589 schools in Europe and 209 elsewhere in the world. Yet at the same time, these schools were under pressure from national curricula and state requirements for teacher education. In some countries, such as Sweden and Finland, the integration of Waldorf schools into the mainstream was so complete that they began to lose their identity, and some began to become almost indistinguishable from state schools. In many countries, the age of school entry was being lowered, which is a problem for Waldorf education; at the same time, at the top of the school, new exams penetrated the Waldorf curriculum, leading to a major loss of Waldorf identity in the high school. The trend to early learning was reversed in many countries after the results of the PISA comparisons began to show that an early start does not always lead to school success, though of course the UK and France continue to follow 'the earlier the better' policy that seems enshrined in the cultural consciousness in those countries.

In subsequent years, a major factor was the expansion of the internet, which enabled a globalization of access to information, including information about Waldorf education and Anthroposophy. However, the increased overall control of education by the state over the last 20 years has severely limited Waldorf education where state funding and recognition is dependent on conformity. Thus, in countries like Singapore and in a number of Gulf nations, Waldorf initiatives can only exist officially, or usually unofficially, as home-schooling ventures. On the other hand, the parental reaction against overregulation and testing has led to ever more parents seeing Waldorf as the only safe alternative and so, in most countries – with the exception of the UK – the growth of the number of Waldorf schools has continued at a steady pace. A recent study in Germany (Barz, 2019) shows this clearly. Parents choose Waldorf schools because the emphasis on exams is less dominant, because of the humanistic values of Bildung – that is, a general cultural education – are valued and because they see Waldorf teachers as both more committed to individual pupils and as having more pedagogical freedom. Whilst government policy marches on in the direction of testing and measuring outcomes and controlling schools ever more tightly, parents who have the choice prefer more pedagogical freedom.

From 1996, when the first Waldorf schools in Asia were founded in the Philippines, Thailand, Taiwan, India, Japan and South Korea, this region has seen the most rapid growth with China. This starting point was connected with the travelling exhibition *Waldorf Education Worldwide*, which was first shown at the UNESCO headquarters in Geneva in 1996, staged by the Friends of Waldorf Education at the invitation of the UNESCO General Secretary. I helped translate the book-length exhibition catalogue into English and accompanied the exhibition to Thailand, where I stayed for six weeks working with colleagues in the new Waldorf school in Bangkok while the exhibition was shown there and in Chiang Mai. The exhibition travelled to various cities in Asia and prompted wide interest in Waldorf education.

In South America, the movement continues to grow, and even Africa has seen modest growth mostly in Eastern and Southern Africa. China represents a special case – as in most things. The legal status of many Waldorf schools is ambiguous; some have established themselves within existing state schools, and a few have gained licenses as

private schools under government supervision. It is a precarious existence. The total number of schools and initiatives in China is much larger than the number of those officially recognized. In 2019, the government cracked down on private schools, not just Waldorf schools, and this led to some closures but mainly to rigorous regulations regarding teacher qualifications and various curriculum aspects. The schools that exist are now pushing into the high school, and the need for their pupils to take the state exams is growing. As I write this (in Chengde in Northern China, near the border to Inner Mongolia, teaching in a Waldorf teacher training course), China has just celebrated the 70th anniversary of the People's Republic. Whatever one thinks of this country, its achievements have been extraordinary. The commitment and the will to learn and to work of the Chinese people whom I encounter in the Waldorf community makes working here in Waldorf education an uplifting experience. The future is wide open. Perhaps in five years, these schools will have disappeared; perhaps they will have doubled in number. The China Waldorf Forum lists 71 schools as official members as of December 2020. In 2019 there were over 400 kindergartens. One can only hope that the very positive signals coming from various sources that welcome innovative ideas in education are indicative of the future. What Chinese Waldorf schools certainly combine is a respect for Chinese cultural and above all Confucian traditions with entrepreneurial skills and undeniable 'Waldorf' qualities. From what I have experienced, children and young people in Waldorf schools in China are experiencing as much 'Waldorf' quality as in most other countries.

Waldorf-inspired ideas in the world

Camphill schools and curative education

The Waldorf approach has a variety of applications beyond the Waldorf schools. It has been applied in therapeutic education – for example, in the Camphill schools (Jackson, 2011; Cushing, 2015; Lyons, 2015), which offer special education on the basis of Waldorf education for school-age children and youth. Many such schools are residential, often located with an intentional community in which a significant feature is that the children live in household communities with families, in which the children and young people participate in all kinds of domestic tasks, crafts and practical work (often in the garden or on a farm), as well as having formal schooling. The Waldorf curriculum and approach are applied, though the tasks the children are given are related to their learning abilities. The emphasis on ritual, rhythms, narrative and the arts are therefore prominent in the practice. Under recent legislation covering the care of people with disabilities, there has been a move by social services agencies to favour the integration of adults with disabilities back into the 'public community', i.e. towns and cities, and of children with disabilities back into regular schools. This has put public funding for Camphill and similar therapeutic schools and communities in difficulties, in effect allowing only people whose parents can afford 'private' provision. This has had a fairly major and negative effect on this field. Space does not permit a discussion of this, though the literature cited previously offers an orientation.

A recent evaluation of three Camphill schools and communities in Vietnam, Lebanon and Kyrgyzstan (Rawson, 2017a, 2017b) using social practice and salutogenesis criteria to explore how this education meets the needs of young people both confirmed the efficacy of the approach and showed that these are useful theoretical models to

research Waldorf education. The study highlighted how enabling participation in meaningful social practices embedded in intentional communities is the key to both the integration of students and enabling learning and development. The work of special education in a wide variety of forms has been coordinated by the Friends of Waldorf Education (www.freunde-waldorf.de/en/), and much of the work is funded by the German Ministry for Economic Cooperation and Development, who indeed commissioned the research referred to earlier. In all three Camphill communities in the study mentioned previously, I met young people from Germany, whose own learning process in such situations is highly significant.

Beyond the Camphill movement, there are many other institutions that apply Waldorf principles to what is usually referred to as curative education. One such example, from many, is the Mulberry Center for Curative Education in Santa Rosa, California, founded by Robyn Brown (2016), who has written an important book on her approach which uses Steiner's concept of the life processes that I discussed in the section on learning. In this book, she shows how Waldorf education seeks to meet the child on her own terms and offers her a way to find calmer, more productive learning spaces in which she can develop, freed from pressure to achieve, in areas in which she really can achieve something. Brown works with animals and includes riding therapy. She says of educators:

> learning to observe without judgement [which] is learning to breathe. It is learning to connect within what is without with what is within. It creates a subtle cord between the child and the teacher, where understanding can begin to flow between the two. This is where we begin the work of healing. We know that the difficulties the child experiences are identical to difficulties we have had ourselves.
>
> (2016, p. 22)

Emergency education

The Friends of Waldorf Education also support emergency education interventions following war or natural disaster, in which educational provision breaks down and children and adults are traumatized. In particular, Bernd Ruf (2013) has developed an approach aimed at applying many methods developed in Waldorf education to help children and adolescents in the first stage of trauma by helping them find ways of processing their experiences during the acute phase of shock and reduce the long-term effects of post-traumatic stress disorders on the learning and development of preschool children, school-age children, and young people. The teams from the Friends' Emergency Education Programme work closely with the local community and other aid agencies. The teams comprise medical staff, educators, therapists, special education specialists, project managers and translators.

The key elements in this emergency education programme are encouraging speaking and listening; enabling children to express their feelings through narrative, play, the arts and music; helping them to relax through showing children (and their carers) how to rest, do breathing and concentration exercises, rhythmical massage, bodily contact, dance, movement and social games; establishing regular life rhythms and rituals to structure the day and provide a sense of security; and making plans about next steps. The teams also try to enable the carers and family members to benefit from these activities and learn how to continue them after the intervention is over.

Another project that has grown out of Waldorf education is the Caritas Essence of Learning (EoL) Project, which offers learning support in crisis contexts (Rutishauser-Ramm & Stolz, 2018). This project, which was developed by former Swiss Waldorf teacher Beatrice Rutishauser-Ramm, was recently assessed by Harvard University and identified as one of five Humanitarian Education Accelerators in an initiative sponsored by the United Nations Children's Fund (UNICEF), the United Nations High Commissioner for Refugees (UNHCR) and the UK Department of International Development (www.hea.globalinnovationexchange.org). She developed the EoL approach in projects in Kosovo, Chechnya, Romania and Gaza.

I had a modest involvement in the origins of this project, when I was brought by Beatrice into a Caritas Preschool Peace Education project in Kosovo following the Kosovo War in 1999. My role was to help develop the training programme for the teachers – mostly traumatized young women from mainly rural communities, many of whom lacked formal education – and later to develop ways of evaluating the learning process within the teacher education. My involvement ended in 2010. By then, there were some 30 peace-education classes in municipal schools in Kosovo, both in rural villages and in the towns, and the project, including the teacher training, was largely in the hands of a local Kosovan team. Today the project is thriving, having gained academic recognition for the training and now has some 250 classes based on Waldorf educational principles in the preschool transition to primary schooling.

The Essence of Learning programme is based on the insight (empirically supported) that a pedagogical approach based on a model of child development can be applied at any age. In crisis situations, development is often retarded or falls back to an earlier stage, and the children and young people are thus often overwhelmed by educational approaches that are based on their actual age of grade at school. By enabling children in various ways to recapitulate the 'lost' or 'hidden' stages of development, they can catch up and return to something like normal development. It is a model that needs to be fine-tuned at the local stage and involves assessing children's emotional, cognitive, motor, language, relational and emotional development, and this process uses indicators developed in Waldorf education. As the core framework points out, EoL

> caters to the developmental status of the child as well as the corresponding thinking abilities and crisis-induced behavioural changes without pathologizing them [and] assists children in the elimination of learning difficulties by repeating previously completed learning steps and helping them (re)acquire their own learning abilities.
>
> (Rutishauser-Ramm & Stolz, 2018, p. 10)

The programme seeks to strengthen the child's resilience in several ways but centrally by strengthening their sensory development. In this respect, the EoL programme gives a detailed account of Steiner's 12 senses (ibid., pp. 22–26) and the more recent developments of this approach, including Antonovsky's salutogenetic approach, highlighting the relationship between the senses and the child's overall development.

An example is the vital sense of bodily well-being. The programme offers the explanation, "I have an understanding of myself, because only I know if I am tired, hungry or happy. I am at home in my body." This condition can be expressed. In salutogenesis, this sense is called "sense of coherence". In the corresponding section on competencies, EoL offers the following diagnostic statements:

> I know what I require. I can provide for myself. It is like balancing a set of scales. Well-being is somewhere between shortage and satisfaction or between too much and too little. "If a child learns to endure the shortage, the exertion, the tiredness and exhaustion, this will enable her to discover, step by step, the potential of her body and the limits of her stress and performance capability, from which she develops self-confidence".
>
> (Auer, 2007, p. 34)

Each of Steiner's 12 senses are interpreted in this way, with the handbook offering brief explanations and diagnostic descriptions. This is probably the most significant application of Steiner's sense theory and Waldorf educational generative principles outside of Waldorf education, in a programme that has already helped many thousands of children – and now that it has been recognized by the UNICEF, UNHCR and other transnational aid agencies, it will be implemented to the benefit of millions of children.

References

Auer, W.-M. (2007). *Sinnes-Welten. Die Sinne entwicklen. Wahrnehmung schulen. Mit Freude leben*. Munich: Kösel-Verlag.

Barz, H. (2019). *Bildung und Schule- Elternstudie 2019. Einstellungen von Eltern in Deutschland zur Schulpolitik*. Münster: Waxmann Verlag.

Brown, R. M. (2016). *A practical guide to curative education: The ladder of the seven life processes*. Great Barrington, MA: Lindesfarne Press.

Cushing, P. (2015). What counts as a community? Alternative approaches to inclusion and developmental disability. *International Journal of Developmental Disabilities*, *61*, 83–92.

Frielingsdorf, V. (2019). *Geschichte der Waldorfpädagogik. Von ihren Ursprung bis zur Gegenwart*. Weinheim Basel: Beltz Verlag.

Göbel, N. (2019a). *Die Waldorfschule und ihre menschen Weltweit. Geschichte und Geschichten 1919 bis 2019*. Stuttgart: Verlag Freies Geistesleben.

Göbel, N. (2019b). The diversity of waldorf schools in the cultures of the world. In T. Zdrazil & H. Kullak-Ublick (Eds.), *The understanding of the human being in Waldorf education across the world's diverse cultures* (pp. 17–29). Stuttgart: Edition Waldorf.

Göbel, N. (2019c). Waldorfpädagogik international. In A. Wiehl (Ed.), *Studienbuch Waldorf-Schulpädagogik* (pp. 13–24). Bad Heilbrunn: Klinkhardt.

Göbel, N. (2019d). Weltweite Herausforderungen für die waldorfpädagogische Bewegung. *die Drei*, *9*, 13–24.

Göbel, N. (2020). *The history of Waldorf education worldwide 1919–1945* (J. Kees, Trans., Vol. 1). Hudson, NY: Waldorf Publications.

Göbel, N., & Reinthal, C. (2019). *100 Jahre Erziehung zur Freiheit. Waldorfpädagogik in den Ländern der Welt*. Stuttgart: Verlag Freies Geistesleben.

Hofmann, U., Prümmer, C., Weidner, D., & Vier, B. (1981). *Forschungsbericht über Bildungslebensläufe ehemaliger Waldorfschüler. Eine Untersuchung der Geburtsjahrgänge 1946 und 1947*. Stuttgart: Pädagogische Forschungsstelle beim Bund der Freien Waldorfschulen.

Jackson, R. (2011). The origin of Camphill and the social pedagogic impulse. *Educational Review*, *63*(1), 95–104.

Lyons, N. (2015). *Re-thinking community care. The camphill village model*. Retrieved from www.camphillresearch.com/content-stuff/uploads/2015/11/Re-Thinking-Community-Care.pdf

Oberman, I. (1997). *Waldorf history: A case study of institutional memory*. Paper presented at the American Educational Research Association Annual Meeting Chicago, Chicago.

Rawson, M. (2017a). Lernen als Partizipation in Gemeinschaften der Praxis in heilpädagogischen Einrichtungen. *Seelenpflege in Heilpädagogik und Sozialtherapie*, *2017*(4), 37–53.

Rawson, M. (2017b). *Lernen als Partizipation in Gemeinschaften der Praxis: Ein Bericht für das ministerium für wirtschaftliche Zusammenarbeit und Kooperation und Freunde der Erziehungskunst Rudolf Steiners (Learning as participation in communities of practice. A report for the German ministry of economic cooperation and the freiends of Waldorf education)*. Berlin: Friends of Waldorf Education/Ministry of Economic Cooperation.

Ruf, B. (2013). *Educating traumatized children. Waldorf education in crisis intervention*. Great Barrington, MA: Lindesfarne Books.

Rutishauser, B., & Stolz, N. (2018). *Essence of learning. Learning support in crisis contexts. A practical approach to education in emergencies*. Lucerne: Caritas Switzerland.

Sagarin, S. (2011). *The story of Waldorf education in the united states*. Great Barrington, MA: Steiner Books.

Zdrazil, T. (2019). *Freie Waldorfschule in Stuttgart 1919–1925. Rudolf Steiner-das Kollegium-die Pädagogik*. Stuttgart: Verlag Freies Geistesleben.

Conclusions

In the context of reviewing a series of new books on Waldorf education and Rudolf Steiner, Dhondt, Van de Vijver and Verstraete (2015) question whether it is actually possible to have an unbiased history of Waldorf education. Accounts of Steiner and his education, they say, are either partisan and mythologizing (by which the authors mean something widely believed but untrue) or take the form of defamatory leaflets. They suggest that a real de-mythologizing has not occurred because of the widespread success of the Waldorf movement and the strict control it exercises over its institutions and because there are too many interests involved. I would contest that *anyone* has much control over the Waldorf movement – occasionally, I wish it were so (like when some schools fail to implement policies that have been democratically agreed upon!). What is certainly the case is that there are strongly polarized positions between adherents and critics.

I acknowledge that some may judge that this book fails to offer an unbiased history of Waldorf or to de-mythologize it. I did not set out to create such a history. I have not attempted to de-mythologize it, but rather I have attempted to re-mythologize it. The Greek word *mythos*, as used by Aristotle, means speech, thought, story, dramatic plot, underlying meaning and discourse, so in this sense I have attempted to rearticulate the Waldorf discourse as it is today, especially for those who want to engage in dialogue with other educationalists.

In terms of de-mythologizing, in the sense of exposing widely held beliefs as untrue or not authentic or original, Stephen Sagarin (2007) has identified a number of Waldorf practices (he counts 20) that are not 'original' in the sense of going back to Steiner and the Foundations. These include a bewildering range of 'beliefs' such as the adoption of circle time, collegial decision-making by consensus, much kindergarten practice, a confusion of artistic teaching with teaching art, the prohibition of the colour black and so on. The point is not whether these ideas are pedagogically valid but that their justification is their assumed authorship and whether they belong to the canon of Steiner's indications. Sagarin's point is that "Steiner said" is not an adequate reason for doing anything. My attempted re-mythologizing aims at precisely that. The generative principles I have formulated have their foundation in theory and critical practice, and their validity can only be sought through research.

I have sought to unpack and open up some taken-for-granted notions, such as the idea of *The Waldorf Curriculum* and developmental phases, replacing these concepts with somewhat more nuanced understandings. In formulating generative principles, I have attempted to articulate this pedagogy as it is practiced today and how its leading theorists frame their accounts, and I have chosen to give this a framework that

can be used in research. I have sought to explain how necessary it is to grasp Steiner's attempt to create a new theory of knowledge, one that puts the human being back into active and conscious participation with the world, and to move education away from reductive notions of learning, being and becoming. My point is not to say, "look, Steiner was right" but rather to say that he was a pioneer directing our attention towards what I think is the right direction. I have never imagined that Steiner's was the ultimate position to take, that we need to catch up with him or get back to where he stood. For me, Steiner devoted his life to opening doors and windows to reality, offering us a range of tools to get hold of a situation that was then and remains critical, in order to avoid further disasters. He was not alone in that. He was no expert in education, but he applied his understanding of the human being to suggest how things could be otherwise. He was committed to an *other education* than the one he knew and anticipated would emerge. I don't think he ever claimed that this was the only possible way it can be. Now, over a century later, we can see that what has emerged has been fruitful and is still very much a valid and viable alternative.

Being a Waldorf teacher is hugely demanding. It is usually materially unrewarding. But it is deeply, spiritually rewarding because in spite of all the challenges and difficulties, in spite of all the setbacks, the resistance and obstruction that the state sometimes puts in our way, the attacks from ideologues and obsessives, our own incompetence and folly, this education works in ways that cannot be measured (and, of course, in many ways that can). It doesn't work *in spite* of its dubious and esoteric ideas; it works *because* these ideas offer a coherence, perhaps not always fully understood, even by its practitioners. Dirk Randoll's (2013) study of Waldorf teachers' experiences in Germany is no doubt culturally coloured but probably reflects something true everywhere. Most Waldorf educators are motivated by a vision of doing something meaningful that makes a difference. One can interpret this with suspicion or with empathy. What argues against the suspicious stance is the sheer diversity of people who feel like this and the manifest lack of structural or personal pressure on individuals to believe in the vision. The pressure of working in a strictly regulated environment of performance measurements, surveillance and top-down imperatives to conform is mostly absent in any Waldorf environment (except when this environment is located in a public-school setting, but even there, Waldorf is most effective when it is *Waldorf*). In some ways, the beauty (and occasional frustration) of Waldorf practice is that no one is telling you what to do. Obviously, there is a degree of peer pressure but in my experience, there is at least as much genuine peer support to do what you think is right. After all, it is in all our interests that it works. Rather, it is the case that one puts oneself under pressure to do one's best for the children in one's care. Sometimes parents are demanding, but all schools have that.

It is true that as a Waldorf teacher, I probably do not fully understand every aspect of Waldorf theory – and some readers of this book will no doubt point this out to me. But I do live with the feeling that it *can be* understood, that it *can be* practically realized and that it does feel personally meaningful to work with it. It is coherent, yet coherence is not a static quality but a dynamic one. That is why I think Jennifer Gidley's postformal approach is so important. I think most Waldorf teachers feel that way about it. There really is always more to learn and understand, and we do need ways of understanding how to do this. I can always push my boundary knowledge a bit further and in so doing, adjust my entire understanding because like climbing a mountain, the view changes with each step. On top of that, what I am trying to understand doesn't

stand still but is itself changing. My insider critique as a practitioner with a little over 40 years' experience as a Waldorf teacher and teacher educator is that we need to be more effective in understanding what we do and why. We Waldorf practitioners need to use systematic methods of dialogue, critical reflection and practitioner research.

Whole cultural weather systems are shifting, and this impacts on every one of us in the classroom. Intellectual abstract reasoning is not the tool to use when trying to understand complexity, for the simple reason that it tends to reduce it and in doing so, deadens the life out of what it wants to understand. We need intuitive, postformal, critical, postcolonial, feminist, queer, ecological, rhizomic perspectives and others I am not capable of grasping – but somewhere out there, there are a lot of young people who can. Apart from Jennifer Gidley's (2016) book on postformal education, one of the few genuinely new perspectives on Waldorf education is Paul Houghton's (2012) *Dialogues of Destiny*, a postmodern analysis of Waldorf education through the voices of six experienced practitioners. The way he contextualizes Waldorf education 100 years on and the methods he uses open new perspectives on the Waldorf narrative.

As Dhondt et al. (2015) point out, in order to write a history of the movement, one would need to take a Foucauldian methodological approach to look at the genealogy of the present, "focusing not so much on the question 'what is Waldorf education?', but rather on issues like 'how did it develop?' and 'how have these ideas been reinterpreted by different generations?'" (2015, p. 646). Accounting for the journey from Stuttgart in 1919 to 2019 would require a different kind of research than has been done to date. Without extensive research on practice and perceptions by practitioners and others in different settings, it is very hard to get a grasp of what is really happening in the heterogenous and in many ways loose network of the Waldorf movement. To outsiders, it may seem that Waldorf is cut from a single block, even a sectarian group with clearly defined beliefs. The closer one gets to the reality, the more complex and messier – or more lively – it gets. It is hard enough to identify commonality of belief, purpose and practice in a single school, or across the Waldorf schools in a city like Hamburg, let alone internationally. And yet when I sit in classrooms in Beijing or Lebanon, or in a Californian Waldorf public school, I know that I am in a *Waldorf* school. It would be extremely interesting and helpful to understand cross-cultural aspects of educational ideas if comparative pedagogy methods (e.g. Kelly, 2014) could be used to research this.

There are interesting and as yet unresolved gaps in our understanding of the spread of Waldorf education. Ida Oberman (1997) addresses what she calls the 'mystery of dissemination' referring to the remarkable fact of an educational model from Stuttgart in Germany crossing historic, social, cultural and geographical boundaries. She describes a process in which a 'remembering of the past' is achieved by teachers intending to follow the same unmistakable Waldorf rituals with the same ethos and curriculum. This is something of an exaggeration, typical at most Waldorf early-years settings but increasingly less typical as one moves up the school. She also makes the point that no educational bureaucracy nor organ oversees Waldorf schools to ensure the survival of its basic forms. She writes that the forms of Waldorf propagation are not apparent or less apparent than those of other educational models.

Oberman argues that the reason for the remarkable transmission lies in certain semiotic features of Waldorf schools that create consistency and allow adaptation, and it is these that account for Waldorf's sustained identity. Central to these semiotic messages is a common foundation story and how this is retold to generations of new

Waldorf teachers and parents. Along with it go stories of the founder, Rudolf Steiner, and the first teachers. Local founding stories do not have the same valency as the original founding and its Foundations (in the sense I have used the word in this book). Other key semiotic messages include the rituals around which the Waldorf community gathers, the festivals, the first school day, the Advent spiral, architecture, the colours of the classrooms, blackboard drawings and the morning verse. Teacher education plays an important role in reiterating the collective cultural memory and disseminating it. Interestingly, she identifies the important role of Stuttgart birthdays (50th, 75th and of course, since her original text, the big one, Waldorf 100) in the Waldorf collective consciousness in the United States, even though it is so remote in time and space! Scriptural fidelity is another aspect, in the sense that the meaning of the text is less powerful than the 'text' in the hearts of the community, referring to the reverence towards Steiner's texts. In effect, what Oberman is describing is a form of charisma, adhering not just to a person but to a set of semiotic messages and perhaps to the ideas behind them. The value of this charisma has been to prevent the disintegration of Waldorf in the United States and most other places.

Now, over 20 years later and as the resonance of the Waldorf 100 celebrations is fading, how much of Oberman's analysis is still true? Probably a lot. Witness the extraordinary number of serious publications of texts explaining the Foundations and the retelling of the founding stories. Perhaps the sense that it is (still) so necessary tells us more about the semiotic message of Waldorf 100. There is a palpable concern that Waldorf is hollowing out and only the husks of rituals remain, not the 'irrelevant' theory that is not worthy of serious academic consideration. It's hard to avoid the message that Stuttgart is telling us, "go back to basics". Yes, but how? Not, I believe, by reminding us again about what Steiner really meant. My counter position is to say: this is what Waldorf practice today does and why – now let us start researching what happens in practice. I suggest that to Ida Oberman's (1997) genial analysis of the mysterious transmission of Waldorf by way of its rituals and stories, we must add the notion of generative principles that travel across time, social and cultural space. Generative principles are not rituals that are passed on, though one generative principle certainly points to the value of ritual and rhythm for children, but for adults, the generative principle here is reflexivity and criticality.

Fortunately, there are many things that Steiner never referred to and could not have referred to. There are three obvious examples of this: Edwin Hübner's (2015) Waldorf-based approach to media pedagogy, the applications of Waldorf principles in the fields of emergency and crisis education (Ruf, 2013; Rutishauser & Stolz, 2018) and perhaps less radical, the more recent developments in second-language learning (Lutzker, 2017; Sievers, 2017) that build on a few ideas of Steiner but have gone much further and have developed a whole new approach with a strong scientific basis. There are many other examples. I hope that a theory of learning that builds on core ideas of Steiner's but integrates a much wider range of perspectives will emerge in the next few years, just indeed as Otto Scharmer's (2016) Theory U approach grew out of Steiner's work, among others.

I hope this book has shown that Waldorf education today can be taken seriously as a legitimate example of counter-practice, that it is practiced by seriously minded and critical people and that, of course, its practice is far from perfect. My appeal to the Waldorf world, and to the non-Waldorf world interested in studying and critiquing it, is that we can drop the personalization of the name Steiner as in Steiner education/

Steiner school and perhaps even move on from the association with Waldorf (the movement with the cigarette factory past) and look at the practice that we have today. Perhaps the movement is not yet ready to adopt Jennifer Gidley's term *postformal*, which requires quite a complex explanation relating to Piaget. It is not my task to find a new name, and I will win no friends inside or outside the movement trying to create one. My point is that we can and should have a serious study of the education in Waldorf schools today, because that is the only way to understand what potential and need there is for ongoing change and modification. Given the dramatic social changes we are going through, there is a real need for better, more grown-up (Biesta, 2013), more relevant educational approaches for the rising generation. We could adopt the term 'grown-up education', but it probably would not be seen as very grown-up!

References

Biesta, G. J. J. (2013). *The beautiful risk of education*. Boulder, CO: Paradigm Publisher.

Dhondt, P., Van de Vijver, N., & Verstraete, P. (2015). The possibility of an unbiased history of Steiner/Waldorf education? *Contemporary European History*, *24*, 639–649.

Gidley, J. M. (2016). *Postformal education. A philosophy for complex futures*. Weinheim Basel: Springer.

Houghton, P. (2012). *Dialogues of destiny: A postmodern appreciation of Waldorf education*. Malvern Hills, UK: Sylvan Associates.

Hübner, E. (2015). *Medien und Pädagogik: Gesichtspunkte zum Verständnis der medien, Grundlagen einer anthroposophisch-anthropologischen Medienpädagogik*. Stuttgart: Pädagogische Forschungsstelle beim Bund der Freien Waldorfschulen.

Kelly, P. (2014). Intercultural comparative research: Rethinking insider and outsider perspectives. *Oxford Review of Education*, *40*(2), 246–265.

Lutzker, P. (2017). *Der Sprachsinn: Sprachwahrnehmung als Sinnesvorgang* (2nd ed.). Stuttgart: Verlag Freies Geistesleben.

Oberman, I. (1997). *The Mystery of Waldorf: A turn-of-the-century German experiment on Today's American soil*. Paper presented at the Annual Meeting of the American Education Research Association, Chicago March 24-28. https://eric.ed.gov/?id=ED409988

Randoll, D. (2013). *Ich bin Waldorflehrer: Einstellungen, Erfahrungen, Diskussionspunkt- Eine Befragungsstudie (I am a waldorf teacher: Attitudes, experiences and points of discussion- a survey)*. Wiesbaden: VS Verlag.

Ruf, B. (2013). *Educating traumatized children. Waldorf education in crisis intervention*. Great Barrington, MA: Lindesfarne Books.

Rutishauser, B., & Stolz, N. (2018). *Essence of learning. Learning support in crisis contexts. A practical approach to education in emergencies*. Lucerne: Caritas Switzerland.

Sagarin, S. (2007). Playing 'SteinerSays': Twenty myths about Waldorf education. *Research Bulletin for Waldorf Educaion*, *12*(2), 37–44.

Scharmer, O. (2016). *Theory U: Leading from the future as it emerges- the social theory of presencing* (Rev. 2nd ed.). San Francisco, CA: Berrett-Koehler.

Sievers, U. (2017). *Creative teaching- sustainable learning*. Norderstedt: BOD.

Subject Index

ability, abilities (skills) 33, 36, 38, 67, 74, 83–85, 92–94, 98, 120, 143–145, 186
academic critique of Waldorf education 4, 162–164; academic reception of Waldorf education 6, 158–159, 161; academic studies of Waldorf alumni 42, 75, 77, 116, 171–173
Alanus University 159
alienation 34, 78, 89, 114
alumni research 43, 75, 77, 116, 158, 171–173
anthroposophy 2–3, 5, 14, 18, 19, 65, 71, 159–164, 172–173, 185–186
Anthroposophical Society 15
Arbeiterbildungsschule (Workers' Education Institute in Berlin) 34, 35
assessment for learning 93, 121, 134–138
attunement 11, 82–83
autopoiesis, self-organisation 41, 114

Bildung (self-formation) 37, 93, 114, 163, 186
biographical learning 93
biographical mythos 11, 14, 29, 69, 123
block teaching 51, 93, 98–99, 101–102, 115, 145
body, embodied, embodiment 22, 25, 39, 41–43, 59, 62, 86–87, 92–93, 99, 103–104, 107, 110–113, 115, 117, 122–125, 134, 140, 147, 160–161, 165, 167, 171, 189–190
boundary ideas, boundary knowledge 29–32, 72, 193

capitalism 11, 16, 34, 65, 185
childscape 38
China 2, 148, 151, 186–187
chronobiology 41
college of teachers 101, 134, 139–140
communities of learning 132–134, 152, 154
communities of practice 133
concept, conception, conceptual, conceptualize 6, 13, 22–25, 26–35, 43, 59, 66, 69, 73–74, 78, 83, 90–92, 95, 97, 100, 103–113, 123, 142, 150, 160, 168–171; living concepts 83, 91–97, 105–107, 142, 165
contemplative meditation 31–32
counter practice 2, 113, 140, 195
curriculum 4, 36, 50–51, 67, 69–70, 94, 115, 139, 142–152, 158, 165, 167–168, 171, 184, 188, 192–193

developmental tasks 38, 50, 78, 94, 115, 142–147
disposition, habits of mind 36–38, 41–42, 51, 83–87, 90, 92–93, 98–102, 133, 137, 139, 143–144, 152–153, 168
Dornach, Switzerland 5, 15–16, 184

Education of the Child (book) 33, 37–38
epistemology/theory of knowledge (Steiner's) 19, 22, 27–28, 67, 106, 166
etheric body, life-body, body of formative life forces, habit body 31, 38–45, 66, 89, 161, 165
ethics, ethical 26–28, 38, 71, 141, 143, 164

feminism, feminist 18, 27–28
forgetting (as part of learning process) 83–84, 86–88, 93, 99
Foundations, the 21, 30–32, 50–51, 72, 86, 107, 122, 144, 152–154, 192, 195
Freie Hochschule Stuttgart 141, 158

generative principles 5, 7, 17, 30, 50–52; Part 2, 132, 135, 138–140, 142, 144, 151, 154, 161, 167, 171, 175, 190, 193, 195
Goetheanum 15, 37
Goethe-Schiller Archive, Weimar 15, 34

habitus 41, 114

I, the, self, spiritual core of human being, subject 18, 20, 26, 37, 44, 66, 69, 84–86, 88, 94, 104, 106, 110, 112, 113, 115, 123–124
ideal-type 145

identity 12, 29, 62–64, 71, 94, 113, 121, 132, 150, 162, 164, 171, 185–186, 194
imagination, imaginative 14, 32–33, 43, 52, 67–68, 82, 99–102, 103–105, 165, 167–169, 171, 174
individuality 10, 28, 37–38, 66, 69–72, 88, 93–94
intuition 16, 23, 28, 31–32, 44, 67, 69, 71, 117, 119, 154, 167

knowledge 3, 5, 6, 10, 12–14, 16, 18–19, 20–29, 39, 50, 62, 68–70, 98–99, 102, 132, 135, 137, 138, 152, 164, 167, 171, 175; esoteric knowledge 160; generating knowledge, growth of knowledge 26, 32, 51, 83–85, 86, 89–93, 105–116, 142, 143, 165, 169; knowing-in-practice 25, 31; knowledgeable action with purpose, skills 145; theory of knowledge 16, 29, 66–67, 106, 193; *Truth and Knowledge* (book by Steiner) 15, 34

language 4, 6, 18, 23, 24–25, 32, 38, 40–42, 52, 88, 125, 134, 145; English 150, 158–159; German 149–159; learning, foreign, second 62–64, 70, 77, 84, 91–92, 101, 107, 147, 170, 172, 195; other 147; perception, sense of 87, 111, 123–124; skepticism (Steiner's) 17
learning, learning process 5, 13, 19, 24–26, 29, 30, 33, 38, 42–45, 50–51, 57, 58, 60–63, 65–66, 71, 75–77, 94–98, 102, 165, 171, 188, 189, 193; collegial 138–142; community, culture 78, 101, 118, 119, 133–135, 145, 166; cultivation of senses 122–126; dispositions 98–100; enjoyment 174; Essence of Learning (EoL) Project 189; expansive/defensive learning 88–89, 115; healthy 72–74, 101; mimetic 42–43, 85; outcomes, goals 144–146, 148; phases/stages 83–94, 95–98; phenomenologically 105–116; readiness 79–83; rhythms, rhythmical process 83–94, 169, 171; self-activity 119–122; social 71; strategies, situation, pathways, environment 75, 116, 142, 171; teacher 152–154; through direct experience 102–103; through imagination 103–105; time to 79–83, 89; transformative 83, 85, 92, 93–94, 114–115, 132, 152–154; unconscious 168
lesson(s) 21, 32, 44, 51, 66, 69, 73, 76; aesthetic performative perspective 114; artistic element 116–119; evaluation 78; main, books 54, 137; main morning lesson (examples of) 51–63, 98–101, 166; preparation, planning 76–77, 136, 138, 153; resonance 114–115; rhythmical part of 81–82; scripts or plans 76–79, 108; start 89; structure 116; subject 101

metabolic-limb system 40
moving classroom 81, 126

narrative 11, 17, 44, 66, 99–110, 132, 134, 150, 189, 194; competence 99; curriculum, story-telling 44, 69–70, 90; grand 17–18; story-telling 13, 74, 169–170
nerve-sense system 34, 40, 67

percept/perception 13, 19, 22–29, 30–33, 41, 44, 66, 83, 86, 89, 106, 110, 117, 122, 125, 140, 169; social 153, 165; spiritual 29, 31
phenomenological approach 30, 51, 105–116, 146, 161, 168
Philosophy of Freedom, The (Philosophy of Spiritual Activity), Steiner book 12, 14, 15, 16, 22, 28
phronesis (practical wisdom) 11, 39
postformal education 16, 69, 71, 193–194
propaedeutic 50

recall (part of learning process) 31, 41, 54, 61, 78, 81, 82, 84, 85, 89–92, 95–99, 105, 113, 117, 136, 152
rhythmic system 40

salutogenesis 72–73, 114, 187, 189
school leadership 139–142
self-activity 51, 63, 71, 75, 79, 82, 103, 119–122
self-formation 38, 93, 109, 119, 164
senses 10, 12, 14, 18–23, 43, 45, 70, 80–82, 85–87, 106, 189–190; sense of coherence 72–75, 114–115, 143, 145, 189; Steiner's theory of 122–126
sentient body 37–39, 45, 110; sentient soul 124
soul/psyche 12, 17–18, 22, 29, 31, 36–39, 42, 67, 79, 83–86, 88, 93, 103, 105, 107, 110–112, 115, 124, 139; economy 72–75
spirit, spiritual(ity) 3, 5–6, 12–14, 25, 27–32, 64–72, 87–88, 107, 110, 124, 140, 143, 193; anthropology 37–44, 83, 85, 115, 159, 161; core of being, I, human spirit 20, 22, 26, 34, 37–38, 59, 84, 89, 94, 110–112, 167–169; in leadership 140–141; modern 10, 16–18; productivity 36; research 169; science 19, 44; world 38
symptomatology 106

tact, pedagogical 31–32, 44, 76, 152
teacher/teaching: authority 43–45, 76–77, 107, 167; artistry, art of teaching 116–119; education 30, 142, 152; methods 119, 160, 167, 168–169, 171, 190; preparation 75–79
Theory U 31, 195
Theosophy (Steiner book) 36–37, 39, 83, 124
threefold social structures 35–36
Waldorf Astoria Cigarette factory 7
Waldorf method 50, 92, 106, 115, 152, 172; method school 50
Waldorf myths 147
Waldorf School, the 1–2, 14, 21, 33, 50, 52, 104, 139, 146, 160, 182
World List of Waldorf Schools 7

Index of Persons

Antonovsky, A. 72–73
Appiah, K.A. 38, 150

Barfield, O. 37, 38, 164
Beuys, J. 17
Biesta, G.J.J. 2, 77, 93, 108–109, 143, 148, 196
Boland, N. 148, 151
Breuer, J. 34

Cassirer, E. 24, 33

Dahlin, B. 23, 35–36, 143, 163, 174
Darwin, C. 14, 19

Easton, F. 166–167
Eisner, E. 167

Fischer-Lichte, E. 114–115
Freud, S. 16, 34, 162–163

Gadamer, H-G. 15, 24, 30–31, 161
Gidley, J. 7, 11, 15, 16, 27, 69–71, 174, 193–194
Goethe, J.W. von 15, 22–23, 34, 106, 163

Haeckel, E. 14
Hattie, J.A.C. 77, 92, 115, 136
Humboldt, W, von 163

Ingold, T. 41, 104, 117–118, 125

Kaiser, U. 17, 37
Kiersch, J. 4, 6, 30, 160
Koguzski, F. 13, 15

Lutzker, P. 70, 90, 123, 153
Luxemburg, R. 14, 15–16, 34–35

Mayreder, R. 10, 14
McDermott, R. 160, 165, 169
McMillan, M. 14, 32–33, 37

Nicholson, D. 167–168
Nielsen, T. 168–169
Nietzsche, F. 15, 162–163

Oberman, I. 144, 170–172, 194–195
Oberski, I. 170

Paul, J. 15
Pountney, R. 171

Rittelmeyer, C. 4, 161

Schad, W. 40, 41
Scharmer, K-O. 31, 118, 141, 195
Schieren, J. 19, 24, 65, 67, 159, 160–161
Schiller, F. 33, 163–164
Schröer, K. 32
Soetebeer, J. 153
Sommer, W. 97–98, 112–114. 116
Specht, P. & Specht, L. 33–34, 72
Steiner, R. Part 1 (his life, philosophy and educational ideas): accusations of racism 164–165; citing as a justification for practice 192–195; learning theory 66–68, 72–74, 76, 83–89, 98, 100–107, 110–112, 115–117, 119, 168–170; pedagogical anthropology 50–51, 86–88, 160, 161; theory of the senses 122–125; threefold social structure 120, 139–141

Uhrmacher, P. B. 165
Ullrich, H. 4, 6, 159–160

Wenger, E. 132–133, 160
Wiechert, C. 117
Wiehl, A. 19, 37, 50, 88, 92, 153
Woods, G. & Woods, P. 140–142
Wright, P. 169

Zajonc, A. 31
Zech, M.M. 36, 88, 94, 97, 100, 107, 145
Zimmermann, H. 116